

CyberLab Reference Guide

Field quick-reference for the kit
capability → tool → command → every option explained

Generated from the repository. Markdown is the source of truth.
github.com/project-cyberlab/cyberlab-reference-guide

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CyberLab Reference Guide

A field quick-reference for the kit. You open it mid-mission or mid-exercise when you need a **capability** and do not remember which tool provides it — it routes you to the tool, the exact command, and **every option, explained**.

Start here

You are...	Go to
Facing a problem — <i>"I need to carve deleted files"</i>	Capability Index — 46 capabilities
Asking what the kit even has	Kit Tool List — 1,006 tools, VM → category → tool
Wanting a finished page	fls · vol · yara
Carrying it into the field	PDF — 372 pages
Wanting to know how pages are built and why	The Format

Where it stands

Kit inventory	1,006 tools, 5 platforms, 54 categories
Captured from a real binary	945 commands
Capabilities	46 , across 10 investigation phases
Tool pages	135 , every one with a complete options table
Flags with usage guidance	153 across 20 reviewed tools
Mined invocations	360 real commands from cyberlab
Citations checked	255 URLs — 245 live
Linter	0 errors , 269 warnings (tracked debt)

What makes this different

Every DFIR quick-reference in the field either organises by tool (so you must already know the tool) or skips flag documentation. The most complete one in public circulation states outright that it *"assumes operator familiarity."*

This guide inverts that:

- **Capability-first.** You search by what you need to *do*, not what the binary is called.
- **Options completely listed out** — every flag, with *what it does* and *when you'd use it*.
- **Tools attached to commands**, each tagged with the VM that carries it, so you know whether you can run it on the box in front of you.

Scope is binding

Only tools in `catalog/kit-tools.json` may be documented. It is generated from the kit VMs' own upstream manifests:

Platform	Tools	Source of truth
Kali Linux	403	<code>kali-meta</code> <code>debian/control</code> (<code>kali-tools-*</code>)
REMnux	268	<code>REMnux/docs</code> <code>discover-the-tools</code>
SIFT Workstation	162	<code>teamdfir/sift-saltstack</code>
FLARE-VM	137	<code>mandiant/flare-vm</code> <code>config.xml</code>
Security Onion	36	kit tracker CSV

Every flag is real

Options are **captured from the running binary** (`--help` / `man`, inside the kit container), stored raw under `capture/`, and enforced by `scripts/lint.py`: a documented flag absent from a capture is a **build error**.

This answers a measured failure in the predecessor project, where fabricated CLI flags reached ~44 of 61 modules. Rationale in `docs/FORMAT.md`.

Citations are liveness-checked too — see `capture/SOURCES.md`.

Rebuild everything

```
python scripts/build_kit_list.py      # kit inventory from upstream manifests
python scripts/candidates.py          # candidate command names
# scripts/probe_container.sh runs inside the kit container on rick
python scripts/merge_coverage.py      # merge probe results -> coverage report
python scripts/mine_cyberlab.py       # candidate invocations (read-only)
python scripts/generate_pages.py      # tool pages (never overwrites hand-written)
python scripts/build_index.py         # capability index
python scripts/lint.py               # gate: 0 errors required
python scripts/validate_sources.py    # citation liveness
python scripts/build_pdf.py           # print render
```

Relationship to cyberlab

Separate repository, deliberately. The [cyberlab training lab](#) is **on hold** and receives no changes from this work; it is read-only input for mining.

Layout

```
catalog/    kit tool list - the binding scope (generated)
reference/  the guide: INDEX.md + one page per tool, grouped by capability
capture/    raw --help/man output, committed as evidence
docs/       FORMAT.md (how pages are built), PLAN.md (roadmap)
scripts/    build, probe, mine, generate, lint, validate, render
```

Capability Index

Start here when you have a problem. Find what you need to *do*; it points you at the tool and its page.

Phases follow [NIST SP 800-86](#) (collect → examine → analyse → report), ordered the way an investigation actually runs.

Legend: **bold** = captured from a real binary, page can be written · plain = in the kit but not captured yet (needs a booted VM or has no CLI).

Phases

- [Acquire & preserve](#)
- [Examine the filesystem](#)
- [Build the timeline](#)
- [Windows artifacts](#)
- [Memory forensics](#)
- [Network analysis](#)
- [Malware triage — static](#)
- [Malware triage — documents](#)
- [Reverse engineering](#)
- [Decode & deobfuscate](#)
- [Report & support](#)

Image a disk, volume or device

[dc3dd](#), [dcfldd](#), [dd](#), [ewfacquire](#), [affconvert](#), [guymager](#)

Verify evidence integrity with hashes

[rahash2](#), [ssdeep](#), [sha256sum](#), [md5sum](#), [sigtool](#), [hashdeep](#), [md5deep](#), [sha256deep](#)

Inspect or mount a forensic image container

[ewfinfo](#), [ewfmount](#), [ewfverify](#), [ewfexport](#), [affinfo](#), [affcat](#), [img_stat](#), [ntfs-3g](#), [vshadowinfo](#), [bdeinfo](#)

Capture live network traffic

[dumpcap](#), [tshark](#), [tcpdump](#)

See the partition and volume layout

[mmls](#), [fsstat](#), [img_stat](#), [testdisk](#)

List files and directories, including deleted ones

[fls](#), [ffind](#), [ils](#), [tsk_recover](#)

Recover deleted or lost files

[tsk_recover](#), [icat](#), [photorec](#), [testdisk](#), [blkls](#), [ext4magic](#), [extundelete](#), [ext3grep](#)

Carve files out of unstructured data

[foremost](#), [scalpel](#), [binwalk](#), [bulk_extractor](#), [tcpxtract](#), [magicrescue](#)

Inspect metadata for one file or inode

[istat](#), [ils](#), [file](#), [stat](#), [exiftool](#), [trid](#), [magika](#)

Search raw data for a pattern

[rafind2](#), [strings](#), [grep](#), [xxd](#), [bulk_extractor](#), [lightgrep](#)

Build a super-timeline from many artifact sources

[log2timeline.py](#), [psort.py](#), [pinfo.py](#), [psteal.py](#)

Build a filesystem MAC-time timeline

[fls](#), [mactime](#), [tsk_gettimes](#)

Parse registry hives

[rip.pl](#), [regripper](#), [hivexsh](#), [regfexport](#), [regfinfo](#), [regfmount](#), [regipy-dump](#), [regipy-parse-header](#), [regipy-plugins-run](#), [regipy-diff](#), [RECcmd](#), [hivexget](#), [hivexml](#)

Parse Windows event logs

[evtexport](#), [evtinfo](#), [EvtxECmd](#), [chainsaw](#), [hayabusa](#), [evtx_dump](#)

Parse ESE / SRUM / Amcache databases

[esedbexport](#), [esedbinfo](#), [SrumECmd](#), [AmcacheParser](#)

Parse execution and persistence artifacts

[PECmd](#), [AppCompatCacheParser](#), [MFTECmd](#), [AmcacheParser](#), [analyzeMFT](#)

Parse mail stores

[pffexport](#), [pffinfo](#), [readpst](#)

Analyse a memory image

[vol](#), [volatility3](#), [volshell](#), [vol.py](#), [rekall](#)

Recover encryption keys from memory

[aeskeyfind](#), [rsakeyfind](#), [bulk_extractor](#)

Read and filter packet captures

[tshark](#), [capinfos](#), [ngrep](#), [tcpflow](#)

Split, merge or repair capture files

[editcap](#), [mergcap](#), [reordercap](#)

Extract files and payloads from traffic

[tcpxtract](#), [tcpflow](#), [foremost](#)

Detect intrusions in traffic

[suricata](#), [suricatasc](#), [zeek](#), [zeek-cut](#), [rita](#)

Probe or scan hosts and services

[nmap](#), [nping](#), [arp-scan](#), [ncat](#), [netdiscover](#)

Simulate network services for detonation

[inetsim](#), [fakedns](#), [fakenet](#)

Identify what a file actually is

[file](#), [die](#), [diec](#), [trid](#), [magika](#), [exiftool](#)

Scan with signatures for known-bad

[yara](#), [yarak](#), [clamscan](#), [freshclam](#), [sigtool](#), [clamdscan](#)

Identify capabilities in a binary

[capa](#), [floss](#)

Extract strings, including obfuscated ones

[floss](#), [strings](#), [base64dump.py](#), [numbers-to-string.py](#)

Inspect PE / ELF structure

[rabin2](#), [readelf](#), [readelf.py](#), [objdump](#), [pescan](#), [pecheck.py](#), [peframe](#)

Detect and reverse packing

[upx](#), [die](#), [diec](#), [binwalk](#), [7za](#), [unzip](#)

Compare or cluster samples

[ssdeep](#), [radiff2](#), [bytehist](#)

Analyse Office documents and macros

[olevba](#), [oleid](#), [oledump.py](#), [olemeta](#), [oletimes](#), [olemap](#), [oledir](#), [olebrowse](#), [olefile](#), [oleobj](#), [mraptor](#), [msodde](#), [pcodedmp](#), [pyxswf](#), [xmldeobfuscator](#), [runxldr2.py](#), [vipermonkey](#)

Analyse RTF documents and embedded objects

[rtfobj](#), [oleobj](#), [rtfdump.py](#)

Analyse PDFs

[pdfid](#), [pdfid.py](#), [pdf-parser](#), [pdf-parser.py](#), [peepdf](#), [qpdf](#)

Analyse other container formats

[7za](#), [unzip](#), [msoffcrypto-tool](#), [onenoteanalyzer](#)

Disassemble and explore a binary

[r2](#), [rabin2](#), [rasm2](#), [objdump](#), [vivbin](#), [vdbbin](#), [rizin](#), [ghidra](#), [cutter](#)

Diff two binaries

[radiff2](#), [bindiff](#)

Emulate or instrument execution

[frida](#), [frida-trace](#), [frida-ps](#), [frida-discover](#), [frida-kill](#), [frida-ls-devices](#), [speakeasy](#), [qiling](#), [scdbg](#), [unicorn](#)

Analyse shellcode

[rasm2](#), [xortool](#), [scdbg](#), [shellcode2exe](#)

Decode, decrypt or transform encoded data

[cyberchef](#), [base64dump.py](#), [rax2](#), [xxd](#), [openssl](#), [numbers-to-string.py](#)

Break simple obfuscation

[xortool](#), [floss](#), [xmldeobfuscator](#), [de4dot](#)

Crack passwords and hashes

[hashcat](#), [john](#), [hydra](#), [unshadow](#), [zip2john](#), [rar2john](#), [fcrackzip](#), [hashid](#)

Find hidden data

[binwalk](#), [ssdeep](#), [steghide](#), [stegosuite](#)

Fetch and verify external references

[curl](#), [wget](#)

Inspect files by hand

[xxd](#), [ezhexviewer](#), [less](#), [hexdump](#)

Coverage of this index

- **46 capabilities** across 11 phases
- **147 captured tools** are reachable through a capability
- **5 named kit tools map to no capability** (below)
- **2 capabilities have no captured tool** behind them yet

Capabilities with nothing captured yet

These are real needs the kit may cover with GUI or Windows-only tools that a Linux container cannot capture.

- Windows artifacts / Parse mail stores
- Network analysis / Detect intrusions in traffic

Named kit tools not yet mapped to a capability

These are named in the kit manifests, captured from a real binary, and reach no capability in this index. Each is a taxonomy gap to close or an explicit out-of-scope decision. Listed rather than silently dropped, because a missing capability is otherwise invisible.

`aircrack-ng`, `cabextract`, `ccrypt`, `cryptsetup`, `dislocker`

Container-provided commands (not named in the kit manifests)

Discovered by walking the container's `PATH`. They ship in a kit image but no kit manifest names them, so the great majority is OS plumbing rather than investigative tooling. Kept for completeness only.

► 807 container-provided commands

The Format

How every page in this guide is built, and why. Derived from studying the reference works that already do this well, then fixing what they each get wrong for our use case: **a field guide you open mid-mission when you need a capability and do not remember the tool.**

1. What the research said

Source	What it does well	What it does not give us
tldr-pages style guide	Curated examples (≤ 8), imperative mood, <code>{{placeholder}}</code> syntax, machine-linted	Deliberately incomplete — omits most options by design. No capability index.
POSIX Utility Conventions ch.12 + man-pages(7)	Unambiguous synopsis grammar: <code>[]</code> optional, <code>\ </code> exclusive, <code>...</code>	Reference-shaped, not task-shaped. You must already know the tool's name.
SANS DFIR cheat sheets	Organised by investigative task — the right primary axis	Print-first, not greppable, not exhaustive on flags
Jai Minton DFIR cheatsheet	200+ scannable anchors; artifact paths and ATT&CK IDs as headings	"Most entries lack detailed flag documentation... assumes operator familiarity"
navi / cheat.sh	Machine-parseable, taggable, directly executable entries	No explanation layer at all
Progressive disclosure (Nielsen, 1995)	Layer information: essentials first, depth on demand	—

The gap we fill: every existing DFIR quick-reference assumes you already know the tool and its flags. None of them explains the options. That is exactly the layer this guide exists to provide, and it is why "completely listed out" is a hard requirement rather than a stretch goal.

2. Two entry points, one dataset

You arrive at this guide in one of two states:

1. **"I have a problem."** — *I need to recover deleted files from this image.* → **Capability Index:** capability → tool → command → options. This is the primary path and the reason the guide exists.
2. **"What do I have?"** — *Is there anything in the kit for Go binaries?* → **Kit Index:** VM → category → tool. This is `catalog/KIT-TOOLS.md`, generated from upstream manifests.

Both views are projections of the same JSON, so they can never disagree.

3. Scope is binding

A tool may appear in this guide **only** if it is in `catalog/kit-tools.json`, which is generated from the kit's own VM manifests. If it is not in the kit, it does not get a page — no matter how good the tool is.

Every command entry is tagged with the **VM(s) that carry it**, so in the field you instantly know whether you can actually run it on the box in front of you.

4. The page format

Every tool page has these sections, in this order. Sections are layered by progressive disclosure: a reader in a hurry stops after §3; a reader who needs precision continues to §4.

Header

```
# <tool>
**Kit:** REMnux · SIFT  **Category:** Filesystem analysis  **Version:** 4.11.1
**Docs:** <upstream URL>  **Verified:** 2026-07-29 from `cyberlab-aio:v1`
```

The version and the verification date are not decoration — they are the claim that the options below were read off the real binary on a known date.

1. Purpose — one line, imperative

List files and directory names in a disk image, including deleted entries.

2. Synopsis — POSIX notation, verbatim from the tool

```
fls [-adDFlhpruvV] [-f fstype] [-i imgtype] [-b dev_sector_size] [-m dir/]
    [-o imgoffset] [-z ZONE] [-s seconds] image [images] [inode]
```

3. Common invocations — ≤8, task-titled, copy-paste ready

Each example is titled by **the task it accomplishes**, not by the flag it uses, because the reader is searching by intent. Placeholders use tldr syntax so they are visually obvious and machine-checkable:

```
# List deleted files in a partition, recursively, with full paths
fls -rpd -o {{2048}} {{path/to/image.dd}}
```

4. Options — COMPLETE, one row per flag

Every flag the binary accepts. Not the useful ones — **all** of them.

Flag	Argument	What it does	When you would use it
<code>-d</code>	—	Display deleted entries only	Narrowing a recovery hunt to what was removed

The fourth column is the one that makes this a guide rather than a man page dump. It answers *why would I reach for this*, which is what you actually need under time pressure.

5. Gotchas

Real operational traps, each one earned. Examples already known on this project: timeout-guard `vol`, `binwalk`, `clamscan` and `radare2` in any harness — a synthetic memory image once pinned `vol` at 99% CPU indefinitely.

6. See also

Sibling tools for the same capability, so a dead end reroutes instead of stalling.

5. Verification: why the options can be trusted

The predecessor project shipped **fabricated CLI flags in roughly 44 of its 61 modules** — invented options for `yara`, `clamscan`, `PE-bear`, `scdbg` and others — because content was generated from a language model rather than read off the tools. That failure is the reason this guide exists in its current form.

So the rule here is mechanical, not aspirational:

1. Run the tool's `--help` / `-h` / man page **inside the kit container or VM**.
2. Store the raw output under `capture/<tool>.help.txt`, committed.
3. Generate/check the options table **against that capture**.
4. `scripts/lint.py` fails the build if a documented flag is absent from the capture, or if a captured flag is missing from the page.

A flag that cannot be produced from a real binary cannot ship. This is the same discipline `tldr-lint` applies to formatting, applied instead to truth.

Worked proof. The captured `fls` usage line advertises `[-adDFlhpruvV]`, but the body of the same output documents `-P`, `-B`, `-S` and `-k` as well. Anyone transcribing the synopsis — or working from memory — silently loses four real options. Only a capture catches that.

6. Sources and citation validation

Two different claims need two different kinds of evidence, and they must not be confused:

- **What a flag is** → the binary. Captured, committed, linter-enforced (§5). No online source overrides a real `--help`.
- **What a tool is for, and how practitioners actually use it** → authoritative upstream documentation and reputable DFIR publishers.

Acceptable sources for the second kind, in preference order: the project's own docs/repo (sleuthkit.org, volatilityfoundation.org, docs.remnux.org, ericzimmerman.github.io, Microsoft Learn), then recognised practitioner sources (SANS, thedfirreport.com, Red Canary, Mandiant/Unit42), then MITRE ATT&CK for technique mapping.

Every cited URL is liveness-checked before it ships. The predecessor project shipped hallucinated citations that looked plausible and 404'd; it already has a hardened, allowlisted, deadline-guarded fetcher (`cyberlab_retrieval.py`, with `verify_url`) built precisely for this, and this guide reuses that discipline. A dead or unreachable citation fails the build rather than degrading quietly.

Version skew is a real hazard: upstream docs describe the *current* release, while the kit runs pinned versions (TSK 4.11.1 in the container vs 4.15.0 in the tracker, for example). Where docs and capture disagree, **the capture wins** and the discrepancy is worth a line in Gotchas.

7. Output

- **Markdown** is the source of truth: greppable, diffable, reviewable, renders on GitHub.
- **PDF/print render** is generated from the markdown for field use. It is a build artifact, never hand-edited, so the two can never drift.

Implementation Plan

Goal: a field quick-reference covering the kit's tools, capability-indexed, with every command's options completely listed and explained, verified against real binaries.

Status legend: ☒ done ·  next · ☐ pending

Phase 0 — Foundations ☒

- ☒ **Scope fixed.** `catalog/kit-tools.json` + `KIT-TOOLS.md` : 1,006 tools, 5 platforms, 54 categories, each from an upstream machine-readable manifest.
- ☒ **Format designed** from six reference works — `docs/FORMAT.md`.
- ☒ **Verification proven.** `cyberlab-aio:v1` on rick has `fls`, `clamscan`, `binwalk`, `yara`, `vol`, `tshark`, `exiftool`, `log2timeline.py`.
- ☒ **Worked example** — `reference/examine-the-filesystem/fls.md`, 22 options from a real capture.

Phase 1 — Capability taxonomy ☒

The primary index. Draft ~40–60 capabilities phrased as **problems**, not tool names, e.g. "Recover deleted files from an image", "Find what persisted on a host", "Decode an obfuscated payload", "Prove lateral movement".

- Seed from the categories already carried by upstream: REMnux's 11 capability categories, Kali's 12 `kali-tools-*` groups, Security Onion's 9, and SANS investigative-task ordering.
- Map every one of the 1,006 kit tools to ≥ 1 capability. Tools that map to none are reported, not silently dropped — an unmapped tool is either a taxonomy gap or genuinely out of scope, and both need a decision.
- **Deliverable:** `reference/INDEX.md` — capability → tools → page links.

Open question for review: should the capability taxonomy follow an existing public framework (NIST/DFIR phases, MITRE ATT&CK tactics) for familiarity, or a kit-native one tuned to what we actually carry? Recommendation: kit-native primary, with ATT&CK IDs as secondary tags where they genuinely apply — forcing forensic tools into attacker tactics distorts both.

Phase 1b — Mine cyberlab as a source ☒

The cyberlab training lab already contains a large body of commands and prior exhaustive research. It is **read-only input** here — cyberlab is on hold and receives no changes.

- `scripts/mine_cyberlab.py` : extract every fenced command block from the 61 modules, plus `docs/CURRICULUM_ROADMAP.md`, `docs/TEST_REPORT.md` and `docs/CONTENT_QUALITY_AUDIT.md`, and the `catalog/` research.
- Produce `capture/cyberlab-candidates.json` : tool → observed invocations, with the module each came from.
- These become **candidate invocations** for the "Common invocations" section — real workflows someone already thought through, which is exactly the part that is expensive to invent.

Mine the cleaned branch, not `main`. The content-quality audit found fabricated CLI flags in ~44 of 61 modules, concentrated in the generated "enrichment tail". Branch `audit/content-quality-p0-p1` (PR #1) already removed that tail — ~11,000 lines — so it is the higher-signal source. Modules 57 (acquisition) and 58 (eventlog) were rated the cleanest and are worth reading first.

Candidates are never trusted. Every mined invocation still has to pass the Phase 3 linter against a real capture. Mining saves the *thinking*, not the *verification* — a mined command with an invented flag must fail exactly as loudly as one written from scratch.

Phase 2 — Capture pipeline ☒

`scripts/capture_help.py` : for each in-scope tool, run `--help / -h / man` in the right container, store raw output under `capture/<tool>.help.txt`.

- Containers: `cyberlab-aio:v1` and `dfir-aio:v4` on rick (Docker active, 830 GB free).
- **Always timeout-guard** — `vol` once pinned 99% CPU indefinitely on a synthetic image. Hard per-tool timeout, failure recorded not retried blindly.
- Emit a coverage report: which kit tools are covered by a container, which need a booted VM (Kali, FLARE-VM Windows guests), which are GUI-only and have no CLI surface to document.

Reality check: the 1,006-tool list includes GUI-only tools (Autopsy, Wireshark GUI, PE-bear) and appliance services (Elasticsearch, Kibana) with no meaningful command surface. Expect the genuinely CLI-documentable set to be substantially smaller. Phase 2's report turns that from a guess into a number before any page is written.

Phase 3 — Page generation + linter ☒

- Generate a page skeleton per tool from its capture: header, synopsis, complete options table. Mechanical, deterministic — no model invention.
- Human/model adds only the *judgement* columns: purpose, "when you'd use it", invocations, gotchas.
- `scripts/lint.py` gates the build:
- every documented flag exists in the capture;
- every captured flag appears in the page (**completeness**, the hard requirement);

- every tool is in `kit-tools.json`;
- format rules hold (imperative mood, ≤8 examples, placeholder syntax).

Phase 4 — Print render

Markdown → PDF for field carry. Build artifact only, never hand-edited. Per-capability booklets rather than one 1,000-page brick.

Phase 5 — Prioritised content ► (20 of 135 tools reviewed)

Do not attempt all tools at once. Order by field value:

1. Core forensics: TSK suite, Volatility, Plaso, RegRipper, bulk_extractor
2. Network: Zeek, Suricata, tshark, tcpdump
3. Malware triage: YARA, ClamAV, capa, FLOSS, Detect-It-Easy
4. Live response / hunting: Velociraptor, osquery, Hayabusa, Chainsaw
5. Offensive-side as needed for exercises

Risks

Risk	Mitigation
Fabricated flags (killed the predecessor project)	Capture-or-it-does-not-ship, enforced by linter
Scale — 1,006 tools is a lot of pages	Phase 2 coverage report right-sizes it; Phase 5 prioritises
Kit VMs powered down (<code>kali</code> , <code>range-linux-web</code> , <code>redinfra01</code>)	Containers cover most Linux tooling; boot VMs only for the remainder
Version drift between capture and deployed kit	Version + capture date in every page header
GUI-only tools with no CLI	Identify in Phase 2; document workflow, not flags

Kit Tool List

The binding scope for the quick-reference guide. A tool absent from this list is **not in the kit** and must not be documented in the guide.

Every entry is derived from an upstream machine-readable manifest; none is written from memory. Sources are listed at the end.

Contents

- [REMnux](#) — 268 tools, 31 categories
- [Kali Linux](#) — 403 tools, 12 categories
- [FLARE-VM](#) — 137 tools, 1 categories
- [SIFT Workstation](#) — 162 tools, 1 categories
- [Security Onion](#) — 36 tools, 9 categories

Total: 1006 tools across 5 platforms.

Analyze Documents / Email Messages

Tool	Command(s)	Purpose
emldump.py	—	Parse and analyze EML files.
mail-parser	—	Parse raw SMTP and .MSG email messages and generate a parsed object from them.
msg-extractor	<code>extract_msg</code>	Extract emails and attachments from MSG files.
msgconvert	—	Convert MSG files to MBOX files.

Analyze Documents / General

Tool	Command(s)	Purpose
base64dump.py	—	Locate and decode strings encoded in Base64 and other common encodings.
Tesseract OCR	<code>tesseract</code>	Examine images to identify and extract text using optical character recognition (OCR).

Analyze Documents / Microsoft Office

Tool	Command(s)	Purpose
EvilClippy	—	Modify aspects of Microsoft Office documents.
Hachoir	<code>hachoir-grep</code> , <code>hachoir-metadata</code> , <code>hachoir-strip</code> , <code>hachoir-wx</code>	View, edit, and carve contents of various binary file types.
libolecf	<code>olecfexport</code> , <code>olecfinfo</code> , <code>olecfmount</code> , <code>etc</code>	Microsoft Office OLE2 compound documents.
msoffcrypto-crack.py	—	Recover the password of an encrypted Microsoft Office document.
msoffcrypto-tool	—	Decrypt a Microsoft Office file with password, intermediate key, or private key which generated its escrow key.
msoffice-crypt	—	Encrypt and decrypt OOXML Microsoft Office documents.
oledump.py	—	Analyze OLE2 Structured Storage files.
olefile	—	Python package to parse, read and write MS OLE2 files.
oletools	<code>mraptor</code> , <code>msodde</code> , <code>olebrowse</code> , <code>oledir</code> , <code>oleid</code> , <code>olemap</code> , <code>olemeta</code> , <code>oleobj</code> , <code>oletimes</code> , <code>olevba</code> , <code>pyxswf</code> , <code>rtfobj</code> , <code>ezhexviewer</code>	Microsoft Office OLE2 compound documents.
onedump.py	—	Extract and analyze embedded files from OneNote documents.
pcode2code	—	Decompile VBA macro p-code from Microsoft Office documents.
pcodedmp	—	Disassemble VBA p-code.
rtfdump.py	—	Analyze a suspicious RTF file.
SSView	<code>ssview</code>	Analyze OLE2 Structured Storage files.
XLMMacroDeobfuscator	<code>xlmdeobfuscator</code> , <code>runxlrd2.py</code>	Deobfuscate XLM macros (also known as Excel 4.0 macros) from Microsoft Office files.
xmldump.py	—	Extract contents of XML files, in particular OOXML-formatted Microsoft Office documents.
zipdump.py	—	Analyze zip-compressed files.

Analyze Documents / Pdf

Tool	Command(s)	Purpose
Origamindoe	<code>pdfcop</code> , <code>pdfdecompress</code> , <code>pdfdecrypt</code> , <code>pdfextract</code> , <code>etc</code>	Parse, modify, generate PDF files.
pdf-parser.py	—	Examine elements of the PDF file.
pdfid.py	—	Identify suspicious elements of the PDF file.

Tool	Command(s)	Purpose
pdfresurrect	—	Extract previous versions of content from PDF files.
pdftk-java	pdftk	Edit, create, and examine PDF files.
pdftool.py	—	Analyze PDF files to identify incremental updates to the document.
peepdf-3	—	Examine elements of the PDF file.
qpdf	—	Manipulate (merge, convert, transform) PDF files.

Dynamically Reverse-Engineer Code / Elf Files

Tool	Command(s)	Purpose
edb	—	An AArch32/x86/x86-64 debugger, well suited for debugging ELF files.
GNU Project Debugger	—	Multi-language debugger.
ltrace	—	Trace library calls and signals.
strace	—	Trace process' system calls and signals.

Dynamically Reverse-Engineer Code / General

Tool	Command(s)	Purpose
Frida	frida , frida-ps , frida-trace , frida-discover , frida-ls-devices , frida-kill	Trace the execution of a process to analyze its behavior.
r2pipe	—	Examine binary files, including disassembling and debugging.
radare2	r2 , rasm2 , rabin2 , rahash2 , rafind2 , r2ai , decai , pdg	Examine binary files, including disassembling and debugging. Includes r2ai and decai plugins for LLM-powered analysis (A
Wine	wine	Run Windows applications.
x64dbg Automate MCP (OpenCode skills)	—	Drive x64dbg on a remote Windows VM from OpenCode on REMnux, with eight AI commands for tracing, unpacking, state snapsh

Dynamically Reverse-Engineer Code / Scripts

Tool	Command(s)	Purpose
box-js	box-js , box-export	Analyze suspicious JavaScript scripts.
JavaScript Deobfuscator	—	Deobfuscate JavaScript by removing common obfuscation techniques such as string arrays and proxy functions.
js_unshroud	—	Monitor and deobfuscate JavaScript behavior in a headless browser to analyze malicious web pages.
JStillery	jstillery	Deobfuscate JavaScript scripts using AST and Partial Evaluation techniques.
objects.js	—	Emulate common browser and PDF viewer objects, methods, and properties when deobfuscating JavaScript.
PowerShell Core	pwsh	Run PowerShell scripts and commands.
Rhino Debugger	rhino-debugger	GUI JavaScript debugger.
SpiderMonkey	js	Execute and deobfuscate JavaScript using Mozilla's standalone JavaScript engine.
SpiderMonkey (Patched)	js-ascii , js-file	Execute and deobfuscate JavaScript using a patched version of Mozilla's standalone JavaScript engine.
STPyV8	—	Python3 and JavaScript interop engine, fork of the original PyV8 project.
Webcrack	—	Deobfuscate, unminify, and unpack bundled JavaScript, including scripts protected with obfuscator.io.

Dynamically Reverse-Engineer Code / Shellcode

Tool	Command(s)	Purpose
libemu	—	A library for x86 code emulation and shellcode detection.
Qiling	—	Emulate code execution of PE files, shellcode, etc. for a variety of OS and hardware platforms.
runsc	—	Run shellcode to trace and analyze its execution.
sctdbg	—	Analyze shellcode by emulating its execution.
shcode2exe	—	Convert 32 and 64-bit shellcode to a Windows executable file.
shellcode2exe.bat	—	Convert 32 and 64-bit shellcode to a Windows executable file.
Speakeasy	—	Emulate code execution, including shellcode, Windows drivers, and Windows PE files.
XORSearch	xorsearch	Locate and decode strings obfuscated using common techniques.

Examine Static Properties / .Net

Tool	Command(s)	Purpose
dnfile	—	Analyze static properties of .NET files.
dotnetfile	—	Analyze static properties of .NET files.

Tool	Command(s)	Purpose
monodis	—	Disassemble and extract resources from .NET assemblies.

Examine Static Properties / Deobfuscation

Tool	Command(s)	Purpose
1768.py	—	Analyze Cobalt Strike beacons.
Balbuzard	<code>balbuzard</code> , <code>bbcraack</code> , <code>bbharvest</code> , <code>bbtrans</code>	Extract and deobfuscate patterns from suspicious files.
base64dump.py	—	Locate and decode strings encoded in Base64 and other common encodings.
brxor.py	—	Bruteforce XOR'ed strings to find those that are English words.
Chepy	<code>chepy</code>	Decode and otherwise analyze data using this command-line tool and Python library.
Cobalt Strike Configuration Extractor (CSCE) and Parser	<code>csce</code> , <code>list-cs-settings</code>	Analyze Cobalt Strike beacons.
cs-analyze-processdump.py	—	Analyze Cobalt Strike beacon process dumps to detect sleep mask encoding.
cs-decrypt-metadata.py	—	Decrypt Cobalt Strike metadata.
cs-extract-key.py	—	Extract AES and HMAC keys from Cobalt Strike beacon process memory.
cut-bytes.py	—	Cut out a part of a data stream.
CyberChef	<code>cyberchef</code>	Decode and otherwise analyze data using this browser app.
DC3-MWCP	<code>mwcp</code>	Parsing configuration information from malware.
ex_pe_xor.py	—	Search an XOR'ed file for indications of executable binaries.
FLOSS	<code>floss</code>	Extract and deobfuscate strings from PE executables.
format-bytes.py	—	Decompose structured binary data with format strings.
hex-to-bin.py	—	Convert hexadecimal text dumps to binary data.
Malchive	—	Perform static analysis of various aspects of malicious code.
NoMoreXOR.py	—	Help guess a file's 256-byte XOR by using frequency analysis.
numbers-to-string.py	—	Translate number sequences into ASCII characters.
re-search.py	—	Search files using regular expressions from a built-in library or custom patterns.
sets.py	—	Perform set operations on lines or bytes in text files.
strdeob.pl	—	Locate and decode stack strings in executable files.
translate.py	—	Translate bytes according to a Python expression.
unicode	—	Display Unicode character properties.
unXOR	—	Deobfuscate XOR'ed files.
xor-kpa.py	—	Implement a XOR known plaintext attack.
xorBruteForcer.py	—	Bruteforce an XOR-encoded file.
XORSearch	<code>xorsearch</code>	Locate and decode strings obfuscated using common techniques.
xorsearch.py	—	Search for XOR, ROL, ROT, and SHIFT encoded strings with YARA and regex support.
XORStrings	—	Search for XOR encoded strings in a file.
xortool	—	Analyze XOR-encoded data.

Examine Static Properties / Elf Files

Tool	Command(s)	Purpose
pyelftools	<code>readelf.py</code>	Python library for parsing and analyzing ELF files and DWARF debugging information.

Examine Static Properties / General

Tool	Command(s)	Purpose
7-Zip	<code>7za</code>	Compress and decompress files using a variety of algorithms.
binwalk	—	Extract and analyze firmware images.
bulk_extractor	—	Extract interesting strings from binary files.
ClamAV	<code>clamscan</code> , <code>freshclam</code>	Scan files for malware signatures.
Detect-It-Easy	—	Determine types of files and examine file properties.
disitool	<code>disitool.py</code>	Manipulate embedded digital signatures.
Droidlysis	<code>droidlysis</code>	Perform static analysis of Android applications.

Tool	Command(s)	Purpose
ExifTool	<code>exiftool</code>	Tool to read from, write to, and edit EXIF metadata of various file types.
file	—	Identify file type using "magic" numbers.
file-magic.py	—	Identify file types using the Python magic module.
Hachoir	<code>hachoir-grep</code> , <code>hachoir-metadata</code> , <code>hachoir-strip</code> , <code>hachoir-wx</code>	View, edit, and carve contents of various binary file types.
Hash ID	<code>hash-id.py</code>	Identify different types of hashes.
LIEF	—	Parse and analyze PE, ELF, MachO, DEX, OAT, VDEX, ART, and DWARF executable formats.
Magika	—	Identify file type using signatures.
Malcat Lite	—	Analyze binary files using a hex editor, disassembler, and file dissector.
msitools	—	Create, inspect and extract Windows Installer (.msi) files.
Name-That-Hash	<code>nth</code>	Identify different types of hashes.
numbers-to-string.py	—	Convert decimal numbers to strings.
re-search.py	—	Search the file for built-in regular expressions of common suspicious artifacts.
signsrch	—	Find patterns of common encryption, compression, or encoding algorithms.
Sleuth Kit	—	Analyze disk images and recover files from them.
ssdeep	—	Compute Context Triggered Piecewise Hashes (CTPH), also known as fuzzy hashes.
strings.py	—	Extract ASCII and Unicode strings from binary files with length sorting and filtering.
thefuzz	—	Fuzzy String Matching in Python.
TrID	<code>trid</code> , <code>tridupdate</code>	Identify file type using signatures.
wxHexEditor	—	Hex editor.
Yara Rules	—	Scan a file with YARA rules to identify capabilities and behaviors (packer detection, anti-debug, networking).
YARA-Forge Rules	—	Scan files with curated YARA rules from 45+ sources for malware family identification.

Examine Static Properties / Go

Tool	Command(s)	Purpose
GoReSym	<code>GoReSym</code>	Extract metadata and symbols from Go binaries, including stripped ones.
Redress	<code>redress</code>	Analyze stripped Go binaries to recover symbols, types, source structure, and integrate with Radare2.

Examine Static Properties / Pe Files

Tool	Command(s)	Purpose
bearparser	<code>bearcommander</code>	Parse PE file contents.
debloat	—	Remove junk contents from bloated Windows executables.
disitool.py	—	Extract, delete, copy, and inject digital signatures in PE files.
dllcharacteristics.py	—	Read and set DLL characteristics of a PE file.
Manalyze	—	Perform static analysis of suspicious PE files.
PE Tree	<code>pe-tree</code>	Examine contents and structure of PE files.
pecheck.py	—	Analyze static properties of PE files.
pedump	—	Statically analyze PE files and extract their components (e.g., resources).
pefile	—	Python library for analyzing static properties of PE files.
PEframe	<code>peframe</code>	Statically analyze PE and Microsoft Office files.
pev	<code>pestr</code> , <code>readpe</code> , <code>pedis</code> , <code>pehash</code> , <code>pescan</code> , <code>peldd</code> , <code>peres</code>	Analyze PE files and extract strings from them.
PortEx	<code>portex</code>	Statically analyze PE files.
readpe (formerly pev)	<code>readpe</code> , <code>pestr</code> , <code>pedis</code> , <code>pehash</code> , <code>pescan</code> , <code>pesec</code> , <code>peldd</code> , <code>pepack</code> , <code>peres</code> , <code>ofs2rva</code> , <code>rva2ofs</code>	Analyze PE files and extract strings from them.

Explore Network Interactions / Connecting

Tool	Command(s)	Purpose
Anonymy	<code>anomy</code>	A wrapper around wget, ssh, sftp, ftp, and telnet to route these connections through Tor to anonymize your traffic.
cURL	<code>curl</code>	Interact with servers via supported protocols, including HTTP, HTTPS, FTP, IMAP, etc. using this command-line tool.
EPIC IRC Client	<code>epic5</code>	Examine IRC activities with this IRC client.

Tool	Command(s)	Purpose
GNU Wget	<code>wget</code>	Interact with servers via HTTP, HTTPS, FTP, and FTPS using this command-line tool.
netcat	<code>nc</code>	Read and write data across network connections.
thug	—	Examine suspicious website using this low-interaction honeyclient.
tor	—	Obfuscate your origins by routing traffic through a network of anonymizing nodes.
Unfurl	—	Deconstruct and decode data from a URL.
zbarimg	<code>zbarimg</code>	Decode QR codes and barcodes from image files.

Explore Network Interactions / Monitoring

Tool	Command(s)	Purpose
Burp Suite Community Edition	<code>burpsuite</code>	Investigate website interactions using this web proxy.
cs-parse-traffic.py	—	Decrypt and parse Cobalt Strike beacon network traffic.
mitmproxy	<code>mitmproxy</code> , <code>mitmdump</code> , <code>mitmweb</code>	Investigate website interactions using this web proxy.
monitor-network	—	Monitor traffic on the first active network interface using tshark, printing a live summary to the screen or saving it t
Network Miner Free Edition	<code>networkminer</code>	Examine network traffic and carve PCAP capture files.
<code>ngrep</code>	—	Look for patterns in network traffic.
PolarProxy	<code>polarproxy</code>	Intercept and decrypt TLS traffic.
tcpdump	—	Capture and analyze network traffic with this command-line sniffer.
<code>tcpflow</code>	—	Analyze the flow of network traffic.
tcpick	—	Capture and analyze network traffic with this command-line sniffer.
<code>tcpxtract</code>	—	Extract files from network traffic.
<code>tshark</code>	—	Capture and analyze network traffic with this console-based sniffer.
Wireshark	—	Capture and analyze network traffic with this sniffer.

Explore Network Interactions / Services

Tool	Command(s)	Purpose
accept-all-ips	—	Accept connections to all IPv4 and IPv6 addresses and redirect it to the corresponding local port.
dnsresolver.py	—	DNS resolver tool for dynamic analysis with wildcard and tracking support.
fakedns	—	Respond to DNS queries with the specified IP address.
fakemail	—	Intercept and examine SMTP email activity with this fake SMTP server.
FakeNet-NG	—	Emulate common network services and interact with malware.
<code>INetSim</code>	<code>inetsim</code>	Emulate common network services and interact with malware.
inspircd 3	—	Examine IRC activity with this IRC server.
netcat	<code>nc</code>	Read and write data across network connections.
Nginx	—	Web server.

Gather And Analyze Data / General

Tool	Command(s)	Purpose
DeXRAY	<code>dexray</code>	Extract and decode data from antivirus quarantine files.
dissect	—	Perform a variety of forensics and incident response tasks using this DFIR framework and toolset.
dnslib	—	Python library to encode/decode DNS wire-format packets.
ioc_parser	—	Extract IOCs from security report PDFs.
ipwhois	—	Retrieve and parse whois data for IP addresses.
malwoverview	<code>malwoverview</code>	Query public repositories of malware data (e.g., VirusTotal, HybridAnalysis).
nsrllookup	—	Look up MD5 file hashes in the NIST National Software Reference Library (NSRL).
pdnstool	—	Query passive DNS databases for DNS data.
<code>Scalpel</code>	—	Carve contents out of binary files, such as partitions.
time-decode	—	Decode and encode date and timestamps.
virustotal-search	<code>virustotal-search.py</code>	Search VirusTotal for file hashes.
virustotal-submit	<code>virustotal-submit.py</code>	Submit files to VirusTotal.
<code>Yara</code>	<code>yara</code>	Identify and classify malware samples using Yara rules.
YARA-X	—	Scan files using YARA rules, the next generation of YARA written in Rust.

General Utilities / General

Tool	Command(s)	Purpose
7-Zip	<code>7za</code>	Compress and decompress files using a variety of algorithms.
cabextract	—	Extract Microsoft cabinet (cab) files.
cURL	<code>curl</code>	Interact with servers via supported protocols, including HTTP, HTTPS, FTP, IMAP, etc. using this command-line tool.
Docker	—	Run and manage containers.
Firefox	<code>firefox</code>	Web browser.
GNOME Calculator	<code>calculator</code>	Calculator.
IBus	<code>ibus-setup</code>	Adjust input methods for the GUI.
Info-ZIP	<code>zip</code> , <code>unzip</code>	Compress and decompress files using the zip algorithm.
myip	—	Determine the IP address of the default network interface.
myjson-filter.py	—	Filter data formatted using the JSON format used by Didier Stevens' tools.
nasm	—	An x86-64 assembler.
Nautilus	—	Graphical file manager.
OpenSSH	<code>sftp</code> , <code>ssh</code> , etc	Initiate and receive SSH and SFTP connections.
PowerShell Core	<code>powershell</code>	Run PowerShell scripts and commands.
RAR	<code>rar</code>	Compress and decompress files using a variety of algorithms.
REMnux Installer	—	Install and update the REMnux distro.
restrict-egress	—	Restrict outbound network access to an allowlist of domains and CIDRs using an nftables default-deny egress policy. It i
sortcanon.py	—	Sort text files using canonicalization functions built into this tool.
SQLite	<code>sqlite3</code>	Manage and interact with SQL database files.
sshpas	<code>sshpas</code>	Supply a password to SSH non-interactively for automated logins.
texteditor.py	—	Edit text files from the command line using search-and-replace commands.
unrar-free	<code>unrar</code>	Decompress files using a variety of algorithms.
Wine	<code>wine</code>	Run Windows applications.

Investigate System Interactions / General

Tool	Command(s)	Purpose
ProcDOT	<code>procdot</code>	Visualize and examine the output of Process Monitor.
ProcmonMCP	—	MCP server that lets AI assistants analyze Process Monitor (Procmon) XML captures.
sandfly-processdecloak	—	Find hidden processes on the local Linux system.
Unhide	—	Find hidden processes or connections on the local Linux system.

Perform Memory Forensics / General

Tool	Command(s)	Purpose
AESKeyFinder	<code>aeskeyfind</code>	Find 128-bit and 256-bit AES keys in a memory image.
bulk_extractor	—	Extract interesting strings from binary files.
RSKeyFinder	<code>rsakeyfind</code>	Find BER-encoded RSA private keys in a memory image.
Volatility Framework	—	Memory forensics tool and framework.

Statically Analyze Code / .Net

Tool	Command(s)	Purpose
de4dot	—	Deobfuscate and unpack .NET programs.
ILSpy	—	Examine and decompile .NET programs.

Statically Analyze Code / Android

Tool	Command(s)	Purpose
androguard	—	Examine Android files.
AndroidProjectCreator	—	Convert an Android APK application file into an Android Studio project for easier analysis.
APKiD	<code>apkid</code>	Identify compilers, packers, and obfuscators used to protect Android APK and DEX files.
apktool	—	Reverse-engineer Android APK files.
baksmali	—	Disassembler for the dex format used by Dalvik, Android's Java VM implementation.
dex2jar	<code>dex-tools</code> , <code>d2j-dex2jar</code>	Examine Dalvik Executable (dex) files.

Tool	Command(s)	Purpose
Droidlysis	<code>droidlysis</code>	Perform static analysis of Android applications.
JADX	<code>jadx</code> , <code>jadx-gui</code>	Generate Java source code from Dalvik Executable (dex) and Android APK files.

Statically Analyze Code / General

Tool	Command(s)	Purpose
Cutter	—	Reverse engineering platform powered by Rizin.
Detect-It-Easy	—	Determine types of files and examine file properties.
Ghidra	—	Software reverse engineering tool suite.
<code>objdump</code>	—	Disassemble binary files.
Qiling	—	Emulate code execution of PE files, shellcode, etc. for a variety of OS and hardware platforms.
radare2	<code>r2</code> , <code>rasm2</code> , <code>rabin2</code> , <code>rahash2</code> , <code>rafind2</code> , <code>r2ai</code> , <code>decai</code> , <code>pdg</code>	Examine binary files, including disassembling and debugging. Includes r2ai and decal plugins for LLM-powered analysis (A)
Vivisect	<code>vivbin</code> , <code>vdbbin</code>	Statically examine and emulate binary files.

Statically Analyze Code / Java

Tool	Command(s)	Purpose
cfr	—	Java decompiler.
Java ID3 Parser	<code>idx_parser.py</code>	Analyze Java ID3 files.
Javassist	—	Java bytecode engineering toolkit/library.
JD-GUI Java Decompiler	<code>jd-gui</code>	Java decompiler with GUI.
Procyon	<code>procyon</code>	Java decompiler.

Statically Analyze Code / Pe-Files

Tool	Command(s)	Purpose
binee (Binary Emulation Environment)	—	Analyze I/O operations of a suspicious PE file by emulating its execution.
<code>capa</code>	—	Detect suspicious capabilities in PE files.
Malchive	—	Perform static analysis of various aspects of malicious code.
mbcscan	<code>mbcscan.py</code>	Scan a PE file to list the associated Malware Behavior Catalog (MBC) details.
Speakeasy	—	Emulate code execution, including shellcode, Windows drivers, and Windows PE files.

Statically Analyze Code / Python

Tool	Command(s)	Purpose
Decompyle++	<code>pycdas</code> , <code>pycdc</code>	Python bytecode disassembler and decompiler.
PyInstaller Extractor	<code>pyinstxtractor.py</code>	Extract contents of a PyInstaller-generated PE files.
pyinstxtractor-ng	<code>pyinstxtractor-ng</code>	Extract contents of PyInstaller-generated executables without requiring a matching Python version.
uncompyle6	<code>uncompyle6</code>	Python cross-version bytecode decompiler for Python 1.0 through 3.8.

Statically Analyze Code / Scripts

Tool	Command(s)	Purpose
Autolt-Ripper	<code>autoit-ripper</code>	Extract Autolt scripts embedded in PE binaries.
decode-vbe.py	—	Decode encoded VBS scripts (VBE).
GootLoaderAutoJsDecode.py	—	Statically deobfuscate GootLoader (GOOTLOADER) malicious JScript to recover the payload and extract C2 domains.
JS Beautifier	<code>js-beautify</code>	Reformat JavaScript scripts for easier analysis.

Statically Analyze Code / Unpacking

Tool	Command(s)	Purpose
<code>binwalk</code>	—	Extract and analyze firmware images.
Bytehist	<code>bytehist</code>	Generate byte-usage-histograms for all types of files with a focus on PE files.
ClamAV	<code>clamscan</code> , <code>freshclam</code>	Scan files for malware signatures.
TrID	<code>trid</code> , <code>tridupdate</code>	Identify file type using signatures.
UPX	<code>upx</code>	Pack and unpack PE files.

Use Artificial Intelligence / General

Tool	Command(s)	Purpose
GhidraAssistMCP	—	MCP server for AI-assisted reverse engineering in Ghidra.
OpenCode	—	Open-source AI coding agent for the terminal.
ProcmonMCP	—	MCP server that lets AI assistants analyze Process Monitor (Procmon) XML captures.
radare2	<code>r2</code> , <code>rasm2</code> , <code>rabin2</code> , <code>rahash2</code> , <code>rafind2</code> , <code>r2ai</code> , <code>decai</code> , <code>pdg</code>	Examine binary files, including disassembling and debugging. Includes r2ai and decal plugins for LLM-powered analysis (A
REMnux MCP Server	<code>remnux-mcp-server</code>	MCP server for using the REMnux malware analysis toolkit via AI assistants.
x64dbg Automate MCP (OpenCode skills)	—	Drive x64dbg on a remote Windows VM from OpenCode on REMnux, with eight AI commands for tracing, unpacking, state snapsh

View Or Edit Files / General

Tool	Command(s)	Purpose
dos2unix	<code>dos2unix</code> , <code>mac2unix</code> , <code>unix2dos</code> , <code>unix2mac</code>	Convert text files with Windows or macOS line breaks to Unix line breaks and vice versa.
Evince	<code>evince</code>	View documents in a variety of formats, including PDF.
feh	—	View images.
ImageMagick	<code>display</code> , <code>convert</code> , <code>mogrify</code> , <code>etc</code>	View and manipulate image and related files.
Mermaid Viewer	—	View Mermaid diagrams, such as AI-generated code-analysis workflow diagrams, in a local browser.
SciTE	—	Edit text files.
VBinDiff	<code>vbindiff</code>	Compare binary files.
Visual Studio Code	<code>code</code>	Powerful source code editor.
wxHexEditor	—	Hex editor.

Crypto Stego

Tool	Command(s)	Purpose
aesfix	<code>aesfix</code>	Tool for correcting bit errors in an AES key schedule.
aeskeyfind	<code>aeskeyfind</code>	Tool for locating AES keys in a captured memory image.
ccrypt	<code>ccrypt</code>	Secure encryption and decryption of files and streams.
steghide	<code>steghide</code>	Steganography hiding tool.
stegosuite	<code>stegosuite</code>	Steganography tool to hide information in image files.
stegsnow	<code>stegsnow</code>	Steganography using ASCII files.

Database

Tool	Command(s)	Purpose
jsql-injection	<code>jsql-injection</code>	Java tool for automatic database injection.
mbdtools	<code>mbdtools</code>	JET / MS Access database (MDB) tools.
osscanner	<code>osscanner</code>	Oracle assessment framework.
sidguesser	<code>sidguesser</code>	Guesses sids against an Oracle database.
sqldict	<code>sqldict</code>	Dictionary attack tool for SQL Server.
sqlitebrowser	<code>sqlitebrowser</code>	GUI editor for SQLite databases.
sqlmap	<code>sqlmap</code>	Automatic SQL injection tool.
sqlninja	<code>sqlninja</code>	SQL server injection and takeover tool.
sqlsus	<code>sqlsus</code>	MySQL injection tool.
tnscmd10g	<code>tnscmd10g</code>	Tool to prod the oracle tnslsnr process.

Exploitation

Tool	Command(s)	Purpose
armitage	<code>armitage</code>	Cyber attack management for Metasploit.
beef-xss	<code>beef-xss</code>	Browser Exploitation Framework (BeEF).
exploitdb	<code>exploitdb</code>	Searchable Exploit Database archive.
metasploit-framework	<code>metasploit-framework</code>	Framework for exploit development and vulnerability research.
msfpcc	<code>msfpcc</code>	MSFvenom Payload Creator (MSFPC).
set	<code>set</code>	Social-Engineer Toolkit.

Tool	Command(s)	Purpose
shellnoob	shellnoob	Shellcode writing toolkit.
sqlmap	sqlmap	Automatic SQL injection tool.
termineter	termineter	Smart meter testing framework.

Forensics

Tool	Command(s)	Purpose
7zip	7zip	7-Zip file archiver with a high compression ratio.
afflib-tools	afflib-tools	Advanced Forensics Format Library (utilities).
apktool	apktool	Tool for reverse engineering Android apk files.
autopsy	autopsy	Graphical interface to SleuthKit.
binwalk	binwalk	Tool library for analyzing binary blobs and executable code.
binwalk3	binwalk3	Tool library for analyzing binary blobs and executable code.
bulk-extractor	bulk-extractor	
bytecode-viewer	bytecode-viewer	Java 8+ Jar & Android APK Reverse Engineering Suite.
cabextract	cabextract	Microsoft Cabinet file unpacker.
chkrootkit	chkrootkit	Rootkit detector.
creddump7	creddump7	Python tool to extract credentials and secrets from Windows registry hives.
dc3dd	dc3dd	Patched version of GNU dd with forensic features.
dcfldd	dcfldd	Enhanced version of dd for forensics and security.
ddrescue	ddrescue	Data recovery and protection tool.
dumpzilla	dumpzilla	Mozilla browser forensic tool.
edb-debugger	edb-debugger	Cross platform x86/x86-64 debugger.
ewf-tools	ewf-tools	Collection of tools for reading and writing EWF files.
exifprobe	exifprobe	Read metadata from digital pictures.
exiv2	exiv2	EXIF/IPTC/XMP metadata manipulation tool.
ext3grep	ext3grep	Tool to help recover deleted files on ext3 filesystems.
ext4magic	ext4magic	Recover deleted files from ext3 or ext4 partitions.
extundelete	extundelete	Utility to recover deleted files from ext3/ext4 partition.
fcrackzip	fcrackzip	Password cracker for zip archives.
firmware-mod-kit	firmware-mod-kit	Deconstruct and reconstruct firmware images.
foremost	foremost	Forensic program to recover lost files.
forensic-artifacts	forensic-artifacts	Knowledge base of forensic artifacts (data files).
forensics-colorize	forensics-colorize	Show differences between files using color graphics.
galleta	galleta	Internet Explorer cookie forensic analysis tool.
gdb	gdb	GNU Debugger.
gpart	gpart	Guess PC disk partition table, find lost partitions.
gparted	gparted	GNOME partition editor.
grokvt	grokvt	Scripts for reading Microsoft Windows event log files.
guymager	guymager	Forensic imaging tool based on Qt.
hashdeep	hashdeep	Recursively compute hashsums or piecewise hashings.
inetsim	inetsim	Software suite for simulating common internet services.
jadx	jadx	Dex to Java decompiler.
javasnoop	javasnoop	Intercept Java applications locally.
libhivex-bin	libhivex-bin	Utilities for reading and writing Windows Registry hives.
libsmali-java	libsmali-java	Assembler/disassembler for Android's dex format.
lvm2	lvm2	Linux Logical Volume Manager.
lynis	lynis	Security auditing tool for Unix based systems.
mac-robber	mac-robber	Collects data about allocated files in mounted filesystems.
magicrescue	magicrescue	Recover files by looking for magic bytes.
md5deep	md5deep	
mdbtools	mdbtools	JET / MS Access database (MDB) tools.

Tool	Command(s)	Purpose
memdump	memdump	Utility to dump memory contents to standard output.
metacam	metacam	Extract EXIF information from digital camera files.
missidentify	missidentify	Find win32 applications.
myrescue	myrescue	Rescue data from damaged disks.
nasm	nasm	General-purpose x86 assembler.
nasty	nasty	Tool which helps you to recover your GPG passphrase.
ollydbg	ollydbg	32-bit assembler level analysing debugger.
parted	parted	Disk partition manipulator.
pasco	pasco	Internet Explorer cache forensic analysis tool.
pdf-parser	pdf-parser	Parses PDF files to identify fundamental elements.
pdfid	pdfid	Scans PDF files for certain PDF keywords.
plaso	plaso	Super timeline all the things -- metapackage.
polenum	polenum	Extracts the password policy from a Windows system.
pst-utils	pst-utils	Tools for reading Microsoft Outlook PST files.
python3-capstone	python3-capstone	Lightweight multi-architecture disassembly framework - Python bindings.
python3-dfdatetime	python3-dfdatetime	Digital Forensics date and time library for Python 3.
python3-dfvfs	python3-dfvfs	Digital Forensics Virtual File System.
python3-dfwinreg	python3-dfwinreg	Digital Forensics Windows Registry library for Python 3.
python3-distorm3	python3-distorm3	Powerful disassembler library for x86/AMD64 binary streams (Python3 bindings).
radare2	radare2	Free and advanced command line hexadecimal editor.
readpe	readpe	Command-line tools to manipulate Windows PE files.
recoverdm	recoverdm	Recover files on disks with damaged sectors.
recoverjpeg	recoverjpeg	Recover JFIF (JPEG) pictures and MOV movies.
recstudio	recstudio	
reglookup	reglookup	Utility to analysis for Windows NT-based registry.
regripper	regripper	Perform forensic analysis of registry hives.
rephrase	rephrase	Specialized passphrase recovery tool for GnuPG.
rifiuti	rifiuti	MS Windows recycle bin analysis tool.
rifiuti2	rifiuti2	Replacement for rifiuti, a MS Windows recycle bin analysis tool.
rizin-cutter	rizin-cutter	Reverse engineering platform powered by rizin.
rkhunter	rkhunter	Rootkit, backdoor, sniffer and exploit scanner.
rsakeyfind	rsakeyfind	Locates BER-encoded RSA private keys in memory images.
rz-ghidra	rz-ghidra	Ghidra decompiler and sleigh disassembler for rizin.
safecopy	safecopy	Data recovery tool for problematic or damaged media.
samdump2	samdump2	Dump Windows 2k/NT/XP password hashes.
scalpel	scalpel	Fast filesystem-independent file recovery.
scrounge-ntfs	scrounge-ntfs	Data recovery program for NTFS filesystems.
sleuthkit	sleuthkit	Tools for forensics analysis on volume and filesystem data.
sqlitebrowser	sqlitebrowser	GUI editor for SQLite databases.
ssdeep	ssdeep	Recursive piecewise hashing tool.
tcpdump	tcpdump	Command-line network traffic analyzer.
tcpflow	tcpflow	TCP flow recorder.
tcpick	tcpick	TCP stream sniffer and connection tracker.
tcpreplay	tcpreplay	Tool to replay saved tcpdump files at arbitrary speeds.
truecrack	truecrack	Bruteforce password cracker for TrueCrypt volumes.
undbx	undbx	Tool to extract, recover and undelete e-mail messages from .dbx files.
unhide	unhide	Forensic tool to find hidden processes and ports.
unrar	unrar	
upx-ucl	upx-ucl	Efficient live-compressor for executables.
vinetto	vinetto	Forensics tool to examine Thumbs.db files.
wce	wce	

Tool	Command(s)	Purpose
winregfs	<code>winregfs</code>	Windows registry FUSE filesystem.
wireshark	<code>wireshark</code>	Network traffic analyzer - graphical interface.
xmount	<code>xmount</code>	Tool for crossmounting between disk image formats.
xplico	<code>xplico</code>	Network Forensic Analysis Tool (NFAT).
yara	<code>yara</code>	Pattern matching swiss knife for malware researchers.

Fuzzing

Tool	Command(s)	Purpose
afl++	<code>afl++</code>	Instrumentation-driven fuzzer for binary formats.
sfuzz	<code>sfuzz</code>	Black Box testing utilities.
spike	<code>spike</code>	Network protocol fuzzer.
wfuzz	<code>wfuzz</code>	Web application bruteforcer.

Information Gathering

Tool	Command(s)	Purpose
Otrace	<code>0trace</code>	Traceroute tool that can run within an existing TCP connection.
arping	<code>arping</code>	Sends IP and/or ARP pings (to the MAC address).
braa	<code>braa</code>	Mass SNMP scanner.
dmitry	<code>dmitry</code>	Deepmagic Information Gathering Tool.
dnsenum	<code>dnsenum</code>	Tool to enumerate domain DNS information.
dnsmmap	<code>dnsmmap</code>	DNS domain name brute forcing tool.
dnsrecon	<code>dnsrecon</code>	Powerful DNS enumeration script.
dnstracer	<code>dnstracer</code>	Trace DNS queries to the source.
dnswalk	<code>dnswalk</code>	Checks dns zone information using nameserver lookups.
enum4linux	<code>enum4linux</code>	Enumerates info from Windows and Samba systems.
fierce	<code>fierce</code>	Domain DNS scanner.
firewalk	<code>firewalk</code>	Active reconnaissance network security tool.
fping	<code>fping</code>	Sends ICMP ECHO_REQUEST packets to network hosts.
fragrouter	<code>fragrouter</code>	IDS evasion toolkit.
ftester	<code>ftester</code>	Tool for testing firewalls and Intrusion Detection System (IDS).
hping3	<code>hping3</code>	Active Network Smashing Tool.
ike-scan	<code>ike-scan</code>	Discover and fingerprint IKE hosts (IPsec VPN Servers).
intrace	<code>intrace</code>	Traceroute-like application piggybacking on existing TCP connections.
irpas	<code>irpas</code>	
lbd	<code>lbd</code>	Load balancer detector.
legion	<code>legion</code>	Semi-automated network penetration testing tool.
maltego	<code>maltego</code>	
masscan	<code>masscan</code>	TCP port scanner.
metagoofil	<code>metagoofil</code>	Tool designed for extracting metadata of public documents.
nbtscan	<code>nbtscan</code>	Scan networks searching for NetBIOS information.
ncat	<code>ncat</code>	NMAP netcat reimplementatoin.
netdiscover	<code>netdiscover</code>	Active/passive network address scanner using ARP requests.
netmask	<code>netmask</code>	Helps determine network masks.
nmap	<code>nmap</code>	The Network Mapper.
onesixtyone	<code>onesixtyone</code>	Fast and simple SNMP scanner.
p0f	<code>p0f</code>	Passive OS fingerprinting tool.
qsslcaudit	<code>qsslcaudit</code>	Test SSL/TLS clients how secure they are.
recon-ng	<code>recon-ng</code>	Web Reconnaissance framework written in Python.
smbmap	<code>smbmap</code>	Handy SMB enumeration tool.
smtp-user-enum	<code>smtp-user-enum</code>	Username guessing tool for the SMTP service.
snmpcheck	<code>snmpcheck</code>	SNMP service enumeration tool.

Tool	Command(s)	Purpose
ssldump	<code>ssldump</code>	SSLv3/TLS network protocol analyzer.
ssllh	<code>ssllh</code>	Applicative protocol multiplexer.
sslsca	<code>sslsca</code>	Tests SSL/TLS enabled services to discover supported cipher suites.
sslyze	<code>sslyze</code>	Fast and full-featured SSL scanner.
swaks	<code>swaks</code>	SMTP command-line test tool.
thc-ipv6	<code>thc-ipv6</code>	The Hacker Choice's IPv6 Attack Toolkit.
theharvester	<code>theharvester</code>	Tool for gathering e-mail accounts and subdomain names from public sources.
tlssled	<code>tlssled</code>	Evaluates the security of a target SSL/TLS (HTTPS) server.
twofi	<code>twofi</code>	Twitter words of interest.
unicomscan	<code>unicomscan</code>	Userland distributed TCP/IP stack.
urlcrazy	<code>urlcrazy</code>	
wafw00f	<code>wafw00f</code>	Identify and fingerprint Web Application Firewall products.
zenmap	<code>zenmap</code>	

Passwords

Tool	Command(s)	Purpose
cewl	<code>cewl</code>	Custom word list generator.
chntpw	<code>chntpw</code>	NT SAM password recovery utility.
cisco-auditing-tool	<code>cisco-auditing-tool</code>	Scans Cisco routers for vulnerabilities.
cmospwd	<code>cmospwd</code>	Decrypt BIOS passwords from CMOS.
crackle	<code>crackle</code>	Crack and decrypt BLE encryption.
creddump7	<code>creddump7</code>	Python tool to extract credentials and secrets from Windows registry hives.
crunch	<code>crunch</code>	Tool for creating wordlist.
fcrackzip	<code>fcrackzip</code>	Password cracker for zip archives.
freerdp3-x11	<code>freerdp3-x11</code>	RDP client for Windows Terminal Services (X11 client transitional package).
gpp-decrypt	<code>gpp-decrypt</code>	Group Policy Preferences decrypter.
hash-identifier	<code>hash-identifier</code>	Tool to identify hash types.
hashcat	<code>hashcat</code>	World's fastest and most advanced password recovery utility.
hashcat-utils	<code>hashcat-utils</code>	Set of small utilities for advanced password cracking.
hashid	<code>hashid</code>	Identify the different types of hashes used to encrypt data.
hydra	<code>hydra</code>	Very fast network logon cracker.
hydra-gtk	<code>hydra-gtk</code>	Very fast network logon cracker - GTK+ based GUI.
john	<code>john</code>	Active password cracking tool.
johnny	<code>johnny</code>	GUI for John the Ripper.
maskprocessor	<code>maskprocessor</code>	High-performance word generator with a per-position configurable charset.
medusa	<code>medusa</code>	Fast, parallel, modular, login brute-forcer for network services.
mimikatz	<code>mimikatz</code>	Uses admin rights on Windows to display passwords in plaintext.
ncrack	<code>ncrack</code>	High-speed network authentication cracking tool.
onesixtyone	<code>onesixtyone</code>	Fast and simple SNMP scanner.
ophcrack	<code>ophcrack</code>	Microsoft Windows password cracker using rainbow tables (gui).
ophcrack-cli	<code>ophcrack-cli</code>	Microsoft Windows password cracker using rainbow tables (cmdline).
pack	<code>pack</code>	Password analysis and cracking kit.
pack2	<code>pack2</code>	Password analysis and cracking kit 2.
passing-the-hash	<code>passing-the-hash</code>	Patched tools to use password hashes as authentication input.
patator	<code>patator</code>	Multi-purpose brute-forcer.
pdfcrack	<code>pdfcrack</code>	PDF files password cracker.
pipal	<code>pipal</code>	Statistical analysis on password dumps.
polenum	<code>polenum</code>	Extracts the password policy from a Windows system.
rainbowcrack	<code>rainbowcrack</code>	Rainbow table password cracker.
rarcrack	<code>rarcrack</code>	Password cracker for rar archives.
rcracki-mt	<code>rcracki-mt</code>	Version of rcrack that supports hybrid and indexed tables.

Tool	Command(s)	Purpose
rsmangler	<code>rsmangler</code>	Wordlist mangling tool.
samdump2	<code>samdump2</code>	Dump Windows 2k/NT/XP password hashes.
seclists	<code>seclists</code>	Collection of multiple types of security lists.
sipcrack	<code>sipcrack</code>	SIP login dumper/cracker.
sipvicious	<code>sipvicious</code>	Tools to audit SIP based VoIP systems.
smbmap	<code>smbmap</code>	Handy SMB enumeration tool.
sqldict	<code>sqldict</code>	Dictionary attack tool for SQL Server.
statsprocessor	<code>statsprocessor</code>	Word generator based on per-position Markov chains.
sucrack	<code>sucrack</code>	Multithreaded su bruteforcer.
thc-pptp-bruter	<code>thc-pptp-bruter</code>	THC PPTP Brute Force.
truecrack	<code>truecrack</code>	Bruteforce password cracker for TrueCrypt volumes.
twofi	<code>twofi</code>	Twitter words of interest.
wordlists	<code>wordlists</code>	Contains the rockyou wordlist.

Post Exploitation

Tool	Command(s)	Purpose
cymothoa	<code>cymothoa</code>	Stealth backdooring tool.
dbd	<code>dbd</code>	Netcat clone with encryption.
dns2tcp	<code>dns2tcp</code>	TCP-over-DNS tunnel server and client.
exe2hexbat	<code>exe2hexbat</code>	Convert EXE to bat.
iodine	<code>iodine</code>	Tool for tunneling IPv4 data through a DNS server.
laudanum	<code>laudanum</code>	Collection of injectable web files.
mimikatz	<code>mimikatz</code>	Uses admin rights on Windows to display passwords in plaintext.
miredo	<code>miredo</code>	Teredo IPv6 tunneling through NATs.
nishang	<code>nishang</code>	Collection of PowerShell scripts and payloads.
powersploit	<code>powersploit</code>	PowerShell Post-Exploitation Framework.
proxychains4	<code>proxychains4</code>	Redirect connections through socks/http proxies (proxychains-ng).
proxytunnel	<code>proxytunnel</code>	Help SSH and other protocols through HTTP(S) proxies.
ptunnel	<code>ptunnel</code>	Tunnel TCP connections over ICMP packets.
pwnat	<code>pwnat</code>	NAT to NAT client-server communication.
sbd	<code>sbd</code>	Secure backdoor for Linux and Windows.
shellter	<code>shellter</code>	
sslh	<code>sslh</code>	Applicative protocol multiplexer.
stunnel4	<code>stunnel4</code>	Universal SSL tunnel for network daemons - compatibility package.
udptunnel	<code>udptunnel</code>	Tunnel UDP packets over a TCP connection.
veil	<code>veil</code>	Generates payloads to bypass anti-virus solutions.
webacoo	<code>webacoo</code>	Web backdoor cookie script kit.
weeveily	<code>weeveily</code>	Stealth tiny web shell.

Reverse Engineering

Tool	Command(s)	Purpose
apktool	<code>apktool</code>	Tool for reverse engineering Android apk files.
bytecode-viewer	<code>bytecode-viewer</code>	Java 8+ Jar & Android APK Reverse Engineering Suite.
clang	<code>clang</code>	C, C++ and Objective-C compiler (LLVM based), clang binary.
dex2jar	<code>dex2jar</code>	Tools to work with android .dex and java .class files.
edb-debugger	<code>edb-debugger</code>	Cross platform x86/x86-64 debugger.
jadx	<code>jadx</code>	Dex to Java decompiler.
javasnoop	<code>javasnoop</code>	Intercept Java applications locally.
jd-gui	<code>jd-gui</code>	GUI Java .class decompiler.
metasploit-framework	<code>metasploit-framework</code>	Framework for exploit development and vulnerability research.
ollydbg	<code>ollydbg</code>	32-bit assembler level analysing debugger.

Tool	Command(s)	Purpose
radare2	<code>radare2</code>	Free and advanced command line hexadecimal editor.
recstudio	<code>recstudio</code>	
rizin	<code>rizin</code>	Reverse engineering framework and command-line toolset.
rizin-cutter	<code>rizin-cutter</code>	Reverse engineering platform powered by rizin.
rz-ghidra	<code>rz-ghidra</code>	Ghidra decompiler and sleigh disassembler for rizin.

Sniffing Spoofing

Tool	Command(s)	Purpose
above	<code>above</code>	Network security sniffer for finding vulnerabilities in the network.
bettercap	<code>bettercap</code>	Complete, modular, portable and easily extensible MITM framework.
darkstat	<code>darkstat</code>	Network traffic analyzer.
dnscraf	<code>dnscraf</code>	DNS proxy for penetration testers.
driftnet	<code>driftnet</code>	Picks out and displays images from network traffic.
dsniff	<code>dsniff</code>	Various tools to sniff network traffic for cleartext insecurities.
ettercap-graphical	<code>ettercap-graphical</code>	Ettercap GUI-enabled executable.
ferret-sidejack	<code>ferret-sidejack</code>	Monitors data and extracts interesting data.
fiked	<code>fiked</code>	Cisco VPN attack tool.
hamster-sidejack	<code>hamster-sidejack</code>	Sidejacking tool.
hexinject	<code>hexinject</code>	Versatile packet injector and sniffer.
isr-evilgrade	<code>isr-evilgrade</code>	Evilgrade framework.
macchanger	<code>macchanger</code>	Utility for manipulating the MAC address of network interfaces.
mitmproxy	<code>mitmproxy</code>	SSL-capable man-in-the-middle HTTP proxy.
netsniff-ng	<code>netsniff-ng</code>	Linux network packet sniffer toolkit.
rebind	<code>rebind</code>	DNS rebinding tool.
responder	<code>responder</code>	LLMNR/NBT-NS/mDNS Poisoner.
sniffjoke	<code>sniffjoke</code>	Transparent TCP connection scrambler.
sslsniff	<code>sslsniff</code>	SSL/TLS man-in-the-middle attack tool.
sslsplit	<code>sslsplit</code>	Transparent and scalable SSL/TLS interception.
tcpflow	<code>tcpflow</code>	TCP flow recorder.
tcpreplay	<code>tcpreplay</code>	Tool to replay saved tcpdump files at arbitrary speeds.
wifi-honey	<code>wifi-honey</code>	Wi-Fi honeypot.
wireshark	<code>wireshark</code>	Network traffic analyzer - graphical interface.

Vulnerability

Tool	Command(s)	Purpose
afl++	<code>afl++</code>	Instrumentation-driven fuzzer for binary formats.
bed	<code>bed</code>	A network protocol fuzzer.
cisco-auditing-tool	<code>cisco-auditing-tool</code>	Scans Cisco routers for vulnerabilities.
cisco-global-exploiter	<code>cisco-global-exploiter</code>	Simple and fast Cisco exploitation tool.
cisco-ocs	<code>cisco-ocs</code>	Mass Cisco scanner.
cisco-torch	<code>cisco-torch</code>	Cisco device scanner.
copy-router-config	<code>copy-router-config</code>	Copies Cisco configs via SNMP.
dhcpiq	<code>dhcpiq</code>	DHCP exhaustion script using scapy network library.
enumiax	<code>enumiax</code>	IAX protocol username enumerator.
gvm	<code>gvm</code>	Remote network security auditor - metapackage and useful scripts.
iaxflood	<code>iaxflood</code>	VoIP flooder tool.
inviteflood	<code>inviteflood</code>	SIP/SDP INVITE message flooding over UDP/IP.
legion	<code>legion</code>	Semi-automated network penetration testing tool.
lynis	<code>lynis</code>	Security auditing tool for Unix based systems.
nikto	<code>nikto</code>	
nmap	<code>nmap</code>	The Network Mapper.

Tool	Command(s)	Purpose
ohrwurm	<code>ohrwurm</code>	RTP fuzzer.
peass	<code>peass</code>	Privilege Escalation Awesome Scripts SUITE.
protos-sip	<code>protos-sip</code>	SIP test suite.
rtpbreak	<code>rtpbreak</code>	Detects, reconstructs, and analyzes RTP sessions.
rtpflood	<code>rtpflood</code>	Tool to flood any RTP device.
rtpinsertsound	<code>rtpinsertsound</code>	Inserts audio into a specified stream.
rtpmixsound	<code>rtpmixsound</code>	Mixes pre-recorded audio in real-time.
sctpscan	<code>sctpscan</code>	SCTP network scanner for discovery and security.
sfuzz	<code>sfuzz</code>	Black Box testing utilities.
siege	<code>siege</code>	HTTP regression testing and benchmarking utility.
siparmyknife	<code>siparmyknife</code>	SIP fuzzing tool.
sipp	<code>sipp</code>	Traffic generator for the SIP protocol.
sipsak	<code>sipsak</code>	SIP Swiss army knife.
sipvicious	<code>sipvicious</code>	Tools to audit SIP based VoIP systems.
slowhttptest	<code>slowhttptest</code>	Application layer Denial of Service attacks simulation tool.
spike	<code>spike</code>	Network protocol fuzzer.
t50	<code>t50</code>	Multi-protocol packet injector tool.
thc-ssl-dos	<code>thc-ssl-dos</code>	Stress tester for the SSL handshake.
unix-privesc-check	<code>unix-privesc-check</code>	Script to check for simple privilege escalation vectors.
voiphopper	<code>voiphopper</code>	Runs a VLAN hop security test.
yersinia	<code>yersinia</code>	Network vulnerabilities check software.

Web

Tool	Command(s)	Purpose
apache-users	<code>apache-users</code>	Enumerate usernames on systems with Apache UserDir module.
apache2	<code>apache2</code>	Apache HTTP Server.
beef-xss	<code>beef-xss</code>	Browser Exploitation Framework (BeEF).
burpsuite	<code>burpsuite</code>	Platform for security testing of web applications.
cadaver	<code>cadaver</code>	Command-line WebDAV client.
commix	<code>commix</code>	Automated All-in-One OS Command Injection and Exploitation Tool.
cutycapt	<code>cutycapt</code>	Utility to capture WebKit's rendering of a web page.
davtest	<code>davtest</code>	Testing tool for WebDAV servers.
default-mysql-server	<code>default-mysql-server</code>	MySQL database server binaries and system database setup (metapackage).
dirb	<code>dirb</code>	URL bruteforcing tool.
dirbuster	<code>dirbuster</code>	Web server directory brute-forcer.
dotdotpwn	<code>dotdotpwn</code>	Directory Traversal Fuzzer.
eyewitness	<code>eyewitness</code>	Rapid web application triage tool.
ferret-sidejack	<code>ferret-sidejack</code>	Monitors data and extracts interesting data.
ftester	<code>ftester</code>	Tool for testing firewalls and Intrusion Detection System (IDS).
hakrawler	<code>hakrawler</code>	Web crawler designed for easy, quick discovery of endpoints and assets.
hamster-sidejack	<code>hamster-sidejack</code>	Sidejacking tool.
heartleech	<code>heartleech</code>	Scanner detecting systems vulnerable to the heartbleed OpenSSL bug.
httpprint	<code>httpprint</code>	
httrack	<code>httrack</code>	Copy websites to your computer (Offline browser).
hydra	<code>hydra</code>	Very fast network logon cracker.
hydra-gtk	<code>hydra-gtk</code>	Very fast network logon cracker - GTK+ based GUI.
jboss-autopwn	<code>jboss-autopwn</code>	JBoss script for obtaining remote shell access.
joomscan	<code>joomscan</code>	OWASP Joomla Vulnerability Scanner Project.
jsql-injection	<code>jsql-injection</code>	Java tool for automatic database injection.
laudanum	<code>laudanum</code>	Collection of injectable web files.
lbd	<code>lbd</code>	Load balancer detector.

Tool	Command(s)	Purpose
maltego	maltego	
medusa	medusa	Fast, parallel, modular, login brute-forcer for network services.
mitmproxy	mitmproxy	SSL-capable man-in-the-middle HTTP proxy.
ncrack	ncrack	High-speed network authentication cracking tool.
nikto	nikto	
nishang	nishang	Collection of PowerShell scripts and payloads.
nmap	nmap	The Network Mapper.
osscanner	osscanner	Oracle assessment framework.
owasp-mantra-ff	owasp-mantra-ff	
padbuster	padbuster	Script for performing Padding Oracle attacks.
paros	paros	Web application proxy.
patator	patator	Multi-purpose brute-forcer.
php	php	Server-side, HTML-embedded scripting language (default).
php-mysql	php-mysql	MySQL module for PHP [default].
proxychains4	proxychains4	Redirect connections through socks/http proxies (proxychains-ng).
proxytunnel	proxytunnel	Help SSH and other protocols through HTTP(S) proxies.
qssllcaudit	qssllcaudit	Test SSL/TLS clients how secure they are.
redsocks	redsocks	Arbitrary TCP connection redirector to a SOCKS or HTTPS proxy server.
sidguesser	sidguesser	Guesses sids against an Oracle database.
siege	siege	HTTP regression testing and benchmarking utility.
skipfish	skipfish	Fully automated, active web application security reconnaissance tool.
slowhttptest	slowhttptest	Application layer Denial of Service attacks simulation tool.
sqldict	sqldict	Dictionary attack tool for SQL Server.
sqlitebrowser	sqlitebrowser	GUI editor for SQLite databases.
sqlmap	sqlmap	Automatic SQL injection tool.
sqlninja	sqlninja	SQL server injection and takeover tool.
sqlsus	sqlsus	MySQL injection tool.
ssldump	ssldump	SSLv3/TLS network protocol analyzer.
ssllh	ssllh	Applicative protocol multiplexer.
sslscan	sslscan	Tests SSL/TLS enabled services to discover supported cipher suites.
sslsniff	sslsniff	SSL/TLS man-in-the-middle attack tool.
sslsplit	sslsplit	Transparent and scalable SSL/TLS interception.
sslyze	sslyze	Fast and full-featured SSL scanner.
stunnel4	stunnel4	Universal SSL tunnel for network daemons - compatibility package.
thc-ssl-dos	thc-ssl-dos	Stress tester for the SSL handshake.
tlssled	tlssled	Evaluates the security of a target SSL/TLS (HTTPS) server.
tnscmd10g	tnscmd10g	Tool to prod the oracle tnlsnr process.
uniscan	uniscan	LFI, RFI, and RCE vulnerability scanner.
wafw00f	wafw00f	Identify and fingerprint Web Application Firewall products.
wapiti	wapiti	Web application vulnerability scanner.
watobo	watobo	Semi-automated web application scanner.
webacoo	webacoo	Web backdoor cookie script kit.
webscarab	webscarab	Web application review tool.
webshells	webshells	Collection of webshells.
weevely	weevely	Stealth tiny web shell.
wfuzz	wfuzz	Web application bruteforcer.
whatweb	whatweb	Next generation web scanner.
wireshark	wireshark	Network traffic analyzer - graphical interface.
wpscan	wpscan	
xsser	xsser	XSS testing framework.
zapproxy	zapproxy	Testing tool for finding vulnerabilities in web applications.

All Packages (choco)

Tool	Command(s)	Purpose
010editor	<code>010editor</code>	Professional text and hex editor with Binary Templates technology.
7zip	<code>7zip</code>	7-Zip file archiver with a high compression ratio.
advanced-installer	<code>advanced-installer</code>	Advanced Installer is a Windows installer authoring tool that can be used to analyze MSI files.
angr	<code>angr</code>	Angr is a multi-architecture binary analysis toolkit providing features like disassembly, IR lifting, program instr...
apimonitor	<code>apimonitor</code>	API Monitor lets you monitor and control API calls made by applications and services.
apktool	<code>apktool</code>	Tool for reverse engineering Android apk files.
asar	<code>asar</code>	Asar decompresses .asar archives.
autoit-ripper	<code>autoit-ripper</code>	Extracts compiled Autolt scripts from PE executables.
binaryninja	<code>binaryninja</code>	Binary Ninja is an interactive decompiler, disassembler, debugger, and binary analysis platform built by reverse en...
bindiff	<code>bindiff</code>	A comparison tool for binary files that assists in quickly finding differences and similarities in disassembled code.
blobrunner	<code>blobrunner</code>	BlobRunner is a simple tool to quickly debug shellcode extracted during malware analysis.
blobrunner64	<code>blobrunner64</code>	BlobRunner is a simple tool to quickly debug shellcode extracted during malware analysis.
bytecodeviewer	<code>bytecodeviewer</code>	A lightweight user-friendly Java/Android Bytecode Viewer, Decompiler and more.
capa	<code>capa</code>	Capa detects capabilities in executable files. You run it against a PE file or shellcode and it tells you what it t...
capa-explorer-web	<code>capa-explorer-web</code>	Web interface for exploring and understanding capa results.
chrome.extensions	<code>chrome.extensions</code>	A package for multiple useful chrome extensions from the Chrome webstore.
cmdr	<code>cmdr</code>	Metapackage for cmdr.
codetrack	<code>codetrack</code>	A free .NET Performance Profile and Execution Analyzer.
cryptotester	<code>cryptotester</code>	Utility tool for performing cryptanalysis with a focus on ransomware cryptography.
cutter	<code>cutter</code>	Disconnect routed IP connections.
cyberchef	<code>cyberchef</code>	Cyber Swiss Army Knife.
cygwin	<code>cygwin</code>	Wrapper for cygwin and useful cygwin packages.
de4dot-cex	<code>de4dot-cex</code>	A de4dot fork with full support for vanilla ConfuserEx.
dependencywalker	<code>dependencywalker</code>	Scans PE files and builds a hierarchical tree diagram of all dependent modules.
dex2jar	<code>dex2jar</code>	Tools to work with android .dex and java .class files.
didier-stevens-beta	<code>didier-stevens-beta</code>	Beta versions of Didier Stevens's software.
didier-stevens-suite	<code>didier-stevens-suite</code>	Tools collection by Didier Stevens.
die	<code>die</code>	Detect It Easy, or abbreviated "DIE" is a program for determining types of files.
dll-to-exe	<code>dll-to-exe</code>	Converts a DLL into a ready-to-use EXE.
dnlib	<code>dnlib</code>	.NET module/assembly reader/writer library.
dnspyex	<code>dnspyex</code>	DnSpyEx is a unofficial continuation of the dnSpy project which is a debugger and .NET assembly editor. You can use...
dotdumper	<code>dotdumper</code>	An automatic unpacker and logger for DotNet Framework targeting files.
dotnet3.5	<code>dotnet3.5</code>	
exeinfope	<code>exeinfope</code>	Displays metadata for a variety of file types and identifies many executable packers.
explorersuite	<code>explorersuite</code>	A suite of tools including CFF Explorer and a process viewer.
extreme_dumper	<code>extreme_dumper</code>	.NET Assembly Dumper from memory of processes.
ezviewer	<code>ezviewer</code>	Standalone, zero dependency viewer for .doc, .docx, .xls, .xlsx, .txt, .log, .rtf, .otd, .htm, .html, .mht, .csv, a...
fakenet-ng	<code>fakenet-ng</code>	FakeNet-NG is a next generation dynamic network analysis tool for malware analysts and penetration testers.
fiddler	<code>fiddler</code>	Intercepts, decrypts, and analyzes HTTPS traffic.
file	<code>file</code>	Recognize the type of data in a file using "magic" numbers.
floss	<code>floss</code>	FLOSS uses advanced static analysis techniques to automatically deobfuscate strings from malware binaries. You can...
garbageman	<code>garbageman</code>	A set of tools designed for .NET heap analysis.
ghidra	<code>ghidra</code>	Software Reverse Engineering Framework.
goresym	<code>goresym</code>	Go symbol recovery tool.
gostringungarbler	<code>gostringungarbler</code>	GoStringUngarbler deobfuscates strings in Go binaries obfuscated by garble.
hashmyfiles	<code>hashmyfiles</code>	HashMyFiles is small utility that allows you to calculate the MD5 and SHA1 hashes of one or more files in your syst...
hollowshunter	<code>hollowshunter</code>	Scans all running processes. Recognizes and dumps a variety of potentially malicious implants (replaced/implanted P...
hxd	<code>hxd</code>	Freeware hex editor.
ida.plugin.capa	<code>ida.plugin.capa</code>	Capa explorer is an IDAPython plugin that integrates capa with IDA Pro.
ida.plugin.comida	<code>ida.plugin.comida</code>	IDA Plugin that help analyzing modules using COM.

Tool	Command(s)	Purpose
ida.plugin.delphihelper	<code>ida.plugin.delphihelper</code>	DelphiHelper.
ida.plugin.dereferencing	<code>ida.plugin.dereferencing</code>	IDA Pro plugin that implements new registers and stack views.
ida.plugin.diaphora	<code>ida.plugin.diaphora</code>	Diaphora is a program diffing IDA plugin.
ida.plugin.flare	<code>ida.plugin.flare</code>	IDA Pro plugins used by the FLARE team.
ida.plugin.flare-emu	<code>ida.plugin.flare-emu</code>	A user friendly scriptable emulation framework that supports multiple binary analysis tools.
ida.plugin.hashdb	<code>ida.plugin.hashdb</code>	Malware string hash lookup plugin for IDA Pro.
ida.plugin.hrtng	<code>ida.plugin.hrtng</code>	IDA Pro plugin with features such as decryption, automation, deobfuscation, patching, lib code recognition and pseu...
ida.plugin.ifl	<code>ida.plugin.ifl</code>	Interactive Functions List IDA Pro plugin.
ida.plugin.xray	<code>ida.plugin.xray</code>	Hexrays decompiler plugin that colorizes and filters the decompiler's output based on regular expressions.
ida.plugin.xrefer	<code>ida.plugin.xrefer</code>	Custom navigation interface within IDA.
idafree	<code>idafree</code>	Free version of IDA, a powerful Interactive DisAssembler and debugger.
idr	<code>idr</code>	Interactive Delphi Reconstructor.
ifpstools	<code>ifpstools</code>	IFPSTools.NET: tools for working with RemObject PascalScript compiled bytecode files.
ilspy	<code>ilspy</code>	ILSpy is a .NET assembly browser and decompiler.
innoextract	<code>innoextract</code>	Tool for extracting data from an Inno Setup installer.
innounp	<code>innounp</code>	Unpacker for Inno Setup installers.
internet_detector	<code>internet_detector</code>	Tool that changes the background and a taskbar icon if it detects internet connectivity.
ipython	<code>ipython</code>	A powerful interactive Python shell.
isd	<code>isd</code>	Inno Setup Decompiler.
js-beautify	<code>js-beautify</code>	JavaScript beautifier and deobfuscator.
js-deobfuscator	<code>js-deobfuscator</code>	Deobfuscator to remove common JS obfuscation techniques.
keystone	<code>keystone</code>	OpenStack identity service.
libraries.python3	<code>libraries.python3</code>	Metapackage to install common Python 3.9 libraries.
magika	<code>magika</code>	Magika is an AI powered file type detection tool that uses deep learning to provide accurate detection.
malware-jail	<code>malware-jail</code>	Sandbox for semi-automatic Javascript malware analysis, deobfuscation and payload extraction.
map	<code>map</code>	Handful of small utility type applications useful for analyzing malicious code.
microsoft-office	<code>microsoft-office</code>	Microsoft Office ProPlusRetail.
nasm	<code>nasm</code>	General-purpose x86 assembler.
net-reactor-slayer	<code>net-reactor-slayer</code>	NETReactorSlayer is an open source (GPLv3) deobfuscator and unpacker for Eziriz .NET Reactor.
nmap	<code>nmap</code>	The Network Mapper.
notepadplusplus	<code>notepadplusplus</code>	Wrapper for Notepad++.
notepadpp.plugin.compare	<code>notepadpp.plugin.compare</code>	ComparePlus plugin for Notepad++.
notepadpp.plugin.jstool	<code>notepadpp.plugin.jstool</code>	A JavaScript (JSON) tool for Notepad++ (formerly JSMinNpp).
notepadpp.plugin.xmltools	<code>notepadpp.plugin.xmltools</code>	XML Tools plugin for Notepad++.
obfuscator-io-deobfuscator	<code>obfuscator-io-deobfuscator</code>	A deobfuscator for scripts obfuscated by Obfuscator.io.
offvis	<code>offvis</code>	The Microsoft Office Visualization Tool (OffVis) is a tool from Microsoft that helps understanding the Microsoft Of...
onenoteanalyzer	<code>onenoteanalyzer</code>	A C# based tool for analyzing malicious OneNote documents.
pdbresym	<code>pdbresym</code>	Download PDBs.
pdfstreamdumper	<code>pdfstreamdumper</code>	PDFStreamDumper is a free, open source tool to analyze malicious PDF documents.
pe_unmapper	<code>pe_unmapper</code>	Small tool to convert between the PE alignments (raw and virtual).
pebear	<code>pebear</code>	Delivers fast and flexible "first view" for malware analysts.
peid	<code>peid</code>	PEiD detects most common packers, cryptors and compilers for PE files.
pesieve	<code>pesieve</code>	Pe-sieve recognizes and dumps variety of implants within the scanned process.
pestudio	<code>pestudio</code>	The goal of pestudio is to spot artifacts of executable files in order to ease and accelerate Malware Initial Asses...
pkg-unpacker	<code>pkg-unpacker</code>	Unpacker for pkg applications.
pma-labs	<code>pma-labs</code>	Binaries for the book Practical Malware Analysis.
procdot	<code>procdot</code>	Creates visual graphs from procmon output.
processdump	<code>processdump</code>	Process Dump is a Windows reverse-engineering command-line tool to dump malware memory components back to disk for...
psnotify	<code>psnotify</code>	A POC tool to fight .NET anti-dumping tricks.
pycdas	<code>pycdas</code>	Python byte-code disassembler.

Tool	Command(s)	Purpose
pycdc	<code>pycdc</code>	Python decompiler.
pylingual	<code>pylingual</code>	Python decompiler for modern Python versions.
rat-king-parser	<code>rat-king-parser</code>	Multi-family RAT config parser/extractor.
recaf	<code>recaf</code>	Java bytecode editor.
reg_export	<code>reg_export</code>	A CLI that exports the raw content of a registry value to a file.
regcool	<code>regcool</code>	In addition to all the features that you can find in RegEdit and RegEdt32, RegCool adds many powerful features that...
regshot	<code>regshot</code>	Regshot is a small, free and open-source registry compare utility that allows you to quickly take a snapshot of you...
resourcehacker	<code>resourcehacker</code>	Resource Hacker is a resource editor for 32bit and 64bit Windows applications.
rundotnetdll	<code>rundotnetdll</code>	A simple utility to list all methods of a given .NET Assembly and to invoke them.
scdbg	<code>scdbg</code>	Scdbg is an emulation based shellcode API logger and debugger.
sclauncher	<code>sclauncher</code>	A small program to load 32-bit shellcode and allow for execution or debugging. Can also output PE files from shellc...
sclauncher64	<code>sclauncher64</code>	A small program to load 64-bit shellcode and allow for execution or debugging. Can also output PE files from shellc...
sfextract	<code>sfextract</code>	Command-line utility to extract files from single file bundles in .NET.
shellcode_launcher	<code>shellcode_launcher</code>	Shellcode launcher utility.
sysinternals	<code>sysinternals</code>	Sysinternals suite.
systeminformer	<code>systeminformer</code>	A free, powerful, multi-purpose tool that helps you monitor system resources, debug software and detect malware.
ttd	<code>ttd</code>	Time travel debugging command line utility.
uncompyle6	<code>uncompyle6</code>	A decompiler for Python 1.0-3.8.
uniextract2	<code>uniextract2</code>	Universal Extractor 2 is an unofficial updated and extended version of the original UniExtract by Jared Breland.
unpyc3	<code>unpyc3</code>	A decompiler for Python 3.7+.
upx	<code>upx</code>	UPX is a free, secure, portable, extendable, high-performance executable packer for several executable formats.
vb-decompiler-lite	<code>vb-decompiler-lite</code>	VB Decompiler is a decompiler for Visual Basic, VB.NET and C# applications.
vbdec	<code>vbdec</code>	VBDec works as a VB6 disassembler, PCode debugger, structure viewer for all vb6 executables, and can generate IDA s...
vcbuildtools	<code>vcbuildtools</code>	Metapackage that requires the dependencies below: - visualstudio2017buildtools - visualstudio2017-workload-vctools.
vcredist-all	<code>vcredist-all</code>	
vscode	<code>vscode</code>	VSCode is a modern, open-source code editor.
vscode.extension.jupyter	<code>vscode.extension.jupyter</code>	Jupyter notebook support, interactive programming and computing that supports Intellisense, debugging and more.
vscode.extension.python	<code>vscode.extension.python</code>	Python language support with extension access points for IntelliSense (Pylance), Debugging (Python Debugger), linti...
windbg	<code>windbg</code>	WinDbg is a debugger that can be used to analyze crash dumps, debug live user-mode and kernel-mode code, and examin...
windows-terminal	<code>windows-terminal</code>	Windows Terminal is a new, modern, feature-rich, productive terminal application for command-line users.
wireshark	<code>wireshark</code>	Network traffic analyzer - graphical interface.
x64dbg	<code>x64dbg</code>	An open-source x64/x32 debugger for Windows.
x64dbg.plugin.dbgchild	<code>x64dbg.plugin.dbgchild</code>	DbgChild is an x64dbg plugin to automatically attach to spawned child processes.
x64dbg.plugin.ollydumpex	<code>x64dbg.plugin.ollydumpex</code>	This plugin is process memory dumper for OllyDbg and Immunity Debugger. OllyDumpEx = OllyDump + PE Dumper - obsolet...
x64dbg.plugin.scyllahide	<code>x64dbg.plugin.scyllahide</code>	ScyllaHide is an advanced open-source x64/x86 user mode Anti-Anti-Debug library.
x64dbg.plugin.x64dbgpy	<code>x64dbg.plugin.x64dbgpy</code>	Automating x64dbg using Python.
yara	<code>yara</code>	Pattern matching swiss knife for malware researchers.

Forensic Packages

Tool	Command(s)	Purpose
absent	<code>absent</code>	
aeskeyfind	<code>aeskeyfind</code>	Tool for locating AES keys in a captured memory image.
afflib-tools	<code>afflib-tools</code>	Advanced Forensics Format Library (utilities).
aircrack-ng	<code>aircrack-ng</code>	Wireless WEP/WPA cracking utilities.
android-sdk-platform-tools	<code>android-sdk-platform-tools</code>	Tools for interacting with an Android platform.
arp-scan	<code>arp-scan</code>	Arp scanning and fingerprinting tool.
autopsy	<code>autopsy</code>	Graphical interface to SleuthKit.
avfs	<code>avfs</code>	Virtual filesystem to access archives, disk images, remote locations.
aws-cli	<code>aws-cli</code>	
bless	<code>bless</code>	A full featured hexadecimal editor.

Tool	Command(s)	Purpose
blt	<code>blt</code>	Graphics extension library for Tcl/Tk - run-time.
bulk-extractor	<code>bulk-extractor</code>	
cabextract	<code>cabextract</code>	Microsoft Cabinet file unpacker.
ccrypt	<code>ccrypt</code>	Secure encryption and decryption of files and streams.
chromium-browser	<code>chromium-browser</code>	Transitional package - chromium-browser -> chromium snap.
clamav	<code>clamav</code>	Anti-virus utility for Unix - command-line interface.
claude-code	<code>claude-code</code>	
cmospwd	<code>cmospwd</code>	Decrypt BIOS passwords from CMOS.
cryptcat	<code>cryptcat</code>	Lightweight version netcat extended with twofish encryption.
cryptsetup	<code>cryptsetup</code>	Disk encryption support - startup scripts.
<code>dc3dd</code>	<code>dc3dd</code>	Patched version of GNU dd with forensic features.
<code>dcfldd</code>	<code>dcfldd</code>	Enhanced version of dd for forensics and security.
default-jre	<code>default-jre</code>	Standard Java or Java compatible Runtime.
disktype	<code>disktype</code>	Detection of content format of a disk or disk image.
dislocker	<code>dislocker</code>	Read/write encrypted BitLocker volumes.
docker	<code>docker</code>	Transitional package.
dos2unix	<code>dos2unix</code>	Convert text file line endings between CRLF and LF.
dotnet	<code>dotnet</code>	
driftnet	<code>driftnet</code>	Picks out and displays images from network traffic.
dsniff	<code>dsniff</code>	Various tools to sniff network traffic for cleartext insecurities.
e2fsprogs	<code>e2fsprogs</code>	Ext2/ext3/ext4 file system utilities.
ent	<code>ent</code>	Pseudorandom number sequence test program.
epic5	<code>epic5</code>	Epic irc client, version 5.
etherape	<code>etherape</code>	Graphical network monitor.
ettercap-graphical	<code>ettercap-graphical</code>	Ettercap GUI-enabled executable.
ewf-tools	<code>ewf-tools</code>	Collection of tools for reading and writing EWF files.
exfat-extras	<code>exfat-extras</code>	
exfat-fuse	<code>exfat-fuse</code>	Read and write exFAT driver for FUSE.
exif	<code>exif</code>	Command-line utility to show EXIF information in JPEG files.
extundelete	<code>extundelete</code>	Utility to recover deleted files from ext3/ext4 partition.
fdupes	<code>fdupes</code>	Identifies duplicate files within given directories.
feh	<code>feh</code>	Imlib2 based image viewer.
<code>file</code>	<code>file</code>	Recognize the type of data in a file using "magic" numbers.
flex	<code>flex</code>	Fast lexical analyzer generator.
<code>foremost</code>	<code>foremost</code>	Forensic program to recover lost files.
gawk	<code>gawk</code>	GNU awk, a pattern scanning and processing language.
gdb	<code>gdb</code>	GNU Debugger.
gddrescue	<code>gddrescue</code>	GNU data recovery tool.
ghex	<code>ghex</code>	GNOME Hex editor for files.
graphviz	<code>graphviz</code>	Rich set of graph drawing tools.
grepcidr	<code>grepcidr</code>	Filter IP addresses matching IPv4/IPv6 CIDR/network specification.
gthumb	<code>gthumb</code>	Image viewer and browser.
gzrt	<code>gzrt</code>	Gzip recovery toolkit.
hashdeep	<code>hashdeep</code>	Recursively compute hashsums or piecewise hashings.
hexedit	<code>hexedit</code>	Viewer and editor in hexadecimal or ASCII for files or devices.
<code>hydra</code>	<code>hydra</code>	Very fast network logon cracker.
hydra-gtk	<code>hydra-gtk</code>	Very fast network logon cracker - GTK+ based GUI.
init	<code>init</code>	Metapackage ensuring an init system is installed.
ipython3	<code>ipython3</code>	Enhanced interactive Python 3 shell.
jq	<code>jq</code>	Lightweight and flexible command-line JSON processor.
kdiff3	<code>kdiff3</code>	Compares and merges 2 or 3 files or directories.

Tool	Command(s)	Purpose
kpartx	kpartx	Create device mappings for partitions.
lft	lft	Layer-four traceroute.
lvm2	lvm2	Linux Logical Volume Manager.
magnus	magnus	Very simple screen magnifier.
mdadm	mdadm	Tool for managing Linux MD devices (software RAID).
mtd-utils	mtd-utils	Memory Technology Device Utilities.
nbd-client	nbd-client	Network Block Device protocol - client.
nbtscan	nbtscan	Scan networks searching for NetBIOS information.
netcat	netcat	TCP/IP swiss army knife -- transitional package.
netpbm	netpbm	Graphics conversion tools between image formats.
netsted	netsted	Network packet-altering stream editor.
netwox	netwox	Networking utilities.
nfdump	nfdump	Netflow capture daemon.
ngrep	ngrep	Grep for network traffic.
nikto	nikto	
ntfs-3g	ntfs-3g	Read/write NTFS driver for FUSE.
okular	okular	Universal document viewer.
onboard	onboard	Simple On-screen Keyboard.
open-iscsi	open-iscsi	ISCSI initiator tools.
openjdk	openjdk	Metapackage for OpenJDK to ensure all packages use the same OpenJDK version.
ophcrack	ophcrack	Microsoft Windows password cracker using rainbow tables (gui).
ophcrack-cli	ophcrack-cli	Microsoft Windows password cracker using rainbow tables (cmdline).
orca	orca	Scriptable screen reader.
outguess	outguess	Universal steganographic tool.
p0f	p0f	Passive OS fingerprinting tool.
p7zip-full	p7zip-full	7z and 7za file archivers with high compression ratio.
patch	patch	Apply a diff file to an original.
pdftk-java	pdftk-java	Pdftk port to java - a tool for manipulating PDF documents.
perl	perl	Larry Wall's Practical Extraction and Report Language.
pev	pev	Text-based tool to analyze PE files.
pff-tools	pff-tools	Utilities for MS Outlook PAB, PST and OST files.
phonon	phonon	
pkg-config	pkg-config	Manage compile and link flags for libraries (transitional package).
plaso-tools	plaso-tools	
powershell	powershell	PowerShell is an automation and configuration management platform.
pst-utils	pst-utils	Tools for reading Microsoft Outlook PST files.
pv	pv	Shell pipeline element to meter data passing through.
python-flowgrep	python-flowgrep	
python3-debian	python3-debian	Python 3 modules to work with Debian-related data formats.
python3-dfvfs	python3-dfvfs	Digital Forensics Virtual File System.
python3-fuse	python3-fuse	Python bindings for FUSE (Filesystems in Userspace) (Python 3 package).
python3-keyrings-alt	python3-keyrings-alt	
python3-m2crypto	python3-m2crypto	Python wrapper for the OpenSSL library (Python 3 modules).
python3-magic	python3-magic	Python3 interface to the libmagic file type identification library.
python3-pefile	python3-pefile	Portable Executable (PE) parsing module for Python.
python3-plaso	python3-plaso	Super timeline all the things -- Python 3.
python3-pypff	python3-pypff	Python 3 bindings for libpff.
python3-pyqt5	python3-pyqt5	Python 3 bindings for Qt5.
python3-pytsk3	python3-pytsk3	
python3-redis	python3-redis	Python bindings for Redis, a persistent key-value database.
python3-setuptools-rust	python3-setuptools-rust	Setuptools Rust extension plugin.

Tool	Command(s)	Purpose
python3-tk	python3-tk	Tkinter - Writing Tk applications with Python 3.x.
python3-tsk	python3-tsk	Python Bindings for The Sleuth Kit.
python3-virtualenv	python3-virtualenv	Python virtual environment creator.
python3-wxgtk4	python3-wxgtk4	
python3-xlswriter	python3-xlswriter	Python 3 module for creating Excel XLSX files.
python3-yara	python3-yara	Python 3 bindings for YARA.
qemu	qemu	Fast processor emulator, dummy package.
qemu-utils	qemu-utils	QEMU utilities.
radare2	radare2	Free and advanced command line hexadecimal editor.
rar	rar	
rsakeyfind	rsakeyfind	Locates BER-encoded RSA private keys in memory images.
safecopy	safecopy	Data recovery tool for problematic or damaged media.
samdump2	samdump2	Dump Windows 2k/NT/XP password hashes.
scalpel	scalpel	Fast filesystem-independent file recovery.
silversearcher-ag	silversearcher-ag	Very fast grep-like program, alternative to ack.
sleuthkit	sleuthkit	Tools for forensics analysis on volume and filesystem data.
socat	socat	Multipurpose relay for bidirectional data transfer.
squashfs-tools	squashfs-tools	Tool to create and append to squashfs filesystems.
ssdeep	ssdeep	Recursive piecewise hashing tool.
ssldump	ssldump	SSLv3/TLS network protocol analyzer.
sslsniff	sslsniff	SSL/TLS man-in-the-middle attack tool.
stunnel4	stunnel4	Universal SSL tunnel for network daemons - compatibility package.
swig	swig	Generate scripting interfaces to C/C++ code.
tcl	tcl	Tool Command Language (default version) - shell.
tcpflow	tcpflow	TCP flow recorder.
tcpick	tcpick	TCP stream sniffer and connection tracker.
tcpreplay	tcpreplay	Tool to replay saved tcpdump files at arbitrary speeds.
tcpslice	tcpslice	Extract pieces of and/or glue together tcpdump files.
tcpstat	tcpstat	Network interface statistics reporting tool.
tcptrace	tcptrace	Tool for analyzing tcpdump output.
tcptrack	tcptrack	TCP connection tracker, with states and speeds.
tcpxtract	tcpxtract	Extract files from network traffic based on file signatures.
testdisk	testdisk	Partition scanner and disk recovery tool, and PhotoRec file recovery tool.
tofrodo	tofrodo	Converts DOS <-> Unix text files, alias tofromdos.
transmission	transmission	Lightweight BitTorrent client.
ugrep	ugrep	Faster grep with an interactive query UI.
unity-control-center	unity-control-center	Utilities to configure the GNOME desktop.
unrar	unrar	
upx-ucl	upx-ucl	Efficient live-compressor for executables.
vbindiff	vbindiff	Visual binary diff, visually compare binary files.
virtuoso-minimal	virtuoso-minimal	High-performance database - core dependency package.
vmfs-tools	vmfs-tools	Tools to access VMFS filesystems.
winbind	winbind	Service to resolve user and group information from Windows NT servers.
wine	wine	Windows API implementation - standard suite.
wireshark	wireshark	Network traffic analyzer - graphical interface.
xdot	xdot	Interactive viewer for Graphviz dot files.
xfsprogs	xfsprogs	Utilities for managing the XFS filesystem.
xmount	xmount	Tool for crossmounting between disk image formats.
zenity	zenity	Display graphical dialog boxes from shell scripts.
zlib1g-dev	zlib1g-dev	Compression library - development.

DB/Search

Tool	Command(s)	Purpose
Elasticsearch	—	NoSQL index for logs/metadata/alerts
ILM	—	Index Lifecycle Management (aging/rolling/deletion)
Kibana	—	Search analytics & dashboards

Detection

Tool	Command(s)	Purpose
ATT&CK Navigator	—	Visual defensive-coverage matrix
ElastAlert 2	—	Continuous alert-indicator scanner
Playbook	—	Custom detections mapped to MITRE ATT&CK
Sigma / Sigma-CLI	—	Translate YAML detection rules to Elastic queries
so-idstools	—	Manage/update Suricata ET rules

Diagnostics

Tool	Command(s)	Purpose
Grafana	—	Infra/node-health dashboards
InfluxDB	—	Time-series store for hardware metrics
Telegraf	—	Hardware health/CPU/mem/disk agent

File Analysis

Tool	Command(s)	Purpose
DomainStats	—	Domain-age checks to flag malicious sites
FreqServer	—	Entropy analysis to detect DGAs
Strelka	—	Real-time file extraction & YARA/file carving

Host/EDR

Tool	Command(s)	Purpose
Elastic Agent	—	Host log/metric/audit collector
Elastic Fleet	—	Central agent config & deployment
OpenCanary	—	Network honeypot / decoy services
Osquery	—	Live endpoint queries via SQL
Wazuh	—	HIDS - File Integrity Monitoring & rootkit checks

IR/AI

Tool	Command(s)	Purpose
CyberChef	—	Visual data decode/parse/analysis
Onion AI Assistant	—	AI triage assistance for analysts
Security Onion Console (SOC)	—	Web UI for threat hunting & grid management
TheHive	—	Collaborative IR & case management

Identity/Infra

Tool	Command(s)	Purpose
ManagerHype	—	Hypervisor integration for VM mapping
Nginx (so-nginx)	—	Reverse proxy for internal web traffic
Ory Kratos & Dex	—	IAM - SSO/auth/sessions
so-apt-cacher-ng	—	Local package caching proxy
so-firewall	—	iptables/nftables wrapper for inter-node comms

Log Routing

Tool	Command(s)	Purpose
Filebeat	—	Lightweight log shipper
Logstash	—	Data pipeline - ECS normalize & enrich

Tool	Command(s)	Purpose
Redis	—	Log ingestion queue/broker
Rsyslog / Syslog-ng	—	Ingest traditional syslog from appliances

Network Visibility

Tool	Command(s)	Purpose
Sensoroni	—	Backend API agent retrieving/extracting PCAPs
Stenographer	—	High-speed raw full packet capture daemon
Suricata	—	Signature-based IDS/IPS & packet capture
Zeek	—	Network traffic analyzer & protocol metadata logging

Sources

- `remnux_docs` — <https://raw.githubusercontent.com/REMnux/docs/master/discover-the-tools>
- `kali_control` — <https://gitlab.com/kalilinux/packages/kali-meta/-/raw/kali/master/debian/control>
- `flare_config` — <https://raw.githubusercontent.com/mandiant/flare-vm/main/config.xml>

Tool Reference

One page per tool. Every option below was read off the real binary and is checked by the linter.

affcat

Kit: Kali Linux · SIFT Workstation **Capability:** Inspect or mount a forensic image container **Version:** affcat version 3.7.20 **Captured:** cyberlab-aio via --help on 2026-08-02 [raw](#)

[← Capability index](#) · [Kit tool list](#)

Synopsis

```
affcat [options] infile [... more infiles]
```

Options

All 7 options parsed from the captured help text. The final column is filled in by review.

Flag	Argument	What it does	When you would use it
-q	—	--- quiet; don't print to STDERR if a page is skipped	
-n	—	--- noisy; tell when pages are skipped.	
-l	—	--- List all of the segment names	
-L	—	--- List segment names, lengths, and args	
-d	—	--- debug. Print the page numbers to stderr as data goes to stdout	
-b	—	--- Output BADFALG for bad blocks (default is NULLs)	
-v	—	--- Just print the version number and exit.	

See also

[ewfinfo](#) , [ewfmount](#) , [ewfverify](#) , [ewfexport](#) , [affinfo](#) , [img_stat](#) , [ntfs-3g](#) , [vshadowinfo](#)

affconvert

Kit: Kali Linux · SIFT Workstation **Capability:** Image a disk, volume or device **Version:** affconvert version 3.7.20 **Captured:** cyberlab-aio via --help on 2026-08-02 raw

[← Capability index](#) · [Kit tool list](#)

Purpose

Please, see more info in manpage.

Synopsis

```
affconvert [options] file1 [... files]
```

Options

No option definitions could be parsed from this tool's help output. It may be subcommand-driven or have no flags; needs manual review.

See also

[dc3dd](#), [dcfldd](#), [dd](#), [ewfacquire](#)

affinfo

Kit: Kali Linux · SIFT Workstation **Capability:** Inspect or mount a forensic image container **Version:** affinfo version 3.7.20 **Captured:** cyberlab-aio via --help on 2026-08-02 [raw](#)

[← Capability index](#) · [Kit tool list](#)

Purpose

(may be repeated)

Synopsis

```
affinfo [options] infile
-a = print ALL segments (normally data segments are suppressed)
-b = print how many bad blocks in each segment (implies -a)
-i = identify the files, don't do info on them.
-w = wide output; print more than 1 line if necessary.
-s segment = Just print information about 'segment'.
```

Options

No option definitions could be parsed from this tool's help output. It may be subcommand-driven or have no flags; needs manual review.

See also

[ewfinfo](#), [ewfmount](#), [ewfverify](#), [ewfexport](#), [affcat](#), [img_stat](#), [ntfs-3g](#), [vshadowinfo](#)

bdeinfo

Kit: SIFT Workstation (libyal) **Capability:** Inspect or mount a forensic image container **Captured:** cyberlab-aio via --help on 2026-08-02 [raw](#)

[← Capability index](#) · [Kit tool list](#)

Purpose

Invalid argument: bdeinfo

Synopsis

```
bdeinfo [ -k keys ] [ -o offset ] [ -p password ]  
[ -r password ] [ -s filename ] [ -hvV ] source
```

Common invocations

```
# from cyberlab 06-windows-artifact-libs  
bdeinfo -V  
# from cyberlab 06-windows-artifact-libs  
bdeinfo exercise/bitlocker.raw
```

Options

All 8 options parsed from the captured help text. The final column is filled in by review.

Flag	Argument	What it does	When you would use it
-h	—	shows this help	
-k	—	the full volume encryption key and tweak key formatted in base16 and separated by a : character e.g. FVEK:TWEAK	
-o	—	specify the volume offset in bytes	
-p	—	specify the password/passphrase	
-r	—	specify the recovery password	
-s	—	specify the file containing the startup key. typically this file has the extension .BEK	
-v	—	verbose output to stderr	
-V	—	print version	

See also

[ewfinfo](#) , [ewfmount](#) , [ewfverify](#) , [ewfexport](#) , [affinfo](#) , [affcat](#) , [img_stat](#) , [ntfs-3g](#)

dc3dd

Kit: Kali Linux · SIFT Workstation **Capability:** Image a disk, volume or device **Version:** dc3dd (dc3dd) 7.2.646 **Captured:** cyberlab-aio via --help on 2026-08-02 [raw](#)
[← Capability index](#) · [Kit tool list](#)

Purpose

dc3dd [OPTION 1] [OPTION 2] ... [OPTION N]

Synopsis

Common invocations

```
# from cyberlab 57--forensic-acquisition
sudo dc3dd if=/dev/sdX of=/evidence/case01.dd hash=sha256 log=/evidence/case01.log
```

Options

All 3 options parsed from the captured help text. The final column is filled in by review.

Flag	Argument	What it does	When you would use it
--help	—	display this help and exit	
--version	—	output version information and exit	
--flags	—	display compile-time flags and exit	

See also

[dcfldd](#), [dd](#), [ewfacquire](#), [affconvert](#)

dcfldd

Kit: Kali Linux · SIFT Workstation **Capability:** Image a disk, volume or device **Version:** dcfldd (dcfldd) 1.9 **Captured:** cyberlab-aio via --help on 2026-08-02 [raw](#)
[← Capability index](#) · [Kit tool list](#)

Purpose

Enhanced version of dd for forensics and security.

Synopsis

```
dcfldd [OPTION]...
```

Options

All 2 options parsed from the captured help text. The final column is filled in by review.

Flag	Argument	What it does	When you would use it
--help	—	display this help and exit	
--version	—	output version information and exit	

See also

[dc3dd](#), [dd](#), [ewfacquire](#), [affconvert](#)

dd

Kit: Base OS — present on every Linux image **Capability:** Image a disk, volume or device **Version:** dd (coreutils) 9.1 **Captured:** cyberlab-aio via --help on 2026-08-02 [raw](#)

[← Capability index](#) · [Kit tool list](#)

Purpose

or: dd OPTION

Synopsis

```
dd [OPERAND]...
or: dd OPTION
```

Common invocations

```
# from cyberlab 01-disk-forensics
dd if=/dev/zero of=exercise/sample.dd bs=1M count=10
# from cyberlab 51-linux-triage-workflow
dd if=/dev/zero of=trriage_sample.raw bs=1M count=8
```

Options

All 2 options parsed from the captured help text. The final column is filled in by review.

Flag	Argument	What it does	When you would use it
--help	—	display this help and exit	
--version	—	output version information and exit	

See also

[dc3dd](#), [dcfldd](#), [ewfacquire](#), [affconvert](#)

dumpcap

Kit: REMnux · Kali Linux · FLARE-VM · SIFT Workstation **Capability:** Capture live network traffic **Captured:** `cyberlab-aio` via `--help` on 2026-08-02 [raw Docs](#): <https://www.wireshark.org>

← [Capability index](#) · [Kit tool list](#)

Purpose

Capture packets to a file. It does nothing else — which is the point.

Synopsis

```
dumpcap [options] ...
```

Options

All 48 options parsed from the captured help text; 19 reviewed with usage guidance.

Flag	Argument	What it does	When you would use it
<code>-i</code>	interface	name or idx of interface (def: first non-loopback), or for remote capturing, use one of these formats: <code>rpcap:/// TCP@:</code>	Interface to capture from. <code>-D</code> lists what is available.
<code>--interface</code>	interface	name or idx of interface (def: first non-loopback), or for remote capturing, use one of these formats: <code>rpcap:/// TCP@:</code>	Interface to capture from. <code>-D</code> lists what is available.
<code>--ifname</code>	name	name to use in the capture file for a pipe from which we're capturing	
<code>--ifdescr</code>	description	description to use in the capture file for a pipe from which we're capturing	
<code>-f</code>	capture filter	packet filter in libpcap filter syntax	Capture filter in BPF syntax. Applied before writing, so anything it excludes is gone permanently.
<code>-s</code>	snaplen	packet snapshot length (def: appropriate maximum)	Snapshot length: truncate each packet. Headers only, when payload must not be recorded.
<code>--snapshot-length</code>	snaplen	packet snapshot length (def: appropriate maximum)	Snapshot length: truncate each packet. Headers only, when payload must not be recorded.
<code>-p</code>	—	don't capture in promiscuous mode	Do not enter promiscuous mode — only traffic for this host.
<code>--no-promiscuous-mode</code>	—	don't capture in promiscuous mode	Do not enter promiscuous mode — only traffic for this host.
<code>-I</code>	—	capture in monitor mode, if available	Monitor mode, for 802.11 management and control frames.
<code>--monitor-mode</code>	—	capture in monitor mode, if available	Monitor mode, for 802.11 management and control frames.
<code>-B</code>	buffer size	size of kernel buffer in MiB (def: 2MiB)	Kernel buffer size in MiB. Raise it first when a fast link reports drops; the default is small for modern traffic.
<code>--buffer-size</code>	buffer size	size of kernel buffer in MiB (def: 2MiB)	Kernel buffer size in MiB. Raise it first when a fast link reports drops; the default is small for modern traffic.
<code>-y</code>	link type	link layer type (def: first appropriate)	Force the link-layer type.
<code>--linktype</code>	link type	link layer type (def: first appropriate)	Force the link-layer type.
<code>-D</code>	—	print list of interfaces and exit	List interfaces and exit.
<code>--list-interfaces</code>	—	print list of interfaces and exit	List interfaces and exit.
<code>-L</code>	—	print list of link-layer types of iface and exit	List link-layer types for the chosen interface.
<code>--list-data-link-types</code>	—	print list of link-layer types of iface and exit	List link-layer types for the chosen interface.
<code>--list-time-stamp-types</code>	—	print list of timestamp types for iface and exit	
<code>-d</code>	—	print generated BPF code for capture filter	Print the compiled BPF for a filter, to check it means what you think before capturing hours of the wrong thing.
<code>-S</code>	—	print statistics for each interface once per second	Print per-interface packet statistics once a second, for confirming traffic is arriving before committing to a long capture.
<code>-M</code>	—	for -D, -L, and -S, produce machine-readable output	Machine-readable output for <code>-D</code> , <code>-L</code> and <code>-S</code> .
<code>-c</code>	packet count	stop after n packets (def: infinite)	Stop after N packets.

Flag	Argument	What it does	When you would use it
<code>-a</code>	autostop cond.	duration:NUM - stop after NUM seconds filesize:NUM - stop this file after NUM kB files:NUM - stop after NUM files packets:NUM - stop after NUM packets	Autostop condition — duration, filesize or files.
<code>--autostop</code>	autostop cond.	duration:NUM - stop after NUM seconds filesize:NUM - stop this file after NUM kB files:NUM - stop after NUM files packets:NUM - stop after NUM packets	Autostop condition — duration, filesize or files.
<code>-w</code>	filename	name of file to save (def: tempfile)	Output file. Without it, dumpcap writes to a temporary file and tells you where, which is rarely what you meant.
<code>-g</code>	—	enable group read access on the output file(s)	
<code>-b</code>	ringbuffer opt.	duration:NUM - switch to next file after NUM secs filesize:NUM - switch to next file after NUM kB files:NUM - ringbuffer: replace after NUM files packets:NUM - ringbuffer: replace after NUM packets in	Ring buffer: roll to a new file on duration, filesize or count. The difference between a capture that runs overnight and one that fills the disk at 3am.
<code>--ring-buffer</code>	ringbuffer opt.	duration:NUM - switch to next file after NUM secs filesize:NUM - switch to next file after NUM kB files:NUM - ringbuffer: replace after NUM files packets:NUM - ringbuffer: replace after NUM packets in	Ring buffer: roll to a new file on duration, filesize or count. The difference between a capture that runs overnight and one that fills the disk at 3am.
<code>-n</code>	—	use pcapng format instead of pcap (default)	pcapng output (the default).
<code>-P</code>	—	use libpcap format instead of pcapng	Legacy pcap output, for a tool that cannot read pcapng.
<code>--capture-comment</code>	comment	add a capture comment to the output file (only for pcapng)	
<code>--temp-dir</code>	directory	write temporary files to this directory (default: /tmp)	
<code>--log-level</code>	level	sets the active log level ("critical", "warning", etc.)	
<code>--log-fatal</code>	level	sets level to abort the program ("critical" or "warning")	
<code>--log-domains</code>	[!]list	comma separated list of the active log domains	
<code>--log-debug</code>	[!]list	comma separated list of domains with "debug" level	
<code>--log-noisy</code>	[!]list	comma separated list of domains with "noisy" level	
<code>--log-file</code>	path	file to output messages to (in addition to stderr)	
<code>-N</code>	packet_limit	maximum number of packets buffered within dumpcap	
<code>-C</code>	byte_limit	maximum number of bytes used for buffering packets within dumpcap	
<code>-t</code>	—	use a separate thread per interface	
<code>-q</code>	—	don't report packet capture counts	Quiet — no packet-count updates.
<code>-v</code>	—	print version information and exit	
<code>--version</code>	—	print version information and exit	
<code>-h</code>	—	display this help and exit	
<code>--help</code>	—	display this help and exit	

Gotchas

- This is deliberately minimal: it captures and it does not dissect. That is why it is the right thing to run as the privileged process and why `tshark` shells out to it — the analysis code never needs the capture privilege.
- Filtering here is destructive. A capture filter that was slightly wrong cannot be widened afterwards; capture broadly and filter at analysis time whenever the disk allows it.
- Drops are reported at the end, not during. Check the count before treating a capture as complete — a busy link with the default buffer loses packets silently.

See also

`tshark`

ewfacquire

Kit: Kali Linux · SIFT Workstation **Capability:** Image a disk, volume or device **Captured:** cyberlab-aio via --help on 2026-08-02 raw

[← Capability index](#) · [Kit tool list](#)

Purpose

Acquire a disk, volume or device into an EWF/E01 evidence container, with the case metadata and hashes stored inside the image itself.

Synopsis

```
ewfacquire [ -A codepage ] [ -b number_of_sectors ]
[ -B number_of_bytes ] [ -c compression_values ]
[ -C case_number ] [ -d digest_type ] [ -D description ]
[ -e examiner_name ] [ -E evidence_number ] [ -f format ]
[ -g number_of_sectors ] [ -l log_filename ]
[ -m media_type ] [ -M media_flags ] [ -N notes ]
```

Common invocations

```
# from cyberlab 57-forensic-acquisition
sudo ewfacquire -t /evidence/case01 -f encase6 -c deflate -S 2G /dev/sdX
```

Options

All 32 options parsed from the captured help text; 30 reviewed with usage guidance.

Flag	Argument	What it does	When you would use it
-A	—	codepage of header section, options: ascii (default), windows-874, windows-932, windows-936, windows-949, windows-950, windows-1250, windows-1251, windows-1252, windows-1253, windows-1254, windows-125	Header codepage, for non-ASCII metadata.
-b	—	specify the number of sectors to read at once (per chunk), options: 16, 32, 64 (default), 128, 256, 512, 1024, 2048, 4096, 8192, 16384 or 32768	Sectors per chunk. Larger chunks read faster but lose more data to each unreadable sector.
-B	—	specify the number of bytes to acquire (default is all bytes)	Acquire a fixed number of bytes rather than the whole device — with <code>-o</code> , the way to image one region.
-c	—	specify the compression values as: level or method:level compression method options: deflate (default), bzip2 (bzip2 is only supported by EWF2 formats) compression level options: none (default), empty	Trade size against time. <code>best</code> on a slow USB source is often faster overall than <code>none</code> , because the bottleneck is the read, not the CPU.
-C	—	specify the case number (default is case_number).	Case number, stored in the image header.
-d	—	calculate additional digest (hash) types besides md5, options: sha1, sha256	Add sha1 or sha256 alongside the default md5. Worth doing once, at acquisition: md5 alone is increasingly challenged, and rehashing later means reading the whole image again.
-D	—	specify the description (default is description).	Description, stored in the image header.
-e	—	specify the examiner name (default is examiner_name).	Examiner name, stored in the image header.
-E	—	specify the evidence number (default is evidence_number).	Evidence number, stored in the image header.
-f	—	specify the EWF file format to write to, options: ewf, smart, ftk, encase2, encase3, encase4, encase5, encase6 (default), encase7, encase7-v2, linen5, linen6, linen7, ewfx	Choose the EWF variant. <code>encase6</code> is the safe default for interoperability; pick the format the tool that will read it expects, not the newest one available.
-g	—	specify the number of sectors to be used as error granularity	Error granularity — how much data around a bad sector is discarded. Lower values preserve more of a failing disk at the cost of speed.
-h	—	shows this help	
-l	—	logs acquiry errors and the digest (hash) to the log_filename	Write the acquisition log, including errors and hashes, to a file. This is the record you cite later; do not skip it.
-m	—	specify the media type, options: fixed (default), removable, optical, memory	Record the media type (fixed, removable, optical, memory) in the header.
-M	—	specify the media flags, options: logical, physical (default)	Record whether this is a physical device or a logical volume.
-N	—	specify the notes (default is notes).	Free-text notes, stored in the image header. These five metadata flags are the reason to choose E01 over a raw <code>dd</code> image: the container describes its own provenance.
-o	—	specify the offset to start to acquire (default is 0)	Start at an offset rather than sector 0.

Flag	Argument	What it does	When you would use it
<code>-p</code>	—	specify the process buffer size (default is the chunk size)	Process buffer size — a throughput tuning knob.
<code>-P</code>	—	specify the number of bytes per sector (default is 512) (use this to override the automatic bytes per sector detection)	Override bytes-per-sector. Needed on 4Kn drives where the 512-byte assumption is wrong.
<code>-q</code>	—	quiet shows minimal status information	Minimal status output, for logs.
<code>-r</code>	—	specify the number of retries when a read error occurs (default is 2)	Read retries before giving up on a sector. Raise it for a dying drive; lower it when retries are heating a drive that may not survive the acquisition.
<code>-R</code>	—	resume acquiry at a safe point	Resume an interrupted acquisition at a safe point, instead of restarting a multi-hour read.
<code>-s</code>	—	swap byte pairs of the media data (from AB to BA) (use this for big to little endian conversion and vice versa)	Swap byte pairs for a big-endian source. Rare, and wrong unless you know the source endianness differs.
<code>-S</code>	—	specify the segment file size in bytes (default is 1.4 GiB) (minimum is 1.0 MiB, maximum is 7.9 EiB for encase6 and encase7 format and 1.9 GiB for other formats)	Segment size. The 1.4 GiB default suits FAT32 and optical media; raise it on a modern filesystem to avoid hundreds of fragments.
<code>-t</code>	—	specify the target file (without extension) to write to	Set the output name, without extension — <code>ewfacquire</code> appends <code>.E01</code> , <code>.E02</code> ... itself. The one flag you always pass.
<code>-T</code>	—	specify the file containing the table of contents (TOC) of an optical disc. The TOC file must be in the CUE format.	Supply a CUE file when imaging optical media, so the track layout is preserved.
<code>-u</code>	—	unattended mode (disables user interaction)	Unattended — suppresses the interactive prompts. Required for any scripted or headless acquisition.
<code>-v</code>	—	verbose output to stderr	Verbose diagnostics to stderr — worth capturing alongside <code>-l</code> when a source is throwing read errors.
<code>-V</code>	—	print version	
<code>-w</code>	—	zero sectors on read error (mimic EnCase like behavior)	Zero unreadable sectors instead of aborting, the way EnCase does. Keeps offsets aligned so the filesystem still parses.
<code>-x</code>	—	use the chunk data instead of the buffered read and write functions.	Bypass the buffered read/write path.
<code>-2</code>	—	specify the secondary target file (without extension) to write to	Write a second copy in the same pass. Two independent copies for the cost of one read, which matters when the source is failing and may not survive a second acquisition.

Gotchas

- The case metadata flags default to literal placeholder strings — `case_number`, `examiner_name`, `evidence_number`. Omit them and the image ships with those placeholders recorded as fact, which is worse than an empty field because it looks filled in.
- `-t` takes the name **without** an extension. Passing `image.E01` produces `image.E01.E01`.
- Acquisition is not verification. `ewfacquire` records hashes; proving the image still matches them later is `ewfverify`.

See also

`dc3dd`, `dcfldd`, `dd`, `affconvert`

ewfexport

Kit: Kali Linux · SIFT Workstation **Capability:** Inspect or mount a forensic image container **Captured:** cyberlab-ai0 via --help on 2026-08-02 raw

← [Capability index](#) · [Kit tool list](#)

Purpose

Convert an EWF/E01 image to raw, or to another EWF format, including extracting a subset of it.

Synopsis

```
ewfexport [ -A codepage ] [ -b number_of_sectors ]
[ -B number_of_bytes ] [ -c compression_values ]
[ -d digest_type ] [ -f format ] [ -l log_filename ]
[ -o offset ] [ -p process_buffer_size ]
[ -S segment_file_size ] [ -t target ] [ -hqsvVwx ] ewf_files
```

Options

All 19 options parsed from the captured help text; 17 reviewed with usage guidance.

Flag	Argument	What it does	When you would use it
-A	—	codepage of header section, options: ascii (default), windows-874, windows-932, windows-936, windows-949, windows-950, windows-1250, windows-1251, windows-1252, windows-1253, windows-1254, windows-125	Header codepage.
-b	—	specify the number of sectors to read at once (per chunk), options: 16, 32, 64 (default), 128, 256, 512, 1024, 2048, 4096, 8192, 16384 or 32768 (not used for raw and files formats)	Sectors per chunk.
-B	—	specify the number of bytes to export (default is all bytes)	Number of bytes to export — with -o , extracts one partition.
-c	—	specify the compression values as: level or method:level compression method options: deflate (default), bzip2 (bzip2 is only supported by EWF2 formats) compression level options: none (default), empty	Compression for an EWF target.
-d	—	calculate additional digest (hash) types besides md5, options: sha1, sha256 (not used for raw and files format)	Calculate an additional digest over the exported data.
-f	—	specify the output format to write to, options: raw (default), files (restricted to logical volume files), ewf, smart, encase1, encase2, encase3, encase4, encase5, encase6, encase7, encase7-v2, linen5	Output format — raw for tools that cannot read EWF, or another EWF variant for compatibility.
-h	—	shows this help	
-l	—	logs export errors and the digest (hash) to the log_filename	Log the export.
-o	—	specify the offset to start the export (default is 0)	Start offset, to export a region rather than the whole image.
-p	—	specify the process buffer size (default is the chunk size)	Process buffer size.
-q	—	quiet shows minimal status information	Minimal output.
-s	—	swap byte pairs of the media data (from AB to BA) (use this for big to little endian conversion and vice versa)	Swap byte pairs.
-S	—	specify the segment file size in bytes (default is 1.4 GiB) (minimum is 1.0 MiB, maximum is 7.9 EiB for raw, encase6 and encase7 format and 1.9 GiB for other formats) (not used for files format)	Segment size for the output.
-t	—	specify the target file to export to, use - for stdout (default is export) stdout is only supported for the raw format	Target name. - writes to stdout, which lets the image be piped straight into another tool without staging a full raw copy on disk.
-u	—	unattended mode (disables user interaction)	Unattended, for scripted conversion.
-v	—	verbose output to stderr	Verbose diagnostics to stderr.
-V	—	print version	
-w	—	zero sectors on checksum error (mimic EnCase like behavior)	Zero sectors that cannot be read.

Flag	Argument	What it does	When you would use it
<code>-x</code>	—	use the chunk data instead of the buffered read and write functions.	Bypass the buffered read/write path.

Gotchas

- Exporting to raw discards the metadata and hashes that justified using E01. Keep the original: the raw copy is a working artefact, not the evidence.
- A raw export needs the full uncompressed size in free space. E01 compression routinely hides a 2 TB image inside 700 GB.

See also

`ewfinfo`, `ewfmount`, `ewfverify`, `affinfo`, `affcat`, `img_stat`, `ntfs-3g`, `vshadowinfo`

ewfinfo

Kit: Kali Linux · SIFT Workstation **Capability:** Inspect or mount a forensic image container **Captured:** cyberlab-aio via --help on 2026-08-02 raw

[← Capability index](#) · [Kit tool list](#)

Purpose

Show the metadata, hashes and acquisition details recorded inside an EWF/E01 image.

Synopsis

```
ewfinfo [ -A codepage ] [ -d date_format ] [ -f format ]
[ -ehimvVx ] ewf_files
```

Common invocations

```
# from cyberlab 57-forensic-acquisition
ewfinfo /evidence/case01.E01
```

Options

All 9 options parsed from the captured help text; 7 reviewed with usage guidance.

Flag	Argument	What it does	When you would use it
-A	—	codepage of header section, options: ascii (default), windows-874, windows-932, windows-936, windows-949, windows-950, windows-1250, windows-1251, windows-1252, windows-1253, windows-1254, windows-125	Header codepage, for images with non-ASCII metadata.
-d	—	specify the date format, options: ctime (default), dm (day/month), md (month/day), iso8601	Date format. <code>iso8601</code> is the unambiguous choice for a report.
-e	—	only show EWF read error information	Read errors recorded at acquisition. Check this before trusting a clean-looking image: unreadable sectors are noted here, not in the filesystem.
-f	—	specify the output format, options: text (default), dfxml	Emit DFXML instead of text, when the output feeds a tool rather than a person.
-h	—	shows this help	
-i	—	only show EWF acquiry information	Acquisition details only — who imaged it, when, with what.
-m	—	only show EWF media information	Media details only — geometry, sector size, media type.
-v	—	verbose output to stderr	Verbose diagnostics to stderr.
-V	—	print version	

Gotchas

- This reads the header only. It reports the hash that was recorded at acquisition; it does not recompute it, so it cannot tell you the image is still intact. Use `ewfverify` for that.

See also

`ewfmount`, `ewfverify`, `ewfexport`, `affinfo`, `affcat`, `img_stat`, `ntfs-3g`, `vshadowinfo`

ewfmount

Kit: Kali Linux · SIFT Workstation **Capability:** Inspect or mount a forensic image container **Captured:** cyberlab-aio via --help on 2026-08-02 raw

[← Capability index](#) · [Kit tool list](#)

Purpose

Invalid argument: ewfmount

Synopsis

```
ewfmount [ -f format ] [ -X extended_options ] [ -hV ] image mount_point
```

Common invocations

```
# from cyberlab 57--forensic-acquisition
sudo ewfmount /evidence/case01.E01 /mnt/ewf
```

Options

All 5 options parsed from the captured help text. The final column is filled in by review.

Flag	Argument	What it does	When you would use it
-f	—	specify the input format, options: raw (default), files (restricted to logical volume files)	
-h	—	shows this help	
-v	—	verbose output to stderr, while ewfmount will remain running in the foreground	
-V	—	print version	
-X	—	extended options to pass to sub system	

See also

[ewfinfo](#) , [ewfverify](#) , [ewfexport](#) , [affinfo](#) , [affcat](#) , [img_stat](#) , [ntfs-3g](#) , [vshadowinfo](#)

ewfverify

Kit: Kali Linux · SIFT Workstation **Capability:** Inspect or mount a forensic image container **Captured:** cyberlab-aio via --help on 2026-08-02 raw

[← Capability index](#) · [Kit tool list](#)

Purpose

Recompute an EWF/E01 image's hashes and check them against the values stored at acquisition.

Synopsis

```
ewfverify [ -A codepage ] [ -d digest_type ] [ -f format ]
[ -l log_filename ] [ -p process_buffer_size ]
[ -hqVwx ] ewf_files
```

Common invocations

```
# from cyberlab 57-forensic-acquisition
ewfverify /evidence/case01.E01
```

Options

All 11 options parsed from the captured help text; 9 reviewed with usage guidance.

Flag	Argument	What it does	When you would use it
-A	—	codepage of header section, options: ascii (default), windows-874, windows-932, windows-936, windows-949, windows-950, windows-1250, windows-1251, windows-1252, windows-1253, windows-1254, windows-125	Header codepage.
-d	—	calculate additional digest (hash) types besides md5, options: sha1, sha256	Also verify an additional digest such as sha256, when one was recorded at acquisition.
-f	—	specify the input format, options: raw (default), files (restricted to logical volume files)	Output format.
-h	—	shows this help	
-l	—	logs verification errors and the digest (hash) to the log_filename	Write the verification result to a log file — the artefact worth keeping with the case, not just terminal output.
-p	—	specify the process buffer size (default is the chunk size)	Process buffer size.
-q	—	quiet shows minimal status information	Minimal output, for scripted checks.
-v	—	verbose output to stderr	Verbose diagnostics to stderr.
-V	—	print version	
-w	—	zero sectors on checksum error (mimic EnCase like behavior)	Wipe sectors that could not be read.
-x	—	use the chunk data instead of the buffered read and write functions.	Bypass the buffered read/write path.

Gotchas

- This reads every byte, so it takes as long as the acquisition did. Budget for that rather than discovering it mid-deadline.
- A pass proves the image matches what was recorded **at acquisition**. It says nothing about whether the acquisition captured the device correctly — a disk failing mid-image produces a verifiable image of incomplete data.

See also

[ewfinfo](#), [ewfmount](#), [ewfexport](#), [affinfo](#), [affcat](#), [img_stat](#), [ntfs-3g](#), [vshadowinfo](#)

HashMyFiles (GUI)

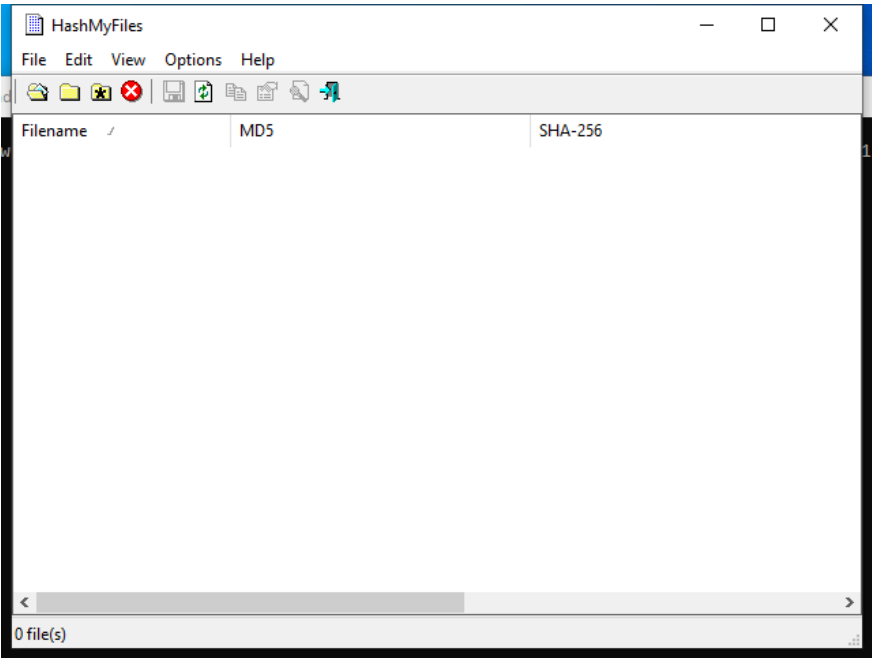
Capability: acquire preserve **Window:** HashMyFiles **Version:** — **Captured:** C:\Tools\HashMyFiles\HashMyFiles.exe on 2026-08-02 — control tree in `capture/gui/HashMyFiles/HashMyFiles.tree.txt`

[← Capability index](#) · [Kit tool list](#)

Purpose

Hash a set of files and compare the results.

Window



Controls

All 4 nodes come from the capture; the 1 interactive controls are listed here.

Control	Type	AutomationId	What it does
0 file(s)	Pane	257	

Using it

1. Drag files, or a folder, onto the window.
2. Read the count in the status pane to confirm everything you expected was taken — it is the only feedback that files were missed.
3. Sort by hash to group identical files; duplicates across directories collapse immediately.
4. Copy the hashes out for the report, or save them as the record of what was collected.

Gotchas

- Hashing here is for triage and deduplication. It is not acquisition — an image's hash of record comes from the acquisition tool that made it.
- It follows what you dropped and nothing more. A folder dropped without recursion silently hashes only the top level.

img_stat

Kit: REMnux · Kali Linux · SIFT Workstation **Capability:** Inspect or mount a forensic image container; See the partition and volume layout **Version:** The Sleuth Kit ver 4.11.1 **Captured:** cyberlab-aio via --help on 2026-08-02 **raw Docs:** <https://www.sleuthkit.org/sleuthkit>

[← Capability index](#) · [Kit tool list](#)

Purpose

Analyze disk images and recover files from them.

Synopsis

```
img_stat [-tvV] [-i imgtype] [-b dev_sector_size] image
```

Options

All 5 options parsed from the captured help text. The final column is filled in by review.

Flag	Argument	What it does	When you would use it
-t	—	display type only	
-i	imgtype	The format of the image file (use '-i list' for list of supported types)	
-b	dev_sector_size	The size (in bytes) of the device sectors	
-v	—	verbose output to stderr	
-V	—	Print version	

See also

[ewfinfo](#), [ewfmount](#), [ewfverify](#), [ewfexport](#), [affinfo](#), [affcat](#), [ntfs-3g](#), [vshadowinfo](#)

md5sum

Kit: Base OS — present on every Linux image **Capability:** Verify evidence integrity with hashes **Version:** md5sum (GNU coreutils) 9.1 **Captured:** cyberlab-aio via --help on 2026-08-02 [raw](#)

[← Capability index](#) · [Kit tool list](#)

Purpose

Print or check MD5 (128-bit) checksums.

Synopsis

```
md5sum [OPTION]... [FILE]...
```

Options

All 17 options parsed from the captured help text. The final column is filled in by review.

Flag	Argument	What it does	When you would use it
-b	—	read in binary mode	
--binary	—	read in binary mode	
-c	—	read checksums from the FILES and check them	
--check	—	read checksums from the FILES and check them	
--tag	—	create a BSD-style checksum	
-t	—	read in text mode (default)	
--text	—	read in text mode (default)	
-z	—	end each output line with NUL, not newline, and disable file name escaping	
--zero	—	end each output line with NUL, not newline, and disable file name escaping	
--ignore-missing	—	don't fail or report status for missing files	
--quiet	—	don't print OK for each successfully verified file	
--status	—	don't output anything, status code shows success	
--strict	—	exit non-zero for improperly formatted checksum lines	
-w	—	warn about improperly formatted checksum lines	
--warn	—	warn about improperly formatted checksum lines	
--help	—	display this help and exit	
--version	—	output version information and exit	

See also

[rahash2](#) , [ssdeep](#) , [sha256sum](#) , [sigtool](#)

ntfs-3g

Kit: SIFT Workstation **Capability:** Inspect or mount a forensic image container **Version:** ntfs-3g 2022.10.3 **Captured:** `cyberlab-aio` via `--help` on 2026-08-02 [raw](#)
[← Capability index](#) · [Kit tool list](#)

Purpose

ntfs-3g 2022.10.3 integrated FUSE 28 - Third Generation NTFS Driver

Synopsis

```
ntfs-3g [-o option[,...]] <device|image_file> <mount_point>
```

Options

No option definitions could be parsed from this tool's help output. It may be subcommand-driven or have no flags; needs manual review.

See also

`ewfinfo`, `ewfmount`, `ewfverify`, `ewfexport`, `affinfo`, `affcat`, `img_stat`, `vshadowinfo`

rahash2

Kit: REMnux **Capability:** Verify evidence integrity with hashes **Version:** rahash2 6.1.9 **Captured:** cyberlab-aio via --help on 2026-08-02 [raw Docs](#): <https://www.radare.org/n/radare2.html>

[← Capability index](#) · [Kit tool list](#)

Purpose

Examine binary files, including disassembling and debugging. Includes r2ai and decal plugins for LLM-powered analysis (API key or local Ollama required), plus the r2ghidra plugin for Ghidra decompilation via the pdg command.

Synopsis

```
rahash2 [-BehjkLqRrvX] [-b S] [-a A] [-c H] [-E A] [-s S] [-f O] [-t O] [file] ...
```

Options

All 23 options parsed from the captured help text. The final column is filled in by review.

Flag	Argument	What it does	When you would use it
-a	algo	comma separated list of algorithms (default is 'sha256')	
-b	bsize	specify the size of the block (instead of full file)	
-B	—	show per-block hash	
-c	hash	compare with this hash	
-e	—	swap endian (use little endian)	
-E	algo	encrypt. Use -S to set key and -I to set IV	
-D	algo	decrypt. Use -S to set key and -I to set IV	
-f	from	start hashing at given address	
-i	num	repeat hash N iterations (f.ex: 3DES)	
-I	iv	use give initialization vector (IV) (hexa or s:string)	
-j	—	output in json	
-J	—	new simplified json output (same as -jj)	
-S	seed	use given seed (hexa or s:string) use ^ to prefix (key for -E) (- will slurp the key from stdin, the @ prefix points to a file	
-k	—	show hash using the openssl's randomkey algorithm	
-q	—	run in quiet mode (-qq to show only the hash)	
-L	—	list muta plugins (combines with -q, used by -a, -E and -D)	
-r	—	output radare commands	
-R	—	output radare2 sdb commands (k file.=...)	
-s	string	hash this string instead of files	
-t	to	stop hashing at given address	
-x	hexstr	hash this hexpair string instead of files	
-X	—	output in hexpairs instead of binary/plain	
-v	—	show version information	

See also

[ssdeep](#) , [sha256sum](#) , [md5sum](#) , [sigtool](#)

sha256sum

Kit: Base OS — present on every Linux image **Capability:** Verify evidence integrity with hashes **Version:** sha256sum (GNU coreutils) 9.1 **Captured:** cyberlab-aio via -help on 2026-08-02 [raw](#)

[← Capability index](#) · [Kit tool list](#)

Purpose

Print or check SHA256 (256-bit) checksums.

Synopsis

```
sha256sum [OPTION]... [FILE]...
```

Common invocations

```
# from cyberlab 01-disk-forensics
sha256sum /tmp/recovered.txt
# from cyberlab 04-registry-analysis
sha256sum exercise/SYSTEM_sample.hive
# from cyberlab 05-file-carving
sha256sum /tmp/ak_tcp/*.jpg
# from cyberlab 10-malicious-documents
sha256sum exercise/sample.doc
# from cyberlab 21-yara-authoring
sha256sum exercise/eicar_sample.txt
# from cyberlab 26-metasploit-workflow
sha256sum exercise/scan.xml
# from cyberlab 32-remnux-static-triage
sha256sum exercise/sample.exe
# from cyberlab 33-binwalk-firmware
sha256sum firmware.bin
```

Options

All 17 options parsed from the captured help text. The final column is filled in by review.

Flag	Argument	What it does	When you would use it
-b	—	read in binary mode	
--binary	—	read in binary mode	
-c	—	read checksums from the FILEs and check them	
--check	—	read checksums from the FILEs and check them	
--tag	—	create a BSD-style checksum	
-t	—	read in text mode (default)	
--text	—	read in text mode (default)	
-z	—	end each output line with NUL, not newline, and disable file name escaping	
--zero	—	end each output line with NUL, not newline, and disable file name escaping	
--ignore-missing	—	don't fail or report status for missing files	
--quiet	—	don't print OK for each successfully verified file	
--status	—	don't output anything, status code shows success	
--strict	—	exit non-zero for improperly formatted checksum lines	
-w	—	warn about improperly formatted checksum lines	
--warn	—	warn about improperly formatted checksum lines	
--help	—	display this help and exit	
--version	—	output version information and exit	

See also

[rahash2](#) , [ssdeep](#) , [md5sum](#) , [sigtool](#)

sigtool

Kit: REMnux · SIFT Workstation **Capability:** Verify evidence integrity with hashes; Scan with signatures for known-bad **Version:** ClamAV 1.4.3 **Captured:** cyberlab-aio
via `--help` on 2026-08-02 **raw Docs:** <https://www.clamav.net>

[← Capability index](#) · [Kit tool list](#)

Purpose

Scan files for malware signatures.

Options

All 35 options parsed from the captured help text. The final column is filled in by review.

Flag	Argument	What it does	When you would use it
<code>--help</code>	—	Show this help	
<code>-h</code>	—	Show this help	
<code>--version</code>	—	Print version number and exit	
<code>-V</code>	—	Print version number and exit	
<code>--quiet</code>	—	Be quiet, output only error messages	
<code>--debug</code>	—	Enable debug messages	
<code>--stdout</code>	—	Write to stdout instead of stderr. Does not affect 'debug' messages.	
<code>--hex-dump</code>	—	Convert data from stdin to a hex string and print it on stdout	
<code>--md5</code>	FILES	Generate MD5 checksum from stdin or MD5 sigs for FILES	
<code>--sha1</code>	FILES	Generate SHA1 checksum from stdin or SHA1 sigs for FILES	
<code>--sha256</code>	FILES	Generate SHA256 checksum from stdin or SHA256 sigs for FILES	
<code>--mdb</code>	FILES	Generate .mdb (section hash) sigs	
<code>--imp</code>	FILES	Generate .imp (import table hash) sigs	
<code>--html-normalise</code>	FILE	Create normalised parts of HTML file	
<code>--ascii-normalise</code>	FILE	Create normalised text file from ascii source	
<code>--utf16-decode</code>	FILE	Decode UTF16 encoded files	
<code>--info</code>	FILE	-i FILE Print database information	
<code>--max-bad-sigs</code>	NUMBER	Maximum number of mismatched signatures When building a CVD. Default: 3000	
<code>--flevel</code>	FLEVEL	Specify a custom flevel. Default: 213	
<code>--cvd-version</code>	NUMBER	Specify the version number to use for the build. Default is to use the value+1 from the current CVD in --datadir. If no datafile is found the default behaviour is to prompt for a version number, this	
<code>--no-cdiff</code>	—	Don't generate .cdiff file	
<code>--unsigned</code>	—	Create unsigned database file (.cud)	
<code>--hybrid</code>	—	Create a hybrid (standard and bytecode) database file	
<code>--print-certs</code>	FILE	Print Authenticode details from a PE	
<code>--server</code>	ADDR	ClamAV Signing Service address	
<code>--datadir</code>	DIR	Use DIR as default database directory	
<code>--unpack</code>	FILE	-u FILE Unpack a CVD/CLD file	
<code>--unpack-current</code>	SHORTNAME	Unpack local CVD/CLD into cwd	
<code>--list-sigs</code>	FILE (optional)	-I[FILE] List signature names	
<code>--find-sigs</code>	REGEX	-fREGEX Find signatures matching REGEX	
<code>--decode-sigs</code>	—	Decode signatures from stdin	
<code>--vba</code>	FILE	Extract VBA/Word6 macro code	
<code>--vba-hex</code>	FILE	Extract Word6 macro code with hex values	
<code>--run-cdiff</code>	FILE	-r FILE Execute update script FILE in cwd	

Flag	Argument	What it does	When you would use it
<code>--tempdir</code>	DIRECTORY	Create temporary files in DIRECTORY	

See also

`rahash2`, `ssdeep`, `sha256sum`, `md5sum`, `yara`, `yarac`, `clamscan`, `freshclam`

ssdeep

Kit: REMnux · Kali Linux · SIFT Workstation **Capability:** Verify evidence integrity with hashes; Compare or cluster samples; Find hidden data **Version:** 2.14.1 **Captured:** cyberlab-aio via -h on 2026-08-02 **raw Docs:** <https://ssdeep-project.github.io/ssdeep/index.html>

[← Capability index](#) · [Kit tool list](#)

Purpose

Compute Context Triggered Piecewise Hashes (CTPH), also known as fuzzy hashes.

Synopsis

```
ssdeep [-m file] [-k file] [-dpgvrsblcxa] [-t val] [-h|-V] [FILES]
```

Common invocations

```
# from cyberlab 08-malware-static-triage
ssdeep -V
# from cyberlab 08-malware-static-triage
ssdeep exercise/sample.bin
# from cyberlab 32-remnux-static-triage
ssdeep sample.exe > baseline.txt
# from cyberlab 32-remnux-static-triage
ssdeep -m baseline.txt sample_mod.exe
# from cyberlab 32-remnux-static-triage
ssdeep exercise/sample.exe > exercise/baseline.txt
# from cyberlab 32-remnux-static-triage
ssdeep -m exercise/baseline.txt exercise/sample_mod.exe
```

Options

No option definitions could be parsed from this tool's help output. It may be subcommand-driven or have no flags; needs manual review.

See also

[rahash2](#), [sha256sum](#), [md5sum](#), [sigtool](#), [radiff2](#), [binwalk](#)

tshark

Kit: REMnux · Kali Linux · FLARE-VM · SIFT Workstation **Capability:** Capture live network traffic; Read and filter packet captures **Captured:** cyberlab-aio via --help on 2026-08-02 **raw Docs:** <https://www.wireshark.org>

← [Capability index](#) · [Kit tool list](#)

Purpose

Wireshark's command line: capture, filter, dissect and export packet data, including fields for a timeline.

Synopsis

```
tshark [options] ...
```

Common invocations

```
# from cyberlab 07-network-pcap
tshark --version | head -n 1
# from cyberlab 07-network-pcap
tshark -r exercise/sample.pcap -q -z io,phs | head -n 30
# from cyberlab 07-network-pcap
tshark -r exercise/sample.pcap -Y 'http.request' -T fields -e ip.dst -e http.host -e http.request.uri
# from cyberlab 07-network-pcap
tshark -r exercise/sample.pcap -Y 'dns.flags.response == 0' -T fields -e dns.qry.name | sort -u
# from cyberlab 07-network-pcap
tshark -r exercise/sample.pcap -Y 'http.request' -T fields -e http.host -e http.request.uri
# from cyberlab 24-wireshark-deep
tshark -r exercise/sample.pcap -c 20
# from cyberlab 24-wireshark-deep
tshark -r exercise/sample.pcap -Y "dns.flags.response == 0" -T fields -e dns.qry.name
# from cyberlab 24-wireshark-deep
tshark -r exercise/sample.pcap -Y "http.request" -T fields -e http.host -e http.request.uri
```

Options

All 75 options parsed from the captured help text; 19 reviewed with usage guidance.

Flag	Argument	What it does	When you would use it
-i	interface	name or idx of interface (def: first non-loopback)	Capture live from an interface instead. Needs capture rights, and on a busy link a live dissect will drop packets.
--interface	interface	name or idx of interface (def: first non-loopback)	Capture live from an interface instead. Needs capture rights, and on a busy link a live dissect will drop packets.
-f	capture filter	packet filter in libpcap filter syntax	Capture filter, in BPF syntax. Applied before packets are written, so what it drops is gone forever.
-s	snaplen	packet snapshot length (def: appropriate maximum)	Snapshot length; truncates each packet as it is captured.
--snapshot-length	snaplen	packet snapshot length (def: appropriate maximum)	Snapshot length; truncates each packet as it is captured.
-p	—	don't capture in promiscuous mode	Do not enter promiscuous mode.
--no-promiscuous-mode	—	don't capture in promiscuous mode	Do not enter promiscuous mode.
-I	—	capture in monitor mode, if available	Monitor mode, for capturing 802.11 management frames.
--monitor-mode	—	capture in monitor mode, if available	Monitor mode, for capturing 802.11 management frames.
-B	buffer size	size of kernel buffer (def: 2MB)	Kernel buffer size. Raise it when a fast link is dropping packets at capture time.
--buffer-size	buffer size	size of kernel buffer (def: 2MB)	Kernel buffer size. Raise it when a fast link is dropping packets at capture time.
-y	link type	link layer type (def: first appropriate)	Force the link-layer type.
--linktype	link type	link layer type (def: first appropriate)	Force the link-layer type.
-D	—	print list of interfaces and exit	List interfaces and exit — how you find the right -i value.

Flag	Argument	What it does	When you would use it
<code>--list-interfaces</code>	—	print list of interfaces and exit	List interfaces and exit — how you find the right <code>-i</code> value.
<code>-L</code>	—	print list of link-layer types of iface and exit	List the link-layer types an interface supports.
<code>--list-data-link-types</code>	—	print list of link-layer types of iface and exit	List the link-layer types an interface supports.
<code>--list-time-stamp-types</code>	—	print list of timestamp types for iface and exit	
<code>-c</code>	packet count	stop after n packets (def: infinite)	Stop after N packets — the fast way to sample a huge file before committing to a full pass.
<code>-a</code>	autostop cond.	duration:NUM - stop after NUM seconds filesize:NUM - stop this file after NUM KB files:NUM - stop after NUM files packets:NUM - stop after NUM packets	Autostop condition for a live capture: duration, filesize or file count.
<code>--autostop</code>	autostop cond.	duration:NUM - stop after NUM seconds filesize:NUM - stop this file after NUM KB files:NUM - stop after NUM files packets:NUM - stop after NUM packets	Autostop condition for a live capture: duration, filesize or file count.
<code>-b</code>	ringbuffer opt.	duration:NUM - switch to next file after NUM secs filesize:NUM - switch to next file after NUM KB files:NUM - ringbuffer: replace after NUM files packets:NUM - switch to next file after NUM packets in	Ring buffer: roll to a new file on time or size, so a long capture cannot fill the disk.
<code>--ring-buffer</code>	ringbuffer opt.	duration:NUM - switch to next file after NUM secs filesize:NUM - switch to next file after NUM KB files:NUM - ringbuffer: replace after NUM files packets:NUM - switch to next file after NUM packets in	Ring buffer: roll to a new file on time or size, so a long capture cannot fill the disk.
<code>-r</code>	infile	set the filename to read from (or '-' for stdin)	Read a capture file. The safe default — analysis needs no privileges and cannot disturb the wire.
<code>--read-file</code>	infile	set the filename to read from (or '-' for stdin)	Read a capture file. The safe default — analysis needs no privileges and cannot disturb the wire.
<code>-2</code>	—	perform a two-pass analysis	Two-pass analysis, so fields that depend on later packets — reassembly, response times, stream indexes — are populated.
<code>-M</code>	packet count	perform session auto reset	Reset dissector state every N packets, to bound memory on a very long capture.
<code>-R</code>	read filter	packet Read filter in Wireshark display filter syntax (requires -2)	Read filter, which needs <code>-2</code> . Prefer <code>-Y</code> unless you know why you want this one.
<code>--read-filter</code>	read filter	packet Read filter in Wireshark display filter syntax (requires -2)	Read filter, which needs <code>-2</code> . Prefer <code>-Y</code> unless you know why you want this one.
<code>-Y</code>	display filter	packet display filter in Wireshark display filter syntax	Display filter, in Wireshark syntax. Applied after capture, so nothing is lost and it can be changed later. Confusing these two is the classic tshark mistake.
<code>--display-filter</code>	display filter	packet display filter in Wireshark display filter syntax	Display filter, in Wireshark syntax. Applied after capture, so nothing is lost and it can be changed later. Confusing these two is the classic tshark mistake.
<code>-n</code>	—	disable all name resolutions (def: "mNd" enabled, or as set in preferences)	
<code>-N</code>	name resolve flags	enable specific name resolution(s): "mnNtdv"	
<code>-H</code>	hosts file	read a list of entries from a hosts file, which will then be written to a capture file. (Implies -W n)	
<code>--enable-protocol</code>	proto_name	enable dissection of proto_name	
<code>--disable-protocol</code>	proto_name	disable dissection of proto_name	
<code>--enable-heuristic</code>	short_name	enable dissection of heuristic protocol	
<code>--disable-heuristic</code>	short_name	disable dissection of heuristic protocol	
<code>-w</code>	outfile -	write packets to a pcapng-format file named "outfile" (or '-' for stdout)	Write packets out rather than dissecting them, which is much faster when you only want a filtered subset.
<code>--capture-comment</code>	comment	add a capture file comment, if supported	
<code>-C</code>	config profile	start with specified configuration profile	
<code>-F</code>	output file type	set the output file type, default is pcapng an empty "-F" option will list the file types	
<code>-V</code>	—	add output of packet tree (Packet Details)	
<code>-O</code>	protocols	Only show packet details of these protocols, comma separated	
<code>-P</code>	—	print packet summary even when writing to a file	
<code>--print</code>	—	print packet summary even when writing to a file	

Flag	Argument	What it does	When you would use it
-S	separator	the line separator to print between packets	
-x	—	add output of hex and ASCII dump (Packet Bytes)	
--hexdump	hexoption	add hexdump, set options for data source and ASCII dump all dump all data sources (-x default) frames dump only frame data source ascii include ASCII dump text (-x default) delimit delimit ASCII dump	
-j	protocolfilter	protocols layers filter if -T ek pdml json selected (e.g. "ip ip.flags text", filter does not expand child nodes, unless child is specified also in the filter)	
-J	protocolfilter	top level protocol filter if -T ek pdml json selected (e.g. "http tcp", filter which expands all child nodes)	
-e	field	field to print if -Tfields selected (e.g. tcp.port, _ws.col.info) this option can be repeated to print multiple fields	Which field to print, repeatable. Only meaningful with -T fields, and the ordering is the column ordering.
-l	—	flush standard output after each packet	
-q	—	be more quiet on stdout (e.g. when using statistics)	
-Q	—	only log true errors to stderr (quieter than -q)	
-g	—	enable group read access on the output file(s)	
-W	n	Save extra information in the file, if supported. n = write network address resolution information	
-U	tap_name	PDU's export mode, see the man page for details	
-z	statistics	various statistics, see the man page for details	
--export-tls-session-keys	keyfile	export TLS Session Keys to a file named "keyfile"	
--color	—	color output text similarly to the Wireshark GUI, requires a terminal with 24-bit color support Also supplies color attributes to pdml and psml formats (Note that attributes are nonstandard)	
--no-duplicate-keys	—	If -T json is specified, merge duplicate keys in an object into a single key with as value a json array containing all values	
--temp-dir	directory	write temporary files to this directory (default: /tmp)	
--log-level	level	sets the active log level ("critical", "warning", etc.)	
--log-fatal	level	sets level to abort the program ("critical" or "warning")	
--log-domains	[!] list	comma separated list of the active log domains	
--log-debug	[!] list	comma separated list of domains with "debug" level	
--log-noisy	[!] list	comma separated list of domains with "noisy" level	
--log-file	path	file to output messages to (in addition to stderr)	
-h	—	display this help and exit	
--help	—	display this help and exit	
-v	—	display version info and exit	
--version	—	display version info and exit	
-K	keytab	keytab file to use for kerberos decryption	
-G	report	dump one of several available reports and exit default report="fields" use "-G help" for more help	

Gotchas

- Capture filters (-f) and display filters (-Y) use **different syntaxes** and apply at different times. -f discards packets permanently; -Y only hides them. Reaching for the wrong one is the most common way to destroy evidence with this tool.
- Dissecting live on a busy link drops packets silently. Capture to a file first, analyse afterwards, whenever completeness matters.
- -T fields prints nothing useful without -e. It is not an error, just empty output, which reads like the filter matched nothing.

See also

[dumppcap](#), [capinfos](#), [ngrep](#), [tcpflow](#)

vshadowinfo

Kit: SIFT Workstation (libyal) **Capability:** Inspect or mount a forensic image container **Captured:** cyberlab-aio via --help on 2026-08-02 [raw](#)

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Purpose

Invalid argument: vshadowinfo

Synopsis

```
vshadowinfo [ -o offset ] [ -ahvV ] source
```

Common invocations

```
# from cyberlab 06-windows-artifact-libs
vshadowinfo -V
# from cyberlab 06-windows-artifact-libs
vshadowinfo exercise/volume.raw
```

Options

All 5 options parsed from the captured help text. The final column is filled in by review.

Flag	Argument	What it does	When you would use it
-a	—	shows allocation information	
-h	—	shows this help	
-o	—	specify the volume offset in bytes	
-v	—	verbose output to stderr	
-V	—	print version	

See also

[ewfinfo](#), [ewfmount](#), [ewfverify](#), [ewfexport](#), [affinfo](#), [affcat](#), [img_stat](#), [ntfs-3g](#)

log2timeline.py

Kit: Kali Linux · SIFT Workstation **Capability:** Build a super-timeline from many artifact sources **Version:** plaso - log2timeline version 20260512 **Captured:** cyberlab-aio via `--help` on 2026-08-02 [raw](#)

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Purpose

Extract timestamped events from evidence into a Plaso storage file, the first half of a super-timeline.

Synopsis

```
log2timeline.py [-h] [--troubles] [-V] [--artifact_definitions PATH]
[--custom_artifact_definitions PATH] [--data PATH]
[--archives TYPES]
[--artifact_filters ARTIFACT_FILTERS]
[--artifact_filters_file PATH]
[--extract_winreg_binary] [--preferred_year YEAR]
```

Common invocations

```
# from cyberlab 03-timeline-analysis
log2timeline.py --version
# from cyberlab 03-timeline-analysis
log2timeline.py --status_view none --storage-file /tmp/case.plaso exercise/sample_fs/
# from cyberlab 49-intrusion-timeline-case
log2timeline.py --storage-file timeline.plaso "$IMAGE"
```

Options

All 100 options parsed from the captured help text; 8 reviewed with usage guidance.

Flag	Argument	What it does	When you would use it
<code>-h</code>	—	Show this help message and exit.	
<code>--help</code>	—	Show this help message and exit.	
<code>--troubles</code>	—	Show troubleshooting information.	
<code>-V</code>	—	Show the version information.	
<code>--version</code>	—	Show the version information.	
<code>--artifact_definitions</code>	PATH	Path to a directory or file containing artifact definitions, which are .yaml files. Artifact definitions can be used to describe and quickly collect data of interest, such as specific files or Windows	
<code>--artifact-definitions</code>	PATH	Path to a directory or file containing artifact definitions, which are .yaml files. Artifact definitions can be used to describe and quickly collect data of interest, such as specific files or Windows	
<code>--custom_artifact_definitions</code>	PATH	Path to a directory or file containing custom artifact definitions, which are .yaml files. Artifact definitions can be used to describe and quickly collect data of interest, such as specific files or	
<code>--custom-artifact-definitions</code>	PATH	Path to a directory or file containing custom artifact definitions, which are .yaml files. Artifact definitions can be used to describe and quickly collect data of interest, such as specific files or	
<code>--data</code>	PATH	Path to a directory containing the data files.	
<code>--archives</code>	TYPES	Define a list of archive and storage media image types for which to process embedded file entries, such as TAR (archive.tar) or ZIP (archive.zip). This is a comma separated list where each entry is th	
<code>--artifact_filters</code>	ARTIFACT_FILTERS	Names of forensic artifact definitions, provided on the command command line (comma separated). Forensic artifacts are stored in .yaml files that are directly pulled from the artifact definitions proj	
<code>--artifact-filters</code>	ARTIFACT_FILTERS	Names of forensic artifact definitions, provided on the command command line (comma separated). Forensic artifacts are stored in .yaml files that are directly pulled from the artifact definitions proj	

Flag	Argument	What it does	When you would use it
<code>--artifact_filters_file</code>	PATH	Names of forensic artifact definitions, provided in a file with one artifact name per line. Forensic artifacts are stored in .yaml files that are directly pulled from the artifact definitions project.	
<code>--artifact-filters_file</code>	PATH	Names of forensic artifact definitions, provided in a file with one artifact name per line. Forensic artifacts are stored in .yaml files that are directly pulled from the artifact definitions project.	
<code>--extract_winreg_binary</code>	—	Extract binary Windows Registry values. WARNING: This can make processing significantly slower.	
<code>--extract-winreg-binary</code>	—	Extract binary Windows Registry values. WARNING: This can make processing significantly slower.	
<code>--preferred_year</code>	YEAR	When a format's timestamp does not include a year, e.g. syslog, use this as the initial year instead of attempting auto-detection.	
<code>--preferred-year</code>	YEAR	When a format's timestamp does not include a year, e.g. syslog, use this as the initial year instead of attempting auto-detection.	
<code>--skip_compressed_streams</code>	—	Skip processing file content within compressed streams, such as syslog.gz and syslog.bz2.	
<code>--skip-compressed-streams</code>	—	Skip processing file content within compressed streams, such as syslog.gz and syslog.bz2.	
<code>-f</code>	FILE_FILTER	List of files to include for targeted collection of files to parse, one line per file path, setup is /path file - where each element can contain either a variable set in the preprocessing stage or a	Use a file filter to limit what gets processed.
<code>--filter-file</code>	FILE_FILTER	List of files to include for targeted collection of files to parse, one line per file path, setup is /path file - where each element can contain either a variable set in the preprocessing stage or a	Use a file filter to limit what gets processed.
<code>--filter_file</code>	FILE_FILTER	List of files to include for targeted collection of files to parse, one line per file path, setup is /path file - where each element can contain either a variable set in the preprocessing stage or a	Use a file filter to limit what gets processed.
<code>--file-filter</code>	FILE_FILTER	List of files to include for targeted collection of files to parse, one line per file path, setup is /path file - where each element can contain either a variable set in the preprocessing stage or a	Use a file filter to limit what gets processed.
<code>--file_filter</code>	FILE_FILTER	List of files to include for targeted collection of files to parse, one line per file path, setup is /path file - where each element can contain either a variable set in the preprocessing stage or a	Use a file filter to limit what gets processed.
<code>--hasher_file_size_limit</code>	SIZE	Define the maximum file size in bytes that hashers should process. Any larger file will be skipped. A size of 0 represents no limit.	
<code>--hasher-file-size-limit</code>	SIZE	Define the maximum file size in bytes that hashers should process. Any larger file will be skipped. A size of 0 represents no limit.	
<code>--hashers</code>	HASHER_LIST	Define a list of hashers to use by the tool. This is a comma separated list where each entry is the name of a hasher, such as "md5,sha256". "all" indicates that all hashers should be enabled. "none" d	Compute hashes during extraction, saving a second pass.
<code>--parsers</code>	PARSER_FILTER_EXPRESSION	Define which presets, parsers and/or plugins to use, or show possible values. The expression is a comma separated string where each element is a preset, parser or plugin name. Each element can be prep	Restrict to specific parsers. A targeted run is minutes instead of hours.
<code>--yara_rules</code>	PATH	Path to a file containing Yara rules definitions.	
<code>--yara-rules</code>	PATH	Path to a file containing Yara rules definitions.	
<code>--partitions</code>	PARTITIONS	Define partitions to be processed. A range of partitions can be defined as: "3..5". Multiple partitions can be defined as: "1,3,5" (a list of comma separated values). Ranges and lists can also be comb	
<code>--partition</code>	PARTITIONS	Define partitions to be processed. A range of partitions can be defined as: "3..5". Multiple partitions can be defined as: "1,3,5" (a list of comma separated values). Ranges and lists can also be comb	
<code>--volumes</code>	VOLUMES	Define volumes to be processed. A range of volumes can be defined as: "3..5". Multiple volumes can be defined as: "1,3,5" (a list of comma separated values). Ranges and lists can also be combined as:	
<code>--volume</code>	VOLUMES	Define volumes to be processed. A range of volumes can be defined as: "3..5". Multiple volumes can be defined as: "1,3,5" (a list of comma separated values). Ranges and lists can also be combined as:	
<code>--codepage</code>	CODEPAGE	The preferred codepage, which is used for decoding single-byte or multi-byte character extracted strings.	
<code>--language</code>	LANGUAGE_TAG	The preferred language, which is used for extracting and formatting Windows EventLog message strings. Use " --language list" to see a list of supported language tags. The en-US (LCID 0x0409) language	

Flag	Argument	What it does	When you would use it
<code>--no-extract-winevt-resources</code>	—	Do not extract Windows EventLog resources such as event message template strings. By default Windows EventLog resources will be extracted when a Windows EventLog parser is enabled.	
<code>--no-extract-winevt-resources</code>	—	Do not extract Windows EventLog resources such as event message template strings. By default Windows EventLog resources will be extracted when a Windows EventLog parser is enabled.	
<code>-z</code>	TIME_ZONE	preferred time zone of extracted date and time values that are stored without a time zone indicator. The time zone is determined based on the source data where possible otherwise it will default to UT	Time zone of the source machine.
<code>--zone</code>	TIME_ZONE	preferred time zone of extracted date and time values that are stored without a time zone indicator. The time zone is determined based on the source data where possible otherwise it will default to UT	Time zone of the source machine.
<code>--timezone</code>	TIME_ZONE	preferred time zone of extracted date and time values that are stored without a time zone indicator. The time zone is determined based on the source data where possible otherwise it will default to UT	Time zone of the source machine.
<code>--no_vss</code>	—	Do not scan for Volume Shadow Snapshots (VSS). This means that Volume Shadow Snapshots (VSS) are not processed. WARNING: this option is deprecated use --vss_stores=none instead.	
<code>--no-vss</code>	—	Do not scan for Volume Shadow Snapshots (VSS). This means that Volume Shadow Snapshots (VSS) are not processed. WARNING: this option is deprecated use --vss_stores=none instead.	
<code>--vss_only</code>	—	Do not process the current volume if Volume Shadow Snapshots (VSS) have been selected.	
<code>--vss-only</code>	—	Do not process the current volume if Volume Shadow Snapshots (VSS) have been selected.	
<code>--vss_stores</code>	VSS_STORES	Define Volume Shadow Snapshots (VSS) (or stores) that need to be processed. A range of snapshots can be defined as: "3..5". Multiple snapshots can be defined as: "1,3,5" (a list of comma separated val	Also process Volume Shadow Copies — often where the pre-attack state survives.
<code>--vss-stores</code>	VSS_STORES	Define Volume Shadow Snapshots (VSS) (or stores) that need to be processed. A range of snapshots can be defined as: "3..5". Multiple snapshots can be defined as: "1,3,5" (a list of comma separated val	Also process Volume Shadow Copies — often where the pre-attack state survives.
<code>-d</code>	—	Enable debug output.	
<code>--debug</code>	—	Enable debug output.	
<code>-q</code>	—	Disable informational output.	
<code>--quiet</code>	—	Disable informational output.	
<code>-u</code>	—	Enable unattended mode and do not ask the user for additional input when needed, but terminate with an error instead.	
<code>--unattended</code>	—	Enable unattended mode and do not ask the user for additional input when needed, but terminate with an error instead.	
<code>--info</code>	—	Print out information about supported plugins and parsers.	
<code>--use_markdown</code>	—	Output lists in Markdown format use in combination with "--hashers list", "--parsers list" or "--timezone list"	
<code>--use-markdown</code>	—	Output lists in Markdown format use in combination with "--hashers list", "--parsers list" or "--timezone list"	
<code>--no_dependencies_check</code>	—	Disable the dependencies check.	
<code>--no-dependencies-check</code>	—	Disable the dependencies check.	
<code>--logfile</code>	FILENAME	Path of the file in which to store log messages, by default this file will be named: "log2timeline- YYYYMMDDThhmmss.log.gz". Note that the file will be gzip compressed if the extension is ".gz".	
<code>--log_file</code>	FILENAME	Path of the file in which to store log messages, by default this file will be named: "log2timeline- YYYYMMDDThhmmss.log.gz". Note that the file will be gzip compressed if the extension is ".gz".	
<code>--log-file</code>	FILENAME	Path of the file in which to store log messages, by default this file will be named: "log2timeline- YYYYMMDDThhmmss.log.gz". Note that the file will be gzip compressed if the extension is ".gz".	
<code>--status_view</code>	TYPE	The processing status view mode: "file", "linear", "none" or "window".	Change progress display; none for clean logs.
<code>--status-view</code>	TYPE	The processing status view mode: "file", "linear", "none" or "window".	Change progress display; none for clean logs.
<code>--status_view_file</code>	PATH	The name of the status view file.	
<code>--status-view-file</code>	PATH	The name of the status view file.	
<code>--status_view_interval</code>	SECONDS	Number of seconds to update the status view.	

Flag	Argument	What it does	When you would use it
<code>--status-view-interval</code>	SECONDS	Number of seconds to update the status view.	
<code>--buffer_size</code>	BUFFER_SIZE	The buffer size for the output (defaults to 196MiB).	
<code>--buffer-size</code>	BUFFER_SIZE	The buffer size for the output (defaults to 196MiB).	
<code>--bs</code>	BUFFER_SIZE	The buffer size for the output (defaults to 196MiB).	
<code>--queue_size</code>	QUEUE_SIZE	The maximum number of queued items per worker (defaults to 125000)	
<code>--queue-size</code>	QUEUE_SIZE	The maximum number of queued items per worker (defaults to 125000)	
<code>--single_process</code>	—	Indicate that the tool should run in a single process.	
<code>--single-process</code>	—	Indicate that the tool should run in a single process.	
<code>--process_memory_limit</code>	SIZE	Maximum amount of memory (data segment) a process is allowed to allocate in bytes, where 0 represents no limit. The default limit is 4294967296 (4 GiB). This applies to both the main (foreman) process	
<code>--process-memory-limit</code>	SIZE	Maximum amount of memory (data segment) a process is allowed to allocate in bytes, where 0 represents no limit. The default limit is 4294967296 (4 GiB). This applies to both the main (foreman) process	
<code>--temporary_directory</code>	DIRECTORY	Path to the directory that should be used to store temporary files created during processing.	
<code>--temporary-directory</code>	DIRECTORY	Path to the directory that should be used to store temporary files created during processing.	
<code>--vfs_back_end</code>	TYPE	The preferred dfvfs back-end: "auto", "fsent", "fsfat", "fshfs", "fsntfs", "tsk" or "vsgpt".	
<code>--vfs-back-end</code>	TYPE	The preferred dfvfs back-end: "auto", "fsent", "fsfat", "fshfs", "fsntfs", "tsk" or "vsgpt".	
<code>--worker_memory_limit</code>	SIZE	Maximum amount of memory (data segment and shared memory) a worker process is allowed to consume in bytes, where 0 represents no limit. The default limit is 2147483648 (2 GiB). If a worker process exc	
<code>--worker-memory-limit</code>	SIZE	Maximum amount of memory (data segment and shared memory) a worker process is allowed to consume in bytes, where 0 represents no limit. The default limit is 2147483648 (2 GiB). If a worker process exc	
<code>--worker_timeout</code>	MINUTES	Number of minutes before a worker process that is not providing status updates is considered inactive. The default timeout is 15.0 minutes. If a worker process exceeds this timeout it is killed by the	
<code>--worker-timeout</code>	MINUTES	Number of minutes before a worker process that is not providing status updates is considered inactive. The default timeout is 15.0 minutes. If a worker process exceeds this timeout it is killed by the	
<code>--workers</code>	WORKERS	Number of worker processes. The default is the number of available system CPUs minus one, for the main (foreman) process.	Number of worker processes; tune to the host's cores.
<code>--sigsegv_handler</code>	—	Enables the SIGSEGV handler. WARNING this functionality is experimental and will a deadlock worker process if a real segfault is caught, but not signal SIGSEGV. This functionality is therefore primari	
<code>--sigsegv-handler</code>	—	Enables the SIGSEGV handler. WARNING this functionality is experimental and will a deadlock worker process if a real segfault is caught, but not signal SIGSEGV. This functionality is therefore primari	
<code>--profilers</code>	PROFILERS_LIST	List of profilers to use by the tool. This is a comma separated list where each entry is the name of a profiler. Use "--profilers list" to list the available profilers.	
<code>--profiling_directory</code>	DIRECTORY	Path to the directory that should be used to store the profiling sample files. By default the sample files are stored in the current working directory.	
<code>--profiling-directory</code>	DIRECTORY	Path to the directory that should be used to store the profiling sample files. By default the sample files are stored in the current working directory.	
<code>--profiling_sample_rate</code>	SAMPLE_RATE	Profiling sample rate (defaults to a sample every 1000 files).	
<code>--profiling-sample-rate</code>	SAMPLE_RATE	Profiling sample rate (defaults to a sample every 1000 files).	
<code>--storage_file</code>	PATH	The path of the storage file. If not specified, one will be made in the form -.plaso	Where the .plaso output goes.
<code>--storage-file</code>	PATH	The path of the storage file. If not specified, one will be made in the form -.plaso	Where the .plaso output goes.
<code>--storage_format</code>	FORMAT	Format of the storage file, the default is: sqlite. Supported options: sqlite	
<code>--storage-format</code>	FORMAT	Format of the storage file, the default is: sqlite. Supported options: sqlite	
<code>--task_storage_format</code>	FORMAT	Format for task storage, the default is: sqlite. Supported options: redis, sqlite	
<code>--task-storage-format</code>	FORMAT	Format for task storage, the default is: sqlite. Supported options: redis, sqlite	

Gotchas

- This produces a .plaso database, **not** a timeline you can read. `psort.py` is the second half; running only this looks like nothing happened.
- A full run on a disk image can take hours and tens of GB. Scope with `--parsers` unless you genuinely need everything.

See also

`psort.py`, `pinfo.py`

mactime

Kit: REMnux · Kali Linux · SIFT Workstation **Capability:** Build a filesystem MAC-time timeline **Captured:** cyberlab-aio via --help on 2026-08-02 [raw Docs](#): <https://www.sleuthkit.org/sleuthkit>

← [Capability index](#) · [Kit tool list](#)

Purpose

Turn a TSK body file into a human-readable chronological timeline.

Common invocations

```
# from cyberlab 01-disk-forensics
mactime -b /tmp/bodyfile.txt -d > /tmp/timeline.csv
# from cyberlab 03-timeline-analysis
mactime -V
# from cyberlab 03-timeline-analysis
mactime -b /tmp/body.txt -d > /tmp/mactime.csv
# from cyberlab 22-sleuthkit-mastery
mactime -b exercise/bodyfile.txt -d > exercise/timeline.csv
# from cyberlab 22-sleuthkit-mastery
mactime -b exercise/bodyfile.txt -d | tail
# from cyberlab 23-plaso-supertimeline
mactime -d -b exercise/bodyfile.txt > timeline.csv
# from cyberlab 23-plaso-supertimeline
mactime -d -b exercise/bodyfile.txt | awk -F',' '{ $3 ~ /b/ } | sort
# from cyberlab 23-plaso-supertimeline
mactime -d -b exercise/bodyfile.txt | awk -F',' '{ $3 ~ /b/ } | sort | head -n 1
```

Options

All 9 options parsed from the captured help text; 8 reviewed with usage guidance.

Flag	Argument	What it does	When you would use it
-b	—	Specifies the body file location, else STDIN is used	Read the body file produced by <code>fls -m</code> — the normal input.
-d	—	Output in comma delimited format	Emit CSV rather than the default text, for spreadsheets or further tooling.
-h	—	Display a header with session information	Produce HTML output for a report.
-y	—	Dates are displayed in ISO 8601 format	Print dates ISO-style (year first), which sorts correctly.
-m	—	Dates have month as number instead of word (does not work with -y)	Print month numerically instead of by name.
-z	—	Specify the timezone the data came from (in the local system format) (does not work with -y)	Time zone of the evidence machine. Getting this wrong shifts the whole timeline.
-g	—	Specifies the group file location, else GIDs are used	Map group IDs to names using a supplied group file.
-p	—	Specifies the password file location, else UIDs are used	Map user IDs to names using a supplied passwd file.
-V	—	Prints the version to STDOUT	

Gotchas

- `mactime` reports in the time zone you give it, not the one embedded in the evidence. An unstated `-z` silently produces a plausible, wrong timeline — state it explicitly every run.
- A date range is passed as a trailing argument (`mactime -b body.txt 2026-01-01..2026-02-01`), not as a flag.

See also

`fls`, `tsk_gettimes`

pinfo.py

Kit: Kali Linux · SIFT Workstation **Capability:** Build a super-timeline from many artifact sources **Version:** plaso - pinfo version 20260512 **Captured:** cyberlab-aio via `--help` on 2026-08-02 [raw](#)

[← Capability index](#) · [Kit tool list](#)

Purpose

Shows information about a Plaso storage file, for example how it was collected, what information was extracted from a source, etc.

Synopsis

```
pinfo.py [-h] [--troubles] [-V] [--logfile FILENAME]
[--process_memory_limit SIZE] [--compare STORAGE_FILE]
[--output_format FORMAT] [--hash TYPE] [--report TYPE]
[--sections SECTIONS_LIST] [-v] [-w OUTPUTFILE]
[PATH]
```

Options

All 20 options parsed from the captured help text. The final column is filled in by review.

Flag	Argument	What it does	When you would use it
<code>-h</code>	—	Show this help message and exit.	
<code>--help</code>	—	Show this help message and exit.	
<code>--troubles</code>	—	Show troubleshooting information.	
<code>-V</code>	—	Show the version information.	
<code>--version</code>	—	Show the version information.	
<code>--logfile</code>	FILENAME	Path of the file in which to store log messages, by default this file will be named: "pinfo- YYYYMMDDThhmmss.log.gz". Note that the file will be gzip compressed if the extension is ".gz".	
<code>--log_file</code>	FILENAME	Path of the file in which to store log messages, by default this file will be named: "pinfo- YYYYMMDDThhmmss.log.gz". Note that the file will be gzip compressed if the extension is ".gz".	
<code>--log-file</code>	FILENAME	Path of the file in which to store log messages, by default this file will be named: "pinfo- YYYYMMDDThhmmss.log.gz". Note that the file will be gzip compressed if the extension is ".gz".	
<code>--process_memory_limit</code>	SIZE	Maximum amount of memory (data segment) a process is allowed to allocate in bytes, where 0 represents no limit. The default limit is 4294967296 (4 GiB). This applies to both the main (foreman) process	
<code>--process-memory-limit</code>	SIZE	Maximum amount of memory (data segment) a process is allowed to allocate in bytes, where 0 represents no limit. The default limit is 4294967296 (4 GiB). This applies to both the main (foreman) process	
<code>--compare</code>	STORAGE_FILE	The path of the storage file to compare against.	
<code>--output_format</code>	FORMAT	Format of the output, the default is: text. Supported options: json, markdown, text.	
<code>--output-format</code>	FORMAT	Format of the output, the default is: text. Supported options: json, markdown, text.	
<code>--hash</code>	TYPE	Type of hash to output in file_hashes report. Supported options: md5, sha1, sha256	
<code>--report</code>	TYPE	Report on specific information. Supported options: browser_search, chrome_extension, environment_variables, file_hashes, list, none, windows_services, winevt_providers	
<code>--sections</code>	SECTIONS_LIST	List of sections to output. This is a comma separated list where each entry is the name of a section. Use " --sections list" to list the available sections and " --sections all" to show all available	
<code>-v</code>	—	Print verbose output.	
<code>--verbose</code>	—	Print verbose output.	
<code>-w</code>	OUTPUTFILE	Output filename.	
<code>--write</code>	OUTPUTFILE	Output filename.	

See also

[log2timeline.py](#) , [psort.py](#)

psort.py

Kit: Kali Linux · SIFT Workstation **Capability:** Build a super-timeline from many artifact sources **Version:** plaso - psort version 20260512 **Captured:** cyberlab-ai0 via `---` help on 2026-08-02 raw

[← Capability index](#) · [Kit tool list](#)

Purpose

Filter, sort and output the events in a Plaso storage file.

Synopsis

```
psort.py [-h] [--troubles] [-V] [--analysis PLUGIN_LIST]
[--process_memory_limit SIZE]
[--temporary_directory DIRECTORY] [--worker_memory_limit SIZE]
[--worker_timeout MINUTES] [--logfile FILENAME] [-d] [-q] [-u]
[--status_view TYPE] [--status_view_file PATH]
[--status_view_interval SECONDS] [--slice DATE_TIME]
```

Common invocations

```
# from cyberlab 03-timeline-analysis
psort.py --version
# from cyberlab 03-timeline-analysis
psort.py -o l2tcsv -w /tmp/timeline.csv /tmp/case.plaso
# from cyberlab 49-intrusion-timeline-case
psort.py -o l2tcsv -w super_timeline.csv timeline.plaso "date > '2024-01-10 00:00:00' AND date < '2024-01-12 00:00:00'"
```

Options

All 59 options parsed from the captured help text; 4 reviewed with usage guidance.

Flag	Argument	What it does	When you would use it
<code>-h</code>	—	Show this help message and exit.	
<code>--help</code>	—	Show this help message and exit.	
<code>--troubles</code>	—	Show troubleshooting information.	
<code>-V</code>	—	Show the version information.	
<code>--version</code>	—	Show the version information.	
<code>--analysis</code>	PLUGIN_LIST	A comma separated list of analysis plugin names to be loaded or "--analysis list" to see a list of available plugins.	Run analysis plugins (tagging, sessionizing) over events.
<code>--process_memory_limit</code>	SIZE	Maximum amount of memory (data segment) a process is allowed to allocate in bytes, where 0 represents no limit. The default limit is 4294967296 (4 GiB). This applies to both the main (foreman) process	
<code>--process-memory-limit</code>	SIZE	Maximum amount of memory (data segment) a process is allowed to allocate in bytes, where 0 represents no limit. The default limit is 4294967296 (4 GiB). This applies to both the main (foreman) process	
<code>--temporary_directory</code>	DIRECTORY	Path to the directory that should be used to store temporary files created during processing.	
<code>--temporary-directory</code>	DIRECTORY	Path to the directory that should be used to store temporary files created during processing.	
<code>--worker_memory_limit</code>	SIZE	Maximum amount of memory (data segment and shared memory) a worker process is allowed to consume in bytes, where 0 represents no limit. The default limit is 2147483648 (2 GiB). If a worker process exc	
<code>--worker-memory-limit</code>	SIZE	Maximum amount of memory (data segment and shared memory) a worker process is allowed to consume in bytes, where 0 represents no limit. The default limit is 2147483648 (2 GiB). If a worker process exc	
<code>--worker_timeout</code>	MINUTES	Number of minutes before a worker process that is not providing status updates is considered inactive. The default timeout is 15.0 minutes. If a worker process exceeds this timeout it is killed by the	
<code>--worker-timeout</code>	MINUTES	Number of minutes before a worker process that is not providing status updates is considered inactive. The default timeout is 15.0 minutes. If a worker process exceeds this timeout it is killed by the	

Flag	Argument	What it does	When you would use it
<code>--logfile</code>	FILENAME	Path of the file in which to store log messages, by default this file will be named: "psort-YYYYMMDDThhmmss.log.gz". Note that the file will be gzip compressed if the extension is ".gz".	
<code>--log_file</code>	FILENAME	Path of the file in which to store log messages, by default this file will be named: "psort-YYYYMMDDThhmmss.log.gz". Note that the file will be gzip compressed if the extension is ".gz".	
<code>--log-file</code>	FILENAME	Path of the file in which to store log messages, by default this file will be named: "psort-YYYYMMDDThhmmss.log.gz". Note that the file will be gzip compressed if the extension is ".gz".	
<code>-d</code>	—	Enable debug output.	
<code>--debug</code>	—	Enable debug output.	
<code>-q</code>	—	Disable informational output.	Quiet, for scripted runs.
<code>--quiet</code>	—	Disable informational output.	Quiet, for scripted runs.
<code>-u</code>	—	Enable unattended mode and do not ask the user for additional input when needed, but terminate with an error instead.	
<code>--unattended</code>	—	Enable unattended mode and do not ask the user for additional input when needed, but terminate with an error instead.	
<code>--status_view</code>	TYPE	The processing status view mode: "file", "linear", "none" or "window".	
<code>--status-view</code>	TYPE	The processing status view mode: "file", "linear", "none" or "window".	
<code>--status_view_file</code>	PATH	The name of the status view file.	
<code>--status-view-file</code>	PATH	The name of the status view file.	
<code>--status_view_interval</code>	SECONDS	Number of seconds to update the status view.	
<code>--status-view-interval</code>	SECONDS	Number of seconds to update the status view.	
<code>--slice</code>	DATE_TIME	Date and time to create a time slice around. This parameter, if defined, will display all events that happened X minutes before and after the defined date, where X is controlled by the --slice_size op	
<code>--slice_size</code>	SLICE_SIZE	Defines the slice size. In the case of a regular time slice it defines the number of minutes the slice size should be. In the case of the --slicer it determines the number of events before and after a	
<code>--slice-size</code>	SLICE_SIZE	Defines the slice size. In the case of a regular time slice it defines the number of minutes the slice size should be. In the case of the --slicer it determines the number of events before and after a	
<code>--slicer</code>	—	Create a time slice around every filter match. This parameter, if defined will save all X events before and after a filter match has been made. X is defined by the --slice_size parameter.	
<code>--data</code>	PATH	Path to a directory containing the data files.	
<code>-a</code>	—	By default the psort removes duplicate entries from the output. This parameter changes that behavior so all events are included.	
<code>--include_all</code>	—	By default the psort removes duplicate entries from the output. This parameter changes that behavior so all events are included.	
<code>--include-all</code>	—	By default the psort removes duplicate entries from the output. This parameter changes that behavior so all events are included.	
<code>--language</code>	LANGUAGE_TAG	The preferred language, which is used for extracting and formatting Windows EventLog message strings. Use " --language list" to see a list of supported language tags. The en-US (LCID 0x0409) language	
<code>--additional_fields</code>	ADDITIONAL_FIELDS	Defines additional fields to be included in the output besides the default fields. Multiple additional field names can be defined as a list of comma separated values. Output formats that support addit	
<code>--additional-fields</code>	ADDITIONAL_FIELDS	Defines additional fields to be included in the output besides the default fields. Multiple additional field names can be defined as a list of comma separated values. Output formats that support addit	
<code>--custom_fields</code>	CUSTOM_FIELDS	Defines custom fields to be included in the output besides the default fields. A custom field is defined as "name:value". Multiple custom field names can be defined as list of comma separated values.	
<code>--custom-fields</code>	CUSTOM_FIELDS	Defines custom fields to be included in the output besides the default fields. A custom field is defined as "name:value". Multiple custom field names can be defined as list of comma separated values.	
<code>--custom_formatter_definitions</code>	PATH	Path to a file containing custom event formatter definitions, which is a .yaml file. Custom event formatter definitions can be used to customize event messages and override the built-in event formatte	
<code>--custom-formatter-definitions</code>	PATH	Path to a file containing custom event formatter definitions, which is a .yaml file. Custom event formatter definitions can be used to customize event messages and override the built-in event formatte	

Flag	Argument	What it does	When you would use it
<code>--dynamic_time</code>	—	Indicate that the output should use dynamic time. Output formats that support dynamic time are: dynamic	
<code>--dynamic-time</code>	—	Indicate that the output should use dynamic time. Output formats that support dynamic time are: dynamic	
<code>--output_time_zone</code>	TIME_ZONE	time zone of date and time values written to the output, if supported by the output format. Use "list" to see a list of available time zones. Output formats that support an output time zone are: dynam	
<code>--output-time-zone</code>	TIME_ZONE	time zone of date and time values written to the output, if supported by the output format. Use "list" to see a list of available time zones. Output formats that support an output time zone are: dynam	
<code>-o</code>	FORMAT	The output format. Use "-o list" to see a list of available output formats.	Output format — <code>l2tcsv</code> , <code>dynamic</code> , <code>json</code> , or a timeline tool.
<code>--output_format</code>	FORMAT	The output format. Use "-o list" to see a list of available output formats.	Output format — <code>l2tcsv</code> , <code>dynamic</code> , <code>json</code> , or a timeline tool.
<code>--output-format</code>	FORMAT	The output format. Use "-o list" to see a list of available output formats.	Output format — <code>l2tcsv</code> , <code>dynamic</code> , <code>json</code> , or a timeline tool.
<code>-w</code>	OUTPUT_FILE	Output filename.	Write output to a file rather than stdout.
<code>--write</code>	OUTPUT_FILE	Output filename.	Write output to a file rather than stdout.
<code>--fields</code>	FIELDS	Defines which fields should be included in the output.	
<code>--profilers</code>	PROFILERS_LIST	List of profilers to use by the tool. This is a comma separated list where each entry is the name of a profiler. Use "--profilers list" to list the available profilers.	
<code>--profiling_directory</code>	DIRECTORY	Path to the directory that should be used to store the profiling sample files. By default the sample files are stored in the current working directory.	
<code>--profiling-directory</code>	DIRECTORY	Path to the directory that should be used to store the profiling sample files. By default the sample files are stored in the current working directory.	
<code>--profiling_sample_rate</code>	SAMPLE_RATE	Profiling sample rate (defaults to a sample every 1000 files).	
<code>--profiling-sample-rate</code>	SAMPLE_RATE	Profiling sample rate (defaults to a sample every 1000 files).	

Gotchas

- The date filter is a positional argument, not a flag. Filtering to the incident window is what makes a multi-million-event timeline usable.
- Output ordering follows the storage file, so always sort or filter explicitly rather than assuming chronology.

See also

`log2timeline.py`, `pinfo.py`

tsk_gettimes

Kit: REMnux · Kali Linux · SIFT Workstation **Capability:** Build a filesystem MAC-time timeline **Version:** The Sleuth Kit ver 4.11.1 **Captured:** cyberlab-aio via --help on 2026-08-02 **raw Docs:** <https://www.sleuthkit.org/sleuthkit>

← [Capability index](#) · [Kit tool list](#)

Purpose

Analyze disk images and recover files from them.

Synopsis

```
tsk_gettimes [-vVm] [-i imgtype] [-b dev_sector_size] [-z zone] [-s seconds] image [image]
```

Options

All 7 options parsed from the captured help text. The final column is filled in by review.

Flag	Argument	What it does	When you would use it
-i	imgtype	The format of the image file (use '-i list' for supported types)	
-b	dev_sector_size	The size (in bytes) of the device sectors	
-m	—	Calculate MD5 hash in output (slow)	
-v	—	verbose output to stderr	
-V	—	Print version	
-z	—	Time zone of original machine (i.e. EST5EDT or GMT) (only useful with -l)	
-s	seconds	Time skew of original machine (in seconds) (only useful with -l & -m)	

See also

[fls](#), [mactime](#)

CryptoTester (GUI)

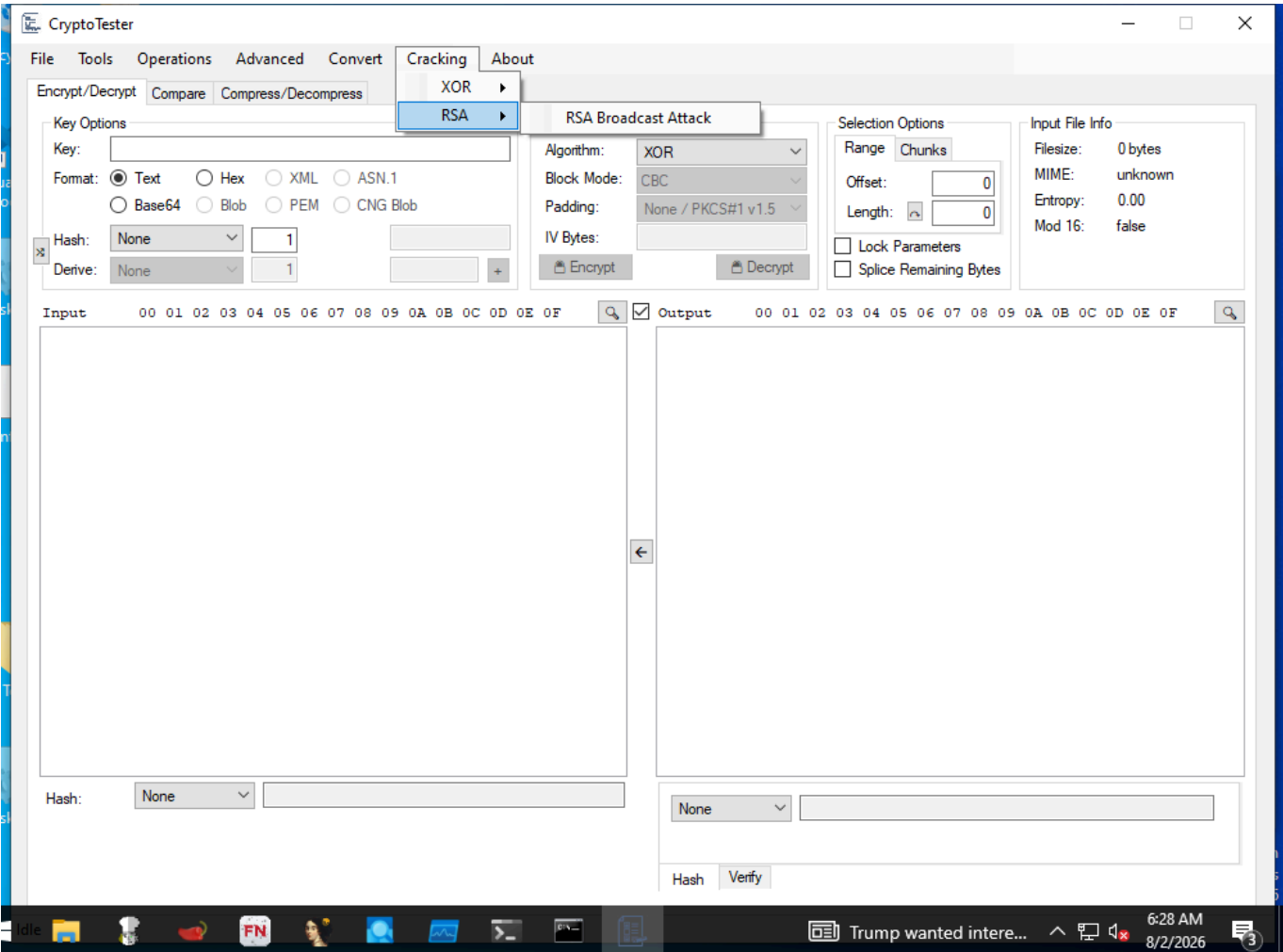
Capability: decode deobfuscate **Window:** WindowsForms10.Window.8.app.0.141b42a_r7_ad1 **Version:** — **Captured:** C:\Tools\CryptoTester\CryptoTester.exe on 2026-08-02 — control tree in [capture/gui/CryptoTester/CryptoTester.tree.txt](#)

← [Capability index](#) · [Kit tool list](#)

Purpose

Try cryptographic operations against a sample.

Window



Controls

All 149 nodes come from the capture; the 126 interactive controls are listed here.

Control	Type	AutomationId	What it does
Idle	Text	-	
File	MenuItem	-	
Open File	MenuItem	-	
Input	MenuItem	-	
Text	MenuItem	-	
Base64	MenuItem	-	
Zero Bytes	MenuItem	-	

Control	Type	AutomationId	What it does
Sequential Bytes	MenuItem	-	
Integer	MenuItem	-	
Chrysanthemum.jpg	MenuItem	-	
Desert.jpg	MenuItem	-	
Save Output	MenuItem	-	
Tools	MenuItem	-	
Blob Analyzer	MenuItem	-	
RSA Calculator	MenuItem	-	
Key Finder	MenuItem	-	
Keystream Finder	MenuItem	-	
RNG Tester	MenuItem	-	
Base Encoder	MenuItem	-	
String Encoder	MenuItem	-	
Unit Tester	MenuItem	-	
ECC Validator	MenuItem	-	
Chunk Viewer	MenuItem	-	
Operations	MenuItem	-	
XOR Files	MenuItem	-	
AND Files	MenuItem	-	
Generate Keystream	MenuItem	-	
Visual Difference	MenuItem	-	
Bruteforce Algorithm	MenuItem	-	
Bruteforce Keys	MenuItem	-	
Attempt Blind Decryption	MenuItem	-	
Advanced	MenuItem	-	
Little Endian	MenuItem	-	
Custom	MenuItem	-	
Rounds	MenuItem	-	
Constant	MenuItem	-	
Position	MenuItem	-	
Matrix	MenuItem	-	
IV	MenuItem	-	
Get IV From Input	MenuItem	-	
Auto Derive IV	MenuItem	-	
Enter Text IV	MenuItem	-	
Recover IV From Plaintext	MenuItem	-	
Presets	MenuItem	-	
HiddenTear	MenuItem	-	
OpenSSL	MenuItem	-	
DiskCryptorVolumeHeader	MenuItem	-	
Convert	MenuItem	-	
DWORD ⇄ Integer	MenuItem	-	
Hex ⇄ Integers	MenuItem	-	
Cracking	MenuItem	-	
XOR	MenuItem	-	
XOR Attack	MenuItem	-	
RSA	MenuItem	-	
RSA Broadcast Attack	MenuItem	-	
About	MenuItem	-	
Encrypt/Decrypt	Pane	393580	
⌕	Pane	590170	

Control	Type	AutomationId	What it does
Input File Info	Group	262444	
Filesize:	Text	459008	
MIME:	Text	917746	
Mod 16:	Text	589930	
Entropy:	Text	720998	
false	Text	131636	
unknown	Text	590202	
0.00	Text	393764	
0 bytes	Text	589906	
Selection Options	Group	786508	
Range	Pane	524354	
Offset:	Text	262752	
Length:	Text	197238	
0	Pane	328284	
	Pane	197208	
Lock Parameters	Pane	131694	
Splice Remaining Bytes	Pane	1049208	
Key Options	Group	328278	
+	Pane	66170	
None	ComboBox	66176	
Open	Button	-	
Derive:	ComboBox	66180	
Derive:	Text	-	
CNG Blob	Pane	66184	
ASN.1	Pane	66186	
1	Pane	66188	
PEM	Pane	66192	
XML	Pane	66202	
Hash:	Text	66204	
Blob	Pane	66206	
Base64	Pane	66208	
Format:	Text	66210	
Hex	Pane	66212	
Text	Pane	66214	
Key:	Text	66216	
Encryption Options	Group	66220	
IV Bytes:	ComboBox	66224	
IV Bytes:	Text	-	
Block Mode:	Text	66230	
Algorithm:	Text	66232	
CBC	ComboBox	66234	
Padding:	Text	66238	
None / PKCS#1 v1.5	ComboBox	66242	
 Encrypt	Pane	66246	
 Decrypt	Pane	66248	
Input	Text	66252	
Output	Text	66254	
00 01 02 03 04 05 06 07 08 09 0A 0B 0C 0D 0E 0F	Text	66256	
	Pane	66260	
←	Pane	66262	
Hash	Pane	66278	

Control	Type	AutomationId	What it does
Public Key:	Text	66288	

Using it

1. Paste or load the ciphertext.
2. Try the obvious first: single-byte XOR and the common block ciphers cover most malware configuration blobs.
3. Use the entropy and pattern views to judge whether the result is plausible plaintext before believing a key.
4. Record the key and mode that worked; a decryption nobody can reproduce is not a finding.

Gotchas

- It is an experiment bench, not an oracle. It tells you a transformation produced output, not that the output is correct.
- Plausible-looking plaintext from a short sample is often coincidence. Confirm against a second sample before concluding.

cyberchef

Kit: REMnux · FLARE-VM · Security Onion **Capability:** Decode, decrypt or transform encoded data **Captured:** cyberlab-aio via --help on 2026-08-02 [raw](#) **Docs:** <https://github.com/gchq/CyberChef/>

[← Capability index](#) · [Kit tool list](#)

Purpose

Decode and otherwise analyze data using this browser app.

Common invocations

```
# from cyberlab 09-deobfuscation
cyberchef &
# from cyberlab 25-cyberchef-recipes
cyberchef --help | head -n 5
# from cyberlab 25-cyberchef-recipes
cyberchef --recipe "From Base64" --input encoded_payload.ps1.txt 2>/dev/null | cyberchef --recipe "XOR({'option':'Hex','value':'2a'})" | cyberchef --1
# from cyberlab 25-cyberchef-recipes
cyberchef --recipe "From Base64" --input encoded_payload.ps1.txt | cyberchef --recipe "XOR({'option':'Hex','value':'2a'})" | cyberchef --recipe "Gunzi
```

Options

No option definitions could be parsed from this tool's help output. It may be subcommand-driven or have no flags; needs manual review.

See also

[base64dump.py](#), [rax2](#), [xxd](#), [openssl](#), [numbers-to-string.py](#)

hashcat

Kit: Kali Linux **Capability:** Crack passwords and hashes **Version:** v6.2.6 **Captured:** cyberlab-aio via --help on 2026-08-02 raw

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Purpose

hashcat (v6.2.6) starting in help mode

Synopsis

```
hashcat [options]... hash[hashfile|hccapxfile [dictionary|mask|directory]]...
```

Common invocations

```
# from cyberlab 11-offensive-kali
hashcat --version
# from cyberlab 11-offensive-kali
hashcat -b -m 0 2>&1 | head -n 8
# from cyberlab 11-offensive-kali
hashcat -m 0 -a 0 exercise/lab_hash.txt /usr/share/wordlists/rockyou.txt
# from cyberlab 11-offensive-kali
hashcat -m 0 --show exercise/lab_hash.txt
# from cyberlab 40-password-cracking
hashcat prints a version string like "v6.2.6".
# from cyberlab 40-password-cracking
hashcat -m 0 -a 0 exercise/hash.txt exercise/wordlist.txt --potfile-path exercise/hc.pot
# from cyberlab 40-password-cracking
hashcat -m 0 --show exercise/hash.txt --potfile-path exercise/hc.pot
# from cyberlab 40-password-cracking
hashcat -m 0 -a 0 -r /usr/share/hashcat/rules/best64.rule exercise/hash.txt exercise/wordlist.txt --potfile-path exercise/hc.pot
```

Options

All 143 options parsed from the captured help text. The final column is filled in by review.

Flag	Argument	What it does	When you would use it
-m	—	Num Hash-type, references below (otherwise autodetect) -m 1000	
--hash-type	—	Num Hash-type, references below (otherwise autodetect) -m 1000	
-a	—	Num Attack-mode, see references below -a 3	
--attack-mode	—	Num Attack-mode, see references below -a 3	
-V	—	Print version	
--version	—	Print version	
-h	—	Print help	
--help	—	Print help	
--quiet	—	Suppress output	
--hex-charset	—	Assume charset is given in hex	
--hex-salt	—	Assume salt is given in hex	
--hex-wordlist	—	Assume words in wordlist are given in hex	
--force	—	Ignore warnings	
--status	—	Enable automatic update of the status screen	
--status-json	—	Enable JSON format for status output	
--status-timer	—	Num Sets seconds between status screen updates to X --status-timer=1	
--stdin-timeout-abort	—	Num Abort if there is no input from stdin for X seconds --stdin-timeout-abort=300	
--machine-readable	—	Display the status view in a machine-readable format	
--keep-guessing	—	Keep guessing the hash after it has been cracked	

Flag	Argument	What it does	When you would use it
<code>--self-test-disable</code>	—	Disable self-test functionality on startup	
<code>--loopback</code>	—	Add new plains to induct directory	
<code>--markov-hcstat2</code>	—	File Specify hcstat2 file to use <code>--markov-hcstat2=my.hcstat2</code>	
<code>--markov-disable</code>	—	Disables markov-chains, emulates classic brute-force	
<code>--markov-classic</code>	—	Enables classic markov-chains, no per-position	
<code>--markov-inverse</code>	—	Enables inverse markov-chains, no per-position	
<code>-t</code>	—	Num Threshold X when to stop accepting new markov-chains <code>-t 50</code>	
<code>--markov-threshold</code>	—	Num Threshold X when to stop accepting new markov-chains <code>-t 50</code>	
<code>--runtime</code>	—	Num Abort session after X seconds of runtime <code>--runtime=10</code>	
<code>--session</code>	—	Str Define specific session name <code>--session=mysession</code>	
<code>--restore</code>	—	Restore session from <code>--session</code>	
<code>--restore-disable</code>	—	Do not write restore file	
<code>--restore-file-path</code>	—	File Specific path to restore file <code>--restore-file-path=x.restore</code>	
<code>-o</code>	—	File Define outfile for recovered hash <code>-o outfile.txt</code>	
<code>--outfile</code>	—	File Define outfile for recovered hash <code>-o outfile.txt</code>	
<code>--outfile-format</code>	—	Str Outfile format to use, separated with commas <code>--outfile-format=1,3</code>	
<code>--outfile-autohex-disable</code>	—	Disable the use of <code>\$HEX[]</code> in output plains	
<code>--outfile-check-timer</code>	—	Num Sets seconds between outfile checks to X <code>--outfile-check-timer=30</code>	
<code>-p</code>	—	Char Separator char for hashlists and outfile <code>-p :</code>	
<code>--separator</code>	—	Char Separator char for hashlists and outfile <code>-p :</code>	
<code>--stdout</code>	—	Do not crack a hash, instead print candidates only	
<code>--show</code>	—	Compare hashlist with potfile; show cracked hashes	
<code>--left</code>	—	Compare hashlist with potfile; show uncracked hashes	
<code>--username</code>	—	Enable ignoring of usernames in hashfile	
<code>--remove</code>	—	Enable removal of hashes once they are cracked	
<code>--remove-timer</code>	—	Num Update input hash file each X seconds <code>--remove-timer=30</code>	
<code>--potfile-disable</code>	—	Do not write potfile	
<code>--potfile-path</code>	—	File Specific path to potfile <code>--potfile-path=my.pot</code>	
<code>--encoding-from</code>	—	Code Force internal wordlist encoding from X <code>--encoding-from=iso-8859-15</code>	
<code>--encoding-to</code>	—	Code Force internal wordlist encoding to X <code>--encoding-to=utf-32le</code>	
<code>--debug-mode</code>	—	Num Defines the debug mode (hybrid only by using rules) <code>--debug-mode=4</code>	
<code>--debug-file</code>	—	File Output file for debugging rules <code>--debug-file=good.log</code>	
<code>--induction-dir</code>	—	Dir Specify the induction directory to use for loopback <code>--induction=inducts</code>	
<code>--outfile-check-dir</code>	—	Dir Specify the outfile directory to monitor for plains <code>--outfile-check-dir=x</code>	
<code>--logfile-disable</code>	—	Disable the logfile	
<code>--hccapx-message-pair</code>	—	Num Load only message pairs from hccapx matching X <code>--hccapx-message-pair=2</code>	
<code>--nonce-error-corrections</code>	—	Num The BF size range to replace AP's nonce last bytes <code>--nonce-error-corrections=16</code>	
<code>--keyboard-layout-mapping</code>	—	File Keyboard layout mapping table for special hash-modes <code>--keyb=german.hckmap</code>	
<code>--truecrypt-keyfiles</code>	—	File Keyfiles to use, separated with commas <code>--truecrypt-keyf=x.png</code>	
<code>--veracrypt-keyfiles</code>	—	File Keyfiles to use, separated with commas <code>--veracrypt-keyf=x.txt</code>	
<code>--veracrypt-pim-start</code>	—	Num VeraCrypt personal iterations multiplier start <code>--veracrypt-pim-start=450</code>	
<code>--veracrypt-pim-stop</code>	—	Num VeraCrypt personal iterations multiplier stop <code>--veracrypt-pim-stop=500</code>	
<code>-b</code>	—	Run benchmark of selected hash-modes	
<code>--benchmark</code>	—	Run benchmark of selected hash-modes	
<code>--benchmark-all</code>	—	Run benchmark of all hash-modes (requires <code>-b</code>)	
<code>--speed-only</code>	—	Return expected speed of the attack, then quit	
<code>--progress-only</code>	—	Return ideal progress step size and time to process	
<code>-c</code>	—	Num Sets size in MB to cache from the wordfile to X <code>-c 32</code>	
<code>--segment-size</code>	—	Num Sets size in MB to cache from the wordfile to X <code>-c 32</code>	
<code>--bitmap-min</code>	—	Num Sets minimum bits allowed for bitmaps to X <code>--bitmap-min=24</code>	
<code>--bitmap-max</code>	—	Num Sets maximum bits allowed for bitmaps to X <code>--bitmap-max=24</code>	

Flag	Argument	What it does	When you would use it
<code>--cpu-affinity</code>	—	Str Locks to CPU devices, separated with commas <code>--cpu-affinity=1,2,3</code>	
<code>--hook-threads</code>	—	Num Sets number of threads for a hook (per compute unit) <code>--hook-threads=8</code>	
<code>--hash-info</code>	—	Show information for each hash-mode	
<code>--example-hashes</code>	—	Alias of <code>--hash-info</code>	
<code>--backend-ignore-cuda</code>	—	Do not try to open CUDA interface on startup	
<code>--backend-ignore-hip</code>	—	Do not try to open HIP interface on startup	
<code>--backend-ignore-metal</code>	—	Do not try to open Metal interface on startup	
<code>--backend-ignore-opengl</code>	—	Do not try to open OpenGL interface on startup	
<code>-I</code>	—	Show system/environment/backend API info <code>-I</code> or <code>-II</code>	
<code>--backend-info</code>	—	Show system/environment/backend API info <code>-I</code> or <code>-II</code>	
<code>-d</code>	—	Str Backend devices to use, separated with commas <code>-d 1</code>	
<code>--backend-devices</code>	—	Str Backend devices to use, separated with commas <code>-d 1</code>	
<code>-D</code>	—	Str OpenGL device-types to use, separated with commas <code>-D 1</code>	
<code>--opengl-device-types</code>	—	Str OpenGL device-types to use, separated with commas <code>-D 1</code>	
<code>-O</code>	—	Enable optimized kernels (limits password length)	
<code>--optimized-kernel-enable</code>	—	Enable optimized kernels (limits password length)	
<code>-M</code>	—	Disable multiply kernel-accel with processor count	
<code>--multiply-accel-disable</code>	—	Disable multiply kernel-accel with processor count	
<code>-w</code>	—	Num Enable a specific workload profile, see pool below <code>-w 3</code>	
<code>--workload-profile</code>	—	Num Enable a specific workload profile, see pool below <code>-w 3</code>	
<code>-n</code>	—	Num Manual workload tuning, set outerloop step size to X <code>-n 64</code>	
<code>--kernel-accel</code>	—	Num Manual workload tuning, set outerloop step size to X <code>-n 64</code>	
<code>-u</code>	—	Num Manual workload tuning, set innerloop step size to X <code>-u 256</code>	
<code>--kernel-loops</code>	—	Num Manual workload tuning, set innerloop step size to X <code>-u 256</code>	
<code>-T</code>	—	Num Manual workload tuning, set thread count to X <code>-T 64</code>	
<code>--kernel-threads</code>	—	Num Manual workload tuning, set thread count to X <code>-T 64</code>	
<code>--backend-vector-width</code>	—	Num Manually override backend vector-width to X <code>--backend-vector=4</code>	
<code>--spin-damp</code>	—	Num Use CPU for device synchronization, in percent <code>--spin-damp=10</code>	
<code>--hwmon-disable</code>	—	Disable temperature and fanspeed reads and triggers	
<code>--hwmon-temp-abort</code>	—	Num Abort if temperature reaches X degrees Celsius <code>--hwmon-temp-abort=100</code>	
<code>--scrypt-tmto</code>	—	Num Manually override TMTO value for scrypt to X <code>--scrypt-tmto=3</code>	
<code>-s</code>	—	Num Skip X words from the start <code>-s 1000000</code>	
<code>--skip</code>	—	Num Skip X words from the start <code>-s 1000000</code>	
<code>-l</code>	—	Num Limit X words from the start + skipped words <code>-l 1000000</code>	
<code>--limit</code>	—	Num Limit X words from the start + skipped words <code>-l 1000000</code>	
<code>--keyspace</code>	—	Show keyspace base:mod values and quit	
<code>-j</code>	—	Rule Single rule applied to each word from left wordlist <code>-j 'c'</code>	
<code>--rule-left</code>	—	Rule Single rule applied to each word from left wordlist <code>-j 'c'</code>	
<code>-k</code>	—	Rule Single rule applied to each word from right wordlist <code>-k '^.'</code>	
<code>--rule-right</code>	—	Rule Single rule applied to each word from right wordlist <code>-k '^.'</code>	
<code>-r</code>	—	File Multiple rules applied to each word from wordlists <code>-r rules/best64.rule</code>	
<code>--rules-file</code>	—	File Multiple rules applied to each word from wordlists <code>-r rules/best64.rule</code>	
<code>-g</code>	—	Num Generate X random rules <code>-g 10000</code>	
<code>--generate-rules</code>	—	Num Generate X random rules <code>-g 10000</code>	
<code>--generate-rules-func-min</code>	—	Num Force min X functions per rule	
<code>--generate-rules-func-max</code>	—	Num Force max X functions per rule	
<code>--generate-rules-func-sel</code>	—	Str Pool of rule operators valid for random rule engine <code>--generate-rules-func-sel=ioTlc</code>	
<code>--generate-rules-seed</code>	—	Num Force RNG seed set to X	
<code>-1</code>	—	CS User-defined charset ?1 <code>-1 ?!d?u</code>	
<code>--custom-charset1</code>	—	CS User-defined charset ?1 <code>-1 ?!d?u</code>	
<code>-2</code>	—	CS User-defined charset ?2 <code>-2 ?!d?s</code>	

Flag	Argument	What it does	When you would use it
<code>--custom-charset2</code>	—	CS User-defined charset ?2 -2 ?!d?s	
<code>-3</code>	—	CS User-defined charset ?3	
<code>--custom-charset3</code>	—	CS User-defined charset ?3	
<code>-4</code>	—	CS User-defined charset ?4	
<code>--custom-charset4</code>	—	CS User-defined charset ?4	
<code>--identify</code>	—	Shows all supported algorithms for input hashes --identify my.hash	
<code>-i</code>	—	Enable mask increment mode	
<code>--increment</code>	—	Enable mask increment mode	
<code>--increment-min</code>	—	Num Start mask incrementing at X --increment-min=4	
<code>--increment-max</code>	—	Num Stop mask incrementing at X --increment-max=8	
<code>-S</code>	—	Enable slower (but advanced) candidate generators	
<code>--slow-candidates</code>	—	Enable slower (but advanced) candidate generators	
<code>--brain-server</code>	—	Enable brain server	
<code>--brain-server-timer</code>	—	Num Update the brain server dump each X seconds (min:60) --brain-server-timer=300	
<code>-z</code>	—	Enable brain client, activates -S	
<code>--brain-client</code>	—	Enable brain client, activates -S	
<code>--brain-client-features</code>	—	Num Define brain client features, see below --brain-client-features=3	
<code>--brain-host</code>	—	Str Brain server host (IP or domain) --brain-host=127.0.0.1	
<code>--brain-port</code>	—	Port Brain server port --brain-port=13743	
<code>--brain-password</code>	—	Str Brain server authentication password --brain-password=bZfhCvGUSjRq	
<code>--brain-session</code>	—	Hex Overrides automatically calculated brain session --brain-session=0x2ae611db	
<code>--brain-session-whitelist</code>	—	Hex Allow given sessions only, separated with commas --brain-session-whitelist=0x2ae611db	

See also

[john](#), [hydra](#)

hydra

Kit: Kali Linux · SIFT Workstation **Capability:** Crack passwords and hashes **Version:** Hydra v9.4 **Captured:** `cyberlab-aio` via `--help` on 2026-08-02 [raw](#)

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Common invocations

```
# from cyberlab 11-offensive-kali
hydra -h 2>&1 | head -n 1
# from cyberlab 11-offensive-kali
hydra -U ssh 2>&1 | head -n 5
```

Options

No option definitions could be parsed from this tool's help output. It may be subcommand-driven or have no flags; needs manual review.

See also

`hashcat` , `john`

john

Kit: Kali Linux **Capability:** Crack passwords and hashes **Version:** stat: version: No such file or directory **Captured:** cyberlab-aio via (no args) on 2026-08-02 [raw](#)

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Purpose

John the Ripper password cracker, version 1.9.0

Synopsis

```
john [OPTIONS] [PASSWORD-FILES]
```

Common invocations

```
# from cyberlab 11-offensive-kali
john --list=build-info 2>&1 | head -n 1
# from cyberlab 11-offensive-kali
john --format=Raw-MD5 --wordlist=/usr/share/wordlists/rockyou.txt exercise/lab_hash.txt
# from cyberlab 11-offensive-kali
john --show --format=Raw-MD5 exercise/lab_hash.txt
# from cyberlab 40-password-cracking
john --version 2>&1 | head -n 1
# from cyberlab 40-password-cracking
john --list=formats | tr ',' '\n' | grep -i -m 5 md5
# from cyberlab 40-password-cracking
john --format=raw-md5 --wordlist=exercise/wordlist.txt exercise/hash.txt
# from cyberlab 40-password-cracking
john --format=raw-md5 --show exercise/hash.txt
```

Options

All 21 options parsed from the captured help text. The final column is filled in by review.

Flag	Argument	What it does	When you would use it
<code>--single</code>	—	"single crack" mode	
<code>--wordlist</code>	FILE	wordlist mode, read words from FILE or stdin	
<code>--stdin</code>	—	wordlist mode, read words from FILE or stdin	
<code>--rules</code>	—	enable word mangling rules for wordlist mode	
<code>--incremental</code>	MODE (optional)	"incremental" mode [using section MODE]	
<code>--external</code>	MODE	external mode or word filter	
<code>--stdout</code>	LENGTH (optional)	just output candidate passwords [cut at LENGTH]	
<code>--restore</code>	NAME (optional)	restore an interrupted session [called NAME]	
<code>--session</code>	NAME	give a new session the NAME	
<code>--status</code>	NAME (optional)	print status of a session [called NAME]	
<code>--make-charset</code>	FILE	make a charset, FILE will be overwritten	
<code>--show</code>	—	show cracked passwords	
<code>--test</code>	TIME (optional)	run tests and benchmarks for TIME seconds each	
<code>--users</code>	[~]LOGIN[UID[...]]	[do not] load this (these) user(s) only	
<code>--groups</code>	[~]GID[...]	load users [not] of this (these) group(s) only	
<code>--shells</code>	[~]SHELL[...]	load users with[out] this (these) shell(s) only	
<code>--salts</code>	[~]N	load salts with[out] at least N passwords only	
<code>--save-memory</code>	LEVEL	enable memory saving, at LEVEL 1..3	
<code>--node</code>	MIN[~MAX]	this node's number range out of TOTAL count	
<code>--fork</code>	N	fork N processes	
<code>--format</code>	NAME	force hash type NAME: descrypt/bsdcrypt/md5crypt/ bcrypt/LM/AFS/tripcode/dummy/crypt	

See also

[hashcat](#) , [hydra](#)

openssl

Kit: Base OS — present on every Linux image **Capability:** Decode, decrypt or transform encoded data **Version:** Invalid command '--version'; type "help" for a list.
Captured: `cyberlab-aio` via `--help` on 2026-08-02 [raw](#)

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Purpose

Standard commands

Options

No option definitions could be parsed from this tool's help output. It may be subcommand-driven or have no flags; needs manual review.

See also

`cyberchef`, `base64dump.py`, `rax2`, `xxd`, `numbers-to-string.py`

rax2

Kit: REMnux · Kali Linux · SIFT Workstation **Capability:** Decode, decrypt or transform encoded data **Version:** rax2 6.1.9 **Captured:** cyberlab-aio via --help on 2026-08-02 **raw Docs:** <https://www.radare.org/n/radare2.html>

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Purpose

Examine binary files, including disassembling and debugging. Includes r2ai and decal plugins for LLM-powered analysis (API key or local Ollama required), plus the r2ghidra plugin for Ghidra decompilation via the pdg command.

Synopsis

```
rax2 [-h|...] [- | expr ...] # convert between numeric bases
int      -> hex          ; rax2 10
hex      -> int          ; rax2 0xa
-int     -> hex          ; rax2 -77
-hex     -> int          ; rax2 0xffffffffb3
int      -> bin          ; rax2 b30
```

Options

All 30 options parsed from the captured help text. The final column is filled in by review.

Flag	Argument	What it does	When you would use it
-a	—	show ascii table ; rax2 -a	
-b	base	output in ; rax2 -b 10 0x46	
-c	—	output in C string ; rax2 -c 0x1234 # \x34\x12\x00\x00	
-C	—	dump as C byte array ; rax2 -C < bytes	
-d	—	force integer ; rax2 -d 3 -> 3 instead of 0x3	
-e	—	swap endianness ; rax2 -e 0x33	
-D	—	base64 decode ; rax2 -D "aGVsbG8="	
-E	—	base64 encode ; rax2 -E "hello"	
-f	—	floating point ; rax2 -f 6.3+2.1	
-F	—	stdin slurp code hex ; rax2 -F < shellcode.c/py/js]	
-h	—	help ; rax2 -h	
-H	—	hash string ; rax2 -H linux osx	
-i	—	IP address <-> LONG ; rax2 -i 3530468537	
-j	—	json format output ; rax2 -j 0x1234 # same as r2 -c '?j 0x1234'	
-k	—	keep base ; rax2 -k 33+3 -> 36	
-K	—	randomart ; rax2 -K 0x34 1020304050	
-n	—	newline ; append newline to output (for -E/-D/-r/.)	
-o	—	octalstr -> raw ; rax2 -o \162 \62 # r2	
-q	—	quiet mode ; rax2 -qC < /etc/hosts # be quiet	
-r	—	r2 style output ; rax2 -r 0x1234 # same as r2 -c '? 0x1234'	
-s	—	hexstr -> raw ; rax2 -s 43 4a 50	
-S	—	raw -> hexstr ; rax2 -S < /bin/ls > ls.hex	
-t	—	tstamp -> str ; rax2 -t 1234567890	
-u	—	units ; rax2 -u 389289238 # 317.0M	
-v	—	version ; rax2 -v	
-w	—	signed word ; rax2 -w 0xffff 0xffff_ffff '0xff&0xffff'	
-x	—	output in hexpairs ; rax2 -x 0x1234 # 34120000	
-X	—	bin -> hex(bignum) ; rax2 -X 111111111 # 0x1ff	
-z	—	str -> bin ; rax2 -z hello	

Flag	Argument	What it does	When you would use it
<code>-Z</code>	—	bin -> str ; rax2 -Z 01000101 01110110	

See also

[cyberchef](#) , [base64dump.py](#) , [xxd](#) , [openssl](#) , [numbers-to-string.py](#)

binwalk

Kit: REMnux · Kali Linux **Capability:** Carve files out of unstructured data; Detect and reverse packing; Find hidden data **Captured:** cyberlab-aio via --help on 2026-08-02 **raw Docs:** <https://github.com/ReFirmLabs/binwalk>

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Purpose

Find and extract embedded files and filesystems inside a binary blob or firmware image.

Synopsis

```
binwalk [OPTIONS] [FILE1] [FILE2] [FILE3] ...
```

Common invocations

```
# from cyberlab 33-binwalk-firmware
binwalk --help | head -n 3
# from cyberlab 33-binwalk-firmware
binwalk firmware.bin
# from cyberlab 33-binwalk-firmware
binwalk -e firmware.bin
# from cyberlab 33-binwalk-firmware
binwalk -E firmware.bin
```

Options

All 102 options parsed from the captured help text; 8 reviewed with usage guidance.

Flag	Argument	What it does	When you would use it
-B	—	Scan target file(s) for common file signatures	
--signature	—	Scan target file(s) for common file signatures	
-R	str	Scan target file(s) for the specified sequence of bytes	
--raw	str	Scan target file(s) for the specified sequence of bytes	
-A	—	Scan target file(s) for common executable opcode signatures	Scan for executable opcodes to identify architecture.
--opcodes	—	Scan target file(s) for common executable opcode signatures	Scan for executable opcodes to identify architecture.
-m	file	Specify a custom magic file to use	
--magic	file	Specify a custom magic file to use	
-b	—	Disable smart signature keywords	
--dumb	—	Disable smart signature keywords	
-I	—	Show results marked as invalid	
--invalid	—	Show results marked as invalid	
-x	str	Exclude results that match	Exclude signatures matching this string, to cut false hits.
--exclude	str	Exclude results that match	Exclude signatures matching this string, to cut false hits.
-y	str	Only show results that match	Only report signatures matching this string.
--include	str	Only show results that match	Only report signatures matching this string.
-e	—	Automatically extract known file types	Extract what is found, rather than only listing it.
--extract	—	Automatically extract known file types	Extract what is found, rather than only listing it.
-D	type[:ext[:cmd]]	Extract signatures (regular expression), give the files an extension of , and execute	
--dd	type[:ext[:cmd]]	Extract signatures (regular expression), give the files an extension of , and execute	
-M	—	Recursively scan extracted files	Recurse into extracted files (matryoshka) — for nested firmware.
--matryoshka	—	Recursively scan extracted files	Recurse into extracted files (matryoshka) — for nested firmware.
-d	int	Limit matryoshka recursion depth (default: 8 levels deep)	Limit recursion depth; unbounded -M can explode.

Flag	Argument	What it does	When you would use it
<code>--depth</code>	int	Limit matryoshka recursion depth (default: 8 levels deep)	Limit recursion depth; unbounded <code>-M</code> can explode.
<code>-C</code>	str	Extract files/folders to a custom directory (default: current working directory)	Choose the output directory for extractions.
<code>--directory</code>	str	Extract files/folders to a custom directory (default: current working directory)	Choose the output directory for extractions.
<code>-j</code>	int	Limit the size of each extracted file	
<code>--size</code>	int	Limit the size of each extracted file	
<code>-n</code>	int	Limit the number of extracted files	
<code>--count</code>	int	Limit the number of extracted files	
<code>-0</code>	str	Execute external extraction utilities with the specified user's privileges	
<code>--run-as</code>	str	Execute external extraction utilities with the specified user's privileges	
<code>-1</code>	—	Do not sanitize extracted symlinks that point outside the extraction directory (dangerous)	
<code>--preserve-symlinks</code>	—	Do not sanitize extracted symlinks that point outside the extraction directory (dangerous)	
<code>-r</code>	—	Delete carved files after extraction	
<code>--rm</code>	—	Delete carved files after extraction	
<code>-z</code>	—	Carve data from files, but don't execute extraction utilities	
<code>--carve</code>	—	Carve data from files, but don't execute extraction utilities	
<code>-V</code>	—	Extract into sub-directories named by the offset	
<code>--subdirs</code>	—	Extract into sub-directories named by the offset	
<code>-E</code>	—	Calculate file entropy	Entropy analysis — the fast way to spot encryption or compression.
<code>--entropy</code>	—	Calculate file entropy	Entropy analysis — the fast way to spot encryption or compression.
<code>-F</code>	—	Use faster, but less detailed, entropy analysis	
<code>--fast</code>	—	Use faster, but less detailed, entropy analysis	
<code>-J</code>	—	Save plot as a PNG	
<code>--save</code>	—	Save plot as a PNG	
<code>-Q</code>	—	Omit the legend from the entropy plot graph	
<code>--nlegend</code>	—	Omit the legend from the entropy plot graph	
<code>-N</code>	—	Do not generate an entropy plot graph	
<code>--nplot</code>	—	Do not generate an entropy plot graph	
<code>-H</code>	float	Set the rising edge entropy trigger threshold (default: 0.95)	
<code>--high</code>	float	Set the rising edge entropy trigger threshold (default: 0.95)	
<code>-L</code>	float	Set the falling edge entropy trigger threshold (default: 0.85)	
<code>--low</code>	float	Set the falling edge entropy trigger threshold (default: 0.85)	
<code>-W</code>	—	Perform a hexdump / diff of a file or files	
<code>--hexdump</code>	—	Perform a hexdump / diff of a file or files	
<code>-G</code>	—	Only show lines containing bytes that are the same among all files	
<code>--green</code>	—	Only show lines containing bytes that are the same among all files	
<code>-i</code>	—	Only show lines containing bytes that are different among all files	
<code>--red</code>	—	Only show lines containing bytes that are different among all files	
<code>-U</code>	—	Only show lines containing bytes that are different among some files	
<code>--blue</code>	—	Only show lines containing bytes that are different among some files	
<code>-u</code>	—	Only display lines that are the same between all files	
<code>--similar</code>	—	Only display lines that are the same between all files	
<code>-w</code>	—	Diff all files, but only display a hex dump of the first file	
<code>--terse</code>	—	Diff all files, but only display a hex dump of the first file	
<code>-X</code>	—	Scan for raw deflate compression streams	
<code>--deflate</code>	—	Scan for raw deflate compression streams	
<code>-Z</code>	—	Scan for raw LZMA compression streams	
<code>--lzma</code>	—	Scan for raw LZMA compression streams	
<code>-P</code>	—	Perform a superficial, but faster, scan	

Flag	Argument	What it does	When you would use it
<code>--partial</code>	—	Perform a superficial, but faster, scan	
<code>-S</code>	—	Stop after the first result	
<code>--stop</code>	—	Stop after the first result	
<code>-l</code>	int	Number of bytes to scan	
<code>--length</code>	int	Number of bytes to scan	
<code>-o</code>	int	Start scan at this file offset	
<code>--offset</code>	int	Start scan at this file offset	
<code>-O</code>	int	Add a base address to all printed offsets	
<code>--base</code>	int	Add a base address to all printed offsets	
<code>-K</code>	int	Set file block size	
<code>--block</code>	int	Set file block size	
<code>-g</code>	int	Reverse every n bytes before scanning	
<code>--swap</code>	int	Reverse every n bytes before scanning	
<code>-f</code>	file	Log results to file	
<code>--log</code>	file	Log results to file	
<code>-c</code>	—	Log results to file in CSV format	
<code>--csv</code>	—	Log results to file in CSV format	
<code>-t</code>	—	Format output to fit the terminal window	
<code>--term</code>	—	Format output to fit the terminal window	
<code>-q</code>	—	Suppress output to stdout	
<code>--quiet</code>	—	Suppress output to stdout	
<code>-v</code>	—	Enable verbose output	
<code>--verbose</code>	—	Enable verbose output	
<code>-h</code>	—	Show help output	
<code>--help</code>	—	Show help output	
<code>-a</code>	str	Only scan files whose names match this regex	
<code>--findinclude</code>	str	Only scan files whose names match this regex	
<code>-p</code>	str	Do not scan files whose names match this regex	
<code>--fexclude</code>	str	Do not scan files whose names match this regex	
<code>-s</code>	int	Enable the status server on the specified port	
<code>--status</code>	int	Enable the status server on the specified port	

Gotchas

- **Always timeout-guard binwalk in a harness.** Signature scanning on a large or synthetic image can run effectively forever; this project has been bitten by exactly that.
- `-e -M` on an unknown blob can produce an enormous directory tree. Set `-d` and carve into a scratch filesystem, not your case directory.
- A signature hit is a guess based on magic bytes. Confirm with `file` or entropy before reporting an embedded filesystem as fact.

See also

`foremost`, `scalpel`, `bulk_extractor`, `tcpxtract`, `upx`, `die`, `diec`, `7za`

blkls

Kit: REMnux · Kali Linux · SIFT Workstation **Capability:** Recover deleted or lost files **Version:** The Sleuth Kit ver 4.11.1 **Captured:** cyberlab-aio via --help on 2026-08-02 **raw Docs:** <https://www.sleuthkit.org/sleuthkit>

← [Capability index](#) · [Kit tool list](#)

Purpose

Extract filesystem blocks — by default the unallocated ones, which is the input a carver wants.

Synopsis

```
blkls [-aAeLvV] [-f fstype] [-i imgtype] [-b dev_sector_size] [-o imgoffset] [-P pooltype] [-B pool_volume_block] image [images] [start-stop]
```

Options

All 13 options parsed from the captured help text; 12 reviewed with usage guidance.

Flag	Argument	What it does	When you would use it
-e	—	every block (including file system metadata blocks)	Every block, including filesystem metadata.
-l	—	print details in time machine list format	List block details rather than emitting their contents.
-a	—	Display allocated blocks	Allocated blocks only — the inverse, when isolating live data.
-A	—	Display unallocated blocks	Unallocated blocks. The default and the usual intent: pipe this into <code>foremost</code> or <code>scalpel</code> so the carver reads only free space instead of the whole image.
-f	fstype	File system type (use '-f list' for supported types)	Force the filesystem type (<code>-f list</code> shows the options).
-i	imgtype	The format of the image file (use '-i list' for supported types)	Image format for non-raw evidence such as E01 or AFF.
-b	dev_sector_size	The size (in bytes) of the device sectors	Device sector size; required on 4Kn drives.
-o	imgoffset	The offset of the file system in the image (in sectors)	Offset, in sectors, from <code>mm1s</code> .
-P	pooltype	Pool container type (use '-P list' for supported types)	Pool type, for APFS or LVM containers.
-B	pool_volume_block	Starting block (for pool volumes only)	Starting block within a pool volume.
-s	—	print slack space only (other flags are ignored)	Slack space only: the tail of the last block of each file, where fragments of previous contents survive. A distinct hunt from carving free space, and it ignores the other flags.
-v	—	verbose to stderr	Verbose diagnostics to stderr.
-V	—	print version	

Gotchas

- Output goes to stdout and is the size of the free space — redirect it to a file on a volume that can hold it, not into a pager.
- Block offsets in the extracted stream do **not** match offsets in the original image, because only unallocated blocks were written. Use `-l` if you need to map a hit back to its real location.

See also

`tsk_recover`, `icat`, `photorec`, `testdisk`

bulk_extractor

Kit: REMnux · Kali Linux · SIFT Workstation **Capability:** Carve files out of unstructured data; Search raw data for a pattern; Recover encryption keys from memory **Version:** bulk_extractor 2.2.0 **Captured:** cyberlab-aio via --help on 2026-08-02 **raw Docs:** https://github.com/simsong/bulk_extractor/

← [Capability index](#) · [Kit tool list](#)

Purpose

Scan an image for features — email addresses, URLs, credit card numbers, EXIF, keys — without parsing the filesystem at all, so deleted and unallocated content is included.

Synopsis

```
bulk_extractor [OPTION...] image_name
```

Common invocations

```
# from cyberlab 02-memory-forensics
bulk_extractor -V
# from cyberlab 02-memory-forensics
bulk_extractor -o be_out sample.mem
# from cyberlab 05-file-carving
bulk_extractor -o /tmp/bulk_out exercise/sample.dd
# from cyberlab 05-file-carving
bulk_extractor -o /tmp/ak_bulk exercise/sample.dd
# from cyberlab 20-volatility-deep
bulk_extractor -o be_out exercise/memdump.raw
# from cyberlab 47-ransomware-memory-case
bulk_extractor -E aes -E zip -o be_out memory.raw
# from cyberlab 47-ransomware-memory-case
bulk_extractor -E aes -o exercise/be_out exercise/memory.raw
# from cyberlab 51-linux-triage-workflow
bulk_extractor -o be_out disk.raw
```

Options

All 69 options parsed from the captured help text; 58 reviewed with usage guidance.

Flag	Argument	What it does	When you would use it
-A	arg	Offset added (in bytes) to feature locations (default: 0)	Add an offset to reported feature locations — use it when scanning a carved fragment so offsets still refer to the original image.
--offset_add	arg	Offset added (in bytes) to feature locations (default: 0)	Add an offset to reported feature locations.
-b	arg	Path of file whose contents are prepended to top of all feature files	Prepend a banner file to every feature file, e.g. a case header.
--banner_file	arg	Path of file whose contents are prepended to top of all feature files	Prepend a banner file to every feature file.
-C	arg	Size of context window reported in bytes (default: 16)	Bytes of context stored around each hit. Raise it when a bare match is not enough to judge relevance.
--context_window	arg	Size of context window reported in bytes (default: 16)	Bytes of context stored around each hit.
-d	—	enable debug-level diagnostic logging	Debug-level diagnostic logging.
--debug	—	enable debug-level diagnostic logging	Debug-level diagnostic logging.
-E	arg	disable all scanners except the one specified. Same as -x all -E scanner.	Run exactly one scanner and disable the rest. The fastest way to answer a single question instead of a full sweep.
--enable_exclusive	arg	disable all scanners except the one specified. Same as -x all -E scanner.	Run exactly one scanner and disable the rest.
-e	arg	enable a scanner (can be repeated)	Enable a scanner that is off by default, repeatable.
--enable	arg	enable a scanner (can be repeated)	Enable a scanner that is off by default, repeatable.
-x	arg	disable a scanner (can be repeated)	Disable a scanner, repeatable. Turning off the noisy ones is usually a bigger speed win than adding threads.
--disable	arg	disable a scanner (can be repeated)	Disable a scanner, repeatable.

Flag	Argument	What it does	When you would use it
<code>-f</code>	arg	search for a pattern (can be repeated)	Search for a regex pattern, repeatable.
<code>--find</code>	arg	search for a pattern (can be repeated)	Search for a regex pattern, repeatable.
<code>-F</code>	arg	read patterns to search from a file (can be repeated)	Read search patterns from a file — the practical form when hunting a list of indicators.
<code>--find_file</code>	arg	read patterns to search from a file (can be repeated)	Read search patterns from a file.
<code>--find-case-sensitive</code>	—	make <code>-f</code> and <code>-F</code> patterns case-sensitive	Make <code>-f</code> / <code>-F</code> case-sensitive; they are not by default.
<code>-G</code>	arg	page size in bytes (default: 16777216)	Page size read per pass.
<code>--pagesize</code>	arg	page size in bytes (default: 16777216)	Page size read per pass.
<code>-g</code>	arg	margin size in bytes (default: 4194304)	Margin carried between pages, so a feature spanning a page boundary is still found.
<code>--marginsize</code>	arg	margin size in bytes (default: 4194304)	Margin carried between pages, so features spanning a page boundary are still found.
<code>-j</code>	arg	number of threads (default: 12)	Thread count. Defaults to the core count; lower it when the scan is competing with other work.
<code>--threads</code>	arg	number of threads (default: 12)	Thread count.
<code>-J</code>	—	read and process data in the primary thread	Single-threaded. Slow, but the first thing to try when a scan crashes or results look non-deterministic.
<code>--no_threads</code>	—	read and process data in the primary thread	Single-threaded — use when debugging a crash.
<code>-M</code>	arg	max recursion depth (default: 12)	Maximum recursion depth into nested containers. Lower it if a zip bomb or deeply nested archive stalls the scan.
<code>--max_depth</code>	arg	max recursion depth (default: 12)	Maximum recursion depth into nested containers.
<code>--max_bad_alloc_errors</code>	arg	max bad allocation errors (default: 3)	Allocation failures tolerated before aborting.
<code>--max_minute_wait</code>	arg	maximum number of minutes to wait until all data are read (default: 60)	How long to wait for all data to be read before giving up — raise it for slow or failing media.
<code>--log-level</code>	arg	diagnostic log level: trace, debug, info, warning, error, critical, or off	Diagnostic log level.
<code>--log-file</code>	arg	diagnostic log file (default: <code>/bulk_extractor.log</code>)	Diagnostic log file; defaults inside the output directory.
<code>--notify_main_thread</code>	—	Display notifications in the main thread after phase1 completes. Useful for running with ThreadSanitizer	
<code>--notify_async</code>	—	Display notificaitons asynchronously (default)	
<code>-o</code>	arg	output directory [REQUIRED]	Output directory. Required, and it must not already exist unless you also pass <code>-Z</code> .
<code>--outdir</code>	arg	output directory [REQUIRED]	Output directory. Required.
<code>-P</code>	arg	directories for scanner shared libraries (can be repeated). Default directories include <code>/usr/local/lib/bulk_extractor</code> , <code>/usr/lib/bulk_extractor</code> and any directories specified in the <code>BE_PATH</code> environment	Additional directories to load scanner plugins from.
<code>--scanner_dir</code>	arg	directories for scanner shared libraries (can be repeated). Default directories include <code>/usr/local/lib/bulk_extractor</code> , <code>/usr/lib/bulk_extractor</code> and any directories specified in the <code>BE_PATH</code> environment	Additional directories to load scanner plugins from.
<code>-p</code>	arg	print the value of <code>[:length][:h][:r]</code> with optional length, hex output, or raw output.	Print the value at a path, with optional length and hex or raw output — inspection rather than scanning.
<code>--path</code>	arg	print the value of <code>[:length][:h][:r]</code> with optional length, hex output, or raw output.	Print the value at a path, for inspection rather than scanning.
<code>-q</code>	—	no status or performance output	Suppress status and performance output.
<code>--quit</code>	—	no status or performance output	Suppress status and performance output.
<code>-r</code>	arg	file to read alert list from	Alert list: features to flag prominently — the inverse of a stop list, for known-bad indicators.
<code>--alert_list</code>	arg	file to read alert list from	Alert list of features to flag prominently.
<code>-R</code>	—	treat image file as a directory to recursively explore	Treat the input as a directory and recurse it, rather than as a disk image.
<code>--recurse</code>	—	treat image file as a directory to recursively explore	Treat the input as a directory and recurse it.
<code>-S</code>	arg	set a <code>name=value</code> option (can be repeated)	Set a scanner option as <code>name=value</code> , repeatable.
<code>--set</code>	arg	set a <code>name=value</code> option (can be repeated)	Set a scanner option as <code>name=value</code> , repeatable.
<code>-s</code>	arg	random sampling parameter <code>frac[:passes]</code>	Random sampling as <code>frac[:passes]</code> . Scans a fraction of a huge image to judge whether a full run is worth the hours.
<code>--sampling</code>	arg	random sampling parameter <code>frac[:passes]</code>	Random sampling as <code>frac[:passes]</code> .
<code>-V</code>	—	Display <code>PACKAGE_VERSION</code> (currently 2.2.0-DEVELOP)	

Flag	Argument	What it does	When you would use it
<code>--version</code>	—	Display PACKAGE_VERSION (currently) 2.2.0-DEVELOP	
<code>-w</code>	arg	file to read stop list from	Stop list: features to suppress. This is how you cut the known-good noise that otherwise buries the findings.
<code>--stop_list</code>	arg	file to read stop list from	Stop list of features to suppress.
<code>-Y</code>	arg	specify [-end] of area on disk to scan	Restrict the scan to a byte range, when <code>mmls</code> already told you which region matters.
<code>--scan</code>	arg	specify [-end] of area on disk to scan	Restrict the scan to a <code><start>[-end]</code> byte range.
<code>-z</code>	arg	specify a starting page number	Start at a given page number, to resume a long scan.
<code>--page_start</code>	arg	specify a starting page number	Start at a given page number.
<code>-Z</code>	—	wipe the output directory (recursively) before starting	Wipe the output directory first. Convenient for reruns and destructive by definition — never point it at a directory holding results you still need.
<code>--zap</code>	—	wipe the output directory (recursively) before starting	Wipe the output directory first — destructive.
<code>-0</code>	—	disable real-time notification	
<code>--no_notify</code>	—	disable real-time notification	
<code>-1</code>	—	version 1.0 notification (console-output)	
<code>--version1</code>	—	version 1.0 notification (console-output)	
<code>-H</code>	—	report information about each scanner	Report what each scanner does. Worth reading once; the scanner set is the tool.
<code>--info_scanners</code>	—	report information about each scanner	Report what each scanner does.
<code>-h</code>	—	print help screen	
<code>--help</code>	—	print help screen	

Gotchas

- It ignores the filesystem completely. That is the point — it finds content in unallocated space and slack that a filesystem-aware tool cannot reach — but it also means a hit carries no filename and no timestamp. Map the offset back with `ffind` / `istat` before naming a file in a report.
- Feature files are raw pattern matches, not verified findings. The credit-card scanner in particular flags anything passing a Luhn check, which includes plenty of ordinary numbers.
- Output is large and the scan is long. Sample with `-s` on a multi-terabyte image before committing to a full pass.

See also

`foremost`, `scalpel`, `binwalk`, `tcpxtract`, `rafind2`, `strings`, `grep`, `xxd`

ffind

Kit: REMnux · Kali Linux · SIFT Workstation **Capability:** List files and directories, including deleted ones **Version:** The Sleuth Kit ver 4.11.1 **Captured:** cyberlab-aio via `-help` on 2026-08-02 **raw Docs:** <https://www.sleuthkit.org/sleuthkit>

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Purpose

Find the file name that points at a given inode — the reverse of a directory lookup.

Synopsis

```
ffind [-aduv] [-f fstype] [-i imgtype] [-b dev_sector_size] [-o imgoffset] [-P pooltype] [-B pool_volume_block] image [images] inode
```

Options

All 11 options parsed from the captured help text; 10 reviewed with usage guidance.

Flag	Argument	What it does	When you would use it
<code>-a</code>	—	Find all occurrences	Show every name for the inode. Hard-linked files have more than one, and stopping at the first hides that.
<code>-d</code>	—	Find deleted entries ONLY	Deleted names only. The direct answer to "what was this inode called before it was removed?"
<code>-u</code>	—	Find undeleted entries ONLY	Undeleted names only, when a recycled inode is returning stale hits.
<code>-f</code>	fstype	Image file system type (use '-f list' for supported types)	Force the filesystem type (<code>-f list</code> shows the options).
<code>-i</code>	imgtype	The format of the image file (use '-i list' for supported types)	Image format for non-raw evidence such as E01 or AFF.
<code>-b</code>	dev_sector_size	The size (in bytes) of the device sectors	Device sector size; required on 4Kn drives.
<code>-o</code>	imgoffset	The offset of the file system in the image (in sectors)	Offset, in sectors, from <code>mm1s</code> .
<code>-P</code>	pooltype	Pool container type (use '-p list' for supported types)	Pool type, for APFS or LVM containers.
<code>-B</code>	pool_volume_block	Starting block (for pool volumes only)	Starting block within a pool volume.
<code>-v</code>	—	Verbose output to stderr	Verbose diagnostics to stderr.
<code>-V</code>	—	Print version	

Gotchas

- Inodes are reused. A name returned for a deleted inode may belong to whatever claimed it next, not to the file you are chasing — corroborate with `istat` timestamps before naming it in a report.
- This is the tool for the question `icat` provokes: you carved data out by inode and now need to say what it was called.

See also

`fls`, `ils`, `tsk_recover`

file

Kit: REMnux · FLARE-VM · SIFT Workstation **Capability:** Inspect metadata for one file or inode; Identify what a file actually is **Version:** file-5.44 **Captured:** cyberlab-aio
via `--help` on 2026-08-02 **raw Docs:** <https://github.com/file/file>

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Purpose

Identify a file's type from its contents rather than its name, using magic signatures.

Synopsis

```
file [OPTION...] [FILE...]
```

Common invocations

```
# from cyberlab 33-binwalk-firmware
file foremost_out/jpg/*.jpg 2>/dev/null || file foremost_out/*/*
# from cyberlab 35-radare2-intro
file exercise/hello
# from cyberlab 38-network-emulation
file /tmp/beacon_reply.bin # -> data / HTML (INetSim default object)
```

Options

All 54 options parsed from the captured help text; 20 reviewed with usage guidance.

Flag	Argument	What it does	When you would use it
<code>--help</code>	—	display this help and exit	
<code>-v</code>	—	output version information and exit	
<code>--version</code>	—	output version information and exit	
<code>-m</code>	LIST	use LIST as a colon-separated list of magic number files	Use an alternative magic database.
<code>--magic-file</code>	LIST	use LIST as a colon-separated list of magic number files	Use an alternative magic database.
<code>-z</code>	—	try to look inside compressed files	Look inside compressed files and report what they contain rather than reporting a compressed stream.
<code>--uncompress</code>	—	try to look inside compressed files	Look inside compressed files and report what they contain rather than reporting a compressed stream.
<code>-Z</code>	—	only print the contents of compressed files	Same, without reporting the compression itself.
<code>--uncompress-noreport</code>	—	only print the contents of compressed files	Same, without reporting the compression itself.
<code>-b</code>	—	do not prepend filenames to output lines	Omit the filename, leaving just the type — for pipelines and for-loops.
<code>--brief</code>	—	do not prepend filenames to output lines	Omit the filename, leaving just the type — for pipelines and for-loops.
<code>-c</code>	—	print the parsed form of the magic file, use in conjunction with <code>-m</code> to debug a new magic file before installing it	
<code>--checking-printout</code>	—	print the parsed form of the magic file, use in conjunction with <code>-m</code> to debug a new magic file before installing it	
<code>-e</code>	TEST	exclude TEST from the list of test to be performed for file. Valid tests are: apptype, ascii, cdf, compress, csv, elf, encoding, soft, tar, json, text, tokens	Exclude a test type, when one is misfiring on a corpus.
<code>--exclude</code>	TEST	exclude TEST from the list of test to be performed for file. Valid tests are: apptype, ascii, cdf, compress, csv, elf, encoding, soft, tar, json, text, tokens	Exclude a test type, when one is misfiring on a corpus.
<code>--exclude-quiet</code>	TEST	like exclude, but ignore unknown tests	
<code>-f</code>	FILE	read the filenames to be examined from FILE	Read the list of files to test from a file, for a corpus.
<code>--files-from</code>	FILE	read the filenames to be examined from FILE	Read the list of files to test from a file, for a corpus.
<code>-F</code>	STRING	use string as separator instead of `:`	
<code>--separator</code>	STRING	use string as separator instead of `:`	

Flag	Argument	What it does	When you would use it
<code>-i</code>	—	output MIME type strings (--mime-type and --mime-encoding)	Print a MIME type instead of prose. The form to use when the output feeds a script rather than a person.
<code>--mime</code>	—	output MIME type strings (--mime-type and --mime-encoding)	Print a MIME type instead of prose. The form to use when the output feeds a script rather than a person.
<code>--apple</code>	—	output the Apple CREATOR/TYPE	
<code>--extension</code>	—	output a slash-separated list of extensions	
<code>--mime-type</code>	—	output the MIME type	MIME type only, without the encoding.
<code>--mime-encoding</code>	—	output the MIME encoding	Character encoding only.
<code>-k</code>	—	don't stop at the first match	Keep going after the first match and print every rule that fired. Files crafted to defeat identification often match more than one signature, and the disagreement is the finding.
<code>--keep-going</code>	—	don't stop at the first match	Keep going after the first match and print every rule that fired. Files crafted to defeat identification often match more than one signature, and the disagreement is the finding.
<code>-l</code>	—	list magic strength	
<code>--list</code>	—	list magic strength	
<code>-L</code>	—	follow symlinks (default if POSIXLY_CORRECT is set)	Follow symlinks and report the target.
<code>--dereference</code>	—	follow symlinks (default if POSIXLY_CORRECT is set)	Follow symlinks and report the target.
<code>-h</code>	—	don't follow symlinks (default if POSIXLY_CORRECT is not set) (default)	Do not follow symlinks — report the link itself.
<code>--no-dereference</code>	—	don't follow symlinks (default if POSIXLY_CORRECT is not set) (default)	Do not follow symlinks — report the link itself.
<code>-n</code>	—	do not buffer output	
<code>--no-buffer</code>	—	do not buffer output	
<code>-N</code>	—	do not pad output	Do not pad filenames to align the output.
<code>--no-pad</code>	—	do not pad output	Do not pad filenames to align the output.
<code>-0</code>	—	terminate filenames with ASCII NUL	Print a NUL after the filename, for safe piping into <code>xargs -0</code> .
<code>--print0</code>	—	terminate filenames with ASCII NUL	Print a NUL after the filename, for safe piping into <code>xargs -0</code> .
<code>-p</code>	—	preserve access times on files	Preserve the access time on the files examined — the flag that stops a triage sweep from rewriting timestamps across the evidence.
<code>--preserve-date</code>	—	preserve access times on files	Preserve the access time on the files examined — the flag that stops a triage sweep from rewriting timestamps across the evidence.
<code>-P</code>	—	set file engine parameter limits bytes 7340032 max bytes to look inside file elf_notes 256 max ELF notes processed elf_phnum 2048 max ELF prog sections processed elf_shnum 32768 max ELF sections proce	Tune a parser limit, e.g. <code>bytes</code> or <code>indir</code> .
<code>--parameter</code>	—	set file engine parameter limits bytes 7340032 max bytes to look inside file elf_notes 256 max ELF notes processed elf_phnum 2048 max ELF prog sections processed elf_shnum 32768 max ELF sections proce	Tune a parser limit, e.g. <code>bytes</code> or <code>indir</code> .
<code>-r</code>	—	don't translate unprintable chars to \ooo	Print raw bytes rather than escaping unprintables.
<code>--raw</code>	—	don't translate unprintable chars to \ooo	Print raw bytes rather than escaping unprintables.
<code>-s</code>	—	treat special (block/char devices) files as ordinary ones	Read block and character devices too. Needed to type a raw disk, which <code>file</code> otherwise refuses.
<code>--special-files</code>	—	treat special (block/char devices) files as ordinary ones	Read block and character devices too. Needed to type a raw disk, which <code>file</code> otherwise refuses.
<code>-S</code>	—	disable system call sandboxing	Disable the sandbox. Only when seccomp blocks a legitimate test, and never on untrusted input if avoidable.
<code>--no-sandbox</code>	—	disable system call sandboxing	Disable the sandbox. Only when seccomp blocks a legitimate test, and never on untrusted input if avoidable.
<code>-C</code>	—	compile file specified by -m	Compile a magic file to its indexed form.
<code>--compile</code>	—	compile file specified by -m	Compile a magic file to its indexed form.
<code>-d</code>	—	print debugging messages	
<code>--debug</code>	—	print debugging messages	

Gotchas

- Magic reads the first few hundred bytes. Prepend a valid header to anything and `file` will report the header — it is a fast triage signal, not an authority on content.

- `-p` preserves atime. Without it, typing a directory tree updates access times across the evidence, which is the kind of avoidable contamination that gets noticed later.
- An 'ASCII text' verdict on something you expected to be binary usually means it is base64 or hex, not that it is harmless.

See also

`istat`, `ils`, `stat`, `die`, `diec`

fls

Kit: SIFT Workstation · REMnux · Kali (`sleuthkit`) **Category:** Filesystem analysis **Version:** The Sleuth Kit 4.11.1 **Docs:** <https://sleuthkit.org/sleuthkit/man/fls.html>
Verified: 2026-07-29 from `cyberlab-aio:v1` — raw output in `capture/cyberlab-aio/help/fls.help.txt`

Worked example of the format. Every option below was read off the real binary, not written from memory. See [docs/FORMAT.md](#).

Purpose

List file and directory names in a disk image, including deleted entries.

Synopsis

```
fls [-adDFlhpruvV] [-f fstype] [-i imgtype] [-b dev_sector_size] [-m dir/]
    [-o imgoffset] [-z ZONE] [-s seconds] image [images] [inode]
```

If `[inode]` is not given, the root directory is used.

Note: the usage line above advertises only `[-adDFlhpruvV]`, but the tool also accepts `-P`, `-B`, `-S` and `-k`. They are documented below.

Common invocations

```
# List the root directory of an image
fls {{path/to/image.dd}}

# List a partition that starts at a sector offset (the usual case for a full disk)
fls -o {{2048}} {{path/to/image.dd}}

# Show deleted entries only - the fast answer to "what was removed?"
fls -d -o {{2048}} {{path/to/image.dd}}

# Recurse the whole filesystem showing full paths, deleted files included
fls -rp -o {{2048}} {{path/to/image.dd}}

# Recurse showing ONLY deleted files with full paths
fls -rpd -o {{2048}} {{path/to/image.dd}}

# Long listing (sizes, MAC times) like ls -l, in a named time zone
fls -l -z {{EST5EDT}} -o {{2048}} {{path/to/image.dd}}

# Emit a body file for timelining, then build the timeline with mactime
fls -r -m {{/}} -o {{2048}} {{path/to/image.dd}} > {{path/to/body.txt}}

# List the contents of one directory by inode number
fls -o {{2048}} {{path/to/image.dd}} {{inode}}
```

Options — complete

Flag	Argument	What it does	When you would use it
<code>-a</code>	—	Display <code>.</code> and <code>..</code> entries	Rarely; they are hidden by default as noise
<code>-d</code>	—	Display deleted entries only	Recovery hunts — isolates what was removed
<code>-D</code>	—	Display only directories	Mapping structure without file noise
<code>-F</code>	—	Display only files	Excluding directories from a listing
<code>-l</code>	—	Display long version (like <code>ls -l</code>)	When you need sizes, UID/GID and MAC times
<code>-p</code>	—	Display full path for each file	Almost always with <code>-r</code> ; otherwise output is ambiguous
<code>-r</code>	—	Recurse on directory entries	Walking a whole filesystem rather than one directory

Flag	Argument	What it does	When you would use it
<code>-u</code>	—	Display undeleted entries only	The inverse of <code>-d</code> ; filtering live files
<code>-m</code>	<code>dir/</code>	Output in mactime input format, with <code>dir/</code> as the mount point	Building a super-timeline; feeds <code>mactime</code>
<code>-h</code>	—	Include MD5 checksum hash in mactime output	Timeline work where you also need file hashes
<code>-f</code>	<code>fstype</code>	Filesystem type (<code>-f list</code> for supported types)	When auto-detection fails or is wrong
<code>-i</code>	<code>imgtype</code>	Image format (<code>-i list</code> for supported types)	Non-raw images: E01, AFF, etc.
<code>-b</code>	<code>dev_sector_size</code>	Device sector size in bytes	4Kn drives, where the 512-byte assumption breaks
<code>-o</code>	<code>imgoffset</code>	Offset into the image, in sectors	The most common flag — targeting a partition in a full-disk image
<code>-P</code>	<code>pooltype</code>	Pool container type (<code>-P list</code> for supported types)	APFS / logical volume containers
<code>-B</code>	<code>pool_volume_block</code>	Starting block, pool volumes only	Selecting a volume inside a pool
<code>-S</code>	<code>snap_id</code>	Snapshot ID (APFS only)	Examining a specific APFS snapshot
<code>-k</code>	<code>password</code>	Decryption password for encrypted volumes	BitLocker/FileVault-style encrypted volumes
<code>-z</code>	<code>ZONE</code>	Time zone of the original machine (e.g. <code>EST5EDT</code> , <code>GMT</code>)	Only meaningful with <code>-l</code> ; wrong TZ silently skews every timestamp
<code>-s</code>	<code>seconds</code>	Time skew of the original machine, in seconds	Only with <code>-l</code> / <code>-m</code> ; correcting a known-bad system clock
<code>-v</code>	—	Verbose output to stderr	Debugging why an image will not parse
<code>-V</code>	—	Print version	Recording tool provenance in your notes

Gotchas

- `-o` **is in sectors, not bytes**. Get it from `mmfs` first. A wrong offset usually yields "Cannot determine file system type", not an obvious error.
- `-z` **only affects `-l` output**. Setting it without `-l` changes nothing, which makes a wrong-timezone timeline easy to produce and hard to notice.
- **Deleted entries may have unrecoverable content**. `fls -d` listing a name does not mean `icat` will return its data — the blocks may be reallocated.
- `-r` **on a large image is slow and noisy**. Pipe to a file, then grep it, rather than scrolling.

See also

- `mmfs` — partition table; where you get the `-o` value
- `icat` — extract the content of an inode `fls` found
- `istat` — detailed metadata for one inode
- `mactime` — consumes the `-m` body file to build the timeline
- `tsk_recover` — bulk export of allocated/deleted files

foremost

Kit: Kali Linux · SIFT Workstation **Capability:** Carve files out of unstructured data; Extract files and payloads from traffic **Version:** 1.5.7 **Captured:** cyberlab-aio via -h on 2026-08-02 [raw](#)

[← Capability index](#) · [Kit tool list](#)

Purpose

Carve files out of an image or raw data by header and footer signatures.

Common invocations

```
# from cyberlab 05-file-carving
foremost -V
# from cyberlab 05-file-carving
foremost -t jpg,pdf -i exercise/sample.dd -o /tmp/foremost_out
# from cyberlab 05-file-carving
foremost -t jpg,pdf -i exercise/sample.dd -o /tmp/ak_foremost
# from cyberlab 33-binwalk-firmware
foremost -i firmware.bin -o foremost_out
```

Options

All 11 options parsed from the captured help text; 8 reviewed with usage guidance.

Flag	Argument	What it does	When you would use it
-V	—	- display copyright information and exit	
-t	—	- specify file type. (-t jpeg,pdf ...)	Restrict carving to given types (<code>jpg</code> , <code>pdf</code> , <code>all</code>). Narrowing this is the difference between a usable result and 40,000 files.
-d	—	- turn on indirect block detection (for UNIX file-systems)	
-i	—	- specify input file (default is stdin)	Input file or device to carve from.
-a	—	- Write all headers, perform no error detection (corrupted files)	Write all headers found, even without a valid footer.
-w	—	- Only write the audit file, do not write any detected files to the disk	Write the audit file only, carving nothing — a cheap dry run.
-o	—	- set output directory (defaults to output)	Output directory — must be empty or foremost refuses to run.
-c	—	- set configuration file to use (defaults to foremost.conf)	Use a custom configuration file to add signatures.
-q	—	- enables quick mode. Search are performed on 512 byte boundaries.	Quick mode: scan only sector boundaries. Much faster, misses embedded files.
-Q	—	- enables quiet mode. Suppress output messages.	
-v	—	- verbose mode. Logs all messages to screen	Verbose output.

Gotchas

- Carving recovers content but **not filenames or timestamps** — those live in filesystem metadata that carving bypasses. Use `tsk_recover` when the filesystem is intact and carve only what it cannot reach.
- The output directory must be empty; foremost aborts otherwise. This trips scripted reruns constantly.

See also

`scalpel` , `binwalk` , `bulk_extractor` , `tcpxtract` , `tcpflow`

fsstat

Kit: REMnux · Kali Linux · SIFT Workstation **Capability:** See the partition and volume layout **Version:** The Sleuth Kit ver 4.11.1 **Captured:** cyberlab-aio via --help on 2026-08-02 **raw Docs:** <https://www.sleuthkit.org/sleuthkit>

← [Capability index](#) · [Kit tool list](#)

Purpose

Report a filesystem's layout and parameters: type, block size, inode range, and the geometry every other TSK tool needs.

Synopsis

```
fsstat [-tvV] [-f fstype] [-i imgtype] [-b dev_sector_size] [-o imgoffset] image
```

Common invocations

```
# from cyberlab 01-disk-forensics
fsstat -V
# from cyberlab 01-disk-forensics
fsstat -o 2048 exercise/sample.dd
# from cyberlab 01-disk-forensics
fsstat -o 0 exercise/sample.dd | grep -Ei "file system type|sector size|cluster size"
# from cyberlab 22-sleuthkit-mastery
fsstat -o 2048 exercise/practice.dd
# from cyberlab 51-linux-triage-workflow
fsstat -o 2048 disk.raw
```

Options

All 10 options parsed from the captured help text; 9 reviewed with usage guidance.

Flag	Argument	What it does	When you would use it
-t	—	display type only	Print only the filesystem type. The scriptable form when all you need is a yes/no on what this volume is.
-i	imgtype	The format of the image file (use '-i list' for supported types)	Set the image format for non-raw evidence such as E01 or AFF.
-b	dev_sector_size	The size (in bytes) of the device sectors	Device sector size. Needed on 4Kn drives, where the 512-byte default silently computes every offset wrong.
-f	fstype	File system type (use '-f list' for supported types)	Force the filesystem type when detection is wrong or the superblock is damaged (<code>-f list</code> shows the options).
-o	imgoffset	The offset of the file system in the image (in sectors)	Offset, in sectors , of the filesystem inside the image. Take it from <code>mm1s</code> ; this is the flag that ties the two tools together.
-P	pooltype	Pool container type (use '-P list' for supported types)	Pool type, for APFS or LVM containers that hold the volume.
-B	pool_volume_block	Starting block (for pool volumes only)	Starting block within a pool volume.
-v	—	verbose output to stderr	Verbose diagnostics to stderr, useful when detection fails.
-V	—	Print version	
-k	password	Decryption password for encrypted volumes	Password for an encrypted volume.

Gotchas

- Run this first. Block size and inode range from `fsstat` are what make `blkls`, `icat` and `ils` output interpretable — starting anywhere else means guessing at the numbers they print.
- If it reports the wrong type or refuses the volume, the `-o` offset is wrong far more often than the image is corrupt.

See also

`mmfs`, `img_stat`, `testdisk`

icat

Kit: REMnux · Kali Linux · SIFT Workstation **Capability:** Recover deleted or lost files **Version:** The Sleuth Kit ver 4.11.1 **Captured:** cyberlab-aio via --help on 2026-08-02 **raw Docs:** <https://www.sleuthkit.org/sleuthkit>

← Capability index · Kit tool list

Purpose

Extract the contents of a file by inode, including deleted files.

Synopsis

```
icat [-hrRsvV] [-f fstype] [-i imgtype] [-b dev_sector_size] [-o imgoffset] image [images] inum[-typ[-id]]
```

Common invocations

```
# from cyberlab 01-disk-forensics
icat -o 2048 exercise/sample.dd 5 > /tmp/recovered_file.bin
# from cyberlab 01-disk-forensics
icat -o 0 exercise/sample.dd 5 > /tmp/recovered.txt
# from cyberlab 22-sleuthkit-mastery
icat -o 2048 exercise/practice.dd 4 | head
# from cyberlab 22-sleuthkit-mastery
icat -o 2048 exercise/practice.dd 6
# from cyberlab 51-linux-triage-workflow
icat -o 2048 disk.raw 5 > recovered_file.bin
# from cyberlab 51-linux-triage-workflow
icat exercise/triage_sample.raw $(fls -p exercise/triage_sample.raw | awk 'eicar.com/{gsub(/:\/,"",$2);print $2}') > exercise/extract/eicar.com
```

Options

All 14 options parsed from the captured help text; 8 reviewed with usage guidance.

Flag	Argument	What it does	When you would use it
-h	—	Do not display holes in sparse files	Skip holes in sparse files so the output is not padded with zeroes.
-r	—	Recover deleted file	Attempt recovery of a deleted file — the reason you usually reach for icat.
-R	—	Recover deleted file and suppress recovery errors	Recover with slack space included, when you want everything the blocks still hold.
-s	—	Display slack space at end of file	Include slack space in the output.
-i	imgtype	The format of the image file (use '-i list' for supported types)	Image format for non-raw evidence.
-b	dev_sector_size	The size (in bytes) of the device sectors	
-f	fstype	File system type (use '-f list' for supported types)	Force the filesystem type when detection is wrong.
-o	imgoffset	The offset of the file system in the image (in sectors)	Partition offset in sectors, from <code>mm1s</code> .
-P	pooltype	Pool container type (use '-P list' for supported types)	
-B	pool_volume_block	Starting block (for pool volumes only)	
-S	snap_id	Snapshot ID (for APFS only)	
-v	—	verbose to stderr	
-V	—	Print version	
-k	password	Decryption password for encrypted volumes	Supply a decryption password for an encrypted volume.

Gotchas

- Always redirect to a file (`icat ... > out.bin`). Binary content dumped to a terminal will corrupt your session.
- A deleted inode listed by `fls -d` may return nothing or garbage: the metadata survived but the blocks were reallocated. Empty output is evidence, not a tool failure.

See also

`tsk_recover`, `photorec`, `testdisk`, `blkls`

ils

Kit: REMnux · Kali Linux · SIFT Workstation **Capability:** List files and directories, including deleted ones; Inspect metadata for one file or inode **Version:** The Sleuth Kit ver 4.11.1 **Captured:** cyberlab-aio via `--help` on 2026-08-02 **raw Docs:** <https://www.sleuthkit.org/sleuthkit>

[← Capability index](#) · [Kit tool list](#)

Purpose

List inode metadata, including inodes that no longer have a name pointing at them.

Synopsis

```
ils [-emOpvV] [-aAlLzZ] [-f fstype] [-i imgtype] [-b dev_sector_size] [-o imgoffset] [-P pooltype] [-B pool_volume_block] [-s seconds] image [images]
```

Options

All 19 options parsed from the captured help text; 18 reviewed with usage guidance.

Flag	Argument	What it does	When you would use it
<code>-e</code>	—	Display all inodes	Every inode, allocated or not.
<code>-m</code>	—	Display output in the mactime format	mactime format — the form <code>mactime</code> consumes to build a timeline. This is how deleted-file metadata reaches the timeline at all.
<code>-O</code>	—	Display inodes that are unallocated, but were still open (UFS/ExtX only)	Unallocated inodes that were still open at the time of imaging (UFS/ExtX). The same trick, caught mid-deletion.
<code>-p</code>	—	Display orphan inodes (unallocated with no file name)	Orphan inodes — allocated content with no directory entry. Files that were unlinked while still open, and a standard hiding place worth checking explicitly.
<code>-s</code>	seconds	Time skew of original machine (in seconds)	Correct for a known clock skew on the source machine, in seconds, so times line up with other evidence.
<code>-a</code>	—	Allocated inodes	Allocated inodes only.
<code>-A</code>	—	Unallocated inodes	Unallocated inodes only — deleted file metadata that often survives after the name is gone.
<code>-l</code>	—	Linked inodes	Linked inodes (a name still points at them).
<code>-L</code>	—	Unlinked inodes	Unlinked inodes (nothing does).
<code>-z</code>	—	Unused inodes	Unused inodes.
<code>-Z</code>	—	Used inodes	Used inodes.
<code>-i</code>	imgtype	The format of the image file (use '-i list' for supported types)	Image format for non-raw evidence such as E01 or AFF.
<code>-b</code>	dev_sector_size	The size (in bytes) of the device sectors	Device sector size; required on 4Kn drives.
<code>-f</code>	fstype	File system type (use '-f list' for supported types)	Force the filesystem type (<code>-f list</code> shows the options).
<code>-o</code>	imgoffset	The offset of the file system in the image (in sectors)	Offset, in sectors, from <code>mmLs</code> .
<code>-P</code>	pooltype	Pool container type (use '-p list' for supported types)	Pool type, for APFS or LVM containers.
<code>-B</code>	pool_volume_block	Starting block (for pool volumes only)	Starting block within a pool volume.
<code>-v</code>	—	verbose output to stderr	Verbose diagnostics to stderr.
<code>-V</code>	—	Display version number	

Gotchas

- `ils` finds metadata with no name; `ffind` turns an inode back into a name; `icat` extracts its content. Deleted-file work is usually all three in sequence.
- An inode surviving does not mean its data did. The blocks it points at may already be reallocated, so `icat` can return another file's contents entirely.

See also

`fls`, `ffind`, `tsk_recover`, `istat`, `file`, `stat`

istat

Kit: REMnux · Kali Linux · SIFT Workstation **Capability:** Inspect metadata for one file or inode **Version:** The Sleuth Kit ver 4.11.1 **Captured:** cyberlab-aio via --help on 2026-08-02 **raw Docs:** <https://www.sleuthkit.org/sleuthkit>

← [Capability index](#) · [Kit tool list](#)

Purpose

Show the full metadata for one inode: times, size, and the blocks it occupies.

Synopsis

```
istat [-N num] [-f fstype] [-i imgtype] [-b dev_sector_size] [-o imgoffset] [-P pooltype] [-B pool_volume_block] [-z zone] [-s seconds] [-rvV] image :
```

Common invocations

```
# from cyberlab 22-sleuthkit-mastery
istat -o 2048 exercise/practice.dd 4
```

Options

All 14 options parsed from the captured help text; 5 reviewed with usage guidance.

Flag	Argument	What it does	When you would use it
-N	num	force the display of NUM address of block pointers	Limit how many block addresses are printed for a large file.
-r	—	display run list instead of list of block addresses	Include recovery information for a deleted inode.
-z	zone	time zone of original machine (i.e. EST5EDT or GMT)	Set the time zone for the displayed timestamps.
-s	seconds	Time skew of original machine (in seconds)	Apply a clock skew in seconds for a known-bad system clock.
-i	imgtype	The format of the image file (use '-i list' for supported types)	
-b	dev_sector_size	The size (in bytes) of the device sectors	
-f	fstype	File system type (use '-f list' for supported types)	
-o	imgoffset	The offset of the file system in the image (in sectors)	Partition offset in sectors, from <code>mm1s</code> .
-P	pooltype	Pool container type (use '-p list' for supported types)	
-B	pool_volume_block	Starting block (for pool volumes only)	
-S	snap_id	Snapshot ID (for APFS only)	
-v	—	verbose output to stderr	
-V	—	print version	
-k	password	Decryption password for encrypted volumes	

Gotchas

- The block list is what lets you prove whether a deleted file is still recoverable — cross-check it before promising a recovery.

See also

`ils`, `file`, `stat`

mmls

Kit: REMnux · Kali Linux · SIFT Workstation **Capability:** See the partition and volume layout **Version:** The Sleuth Kit ver 4.11.1 **Captured:** cyberlab-aio via `--help` on 2026-08-02 **raw Docs:** <https://www.sleuthkit.org/sleuthkit>

← [Capability index](#) · [Kit tool list](#)

Purpose

Display the partition layout of a disk image, including unallocated gaps.

Common invocations

```
# from cyberlab 01-disk-forensics
mmls -V
# from cyberlab 01-disk-forensics
mmls exercise/sample.dd
# from cyberlab 22-sleuthkit-mastery
mmls exercise/practice.dd
# from cyberlab 51-linux-triage-workflow
mmls disk.raw
# from cyberlab 57-forensic-acquisition
sudo mmls /mnt/ewf/ewf1
```

Options

All 12 options parsed from the captured help text; 9 reviewed with usage guidance.

Flag	Argument	What it does	When you would use it
-t	vstype	The type of volume system (use '-t list' for list of supported types)	Force the volume-system type when auto-detection guesses wrong (<code>-t list</code> shows the options).
-i	imgtype	The format of the image file (use '-i list' for list supported types)	Set the image format for non-raw evidence such as E01 or AFF.
-b	dev_sector_size	The size (in bytes) of the device sectors	Set the device sector size; required on 4Kn drives where the 512-byte default is wrong.
-o	imgoffset	Offset to the start of the volume that contains the partition system (in sectors)	Read a volume system nested at an offset — rare, but needed for nested containers.
-B	—	print the rounded length in bytes	Print volume sizes in bytes rather than sectors, when reporting.
-r	—	recurse and look for other partition tables in partitions (DOS Only)	Recurse into nested volume systems (e.g. an extended partition).
-v	—	verbose output	Verbose diagnostics to stderr when an image will not parse.
-V	—	print the version	
-a	—	Show allocated volumes	Show allocated volumes only, when the gap entries are noise.
-A	—	Show unallocated volumes	Show unallocated space only — where a hidden or deleted partition would show up.
-m	—	Show metadata volumes	
-M	—	Hide metadata volumes	

Gotchas

- The **Start** column is in sectors. That value is what every other TSK tool wants for `-o`. Multiplying by the sector size here is the single most common mistake in a TSK workflow.
- If `mmls` reports no partition table, the image may be a single volume rather than a whole disk — try `fsstat` on it directly at offset 0 before assuming the image is corrupt.

See also

[fsstat](#), [img_stat](#), [testdisk](#)

photorec

Kit: SIFT Workstation **Capability:** Recover deleted or lost files **Version:** PhotoRec 7.1 **Captured:** cyberlab-aio via --help on 2026-08-02 [raw](#)

[← Capability index](#) · [Kit tool list](#)

Purpose

PhotoRec 7.1, Data Recovery Utility, July 2019

Synopsis

```
photorec [/log] [/debug] [/d recup_dir] [file.dd|file.e01|device]
photorec /version
```

Common invocations

```
# from cyberlab 01-disk-forensics
photorec /version
# from cyberlab 01-disk-forensics
photorec /log /d /tmp/carved /cmd exercise/sample.dd partition_none,options,mode_ext2,fileopt,everything,enable,search
```

Options

All 2 options parsed from the captured help text. The final column is filled in by review.

Flag	Argument	What it does	When you would use it
/log	—	create a photorec.log file	
/debug	—	add debug information	

See also

[tsk_recover](#), [icat](#), [testdisk](#), [blkls](#)

rafind2

Kit: REMnux · Kali Linux · SIFT Workstation **Capability:** Search raw data for a pattern **Version:** rafind2 6.1.9 **Captured:** cyberlab-aio via `--help` on 2026-08-02 [raw](#)
Docs: <https://www.radare.org/n/radare2.html>

[← Capability index](#) · [Kit tool list](#)

Purpose

Examine binary files, including disassembling and debugging. Includes r2ai and decal plugins for LLM-powered analysis (API key or local Ollama required), plus the r2ghidra plugin for Ghidra decompilation via the pdg command.

Synopsis

```
rafind2 [-mBXnzZhqv] [-a align] [-b sz] [-f/t from/to] [-[e|s|S] str] [-x hex] [-R str] [-I str] [-g] -|file|dir ..
```

Options

No option definitions could be parsed from this tool's help output. It may be subcommand-driven or have no flags; needs manual review.

See also

[strings](#), [grep](#), [xxd](#), [bulk_extractor](#)

scalpel

Kit: REMnux · Kali Linux · SIFT Workstation **Capability:** Carve files out of unstructured data **Version:** Scalpel version 1.60 **Captured:** cyberlab-aio via -h on 2026-08-02 **raw Docs:** <https://github.com/sleuthkit/scalpel>

← [Capability index](#) · [Kit tool list](#)

Purpose

Carve contents out of binary files, such as partitions.

Synopsis

```
scalpel [-b] [-c <config file>] [-d] [-h|V] [-i <file>]
[-m blocksize] [-n] [-o <outputdir>] [-O num] [-q clustersize]
[-r] [-s num] [-t <blockmap file>] [-u] [-v]
<imgfile> [<imgfile>] ...
```

Common invocations

```
# from cyberlab 05-file-carving
scalpel -V
# from cyberlab 05-file-carving
scalpel -c /etc/scalpel/scalpel.conf -o /tmp/scalpel_out exercise/sample.dd
# from cyberlab 05-file-carving
scalpel -c /etc/scalpel/scalpel.conf -o /tmp/ak_scalpel exercise/sample.dd
```

Options

All 17 options parsed from the captured help text. The final column is filled in by review.

Flag	Argument	What it does	When you would use it
-b	—	Carve files even if defined footers aren't discovered within maximum carve size for file type [foremost 0.69 compat mode].	
-c	—	Choose configuration file.	
-d	—	Generate header/footer database; will bypass certain optimizations and discover all footers, so performance suffers. Doesn't affect the set of files carved. EXPERIMENTAL	
-h	—	Print this help message and exit.	
-i	—	Read names of disk images from specified file.	
-m	—	Generate/update carve coverage blockmap file. The first 32bit unsigned int in the file identifies the block size. Thereafter each 32bit unsigned int entry in the blockmap file corresponds to one block	
-n	—	Don't add extensions to extracted files.	
-o	—	Set output directory for carved files.	
-O	—	Don't organize carved files by type. Default is to organize carved files into subdirectories.	
-p	—	Perform image file preview; audit log indicates which files would have been carved, but no files are actually carved.	
-q	—	Carve only when header is cluster-aligned.	
-r	—	Find only first of overlapping headers/footers [foremost 0.69 compat mode].	
-s	—	Skip n bytes in each disk image before carving.	
-t	—	Set directory for coverage blockmap. EXPERIMENTAL	

Flag	Argument	What it does	When you would use it
<code>-u</code>	—	Use carve coverage blockmap when carving. Carve only sections of the image whose entries in the blockmap are 0. These areas are treated as contiguous regions. EXPERIMENTAL	
<code>-V</code>	—	Print copyright information and exit.	
<code>-v</code>	—	Verbose mode.	

See also

`foremost`, `binwalk`, `bulk_extractor`, `tcpextract`

strings

Kit: Base OS — present on every Linux image **Capability:** Search raw data for a pattern; Extract strings, including obfuscated ones **Version:** GNU strings (GNU Binutils for Debian) 2.40 **Captured:** `cyberlab-aio` via `--help` on 2026-08-02 [raw](#)

[← Capability index](#) · [Kit tool list](#)

Purpose

Print sequences of printable characters found in a binary file.

Synopsis

```
strings [option(s)] [file(s)]
```

Options

All 14 options parsed from the captured help text; 3 reviewed with usage guidance.

Flag	Argument	What it does	When you would use it
<code>-d</code>	—	Only scan the data sections in the file	
<code>--data</code>	—	Only scan the data sections in the file	
<code>-f</code>	—	Print the name of the file before each string	
<code>--print-file-name</code>	—	Print the name of the file before each string	
<code>-n</code>	number	Locate & print any sequence of at least	Minimum length. Default 4 is noisy; 8–10 cuts most false hits.
<code>--bytes</code>	number	displayable characters. (The default is 4).	
<code>-t</code>	o,d,x	Print the location of the string in base 8, 10 or 16	
<code>--radix</code>	o,d,x	Print the location of the string in base 8, 10 or 16	
<code>-o</code>	—	An alias for <code>--radix=o</code>	Print the byte offset of each string — lets you seek back to it.
<code>-T</code>	—	Specify the binary file format	
<code>--target</code>	BFDNAME	Specify the binary file format	
<code>-U</code>	d s i x e h	Specify how to treat UTF-8 encoded unicode characters	Control how unicode is handled; needed for UTF-16 Windows strings.
<code>-h</code>	—	Display this information	
<code>--help</code>	—	Display this information	

Gotchas

- GNU `strings` reads only initialised, loaded sections by default. Use `-a` to scan the whole file — malware routinely hides outside them.
- It finds ASCII by default and will miss UTF-16LE strings that Windows binaries are full of. When a PE looks empty, that is usually why — reach for `floss` instead.

See also

[rafind2](#), [grep](#), [xxd](#), [bulk_extractor](#), [floss](#), [base64dump.py](#), [numbers-to-string.py](#)

tcpextract

Kit: REMnux · SIFT Workstation **Capability:** Carve files out of unstructured data; Extract files and payloads from traffic **Version:** tcpextract v1.0.1 **Captured:** cyberlab-aio
via `--help` on 2026-08-02 **raw Docs:** <http://tcpextract.sourceforge.net/>

[← Capability index](#) · [Kit tool list](#)

Purpose

Extract files from network traffic.

Synopsis

```
tcpextract [OPTIONS] [[-d <DEVICE>] [-f <FILE>]]
```

Common invocations

```
# from cyberlab 05-file-carving
tcpextract --version 2>&1 | head -n 1
# from cyberlab 05-file-carving
tcpextract -f exercise/sample.pcap -o /tmp/tcpextract_out
# from cyberlab 05-file-carving
tcpextract -f exercise/sample.pcap -o /tmp/ak_tcp
```

Options

All 12 options parsed from the captured help text. The final column is filled in by review.

Flag	Argument	What it does	When you would use it
<code>--file</code>	FILE	to specify an input capture file instead of a device	
<code>-f</code>	FILE	to specify an input capture file instead of a device	
<code>--device</code>	DEVICE	to specify an input device (i.e. eth0)	
<code>-d</code>	DEVICE	to specify an input device (i.e. eth0)	
<code>--config</code>	FILE	use FILE as the config file	
<code>-c</code>	FILE	use FILE as the config file	
<code>--output</code>	DIRECTORY	dump files to DIRECTORY instead of current directory	
<code>-o</code>	DIRECTORY	dump files to DIRECTORY instead of current directory	
<code>--version</code>	—	display the version number of this program	
<code>-v</code>	—	display the version number of this program	
<code>--help</code>	—	display this lovely screen	
<code>-h</code>	—	display this lovely screen	

See also

[foremost](#), [scalpel](#), [binwalk](#), [bulk_extractor](#), [tcpflow](#)

testdisk

Kit: SIFT Workstation **Capability:** See the partition and volume layout; Recover deleted or lost files **Version:** TestDisk 7.1 **Captured:** cyberlab-ai0 via --help on 2026-08-02 [raw](#)

[← Capability index](#) · [Kit tool list](#)

Purpose

TestDisk 7.1, Data Recovery Utility, July 2019

Synopsis

```
testdisk [/log] [/debug] [file.dd|file.e01|device]
testdisk /list [/log] [file.dd|file.e01|device]
testdisk /version
```

Common invocations

```
# from cyberlab 01-disk-forensics
testdisk /version
```

Options

All 3 options parsed from the captured help text. The final column is filled in by review.

Flag	Argument	What it does	When you would use it
/log	—	create a testdisk.log file	
/debug	—	add debug information	
/list	—	display current partitions	

See also

[mmls](#), [fsstat](#), [img_stat](#), [tsk_recover](#), [icat](#), [photorec](#), [blkls](#)

tsk_recover

Kit: REMnux · Kali Linux · SIFT Workstation **Capability:** List files and directories, including deleted ones; Recover deleted or lost files **Version:** The Sleuth Kit ver 4.11.1
Captured: cyberlab-aio via --help on 2026-08-02 **raw Docs:** <https://www.sleuthkit.org/sleuthkit>

← [Capability index](#) · [Kit tool list](#)

Purpose

Bulk-export files from an image to a directory.

Synopsis

```
tsk_recover [-vVae] [-f fstype] [-i imgtype] [-b dev_sector_size] [-o sector_offset] [-P pooltype] [-B pool_volume_block] [-d dir_inum] image [image]
```

Options

All 11 options parsed from the captured help text; 6 reviewed with usage guidance.

Flag	Argument	What it does	When you would use it
-i	imgtype	The format of the image file (use '-i list' for supported types)	Image format for non-raw evidence.
-b	dev_sector_size	The size (in bytes) of the device sectors	
-f	fstype	The file system type (use '-f list' for supported types)	Force the filesystem type.
-v	—	verbose output to stderr	
-V	—	Print version	
-a	—	Recover allocated files only	Recover allocated (live) files only.
-e	—	Recover all files (allocated and unallocated)	Recover every file, allocated and deleted — the usual choice.
-o	sector_offset	sector offset for a volume to recover (recovers only that volume)	Partition offset in sectors, from <code>mmls</code> .
-P	pooltype	Pool container type (use '-P list' for supported types)	
-B	pool_volume_block	Starting block (for pool volumes only)	
-d	dir_inum	Directory inum to recover from (must also specify a specific partition using -o or there must not be a volume system)	Recover from a specified directory inode rather than the root.

Gotchas

- Default behaviour recovers only *deleted* files, which surprises people expecting a full export. Use `-e` for everything.
- This preserves paths and names, unlike carving. Prefer it whenever the filesystem metadata is intact, and carve only what it cannot reach.

See also

`fls`, `ffind`, `ils`, `icat`, `photorec`, `testdisk`, `blkls`

xxd

Kit: Base OS — present on every Linux image **Capability:** Search raw data for a pattern; Decode, decrypt or transform encoded data; Inspect files by hand **Captured:** cyberlab-aio via --help on 2026-08-02 raw

← [Capability index](#) · [Kit tool list](#)

Purpose

xxd [options] [infile [outfile]]

Synopsis

```
xxd [options] [infile [outfile]]
or
xxd -r [-s [-]offset] [-c cols] [-ps] [infile [outfile]]
```

Options

All 16 options parsed from the captured help text. The final column is filled in by review.

Flag	Argument	What it does	When you would use it
-a	—	toggle autoskip: A single '*' replaces nul-lines. Default off.	
-b	—	binary digit dump (incompatible with -ps,-i,-r). Default hex.	
-C	—	capitalize variable names in C include file style (-i).	
-c	cols	format octets per line. Default 16 (-i: 12, -ps: 30).	
-E	—	show characters in EBCDIC. Default ASCII.	
-e	—	little-endian dump (incompatible with -ps,-i,-r).	
-g	bytes	number of octets per group in normal output. Default 2 (-e: 4).	
-h	—	print this summary.	
-i	—	output in C include file style.	
-l	len	stop after octets.	
-n	name	set the variable name used in C include output (-i).	
-o	off	add to the displayed file position.	
-r	—	reverse operation: convert (or patch) hexdump into binary.	
-d	—	show offset in decimal instead of hex.	
-u	—	use upper case hex letters.	
-v	—	show version: "xxd 2022-01-14 by Juergen Weigert et al."	

See also

[rafind2](#) , [strings](#) , [grep](#) , [bulk_extractor](#) , [cyberchef](#) , [base64dump.py](#) , [rax2](#) , [openssl](#)

mraptor

Kit: REMnux **Capability:** Analyse Office documents and macros **Version:** MacroRaptor 0.56.2 **Captured:** cyberlab-aio via --help on 2026-08-02 [raw](#) **Docs:** <https://www.decorage.info/python/oletools>

[← Capability index](#) · [Kit tool list](#)

Purpose

Microsoft Office OLE2 compound documents.

Synopsis

```
mraptor [options] <filename> [filename2 ...]
```

Common invocations

```
# from cyberlab 36-oletools-deep
mraptor --version
# from cyberlab 36-oletools-deep
mraptor sample_macro.doc
# from cyberlab 36-oletools-deep
mraptor sample_macro.doc # automated verdict
```

Options

All 11 options parsed from the captured help text. The final column is filled in by review.

Flag	Argument	What it does	When you would use it
-h	—	show this help message and exit	
--help	—	show this help message and exit	
-r	—	find files recursively in subdirectories.	
-z	ZIP_PASSWORD	if the file is a zip archive, open all files from it, using the provided password (requires Python 2.6+)	
--zip	ZIP_PASSWORD	if the file is a zip archive, open all files from it, using the provided password (requires Python 2.6+)	
-f	ZIP_FNAME	if the file is a zip archive, file(s) to be opened within the zip. Wildcards * and ? are supported. (default:*)	
--zipfname	ZIP_FNAME	if the file is a zip archive, file(s) to be opened within the zip. Wildcards * and ? are supported. (default:*)	
-l	LOGLEVEL	logging level debug/info/warning/error/critical (default=warning)	
--loglevel	LOGLEVEL	logging level debug/info/warning/error/critical (default=warning)	
-m	—	Show matched strings.	
--matches	—	Show matched strings.	

See also

[olevba](#), [oleid](#), [oledump.py](#), [olemeta](#), [oletimes](#), [olemap](#), [oledir](#), [olebrowse](#)

msodde

Kit: REMnux **Capability:** Analyse Office documents and macros **Captured:** cyberlab-aio via --help on 2026-08-02 [raw Docs:](#) <https://www.decalage.info/python/oletools>

[← Capability index](#) · [Kit tool list](#)

Purpose

Microsoft Office OLE2 compound documents.

Synopsis

```
msodde [-h] [-j] [--nounquote] [-l LOGLEVEL] [-p PASSWORD] [-d] [-f]
[-a]
FILE
```

Options

All 15 options parsed from the captured help text. The final column is filled in by review.

Flag	Argument	What it does	When you would use it
-h	—	show this help message and exit	
--help	—	show this help message and exit	
-j	—	Output in json format. Do not use with -ldebug	
--json	—	Output in json format. Do not use with -ldebug	
--nounquote	—	don't unquote values	
-l	LOGLEVEL	logging level debug/info/warning/error/critical (default=warning)	
--loglevel	LOGLEVEL	logging level debug/info/warning/error/critical (default=warning)	
-p	PASSWORD	if encrypted office files are encountered, try decryption with this password. May be repeated.	
--password	PASSWORD	if encrypted office files are encountered, try decryption with this password. May be repeated.	
-d	—	Return only DDE and DDEAUTO fields	
--dde-only	—	Return only DDE and DDEAUTO fields	
-f	—	Return all fields except harmless ones	
--filter	—	Return all fields except harmless ones	
-a	—	Return all fields, irrespective of their contents	
--all-fields	—	Return all fields, irrespective of their contents	

See also

[olevba](#), [oleid](#), [oledump.py](#), [olemeta](#), [oletimes](#), [olemap](#), [oledir](#), [olebrowse](#)

msoffcrypto-tool

Kit: REMnux **Capability:** Analyse other container formats **Captured:** cyberlab-aio via --help on 2026-08-02 **raw Docs:** <https://github.com/nolze/msoffcrypto-tool>
[← Capability index](#) · [Kit tool list](#)

Purpose

Decrypt a Microsoft Office file with password, intermediate key, or private key which generated its escrow key.

Synopsis

```
msoffcrypto-tool [-h] (-p [PASSWORD] | -t) [-e] [-v] [infile] [outfile]
```

Options

All 8 options parsed from the captured help text. The final column is filled in by review.

Flag	Argument	What it does	When you would use it
-h	—	show this help message and exit	
--help	—	show this help message and exit	
-p	PASSWORD	password text	
--password	PASSWORD	password text	
-t	—	test if the file is encrypted	
--test	—	test if the file is encrypted	
-e	—	encryption mode (default is false)	
-v	—	print verbose information	

See also

7za, unzip

OffVis (GUI)

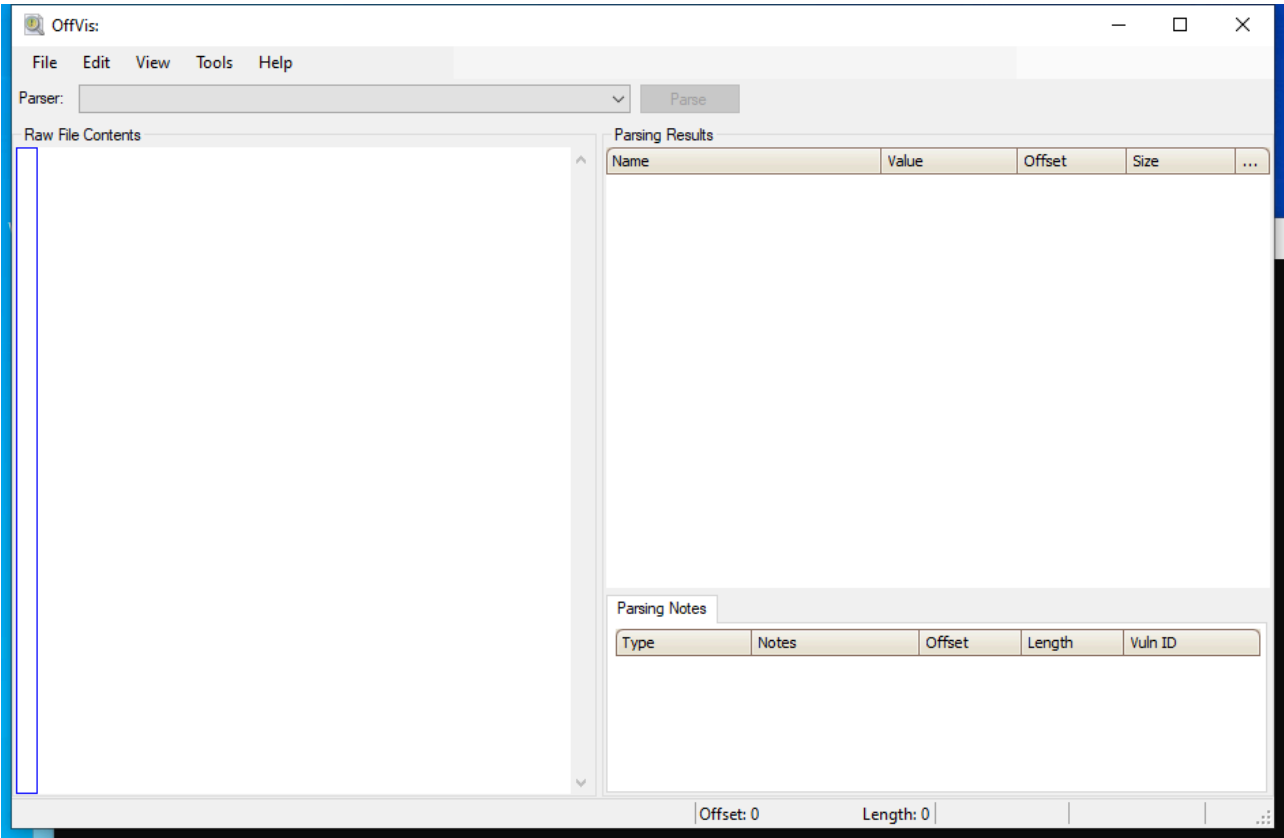
Capability: malware triage documents **Window:** WindowsForms10.Window.8.app.0.378734a **Version:** — **Captured:** C:\Tools\OffVis\OffVis.exe on 2026-08-02 — control tree in [capture/gui/OffVis/OffVis.tree.txt](#)

[← Capability index](#) · [Kit tool list](#)

Purpose

Visualise Office binary file structure.

Window



Controls

All 19 nodes come from the capture; the 7 interactive controls are listed here.

Control	Type	AutomationId	What it does
Raw File Contents	Pane	459370	
Parsing Results	Pane	262750	
Parsing Notes	Pane	262732	
Parse	Pane	197186	
Parser:	Pane	197182	
menuStrip1	Pane	197178	
statusStrip1	Pane	197176	

Using it

1. Open the legacy Office binary — .doc , .xls , .ppt .
2. Read the parsed structure against the raw bytes shown alongside it.
3. Compare what the record headers claim against what is actually present; the mismatch is the exploit.

Gotchas

- It only understands the *binary* formats. OOXML files — `.docx` , `.xlsx` — are ZIP archives and this tool cannot read them; unzip and use `oledump.py` on the extracted `vbaProject.bin` .
- It is a structure visualiser, not a detector. It shows you the record layout so you can see the malformation; it does not tell you one is present.
- The tool is old and unmaintained. Treat a crash as information about the file rather than a reason to stop.

olebrowse

Kit: REMnux **Capability:** Analyse Office documents and macros **Captured:** cyberlab-aio via --help on 2026-08-02 [raw](#) **Docs:** <https://www.decalage.info/python/oletools>

[← Capability index](#) · [Kit tool list](#)

Purpose

Microsoft Office OLE2 compound documents.

Options

No option definitions could be parsed from this tool's help output. It may be subcommand-driven or have no flags; needs manual review.

See also

[olevba](#), [oleid](#), [oledump.py](#), [olemeta](#), [oletimes](#), [olemap](#), [oledir](#), [olefile](#)

oledir

Kit: REMnux **Capability:** Analyse Office documents and macros **Version:** oledir 0.54 **Captured:** cyberlab-aio via --help on 2026-08-02 [raw](#) **Docs:** <https://www.decalage.info/python/oletools>

[← Capability index](#) · [Kit tool list](#)

Purpose

Microsoft Office OLE2 compound documents.

Synopsis

```
oledir [options] <filename> [filename2 ...]
```

Options

All 7 options parsed from the captured help text. The final column is filled in by review.

Flag	Argument	What it does	When you would use it
-h	—	show this help message and exit	
--help	—	show this help message and exit	
-r	—	find files recursively in subdirectories.	
-z	ZIP_PASSWORD	if the file is a zip archive, open all files from it, using the provided password (requires Python 2.6+)	
--zip	ZIP_PASSWORD	if the file is a zip archive, open all files from it, using the provided password (requires Python 2.6+)	
-f	ZIP_FNAME	if the file is a zip archive, file(s) to be opened within the zip. Wildcards * and ? are supported. (default:*)	
--zipfname	ZIP_FNAME	if the file is a zip archive, file(s) to be opened within the zip. Wildcards * and ? are supported. (default:*)	

See also

[olevba](#), [oleid](#), [oledump.py](#), [olemeta](#), [oletimes](#), [olemap](#), [olebrowse](#), [olefile](#)

oledump.py

Kit: REMnux **Capability:** Analyse Office documents and macros **Version:** oledump.py 0.0.85 **Captured:** cyberlab-aio via `--help` on 2026-08-02 [raw Docs: https://www.decorage.info/python/oletools](https://www.decorage.info/python/oletools)

[← Capability index](#) · [Kit tool list](#)

Purpose

List and extract the streams inside an OLE2 file — the container behind legacy Office documents and many malicious attachments.

Synopsis

```
oledump.py [options] [file]
```

Common invocations

```
# from cyberlab 10-malicious-documents
oledump.py -h | head -n 1
# from cyberlab 10-malicious-documents
oledump.py exercise/sample.doc
# from cyberlab 10-malicious-documents
oledump.py -s 3 -v exercise/sample.doc
# from cyberlab 36-oletools-deep
oledump.py --help | head -n 3
# from cyberlab 36-oletools-deep
oledump.py sample_macro.doc
# from cyberlab 36-oletools-deep
oledump.py -s 3 -v sample_macro.doc
# from cyberlab 36-oletools-deep
oledump.py sample_macro.doc # locate the 'M' stream (3)
# from cyberlab 36-oletools-deep
oledump.py -s 3 -v sample_macro.doc # dump VBA source
```

Options

All 64 options parsed from the captured help text; 13 reviewed with usage guidance.

Flag	Argument	What it does	When you would use it
<code>--version</code>	—	show program's version number and exit	
<code>-h</code>	—	show this help message and exit	
<code>--help</code>	—	show this help message and exit	
<code>-m</code>	—	Print manual	Print the full manual, which documents the plugin and selection syntax this summary cannot cover.
<code>--man</code>	—	Print manual	Print the full manual, which documents the plugin and selection syntax this summary cannot cover.
<code>-s</code>	SELECT	select item nr for dumping (a for all)	Select a stream by number, or <code>a</code> for all. Everything else operates on the selection, so this usually comes first.
<code>--select</code>	SELECT	select item nr for dumping (a for all)	Select a stream by number, or <code>a</code> for all. Everything else operates on the selection, so this usually comes first.
<code>-d</code>	—	perform dump	Dump the selected stream raw, for carving an embedded payload out to a file.
<code>--dump</code>	—	perform dump	Dump the selected stream raw, for carving an embedded payload out to a file.
<code>-x</code>	—	perform hex dump	Hex dump, when the stream is binary and you need to see structure.
<code>--hexdump</code>	—	perform hex dump	Hex dump, when the stream is binary and you need to see structure.
<code>-a</code>	—	perform ascii dump	ASCII dump, for a quick look at mostly-text content.
<code>--asciidump</code>	—	perform ascii dump	ASCII dump, for a quick look at mostly-text content.
<code>-A</code>	—	perform ascii dump with RLE	ASCII dump with run-length encoding, which collapses long runs of padding that otherwise bury the content.

Flag	Argument	What it does	When you would use it
<code>--asciidumprle</code>	—	perform ascii dump with RLE	ASCII dump with run-length encoding, which collapses long runs of padding that otherwise bury the content.
<code>-S</code>	—	perform strings dump	Strings dump — the fastest way to see whether a stream holds anything readable.
<code>--strings</code>	—	perform strings dump	Strings dump — the fastest way to see whether a stream holds anything readable.
<code>-T</code>	—	do head & tail	Head and tail only, to glance at a large stream.
<code>--headtail</code>	—	do head & tail	Head and tail only, to glance at a large stream.
<code>-v</code>	—	VBA decompression	Decompress VBA. Macro streams are stored compressed, so without this the source looks like binary noise. This is the flag the tool exists for.
<code>--vbadecompress</code>	—	VBA decompression	Decompress VBA. Macro streams are stored compressed, so without this the source looks like binary noise. This is the flag the tool exists for.
<code>--vbadecompressskipattributes</code>	—	VBA decompression, skipping initial attributes	Skip the attribute preamble and show only the macro body.
<code>--vbadecompresscorrupt</code>	—	VBA decompression, display beginning if corrupted	Decompress as far as possible and show it, for a deliberately corrupted macro stream that defeats a clean decompress.
<code>-r</code>	—	read raw file (use with options -v or -p)	Treat the input as a raw stream rather than an OLE file, for a stream already carved out elsewhere.
<code>--raw</code>	—	read raw file (use with options -v or -p)	Treat the input as a raw stream rather than an OLE file, for a stream already carved out elsewhere.
<code>-t</code>	TRANSLATE	string translation, like utf16 or .decode("utf8")	Apply a translation such as <code>utf16</code> when the stream is wide-character text.
<code>--translate</code>	TRANSLATE	string translation, like utf16 or .decode("utf8")	Apply a translation such as <code>utf16</code> when the stream is wide-character text.
<code>-e</code>	—	extract OLE embedded file	
<code>--extract</code>	—	extract OLE embedded file	
<code>-i</code>	—	print extra info for selected item	
<code>--info</code>	—	print extra info for selected item	
<code>-p</code>	PLUGINS	plugins to load (separate plugins with a comma , ; @file supported)	
<code>--plugins</code>	PLUGINS	plugins to load (separate plugins with a comma , ; @file supported)	
<code>--pluginoptions</code>	PLUGINOPTIONS	options for the plugin	
<code>--plugindir</code>	PLUGINDIR	directory for the plugin	
<code>-q</code>	—	only print output from plugins	
<code>--quiet</code>	—	only print output from plugins	
<code>-y</code>	YARA	YARA rule-file, @file, directory or #rule to check streams (YARA search doesn't work with -s option)	
<code>--yara</code>	YARA	YARA rule-file, @file, directory or #rule to check streams (YARA search doesn't work with -s option)	
<code>-D</code>	DECODERS	decoders to load (separate decoders with a comma , ; @file supported)	
<code>--decoders</code>	DECODERS	decoders to load (separate decoders with a comma , ; @file supported)	
<code>--decoderoptions</code>	DECODEROPTIONS	options for the decoder	
<code>--decoderdir</code>	DECODERDIR	directory for the decoder	
<code>--yarastrings</code>	—	Print YARA strings	
<code>-M</code>	—	Print metadata	
<code>--metadata</code>	—	Print metadata	
<code>-c</code>	—	Add extra calculated data to output, like hashes	
<code>--calc</code>	—	Add extra calculated data to output, like hashes	
<code>--decompress</code>	—	Search for compressed data in the stream and decompress it	
<code>-V</code>	—	verbose output with decoder errors and YARA rules	
<code>--verbose</code>	—	verbose output with decoder errors and YARA rules	
<code>-C</code>	CUT	cut data	
<code>--cut</code>	CUT	cut data	

Flag	Argument	What it does	When you would use it
<code>-E</code>	EXTRA	add extra info (environment variable: OLEDUMP_EXTRA)	
<code>--extra</code>	EXTRA	add extra info (environment variable: OLEDUMP_EXTRA)	
<code>--storages</code>	—	Include storages in report	
<code>-f</code>	FIND	Find D0CF11E0 MAGIC sequence (use l for listing, number for selecting)	
<code>--find</code>	FIND	Find D0CF11E0 MAGIC sequence (use l for listing, number for selecting)	
<code>-j</code>	—	produce json output	
<code>--jsonoutput</code>	—	produce json output	
<code>-u</code>	—	Include unused data after end of stream	
<code>--unuseddata</code>	—	Include unused data after end of stream	
<code>--password</code>	PASSWORD	The ZIP password to be used (default infected)	
<code>--trimnull</code>	—	Trim starting at first null byte	

Gotchas

- The stream letters in the listing are the finding: `M` marks a macro stream and `O` an embedded object. A document with neither is not carrying either, whatever else it contains.
- OOXML files — .docx, .xlsx — are ZIP archives, not OLE2. Unzip first and run this against the extracted `vbaProject.bin`.
- Reading a decompressed macro is not analysing it. `olevba` adds the keyword and IOC pass; this tool gets you the bytes.

See also

`olevba`, `oleid`, `olemeta`, `oletimes`, `olemap`, `oledir`, `olebrowse`, `olefile`

olefile

Kit: REMnux **Capability:** Analyse Office documents and macros **Captured:** cyberlab-aio via --help on 2026-08-02 **raw Docs:** <https://github.com/decalage2/olefile>
← [Capability index](#) · [Kit tool list](#)

Purpose

Python package to parse, read and write MS OLE2 files.

Synopsis

```
olefile [options] <filename> [filename2 ...]
```

Options

All 7 options parsed from the captured help text. The final column is filled in by review.

Flag	Argument	What it does	When you would use it
-h	—	show this help message and exit	
--help	—	show this help message and exit	
-c	—	check all streams (for debugging purposes)	
-p	—	extract all user-defined propertires	
-d	—	debug mode, shortcut for -l debug (displays a lot of debug information, for developers only)	
-l	LOGLEVEL	logging level debug/info/warning/error/critical (default=warning)	
--loglevel	LOGLEVEL	logging level debug/info/warning/error/critical (default=warning)	

See also

[olevba](#), [oleid](#), [oledump.py](#), [olemeta](#), [oletimes](#), [olemap](#), [oledir](#), [olebrowse](#)

oleid

Kit: REMnux **Capability:** Analyse Office documents and macros **Version:** oleid 0.60.1 **Captured:** cyberlab-aio via --help on 2026-08-02 [raw Docs:](#) <https://www.decorage.info/python/oletools>

[← Capability index](#) · [Kit tool list](#)

Purpose

Microsoft Office OLE2 compound documents.

Synopsis

```
oleid [-h] [FILE ...]
```

Common invocations

```
# from cyberlab 10-malicious-documents
oleid exercise/sample.doc
# from cyberlab 36-oletools-deep
oleid -h | head -n 3
# from cyberlab 36-oletools-deep
oleid gives a quick risk indicator table (VBA macros: Yes/No, encryption, etc.)
# from cyberlab 36-oletools-deep
oleid sample_macro.doc
# from cyberlab 48-phishing-doc-case
oleid --version
# from cyberlab 48-phishing-doc-case
oleid exercise/invoice_sample.doc
```

Options

All 2 options parsed from the captured help text. The final column is filled in by review.

Flag	Argument	What it does	When you would use it
-h	—	show this help message and exit	
--help	—	show this help message and exit	

See also

[olevba](#), [oledump.py](#), [olemeta](#), [oletimes](#), [olemap](#), [oledir](#), [olebrowse](#), [olefile](#)

olemap

Kit: REMnux **Capability:** Analyse Office documents and macros **Captured:** cyberlab-aio via --help on 2026-08-02 [raw](#) **Docs:** <https://www.decalage.info/python/oletools>

[← Capability index](#) · [Kit tool list](#)

Purpose

Microsoft Office OLE2 compound documents.

Options

No option definitions could be parsed from this tool's help output. It may be subcommand-driven or have no flags; needs manual review.

See also

[olevba](#), [oleid](#), [oledump.py](#), [olemeta](#), [oletimes](#), [oledir](#), [olebrowse](#), [olefile](#)

olemeta

Kit: REMnux **Capability:** Analyse Office documents and macros **Version:** olemeta 0.54 **Captured:** cyberlab-aio via --help on 2026-08-02 [raw](#) **Docs:** <https://www.decalage.info/python/oletools>

[← Capability index](#) · [Kit tool list](#)

Purpose

Microsoft Office OLE2 compound documents.

Synopsis

```
olemeta [options] <filename> [filename2 ...]
```

Options

All 7 options parsed from the captured help text. The final column is filled in by review.

Flag	Argument	What it does	When you would use it
-h	—	show this help message and exit	
--help	—	show this help message and exit	
-r	—	find files recursively in subdirectories.	
-z	ZIP_PASSWORD	if the file is a zip archive, open all files from it, using the provided password (requires Python 2.6+)	
--zip	ZIP_PASSWORD	if the file is a zip archive, open all files from it, using the provided password (requires Python 2.6+)	
-f	ZIP_FNAME	if the file is a zip archive, file(s) to be opened within the zip. Wildcards * and ? are supported. (default:*)	
--zipfname	ZIP_FNAME	if the file is a zip archive, file(s) to be opened within the zip. Wildcards * and ? are supported. (default:*)	

See also

[olevba](#), [oleid](#), [oledump.py](#), [oletimes](#), [olemap](#), [oledir](#), [olebrowse](#), [olefile](#)

oleobj

Kit: REMnux **Capability:** Analyse Office documents and macros; Analyse RTF documents and embedded objects **Version:** oleobj 0.60.1 **Captured:** cyberlab-aio via --help on 2026-08-02 **raw Docs:** <https://www.decorage.info/python/oletools>

[← Capability index](#) · [Kit tool list](#)

Purpose

Microsoft Office OLE2 compound documents.

Synopsis

```
usage: oleobj [options] <filename> [filename2 ...]
```

Options

All 14 options parsed from the captured help text. The final column is filled in by review.

Flag	Argument	What it does	When you would use it
-h	—	show this help message and exit	
--help	—	show this help message and exit	
-r	—	find files recursively in subdirectories.	
-d	OUTPUT_DIR	use specified directory to output files.	
-z	ZIP_PASSWORD	if the file is a zip archive, open first file from it, using the provided password (requires Python 2.6+)	
--zip	ZIP_PASSWORD	if the file is a zip archive, open first file from it, using the provided password (requires Python 2.6+)	
-f	ZIP_FNAME	if the file is a zip archive, file(s) to be opened within the zip. Wildcards * and ? are supported. (default:*)	
--zipfname	ZIP_FNAME	if the file is a zip archive, file(s) to be opened within the zip. Wildcards * and ? are supported. (default:*)	
-l	LOGLEVEL	logging level debug/info/warning/error/critical (default=warning)	
--loglevel	LOGLEVEL	logging level debug/info/warning/error/critical (default=warning)	
-i	FILE	Additional file to parse (same as positional arguments)	
--more-input	FILE	Additional file to parse (same as positional arguments)	
-v	—	verbose mode, set logging to DEBUG (overwrites -l)	
--verbose	—	verbose mode, set logging to DEBUG (overwrites -l)	

See also

[olevba](#), [oleid](#), [oledump.py](#), [olemeta](#), [oletimes](#), [olemap](#), [oledir](#), [olebrowse](#)

oletimes

Kit: REMnux **Capability:** Analyse Office documents and macros **Version:** oletimes 0.54 **Captured:** cyberlab-aio via --help on 2026-08-02 [raw](#) **Docs:** <https://www.decalage.info/python/oletools>

[← Capability index](#) · [Kit tool list](#)

Purpose

Microsoft Office OLE2 compound documents.

Synopsis

```
oletimes [options] <filename> [filename2 ...]
```

Options

All 7 options parsed from the captured help text. The final column is filled in by review.

Flag	Argument	What it does	When you would use it
-h	—	show this help message and exit	
--help	—	show this help message and exit	
-r	—	find files recursively in subdirectories.	
-z	ZIP_PASSWORD	if the file is a zip archive, open all files from it, using the provided password (requires Python 2.6+)	
--zip	ZIP_PASSWORD	if the file is a zip archive, open all files from it, using the provided password (requires Python 2.6+)	
-f	ZIP_FNAME	if the file is a zip archive, file(s) to be opened within the zip. Wildcards * and ? are supported. (default:*)	
--zipfname	ZIP_FNAME	if the file is a zip archive, file(s) to be opened within the zip. Wildcards * and ? are supported. (default:*)	

See also

[olevba](#), [oleid](#), [oledump.py](#), [olemeta](#), [olemap](#), [oledir](#), [olebrowse](#), [olefile](#)

olevba

Kit: REMnux **Capability:** Analyse Office documents and macros **Version:** olevba 0.60.2 **Captured:** cyberlab-aio via --help on 2026-08-02 [raw Docs:](#) <https://www.decorage.info/python/oletools>

[← Capability index](#) · [Kit tool list](#)

Purpose

Extract and analyse VBA macros from Office documents.

Synopsis

```
usage: olevba [options] <filename> [filename2 ...]
```

Common invocations

```
# from cyberlab 10-malicious-documents
olevba --version
# from cyberlab 10-malicious-documents
olevba --decode exercise/sample.doc
# from cyberlab 10-malicious-documents
olevba exercise/sample.doc | grep -E "AutoExec|AutoOpen|Suspicious"
# from cyberlab 36-oletools-deep
olevba sample_macro.doc
# from cyberlab 36-oletools-deep
olevba sample_macro.doc # keyword analysis table
# from cyberlab 48-phishing-doc-case
olevba --decode exercise/invoice_sample.doc
```

Options

All 29 options parsed from the captured help text; 10 reviewed with usage guidance.

Flag	Argument	What it does	When you would use it
-h	—	show this help message and exit	
--help	—	show this help message and exit	
-r	—	find files recursively in subdirectories.	
-z	ZIP_PASSWORD	if the file is a zip archive, open all files from it, using the provided password.	Read documents from inside a zip archive.
--zip	ZIP_PASSWORD	if the file is a zip archive, open all files from it, using the provided password.	Read documents from inside a zip archive.
-p	PASSWORD	if encrypted office files are encountered, try decryption with this password. May be repeated.	Supply a password for an encrypted document.
--password	PASSWORD	if encrypted office files are encountered, try decryption with this password. May be repeated.	Supply a password for an encrypted document.
-f	ZIP_FNAME	if the file is a zip archive, file(s) to be opened within the zip. Wildcards * and ? are supported. (default: *)	
--zipfname	ZIP_FNAME	if the file is a zip archive, file(s) to be opened within the zip. Wildcards * and ? are supported. (default: *)	
-a	—	display only analysis results, not the macro source code	Show only the analysis results, skipping the macro source.
--analysis	—	display only analysis results, not the macro source code	Show only the analysis results, skipping the macro source.
-c	—	display only VBA source code, do not analyze it	Show only the macro source code.
--code	—	display only VBA source code, do not analyze it	Show only the macro source code.
--decode	—	display all the obfuscated strings with their decoded content (Hex, Base64, StrReverse, Dridex, VBA).	Display all the decoded strings the analyser recovered.
--attr	—	display the attribute lines at the beginning of VBA source code	
--reveal	—	display the macro source code after replacing all the obfuscated strings by their decoded content.	Substitute deobfuscated values back into the source, which is the most readable view of a hostile macro.
-l	LOGLEVEL	logging level debug/info/warning/error/critical (default=warning)	Set log level when diagnosing a parse failure.

Flag	Argument	What it does	When you would use it
<code>--loglevel</code>	LOGLEVEL	logging level debug/info/warning/error/critical (default=warning)	Set log level when diagnosing a parse failure.
<code>--deobf</code>	—	Attempt to deobfuscate VBA expressions (slow)	
<code>--relaxed</code>	—	Do not raise errors if opening of substream fails (this option is now deprecated, enabled by default)	
<code>--show-pcode</code>	—	Show disassembled P-code (using pcodedmp)	Disassemble the stored p-code — catches macros where the VBA source and compiled p-code disagree.
<code>--no-pcode</code>	—	Disable extraction and analysis of pcode	
<code>--no-xlm</code>	—	Do not extract XLM Excel macros. This may speed up analysis of large files.	
<code>-t</code>	—	triage mode, display results as a summary table (default for multiple files)	Triage mode — one compact line per file, ideal for a batch.
<code>--triage</code>	—	triage mode, display results as a summary table (default for multiple files)	Triage mode — one compact line per file, ideal for a batch.
<code>-d</code>	—	detailed mode, display full results (default for single file)	Show the deobfuscated macro source.
<code>--detailed</code>	—	detailed mode, display full results (default for single file)	Show the deobfuscated macro source.
<code>-j</code>	—	json mode, detailed in json format (never default)	
<code>--json</code>	—	json mode, detailed in json format (never default)	

Gotchas

- **VBA source and p-code can differ.** Malware abuses this so the readable source looks benign while the p-code executes. If a document is suspicious but the source is clean, check `--show-pcode`.
- Run this on a copy, in an isolated VM. Nothing here executes the macro, but the sample itself is still live malware.

See also

`oleid`, `oledump.py`, `olemeta`, `oletimes`, `olemap`, `oledir`, `olebrowse`, `olefile`

pcodedmp

Kit: REMnux **Capability:** Analyse Office documents and macros **Version:** pcodedmp version 1.2.6 **Captured:** cyberlab-ai0 via --help on 2026-08-02 **raw Docs:** <https://github.com/bontchev/pcodedmp>

[← Capability index](#) · [Kit tool list](#)

Purpose

Disassemble VBA p-code.

Synopsis

```
pcodedmp [-h] [-v] [-n] [-d] [-b] [-o OUTPUTFILE]
fileOrDir [fileOrDir ...]
```

Options

All 12 options parsed from the captured help text. The final column is filled in by review.

Flag	Argument	What it does	When you would use it
-h	—	show this help message and exit	
--help	—	show this help message and exit	
-v	—	show program's version number and exit	
--version	—	show program's version number and exit	
-n	—	Don't recurse into directories	
--norecurse	—	Don't recurse into directories	
-d	—	Only disassemble, no stream dumps	
--disasmonly	—	Only disassemble, no stream dumps	
-b	—	Dump the stream contents	
--verbose	—	Dump the stream contents	
-o	OUTPUTFILE	Output file name	
--output	OUTPUTFILE	Output file name	

See also

[olevba](#), [oleid](#), [oledump.py](#), [olemeta](#), [oletimes](#), [olemap](#), [oledir](#), [olebrowse](#)

pdf-parser

Kit: Kali Linux **Capability:** Analyse PDFs **Version:** pdf-parser.py 0.7.14 **Captured:** cyberlab-aio via --help on 2026-08-02 [raw](#)

[← Capability index](#) · [Kit tool list](#)

Purpose

pdf-parser, use it to parse a PDF document

Synopsis

```
pdf-parser.py [options] pdf-file|zip-file|url
```

Common invocations

```
# from cyberlab 37-pdf-analysis
pdf-parser --version
# from cyberlab 37-pdf-analysis
pdf-parser --search JavaScript exercise/sample.pdf
# from cyberlab 37-pdf-analysis
pdf-parser --object 4 --filter --raw exercise/sample.pdf
```

Options

All 54 options parsed from the captured help text. The final column is filled in by review.

Flag	Argument	What it does	When you would use it
--version	—	show program's version number and exit	
-h	—	show this help message and exit	
--help	—	show this help message and exit	
-m	—	Print manual	
--man	—	Print manual	
-s	SEARCH	string to search in indirect objects (except streams)	
--search	SEARCH	string to search in indirect objects (except streams)	
-f	—	pass stream object through filters (FlateDecode, ASCIIHexDecode, ASCII85Decode, LZWDecode and RunLengthDecode only)	
--filter	—	pass stream object through filters (FlateDecode, ASCIIHexDecode, ASCII85Decode, LZWDecode and RunLengthDecode only)	
-o	OBJECT	id(s) of indirect object(s) to select, use comma (,) to separate ids (version independent)	
--object	OBJECT	id(s) of indirect object(s) to select, use comma (,) to separate ids (version independent)	
-r	REFERENCE	id of indirect object being referenced (version independent)	
--reference	REFERENCE	id of indirect object being referenced (version independent)	
-e	ELEMENTS	type of elements to select (ctxsi)	
--elements	ELEMENTS	type of elements to select (ctxsi)	
-w	—	raw output for data and filters	
--raw	—	raw output for data and filters	
-a	—	display stats for pdf document	
--stats	—	display stats for pdf document	
-t	TYPE	type of indirect object to select	
--type	TYPE	type of indirect object to select	
-O	—	parse stream of /ObjStm objects	
--objstm	—	parse stream of /ObjStm objects	
-v	—	display malformed PDF elements	
--verbose	—	display malformed PDF elements	

Flag	Argument	What it does	When you would use it
<code>-x</code>	EXTRACT	filename to extract malformed content to	
<code>--extract</code>	EXTRACT	filename to extract malformed content to	
<code>-H</code>	—	display hash of objects	
<code>--hash</code>	—	display hash of objects	
<code>-n</code>	—	do not canonicalize the output	
<code>--nocanonicalizedoutput</code>	—	do not canonicalize the output	
<code>-d</code>	DUMP	filename to dump stream content to	
<code>--dump</code>	DUMP	filename to dump stream content to	
<code>-D</code>	—	display debug info	
<code>--debug</code>	—	display debug info	
<code>-c</code>	—	display the content for objects without streams or with streams without filters	
<code>--content</code>	—	display the content for objects without streams or with streams without filters	
<code>--searchstream</code>	SEARCHSTREAM	string to search in streams	
<code>--unfiltered</code>	—	search in unfiltered streams	
<code>--casesensitive</code>	—	case sensitive search in streams	
<code>--regex</code>	—	use regex to search in streams	
<code>--overridingfilters</code>	OVERRIDINGFILTERS	override filters with given filters (use raw for the raw stream content)	
<code>-g</code>	—	generate a Python program that creates the parsed PDF file	
<code>--generate</code>	—	generate a Python program that creates the parsed PDF file	
<code>--generateembedded</code>	GENERATEEMBEDDED	generate a Python program that embeds the selected indirect object as a file	
<code>-y</code>	YARA	YARA rule (or directory or @file) to check streams (can be used with option --unfiltered)	
<code>--yara</code>	YARA	YARA rule (or directory or @file) to check streams (can be used with option --unfiltered)	
<code>--yarastrings</code>	—	Print YARA strings	
<code>--decoders</code>	DECODERS	decoders to load (separate decoders with a comma , ; @file supported)	
<code>--decoderoptions</code>	DECODEROPTIONS	options for the decoder	
<code>-k</code>	KEY	key to search in dictionaries	
<code>--key</code>	KEY	key to search in dictionaries	
<code>-j</code>	—	produce json output	
<code>--jsonoutput</code>	—	produce json output	

See also

[pdfid](#) , [pdfid.py](#) , [pdf-parser.py](#)

pdf-parser.py

Kit: REMnux · Kali Linux **Capability:** Analyse PDFs **Version:** pdf-parser.py 0.7.14 **Captured:** cyberlab-aio via --help on 2026-08-02 [raw](#) **Docs:** <https://blog.didierstevens.com/programs/pdf-tools/>

[← Capability index](#) · [Kit tool list](#)

Purpose

Examine elements of the PDF file.

Synopsis

```
pdf-parser.py [options] pdf-file|zip-file|url
```

Common invocations

```
# from cyberlab 10-malicious-documents
pdf-parser.py --version
# from cyberlab 10-malicious-documents
pdf-parser.py --search JavaScript exercise/sample.pdf
# from cyberlab 48-phishing-doc-case
pdf-parser.py --search OpenAction exercise/resume_sample.pdf
# from cyberlab 48-phishing-doc-case
pdf-parser.py --type /JavaScript exercise/resume_sample.pdf
```

Options

All 54 options parsed from the captured help text. The final column is filled in by review.

Flag	Argument	What it does	When you would use it
--version	—	show program's version number and exit	
-h	—	show this help message and exit	
--help	—	show this help message and exit	
-m	—	Print manual	
--man	—	Print manual	
-s	SEARCH	string to search in indirect objects (except streams)	
--search	SEARCH	string to search in indirect objects (except streams)	
-f	—	pass stream object through filters (FlateDecode, ASCIIHexDecode, ASCII85Decode, LZWDecode and RunLengthDecode only)	
--filter	—	pass stream object through filters (FlateDecode, ASCIIHexDecode, ASCII85Decode, LZWDecode and RunLengthDecode only)	
-o	OBJECT	id(s) of indirect object(s) to select, use comma (,) to separate ids (version independent)	
--object	OBJECT	id(s) of indirect object(s) to select, use comma (,) to separate ids (version independent)	
-r	REFERENCE	id of indirect object being referenced (version independent)	
--reference	REFERENCE	id of indirect object being referenced (version independent)	
-e	ELEMENTS	type of elements to select (ctxsi)	
--elements	ELEMENTS	type of elements to select (ctxsi)	
-w	—	raw output for data and filters	
--raw	—	raw output for data and filters	
-a	—	display stats for pdf document	
--stats	—	display stats for pdf document	
-t	TYPE	type of indirect object to select	
--type	TYPE	type of indirect object to select	
-O	—	parse stream of /ObjStm objects	

Flag	Argument	What it does	When you would use it
<code>--objstm</code>	—	parse stream of /ObjStm objects	
<code>-v</code>	—	display malformed PDF elements	
<code>--verbose</code>	—	display malformed PDF elements	
<code>-x</code>	EXTRACT	filename to extract malformed content to	
<code>--extract</code>	EXTRACT	filename to extract malformed content to	
<code>-H</code>	—	display hash of objects	
<code>--hash</code>	—	display hash of objects	
<code>-n</code>	—	do not canonicalize the output	
<code>--nocanonicalizedoutput</code>	—	do not canonicalize the output	
<code>-d</code>	DUMP	filename to dump stream content to	
<code>--dump</code>	DUMP	filename to dump stream content to	
<code>-D</code>	—	display debug info	
<code>--debug</code>	—	display debug info	
<code>-c</code>	—	display the content for objects without streams or with streams without filters	
<code>--content</code>	—	display the content for objects without streams or with streams without filters	
<code>--searchstream</code>	SEARCHSTREAM	string to search in streams	
<code>--unfiltered</code>	—	search in unfiltered streams	
<code>--casesensitive</code>	—	case sensitive search in streams	
<code>--regex</code>	—	use regex to search in streams	
<code>--overridingfilters</code>	OVERRIDINGFILTERS	override filters with given filters (use raw for the raw stream content)	
<code>-g</code>	—	generate a Python program that creates the parsed PDF file	
<code>--generate</code>	—	generate a Python program that creates the parsed PDF file	
<code>--generateembedded</code>	GENERATEEMBEDDED	generate a Python program that embeds the selected indirect object as a file	
<code>-y</code>	YARA	YARA rule (or directory or @file) to check streams (can be used with option --unfiltered)	
<code>--yara</code>	YARA	YARA rule (or directory or @file) to check streams (can be used with option --unfiltered)	
<code>--yarastrings</code>	—	Print YARA strings	
<code>--decoders</code>	DECODERS	decoders to load (separate decoders with a comma , ; @file supported)	
<code>--decoderoptions</code>	DECODEROPTIONS	options for the decoder	
<code>-k</code>	KEY	key to search in dictionaries	
<code>--key</code>	KEY	key to search in dictionaries	
<code>-j</code>	—	produce json output	
<code>--jsonoutput</code>	—	produce json output	

See also

[pdfid](#) , [pdfid.py](#) , [pdf-parser](#)

pdfid

Kit: Kali Linux **Capability:** Analyse PDFs **Version:** pdfid.py 0.2.10 **Captured:** cyberlab-aio via --help on 2026-08-02 [raw](#)

[← Capability index](#) · [Kit tool list](#)

Purpose

Triage a PDF by counting the structural keywords that indicate active content.

Synopsis

```
pdfid.py [options] [pdf-file|zip-file|url[@file]] ...
```

Common invocations

```
# from cyberlab 37-pdf-analysis
pdfid --version
# from cyberlab 37-pdf-analysis
pdfid exercise/sample.pdf
```

Options

All 31 options parsed from the captured help text; 7 reviewed with usage guidance.

Flag	Argument	What it does	When you would use it
--version	—	show program's version number and exit	
-h	—	show this help message and exit	
--help	—	show this help message and exit	
-s	—	scan the given directory	Scan a directory of PDFs rather than a single file.
--scan	—	scan the given directory	Scan a directory of PDFs rather than a single file.
-a	—	display all the names	Display all keyword counts, including zero entries.
--all	—	display all the names	Display all keyword counts, including zero entries.
-e	—	display extra data, like dates	Include extra keywords beyond the default set.
--extra	—	display extra data, like dates	Include extra keywords beyond the default set.
-f	—	force the scan of the file, even without proper %PDF header	
--force	—	force the scan of the file, even without proper %PDF header	
-d	—	disable JavaScript and auto launch	Disarm the PDF — neutralise /JS and /JavaScript. Produces a safe-to-open copy; never overwrite the original.
--disarm	—	disable JavaScript and auto launch	Disarm the PDF — neutralise /JS and /JavaScript. Produces a safe-to-open copy; never overwrite the original.
-p	PLUGINS	plugins to load (separate plugins with a comma , ; @file supported)	
--plugins	PLUGINS	plugins to load (separate plugins with a comma , ; @file supported)	
-c	—	output csv data when using plugins	CSV output for batch triage.
--csv	—	output csv data when using plugins	CSV output for batch triage.
-m	MINIMUMSCORE	minimum score for plugin results output	Only report files scoring above a minimum, for a large sweep.
--minimumscore	MINIMUMSCORE	minimum score for plugin results output	Only report files scoring above a minimum, for a large sweep.
-v	—	verbose (will also raise caught exceptions)	
--verbose	—	verbose (will also raise caught exceptions)	
-S	SELECT	selection expression	
--select	SELECT	selection expression	
-n	—	supress output for counts equal to zero	Hide zero-count keywords to shorten the output.

Flag	Argument	What it does	When you would use it
<code>--nozero</code>	—	supress output for counts equal to zero	Hide zero-count keywords to shorten the output.
<code>-o</code>	OUTPUT	output to log file	
<code>--output</code>	OUTPUT	output to log file	
<code>--pluginoptions</code>	PLUGINOPTIONS	options for the plugin	
<code>-l</code>	—	take filenames literally, no wildcard matching	
<code>--literalfilenames</code>	—	take filenames literally, no wildcard matching	
<code>--recursedir</code>	—	Recurse directories (wildcards and here files (@...) allowed)	

Gotchas

- `pdfid` **counts keywords; it does not parse**. A high `/JS` count is a reason to look, not proof of malice — and a crafted PDF can hide content from it. Follow up with `pdf-parser`.
- `/OpenAction` plus `/JS` is the classic auto-execute pair worth prioritising in triage.

See also

`pdfid.py`, `pdf-parser`, `pdf-parser.py`

pdfid.py

Kit: REMnux · Kali Linux **Capability:** Analyse PDFs **Version:** pdfid.py 0.2.10 **Captured:** cyberlab-aio via --help on 2026-08-02 [raw Docs:](#) <https://blog.didierstevens.com/programs/pdf-tools/>

[← Capability index](#) · [Kit tool list](#)

Purpose

Identify suspicious elements of the PDF file.

Synopsis

```
pdfid.py [options] [pdf-file|zip-file|url|@file] ...
```

Common invocations

```
# from cyberlab 10-malicious-documents
pdfid.py --version
# from cyberlab 10-malicious-documents
pdfid.py exercise/sample.pdf
```

Options

All 31 options parsed from the captured help text. The final column is filled in by review.

Flag	Argument	What it does	When you would use it
--version	—	show program's version number and exit	
-h	—	show this help message and exit	
--help	—	show this help message and exit	
-s	—	scan the given directory	
--scan	—	scan the given directory	
-a	—	display all the names	
--all	—	display all the names	
-e	—	display extra data, like dates	
--extra	—	display extra data, like dates	
-f	—	force the scan of the file, even without proper %PDF header	
--force	—	force the scan of the file, even without proper %PDF header	
-d	—	disable JavaScript and auto launch	
--disarm	—	disable JavaScript and auto launch	
-p	PLUGINS	plugins to load (separate plugins with a comma , ; @file supported)	
--plugins	PLUGINS	plugins to load (separate plugins with a comma , ; @file supported)	
-c	—	output csv data when using plugins	
--csv	—	output csv data when using plugins	
-m	MINIMUMSCORE	minimum score for plugin results output	
--minimumscore	MINIMUMSCORE	minimum score for plugin results output	
-v	—	verbose (will also raise caught exceptions)	
--verbose	—	verbose (will also raise caught exceptions)	
-S	SELECT	selection expression	
--select	SELECT	selection expression	
-n	—	supress output for counts equal to zero	
--nozero	—	supress output for counts equal to zero	
-o	OUTPUT	output to log file	
--output	OUTPUT	output to log file	
--pluginoptions	PLUGINOPTIONS	options for the plugin	

Flag	Argument	What it does	When you would use it
<code>-l</code>	—	take filenames literally, no wildcard matching	
<code>--literalfilenames</code>	—	take filenames literally, no wildcard matching	
<code>--recursedir</code>	—	Recurse directories (wildcards and here files (@...) allowed)	

See also

`pdfid`, `pdf-parser`, `pdf-parser.py`

PDFStreamDumper (GUI)

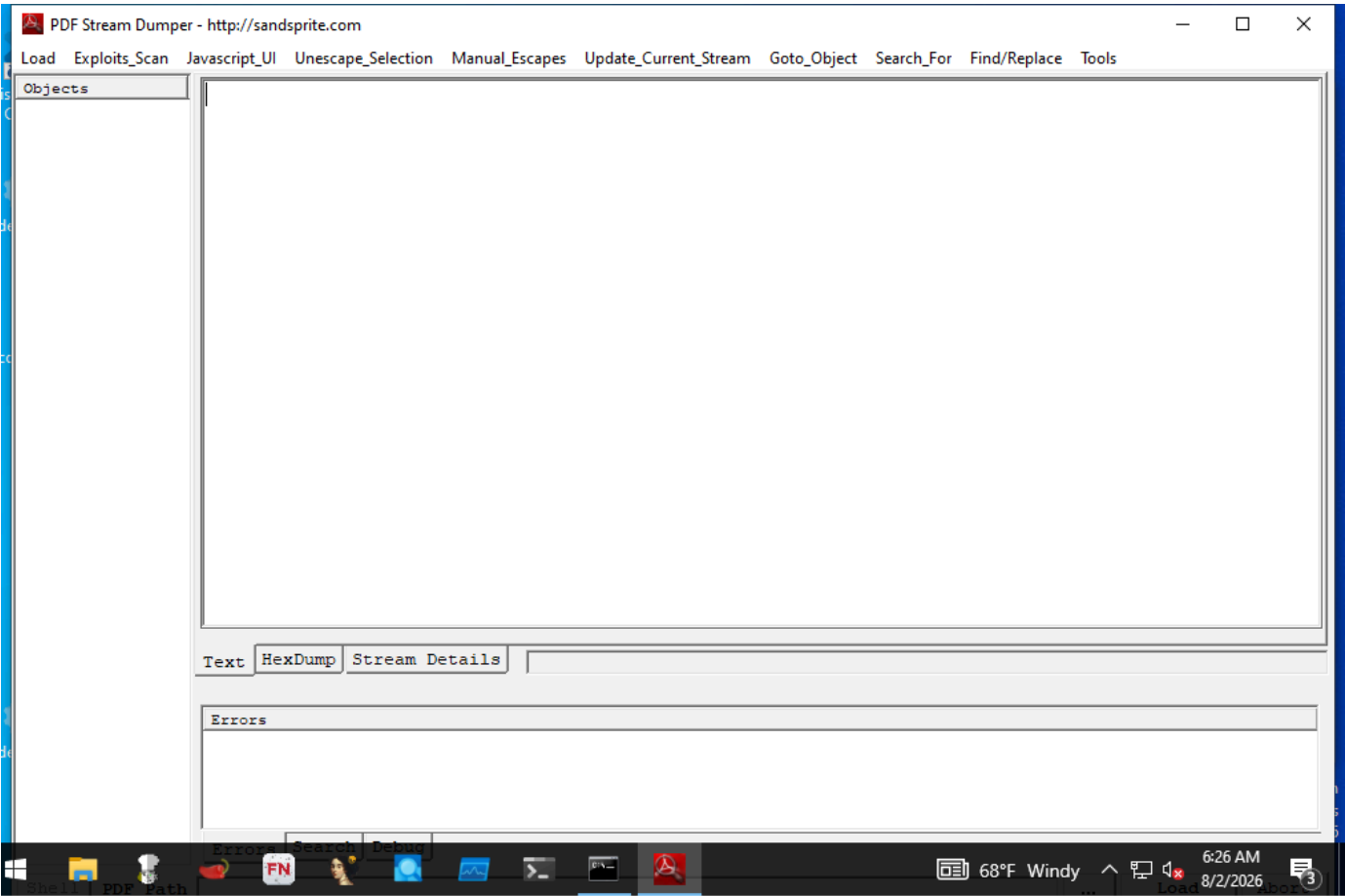
Capability: malware triage documents **Window:** ThunderRT6FormDC **Version:** — **Captured:** C:\Tools\PDFStreamDumper\PDFStreamDumper.exe on 2026-08-02 — control tree in [capture/gui/PDFStreamDumper/PDFStreamDumper.tree.txt](#)

[← Capability index](#) · [Kit tool list](#)

Purpose

Inspect and extract PDF object streams.

Window



Controls

All 16 nodes come from the capture; the 5 interactive controls are listed here.

Control	Type	AutomationId	What it does
Load	Pane	6	
...	Pane	7	
Shell	Pane	8	
Abort	Pane	9	
Progress Bar	Pane	-	

Using it

1. Open the PDF. Objects load into the list on the left.

2. Look for streams with `/JavaScript`, `/OpenAction`, `/Launch` or `/EmbeddedFile` first — those are where behaviour hides.
3. Select a stream to see its decoded content; the tool applies the filters so you do not have to.
4. Extract an embedded file rather than executing anything.

Gotchas

- A PDF that parses cleanly here can still be malicious, and one that fails to parse is often *more* interesting — malformed structure is used deliberately to defeat parsers.
- It decodes streams for you, which means it also decodes hostile content. Run it on the analysis VM with no network path, not on a workstation.
- Object numbers are not stable across a rewrite of the same document, so cite content, not object numbers, in a report.

pyxswf

Kit: REMnux **Capability:** Analyse Office documents and macros **Version:** pyxswf 0.54 **Captured:** cyberlab-aio via --help on 2026-08-02 [raw Docs:](#) <https://www.decalage.info/python/oletools>

[← Capability index](#) · [Kit tool list](#)

Purpose

Microsoft Office OLE2 compound documents.

Synopsis

```
pyxswf.py
```

Options

All 20 options parsed from the captured help text. The final column is filled in by review.

Flag	Argument	What it does	When you would use it
-h	—	show this help message and exit	
--help	—	show this help message and exit	
-x	—	Extracts the embedded SWF(s), names it MD5HASH.swf & saves it in the working dir. No addition args needed	
--extract	—	Extracts the embedded SWF(s), names it MD5HASH.swf & saves it in the working dir. No addition args needed	
-y	—	Scans the SWF(s) with yara. If the SWF(s) is compressed it will be deflated. No addition args needed	
--yara	—	Scans the SWF(s) with yara. If the SWF(s) is compressed it will be deflated. No addition args needed	
-s	—	Scans the SWF(s) for MD5 signatures. Please see func checkMD5 to define hashes. No addition args needed	
--md5scan	—	Scans the SWF(s) for MD5 signatures. Please see func checkMD5 to define hashes. No addition args needed	
-H	—	Displays the SWFs file header. No addition args needed	
--header	—	Displays the SWFs file header. No addition args needed	
-d	—	Deflates compressed SWFS(s)	
--decompress	—	Deflates compressed SWFS(s)	
-r	PATH	Will recursively scan a directory for files that contain SWFs. Must provide path in quotes	
--recdir	PATH	Will recursively scan a directory for files that contain SWFs. Must provide path in quotes	
-c	—	Compresses the SWF using Zlib	
--compress	—	Compresses the SWF using Zlib	
-o	—	Parse an OLE file (e.g. Word, Excel) to look for SWF in each stream	
--ole	—	Parse an OLE file (e.g. Word, Excel) to look for SWF in each stream	
-f	—	Parse an RTF file to look for SWF in each embedded object	
--rtf	—	Parse an RTF file to look for SWF in each embedded object	

See also

[olevba](#), [oleid](#), [oledump.py](#), [olemeta](#), [oletimes](#), [olemap](#), [oledir](#), [olebrowse](#)

rtfobj

Kit: REMnux **Capability:** Analyse RTF documents and embedded objects **Version:** rtfobj 0.60.1 **Captured:** cyberlab-aio via --help on 2026-08-02 [raw Docs:](#) <https://www.decalage.info/python/oletools>

[← Capability index](#) · [Kit tool list](#)

Purpose

Microsoft Office OLE2 compound documents.

Synopsis

```
rtfobj [options] <filename> [filename2 ...]
```

Options

All 12 options parsed from the captured help text. The final column is filled in by review.

Flag	Argument	What it does	When you would use it
-h	—	show this help message and exit	
--help	—	show this help message and exit	
-r	—	find files recursively in subdirectories.	
-z	ZIP_PASSWORD	if the file is a zip archive, open first file from it, using the provided password (requires Python 2.6+)	
--zip	ZIP_PASSWORD	if the file is a zip archive, open first file from it, using the provided password (requires Python 2.6+)	
-f	ZIP_FNAME	if the file is a zip archive, file(s) to be opened within the zip. Wildcards * and ? are supported. (default:*)	
--zipfname	ZIP_FNAME	if the file is a zip archive, file(s) to be opened within the zip. Wildcards * and ? are supported. (default:*)	
-l	LOGLEVEL	logging level debug/info/warning/error/critical (default=warning)	
--loglevel	LOGLEVEL	logging level debug/info/warning/error/critical (default=warning)	
-s	SAVE_OBJECT	Save the object corresponding to the provided number to a file, for example "-s 2". Use "-s all" to save all objects at once.	
--save	SAVE_OBJECT	Save the object corresponding to the provided number to a file, for example "-s 2". Use "-s all" to save all objects at once.	
-d	OUTPUT_DIR	use specified directory to save output files.	

See also

[oleobj](#)

xlmdeobfuscator

Kit: REMnux **Capability:** Analyse Office documents and macros; Break simple obfuscation **Captured:** cyberlab-aio via --help on 2026-08-02 [raw Docs:](#) <https://github.com/DissectMalware/XLMMacroDeobfuscator>

[← Capability index](#) · [Kit tool list](#)

Purpose

Deobfuscate XLM macros (also known as Excel 4.0 macros) from Microsoft Office files.

Synopsis

```
xlmdeobfuscator [-h] [-c FILE_PATH] [-f FILE_PATH] [-n] [-x]
[--sort-formulas] [--defined-names] [-2]
[--with-ms-excel] [-s] [-d DAY]
[--output-formula-format OUTPUT_FORMULA_FORMAT]
[--extract-formula-format EXTRACT_FORMULA_FORMAT]
[--no-indent] [--silent] [--export-json FILE_PATH]
```

Common invocations

```
# from cyberlab 10-malicious-documents
xlmdeobfuscator --help | head -n 1
# from cyberlab 10-malicious-documents
xlmdeobfuscator --file exercise/sample.xls
```

Options

All 30 options parsed from the captured help text. The final column is filled in by review.

Flag	Argument	What it does	When you would use it
-h	—	show this help message and exit	
--help	—	show this help message and exit	
-c	FILE_PATH	Specify a config file (must be a valid JSON file)	
--config-file	FILE_PATH	Specify a config file (must be a valid JSON file)	
-f	FILE_PATH	The path of a XLSM file	
--file	FILE_PATH	The path of a XLSM file	
-n	—	Disable interactive shell	
--noninteractive	—	Disable interactive shell	
-x	—	Only extract cells without any emulation	
--extract-only	—	Only extract cells without any emulation	
--sort-formulas	—	Sort extracted formulas based on their cell address (requires -x)	
--defined-names	—	Extract all defined names	
-2	—	[Deprecated] Do not use MS Excel to process XLS files	
--no-ms-excel	—	[Deprecated] Do not use MS Excel to process XLS files	
--with-ms-excel	—	Use MS Excel to process XLS files	
-s	—	Open an XLM shell before interpreting the macros in the input	
--start-with-shell	—	Open an XLM shell before interpreting the macros in the input	
-d	DAY	Specify the day of month	
--day	DAY	Specify the day of month	
--output-formula-format	OUTPUT_FORMULA_FORMAT	Specify the format for output formulas ([[CELL-ADDR]], [[INT-FORMULA]], and [[STATUS]])	
--extract-formula-format	EXTRACT_FORMULA_FORMAT	Specify the format for extracted formulas ([[CELL- ADDR]], [[CELL-FORMULA]], and [[CELL-VALUE]])	
--no-indent	—	Do not show indent before formulas	

Flag	Argument	What it does	When you would use it
<code>--silent</code>	—	Do not print output	
<code>--export-json</code>	FILE_PATH	Export the output to JSON	
<code>--start-point</code>	CELL_ADDR	Start interpretation from a specific cell address	
<code>-p</code>	PASSWORD	Password to decrypt the protected document	
<code>--password</code>	PASSWORD	Password to decrypt the protected document	
<code>-o</code>	OUTPUT_LEVEL	Set the level of details to be shown (0:all commands, 1: commands no jump 2:important commands 3:strings in important commands).	
<code>--output-level</code>	OUTPUT_LEVEL	Set the level of details to be shown (0:all commands, 1: commands no jump 2:important commands 3:strings in important commands).	
<code>--timeout</code>	N	stop emulation after N seconds (0: not interruption N>0: stop emulation after N seconds)	

See also

[olevba](#), [oleid](#), [oledump.py](#), [olemeta](#), [oletimes](#), [olemap](#), [oledir](#), [olebrowse](#)

7za

Kit: REMnux **Capability:** Detect and reverse packing; Analyse other container formats **Captured:** cyberlab-aio via --help on 2026-08-02 **raw Docs:** <https://www.7zip.org>

[← Capability index](#) · [Kit tool list](#)

Purpose

Compress and decompress files using a variety of algorithms.

Synopsis

```
7za <command> [<switches>...] <archive_name> [<file_names>...] [@listfile]
```

Options

All 1 options parsed from the captured help text. The final column is filled in by review.

Flag	Argument	What it does	When you would use it
-y	—	assume Yes on all queries	

See also

[upx](#) , [die](#) , [diec](#) , [binwalk](#) , [unzip](#) , [msoffcrypto-tool](#)

base64dump.py

Kit: REMnux **Capability:** Extract strings, including obfuscated ones; Decode, decrypt or transform encoded data **Version:** base64dump.py 0.0.30 **Captured:** cyberlab-
aio via `--help` on 2026-08-02 **raw Docs:** <https://blog.didierstevens.com/2020/07/03/update-base64dump-py-version-0-0-12/>

[← Capability index](#) · [Kit tool list](#)

Purpose

Locate and decode strings encoded in Base64 and other common encodings.

Synopsis

```
base64dump.py [options] [file]
```

Common invocations

```
# from cyberlab 09-deobfuscation
base64dump.py --version 2>&1 | head -n 1
# from cyberlab 09-deobfuscation
base64dump.py exercise/encoded_payload.bin
# from cyberlab 09-deobfuscation
base64dump.py -s 1 -d exercise/encoded_payload.bin | head -c 200
# from cyberlab 25-cyberchef-recipes
base64dump.py --version
# from cyberlab 25-cyberchef-recipes
base64dump.py encoded_payload.ps1.txt
# from cyberlab 25-cyberchef-recipes
base64dump.py -s 1 -d encoded_payload.ps1.txt | xxd | head -n 2
```

Options

All 54 options parsed from the captured help text. The final column is filled in by review.

Flag	Argument	What it does	When you would use it
<code>--version</code>	—	show program's version number and exit	
<code>-h</code>	—	show this help message and exit	
<code>--help</code>	—	show this help message and exit	
<code>-m</code>	—	Print manual	
<code>--man</code>	—	Print manual	
<code>-e</code>	ENCODING	select encoding to use, use "all" to try all encodings (default base64)	
<code>--encoding</code>	ENCODING	select encoding to use, use "all" to try all encodings (default base64)	
<code>-s</code>	SELECT	select item nr for dumping (a for all, L for largest)	
<code>--select</code>	SELECT	select item nr for dumping (a for all, L for largest)	
<code>-d</code>	—	perform dump	
<code>--dump</code>	—	perform dump	
<code>-x</code>	—	perform hex dump	
<code>--hexdump</code>	—	perform hex dump	
<code>-a</code>	—	perform ascii dump	
<code>--asciidump</code>	—	perform ascii dump	
<code>-A</code>	—	perform ascii dump with RLE	
<code>--asciidumprle</code>	—	perform ascii dump with RLE	
<code>-S</code>	—	perform strings dump	
<code>--strings</code>	—	perform strings dump	
<code>-t</code>	TRANSLATE	string translation, like utf16 or .decode("utf8")	
<code>--translate</code>	TRANSLATE	string translation, like utf16 or .decode("utf8")	

Flag	Argument	What it does	When you would use it
<code>-n</code>	NUMBER	minimum number of bytes in decoded data	
<code>--number</code>	NUMBER	minimum number of bytes in decoded data	
<code>-c</code>	CUT	cut data	
<code>--cut</code>	CUT	cut data	
<code>-w</code>	—	ignore whitespace	
<code>--ignorewhitespace</code>	—	ignore whitespace	
<code>-u</code>	—	do not repeat identical decoded data	
<code>--unique</code>	—	do not repeat identical decoded data	
<code>-z</code>	—	ignore null (zero) bytes	
<code>--ignorenullbytes</code>	—	ignore null (zero) bytes	
<code>-i</code>	IGNORE	characters to ignore	
<code>--ignore</code>	IGNORE	characters to ignore	
<code>-I</code>	IGNOREHEX	characters to ignore (hexadecimal)	
<code>--ignorehex</code>	IGNOREHEX	characters to ignore (hexadecimal)	
<code>-y</code>	YARA	YARA rule-file, @file, directory or #rule to check data (YARA search doesn't work with -s option)	
<code>--yara</code>	YARA	YARA rule-file, @file, directory or #rule to check data (YARA search doesn't work with -s option)	
<code>-D</code>	DECODERS	decoders to load (separate decoders with a comma , ; @file supported)	
<code>--decoders</code>	DECODERS	decoders to load (separate decoders with a comma , ; @file supported)	
<code>--decoderoptions</code>	DECODEROPTIONS	options for the decoder	
<code>--decoderdir</code>	DECODERDIR	directory for the decoder	
<code>--yarastrings</code>	—	Print YARA strings	
<code>-V</code>	—	verbose output with decoder errors and YARA rules	
<code>--verbose</code>	—	verbose output with decoder errors and YARA rules	
<code>--jsonoutput</code>	—	produce json output	
<code>--jsoninput</code>	—	consume json input	
<code>-T</code>	—	do head & tail	
<code>--headtail</code>	—	do head & tail	
<code>-p</code>	PREPROCESS	Python function to process encodings prior to decoding (like L4, lambda bytes: bytes[:-1], ...)	
<code>--preprocess</code>	PREPROCESS	Python function to process encodings prior to decoding (like L4, lambda bytes: bytes[:-1], ...)	
<code>-P</code>	POSTPROCESS	Python function to post-process decodings	
<code>--postprocess</code>	POSTPROCESS	Python function to post-process decodings	
<code>--sort</code>	SORT	Sort output	
<code>--stats</code>	—	provide statistics for encoded data	

See also

[floss](#), [strings](#), [numbers-to-string.py](#), [cyberchef](#), [rax2](#), [xxd](#), [openssl](#)

capa

Kit: REMnux · FLARE-VM **Capability:** Identify capabilities in a binary **Version:** capa 9.4.0 **Captured:** cyberlab-aio via --help on 2026-08-02 [raw Docs:](#) <https://github.com/mandiant/capa>

← [Capability index](#) · [Kit tool list](#)

Purpose

Identify the capabilities of a binary by matching rules against its disassembly, mapped to MITRE ATT&CK.

Synopsis

```
capa [-h] [--version] [-v] [-vv] [-d] [-q]
[--color {auto,always,never}]
[-f {auto,pe,dotnet,elf,sc32,sc64,cape,drakvuf,vmray,freeze,binexport2,binja_database}]
[-b {auto,vivisect,ida,pefile,binja,dotnet,binexport2,ghidra,freeze,cape,drakvuf,vmray}]
[--restrict-to-functions RESTRICT_TO_FUNCTIONS]
[--restrict-to-processes RESTRICT_TO_PROCESSES]
```

Common invocations

```
# from cyberlab 08-malware-static-triage
capa --version
# from cyberlab 08-malware-static-triage
capa exercise/sample.bin | head -n 40
# from cyberlab 21-yara-authoring
capa -q /bin/ls | head -n 30
# from cyberlab 12-static-re
capa exercise\sample_static.exe
# from cyberlab 12-static-re
capa -v exercise\sample_static.exe
# from cyberlab 42-floss-strings
capa .\exercise\sample.exe
# from cyberlab 42-floss-strings
capa -v .\exercise\sample.exe | Select-String -Pattern "XOR|encode|T1027"
# from cyberlab 46-cutter-windows
capa -v .\exercise\sample.exe
```

Options

All 21 options parsed from the captured help text; 8 reviewed with usage guidance.

Flag	Argument	What it does	When you would use it
-h	—	show this help message and exit	
--help	—	show this help message and exit	
--version	—	show program's version number and exit	
-v	—	enable verbose result document (no effect with --json)	Verbose: show which rules matched and where.
--verbose	—	enable verbose result document (no effect with --json)	Verbose: show which rules matched and where.
-d	—	enable debugging output on STDERR	
--debug	—	enable debugging output on STDERR	
-q	—	disable all output but errors	Quiet output for scripted use.
--quiet	—	disable all output but errors	Quiet output for scripted use.
--color	auto,always,never	enable ANSI color codes in results, default: only during interactive session	
--restrict-to-functions	RESTRICT_TO_FUNCTIONS	provide a list of comma-separated function virtual addresses to analyze (static analysis).	Analyse only the named functions — useful on a large binary you have already triaged.
--restrict-to-processes	RESTRICT_TO_PROCESSES	provide a list of comma-separated process IDs to analyze (dynamic analysis).	

Flag	Argument	What it does	When you would use it
<code>--os</code>	auto,linux,macos,windows	select sample OS: auto (detect OS automatically - default), linux, macos, windows	Force the target OS when auto-detection is wrong.
<code>-r</code>	RULES	path to rule file or directory, use embedded rules by default	Use an alternate rules directory.
<code>--rules</code>	RULES	path to rule file or directory, use embedded rules by default	Use an alternate rules directory.
<code>-s</code>	SIGNATURES	path to .sig/.pat file or directory used to identify library functions, use embedded signatures by default	Point at the library-identification signatures.
<code>--signatures</code>	SIGNATURES	path to .sig/.pat file or directory used to identify library functions, use embedded signatures by default	Point at the library-identification signatures.
<code>-t</code>	TAG	filter on rule meta field values	Only run rules carrying a given tag.
<code>--tag</code>	TAG	filter on rule meta field values	Only run rules carrying a given tag.
<code>-j</code>	—	emit JSON instead of text	JSON output, for feeding a triage pipeline.
<code>--json</code>	—	emit JSON instead of text	JSON output, for feeding a triage pipeline.

Gotchas

- capa reports *capabilities*, not verdicts. 'Can encrypt data' describes a backup tool as readily as ransomware — it narrows where to look.
- Packed samples yield almost nothing useful. Unpack first, or capa will just describe the packer stub.

See also

[floss](#)

CFF-Explorer (GUI)

Capability: malware triage static **Window:** #32770 **Version:** — **Captured:** C:\Tools\Explorer Suite\CFF Explorer.exe on 2026-08-02 — control tree in [capture/gui/CFF-Explorer/CFF-Explorer.tree.txt](#)

← [Capability index](#) · [Kit tool list](#)

Purpose

PE structure editor and viewer.

Window



Controls

All 4 nodes come from the capture; the 1 interactive controls are listed here.

Control	Type	AutomationId	What it does
Tree	Pane	1001	

Using it

1. Drag the binary onto the window, or use the file menu.
2. Work through the **Tree** on the left: headers, sections, imports, resources.
3. Read the import table first — what a binary asks the OS for is the fastest read on what it can do.
4. Compare section virtual size against raw size; a large gap is the classic packed-binary signal.
5. Use the hex editor for a targeted look rather than a general browse.

Gotchas

- It is an *editor*, not just a viewer. It will happily write to the file you opened, so work on a copy — never the evidence.
- It reports the headers as they are written. Malware edits headers freely, so a field saying DLL means the field says DLL.
- The UI is an old Win32 dialog and exposes almost nothing to automation: the whole window comes back as four unnamed panes, which is why this page has a small control table and a screenshot doing the work.

clamscan

Kit: REMnux · SIFT Workstation **Capability:** Scan with signatures for known-bad **Version:** ClamAV 1.4.3 **Captured:** cyberlab-aio via --help on 2026-08-02 [raw Docs: https://www.clamav.net](https://www.clamav.net)

← [Capability index](#) · [Kit tool list](#)

Purpose

Scan files and directories with the ClamAV signature engine.

Common invocations

```
# from cyberlab 34-clamav-scanning
clamscan --version
# from cyberlab 34-clamav-scanning
clamscan -r -i /tmp/samples
# from cyberlab 34-clamav-scanning
clamscan /tmp/eicar.com
# from cyberlab 34-clamav-scanning
clamscan exercise/sample.txt
# from cyberlab 34-clamav-scanning
clamscan --database=exercise/lab34.yar exercise/sample.txt
# from cyberlab 51-linux-triage-workflow
clamscan -r --infected --stdout be_out recovered_file.bin
# from cyberlab 51-linux-triage-workflow
clamscan --infected --stdout exercise/extract/eicar.com
```

Options

All 53 options parsed from the captured help text; 11 reviewed with usage guidance.

Flag	Argument	What it does	When you would use it
--help	—	Show this help	
-h	—	Show this help	
--version	—	Print version number	
-V	—	Print version number	
--verbose	—	Be verbose	
-v	—	Be verbose	
--archive-verbose	—	Show filenames inside scanned archives	
-a	—	Show filenames inside scanned archives	
--debug	—	Enable libclamav's debug messages	
--quiet	—	Only output error messages	Only report errors, for scripted runs.
--stdout	—	Write to stdout instead of stderr. Does not affect 'debug' messages.	
--no-summary	—	Disable summary at end of scanning	Suppress the trailing summary block when parsing output.
--infected	—	Only print infected files	Print infected files only — the flag that makes output usable.
-i	—	Only print infected files	Print infected files only — the flag that makes output usable.
--bell	—	Sound bell on virus detection	Audible alert on detection; useful during a long live scan.
--tempdir	DIRECTORY	Create temporary files in DIRECTORY	Redirect temp extraction; important when scanning large archives on a small root filesystem.
--database	FILE	-d FILE/DIR Load virus database from FILE or load all supported db files from DIR	
--fail-if-cvd-older-than	days	Return with a nonzero error code if virus database outdated.	
--log	FILE	-l FILE Save scan report to FILE	Write results to a log file for the case record.
--file-list	FILE	-f FILE Scan files from FILE	Scan exactly the paths listed in a file.
--move	DIRECTORY	Move infected files into DIRECTORY	Quarantine detections into a directory. Never point this at evidence you must preserve.

Flag	Argument	What it does	When you would use it
<code>--copy</code>	DIRECTORY	Copy infected files into DIRECTORY	Copy rather than move detections, preserving the original.
<code>--exclude</code>	REGEX	Don't scan file names matching REGEX	Skip paths matching a regex — noisy or irrelevant trees.
<code>--exclude-dir</code>	REGEX	Don't scan directories matching REGEX	
<code>--include</code>	REGEX	Only scan file names matching REGEX	Restrict the scan to paths matching a regex.
<code>--include-dir</code>	REGEX	Only scan directories matching REGEX	
<code>--bytecode-timeout</code>	N	Set bytecode timeout (in milliseconds)	
<code>--exclude-pua</code>	CAT	Skip PUA sigs of category CAT	
<code>--include-pua</code>	CAT	Load PUA sigs of category CAT	
<code>--structured-ssn-format</code>	X	SSN format (0=normal,1=stripped,2=both)	
<code>--structured-ssn-count</code>	N	Min SSN count to generate a detect	
<code>--structured-cc-count</code>	N	Min CC count to generate a detect	
<code>--structured-cc-mode</code>	X	CC mode (0=credit debit and private label, 1=credit cards only)	
<code>--nocerts</code>	—	Disable authenticode certificate chain verification in PE files	
<code>--dumpcerts</code>	—	Dump authenticode certificate chain in PE files	
<code>--max-scantime</code>	#n	Scan time longer than this will be skipped and assumed clean (milliseconds)	
<code>--max-filesize</code>	#n	Files larger than this will be skipped and assumed clean	
<code>--max-scansize</code>	#n	The maximum amount of data to scan for each container file (**)	
<code>--max-files</code>	#n	The maximum number of files to scan for each container file (**)	
<code>--max-recursion</code>	#n	Maximum archive recursion level for container file (**)	
<code>--max-dir-recursion</code>	#n	Maximum directory recursion level	
<code>--max-embeddedpe</code>	#n	Maximum size file to check for embedded PE	
<code>--max-htmlnormalize</code>	#n	Maximum size of HTML file to normalize	
<code>--max-htmlnotags</code>	#n	Maximum size of normalized HTML file to scan	
<code>--max-scriptnormalize</code>	#n	Maximum size of script file to normalize	
<code>--max-ziptypercg</code>	#n	Maximum size zip to type reanalyze	
<code>--max-partitions</code>	#n	Maximum number of partitions in disk image to be scanned	
<code>--max-iconspe</code>	#n	Maximum number of icons in PE file to be scanned	
<code>--max-rechwp3</code>	#n	Maximum recursive calls to HWP3 parsing function	
<code>--pcre-match-limit</code>	#n	Maximum calls to the PCRE match function.	
<code>--pcre-recmatch-limit</code>	#n	Maximum recursive calls to the PCRE match function.	
<code>--pcre-max-filesize</code>	#n	Maximum size file to perform PCRE subsig matching.	
<code>--disable-cache</code>	—	Disable caching and cache checks for hash sums of scanned files.	

Gotchas

- Without `-i` the output lists every clean file too, which buries the hits in a large tree.
- `--move` and `--remove` MUTATE evidence. On an investigation, scan a copy or use `--copy` — a quarantine action on original evidence is not defensible.
- Signatures must be current for the result to mean anything; a stale database yields a comfortable and worthless clean result.

See also

`yara`, `yarac`, `freshclam`, `sigtool`

die

Kit: FLARE-VM **Capability:** Identify what a file actually is; Detect and reverse packing **Version:** Detect It Easy v3.10 **Captured:** cyberlab-aio via --help on 2026-08-02 raw

[← Capability index](#) · [Kit tool list](#)

Purpose

qt.qpa.xcb: could not connect to display

Options

No option definitions could be parsed from this tool's help output. It may be subcommand-driven or have no flags; needs manual review.

See also

[file](#), [diec](#), [upx](#), [binwalk](#), [7za](#), [unzip](#)

Detect It Easy (GUI)

Kit: REMnux · FLARE-VM **Capability:** Identify a file's format, compiler and packer **Version:** Detect It Easy v3.10 (x86_64) **CLI counterpart:** `diec` **Captured:** `C:\Tools\die\die.exe` on FLARE-VM, 2026-08-02 — control tree in `capture/gui/die/die.tree.txt`

Worked example of the GUI format. Every control below was read from the application's accessibility tree, not from a screenshot or from memory. See `docs/GUI-FORMAT.md`.

Purpose

Identify what a binary actually is — its format, architecture, compiler, linker and any packer or protector — before committing to a deeper analysis path.

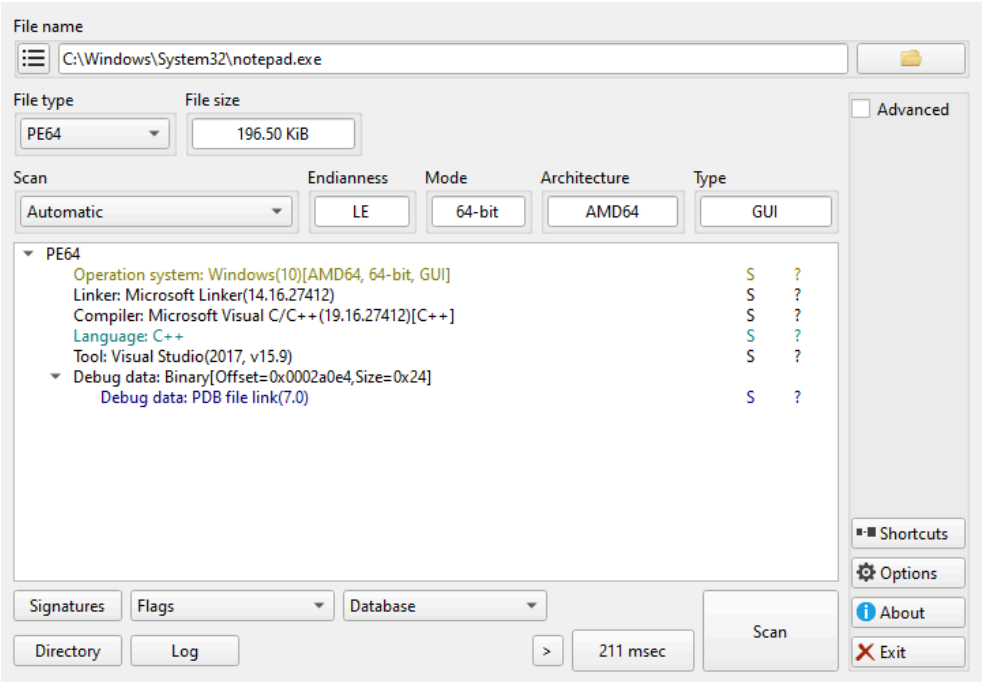
Use the GUI or the CLI?

`diec` does the same detection and is the better choice for scripting, batch work and anything reproducible. The GUI earns its place for three things the CLI cannot do:

- browsing the **Signatures** that produced a verdict, when you need to know *why* something was flagged
- switching detection engines and re-scanning interactively to compare results
- the **Advanced** panes — entropy, section and hex views — which are exploratory by nature

Anything you would put in a report or a pipeline belongs in `diec`.

Window map



Captured against `C:\Windows\System32\notepad.exe` — a benign, signed, universally present PE — so the panes show real output rather than an empty window. The result tree is the point of the tool: operating system, linker, compiler, build tool and debug data, each with the signature that produced it.

One window, `GuiMainWindow`, in three regions:

Region	Automation container	What it is for
File name	<code>groupBoxFileName</code>	Choosing the target file
Formats	<code>widgetFormats</code>	The detection result, and the scan controls
Extra	<code>groupBoxExtra</code>	Advanced views, options and about

Controls

All 72 nodes come from the capture; the leaf controls are listed here.

Control	Type	AutomationId	What it does
File name	Edit	lineEditFileName	Path of the file being examined
...	Button	toolButtonOpenFile	Open the file chooser
>	Button	toolButtonRecentFiles	Recently opened files
File type	ComboBox	comboBoxFileType	Detected container format; selectable when a file has more than one interpretation
File size	Edit	lineEditFileSize	Size of the target, read-only
Scan	ComboBox	comboBoxScanEngine	Which detection engine runs — see below
Endianness	Edit	lineEditEndianness	Byte order, read-only
Mode	Edit	lineEditMode	32- or 64-bit, read-only
Architecture	Edit	lineEditArch	Target CPU, read-only
Type	Edit	lineEditType	Executable, DLL, driver, read-only
Signatures	Button	pushButtonDieSignatures	Open the signature browser — the "why was this flagged" view
Directory	Button	pushButtonDieScanDirectory	Scan a whole directory instead of one file
Log	Button	pushButtonDieLog	Scan log
Scan	Button	pushButtonDieScanStart	Run the scan
211 msec	Button	toolButtonElapsedTime	Elapsed time of the last scan — the label is the timing, so it changes with every run
Advanced	CheckBox	checkBoxAdvanced	Reveal entropy, sections and hex views
Shortcuts	Button	toolButtonShortcuts	Keyboard shortcut list
Options	Button	toolButtonOptions	Settings
About	Button	toolButtonAbout	Version and credits
Exit	Button	toolButtonExit	Close

Two combo boxes carry no label in the tree and sit beside **Signatures**: `comboBoxFlags` and `comboBoxDatabases`, which select the signature flags and the signature database in use.

Scan engine values

Enumerated from `comboBoxScanEngine`, so this list is the actual set of choices in this build:

Value	When to use it
Automatic	The default; picks an engine from the detected format
Detect It Easy (DiE)	DiE's own signature set — the packer and compiler detection it is known for
Nauz File Detector (NFD)	A second opinion from a different signature set, when DiE returns nothing
Yara rules	Run YARA rules instead of signatures, when you have rules of your own

Identify an unknown binary

1. Select the file with ..., or drag it onto **File name**.
2. Leave **Scan** on **Automatic** for a first pass.
3. Press **Scan**.
4. Read **Architecture**, **Mode** and **Type** — these are structural facts, not guesses.
5. Press **Signatures** to see which signature produced the verdict.
6. If nothing is detected, switch **Scan** to **Nauz File Detector (NFD)** and scan again.
7. Tick **Advanced** and check entropy before concluding a file is unpacked.

Gotchas

- A packer verdict is a **signature match, not proof**. Packers are routinely misidentified, and a custom packer will not match anything at all. High entropy with no signature is the interesting case, not the boring one.
- "Nothing detected" is a statement about the signature set, not about the file. Try the other engine before concluding anything.
- The read-only fields are parsed from headers, which malware edits freely. **Type** saying `DLL` means the header says DLL.
- The GUI scans one file at a time unless you use **Directory**. For a corpus, use `diec` — this window is for looking at one sample closely.

diec

Kit: REMnux · FLARE-VM **Capability:** Identify what a file actually is; Detect and reverse packing **Version:** die 3.10 **Captured:** cyberlab-aio via --help on 2026-08-02
raw Docs: <https://github.com/horsicq/Detect-It-Easy>

← [Capability index](#) · [Kit tool list](#)

Purpose

Identify a file's format, compiler, linker and packer from the command line — the scriptable half of Detect It Easy.

Synopsis

```
diec [options] target
```

Common invocations

```
# from cyberlab 08-malware-static-triage
diec --version
# from cyberlab 08-malware-static-triage
diec -j exercise/sample.bin | head -n 40
# from cyberlab 32-remnux-static-triage
diec sample.exe
# from cyberlab 32-remnux-static-triage
diec exercise/sample.exe | grep -iE "compiler|packer|linker"
```

Options

All 47 options parsed from the captured help text; 12 reviewed with usage guidance.

Flag	Argument	What it does	When you would use it
-h	—	Displays help on commandline options.	
--help	—	Displays help on commandline options.	
--help-all	—	Displays help including Qt specific options.	
-v	—	Displays version information.	
--version	—	Displays version information.	
-r	—	Recursive scan.	Recurse a directory. This is the flag that makes diec the right tool for a corpus, where the GUI handles one sample.
--recursivescan	—	Recursive scan.	Recurse a directory. This is the flag that makes diec the right tool for a corpus, where the GUI handles one sample.
-d	—	Deep scan.	Deep scan: look past the entry point for signatures a quick pass misses. Slower, and worth it on anything suspicious.
--deepscan	—	Deep scan.	Deep scan: look past the entry point for signatures a quick pass misses. Slower, and worth it on anything suspicious.
-u	—	Heuristic scan.	Heuristic scan, for packers with no exact signature. Raises false positives, so treat a heuristic-only hit as a lead.
--heuristicscan	—	Heuristic scan.	Heuristic scan, for packers with no exact signature. Raises false positives, so treat a heuristic-only hit as a lead.
-b	—	Verbose.	Verbose output, including which signature matched.
--verbose	—	Verbose.	Verbose output, including which signature matched.
-g	—	Aggressive scan.	Aggressive scan — the most thorough and the noisiest.
--aggressivescan	—	Aggressive scan.	Aggressive scan — the most thorough and the noisiest.
-a	—	Scan all types.	Scan every type rather than stopping at the detected format.
--alltypes	—	Scan all types.	Scan every type rather than stopping at the detected format.
-l	—	Profiling signatures.	Profile signature performance — for tuning a slow scan, not for analysis.
--profiling	—	Profiling signatures.	Profile signature performance — for tuning a slow scan, not for analysis.
-U	—	Hide unknown.	Hide unknown results, to cut noise across a large scan.

Flag	Argument	What it does	When you would use it
<code>--hideunknown</code>	—	Hide unknown.	Hide unknown results, to cut noise across a large scan.
<code>-e</code>	—	Show entropy.	Show entropy. High entropy with no packer signature is the interesting case: something is compressed or encrypted and nothing recognises it.
<code>--entropy</code>	—	Show entropy.	Show entropy. High entropy with no packer signature is the interesting case: something is compressed or encrypted and nothing recognises it.
<code>-i</code>	—	Show file info.	Show file info — size, format, architecture.
<code>--info</code>	—	Show file info.	Show file info — size, format, architecture.
<code>-S</code>	method	Special file info for . For example -S "Hash" or -S "Hash#MD5".	Ask for one specific piece of info, e.g. <code>-S Hash#MD5</code> . The form to use when scripting rather than reading.
<code>--special</code>	method	Special file info for . For example -S "Hash" or -S "Hash#MD5".	Ask for one specific piece of info, e.g. <code>-S Hash#MD5</code> . The form to use when scripting rather than reading.
<code>-x</code>	—	Result as XML.	
<code>--xml</code>	—	Result as XML.	
<code>-j</code>	—	Result as JSON.	JSON output, for a pipeline.
<code>--json</code>	—	Result as JSON.	JSON output, for a pipeline.
<code>-c</code>	—	Result as CSV.	
<code>--csv</code>	—	Result as CSV.	
<code>-t</code>	—	Result as TSV.	
<code>--tsv</code>	—	Result as TSV.	
<code>-p</code>	—	Result as Plain Text.	
<code>--plaintext</code>	—	Result as Plain Text.	
<code>-D</code>	path	Set database.	
<code>--database</code>	path	Set database.	
<code>-E</code>	path	Set extra database.	
<code>--extradatabase</code>	path	Set extra database.	
<code>-C</code>	path	Set custom database.	
<code>--customdatabase</code>	path	Set custom database.	
<code>-s</code>	—	Show database.	
<code>--showdatabase</code>	—	Show database.	
<code>-m</code>	—	Show all special methods for the file.	
<code>--showmethods</code>	—	Show all special methods for the file.	

Gotchas

- `diec` and the GUI share a signature set, so they agree by construction. Use this for corpora and reports; use [the GUI](#) when you need to browse *why* something matched.
- A packer name is a signature match, not proof. Custom and modified packers match nothing, so silence is not the same as clean.
- Deep and aggressive scans cost real time on a large directory. Sample before committing to a full corpus run.

See also

[file](#), [die](#), [upx](#), [binwalk](#), [7za](#), [unzip](#)

floss

Kit: REMnux · FLARE-VM **Capability:** Identify capabilities in a binary; Extract strings, including obfuscated ones; Break simple obfuscation **Version:** floss 3.1.1 **Captured:** cyberlab-aio via --help on 2026-08-02 **raw Docs:** <https://github.com/mandiant/flare-floss>

← [Capability index](#) · [Kit tool list](#)

Purpose

Extract strings from a binary, including obfuscated strings that `strings` cannot see.

Synopsis

```
floss [-h] [-H] [-n MIN_LENGTH]
[--no {static,stack,tight,decoded} [{static,stack,tight,decoded} ...]]
[--only {static,stack,tight,decoded} [{static,stack,tight,decoded} ...]]
[-j] [-v] [-d] [-q] [--color {auto,always,never}]
sample
```

Common invocations

```
# from cyberlab 08-malware-static-triage
floss --version
# from cyberlab 08-malware-static-triage
floss --no-color exercise/sample.bin | head -n 60
# from cyberlab 12-static-re
floss --no-static exercise\sample_static.exe
# from cyberlab 12-static-re
floss exercise\sample_static.exe
# from cyberlab 42-floss-strings
floss --help
# from cyberlab 42-floss-strings
floss .\exercise\sample.exe
# from cyberlab 42-floss-strings
floss --json --only decoded stack .\exercise\sample.exe > .\exercise\floss.json
# from cyberlab 42-floss-strings
floss --only static .\exercise\sample.exe | Select-String "HELLO"
```

Options

All 14 options parsed from the captured help text; 5 reviewed with usage guidance.

Flag	Argument	What it does	When you would use it
<code>-h</code>	—	show this help message and exit	
<code>--help</code>	—	show this help message and exit	
<code>-H</code>	—	show advanced options and exit	
<code>-n</code>	MIN_LENGTH	minimum string length	Minimum string length; raise it to cut noise on large binaries.
<code>--minimum-length</code>	MIN_LENGTH	minimum string length	Minimum string length; raise it to cut noise on large binaries.
<code>-j</code>	—	emit JSON instead of text	Emit JSON, for feeding results into other tooling.
<code>--json</code>	—	emit JSON instead of text	Emit JSON, for feeding results into other tooling.
<code>-v</code>	—	enable verbose results, e.g. including function offsets (does not affect JSON output)	Verbose progress; FLOSS emulation can be slow and silent otherwise.
<code>--verbose</code>	—	enable verbose results, e.g. including function offsets (does not affect JSON output)	Verbose progress; FLOSS emulation can be slow and silent otherwise.
<code>-d</code>	—	enable debugging output on STDERR, specify multiple times to increase verbosity	
<code>--debug</code>	—	enable debugging output on STDERR, specify multiple times to increase verbosity	
<code>-q</code>	—	disable all status output on STDOUT except fatal errors	Quiet mode — just the strings, for piping.

Flag	Argument	What it does	When you would use it
<code>--quiet</code>	—	disable all status output on STDOUT except fatal errors	Quiet mode — just the strings, for piping.
<code>--color</code>	auto,always,never	enable ANSI color codes in results, default: only during interactive session	Control colour output when redirecting to a file.

Gotchas

- FLOSS emulates code to recover decoded strings, so it is far slower than `strings` and can take minutes on a large sample. Budget for it.
- It is built for Windows PE files. Pointing it at an ELF or a script gives you little more than plain `strings` would.

See also

`capa`, `strings`, `base64dump.py`, `numbers-to-string.py`, `xortool`, `xlndeobfuscator`

freshclam

Kit: REMnux · SIFT Workstation **Capability:** Scan with signatures for known-bad **Version:** ClamAV 1.4.3 **Captured:** cyberlab-aio via --help on 2026-08-02 [raw Docs: https://www.clamav.net](https://www.clamav.net)

[← Capability index](#) · [Kit tool list](#)

Purpose

Scan files for malware signatures.

Common invocations

```
# from cyberlab 34-clamav-scanning
freshclam --version
# from cyberlab 34-clamav-scanning
sudo freshclam
```

Options

All 27 options parsed from the captured help text. The final column is filled in by review.

Flag	Argument	What it does	When you would use it
--help	—	Show this help	
-h	—	Show this help	
--version	—	Print version number and exit	
-V	—	Print version number and exit	
--verbose	—	Be verbose	
-v	—	Be verbose	
--debug	—	Enable debug messages	
--quiet	—	Only output error messages	
--no-warnings	—	Don't print and log warnings	
--stdout	—	Write to stdout instead of stderr. Does not affect 'debug' messages.	
--show-progress	—	Show download progress percentage	
--config-file	FILE	Read configuration from FILE.	
--log	FILE	-l FILE Log into FILE	
--daemon	—	Run in daemon mode	
-d	—	Run in daemon mode	
--pid	FILE	-p FILE Write the daemon's pid to FILE	
--foreground	—	Don't fork into background (for use in daemon mode).	
-F	—	Don't fork into background (for use in daemon mode).	
--user	USER	-u USER Run as USER	
--no-dns	—	Force old non-DNS verification method	
--checks	#n	-c #n Number of checks per day, 1 <= n <= 50	
--datadir	DIRECTORY	Download new databases into DIRECTORY NOTE: DIRECTORY must already exist, be an absolute path, and be writeable by freshclam and readable by clamd/clamscan. --daemon-notify[=/path/clamd.conf] Send REL	
--local-address	IP	-a IP Bind to IP for HTTP downloads	
--on-update-execute	COMMAND	Execute COMMAND after successful update. Use EXIT_1 to return 1 after successful database update.	
--on-error-execute	COMMAND	Execute COMMAND if errors occurred	
--on-outdated-execute	COMMAND	Execute COMMAND when software is outdated	
--update-db	DBNAME	Only update database DBNAME	

See also

[yara](#), [yarak](#), [clamscan](#), [sigtool](#)

numbers-to-string.py

Kit: REMnux **Capability:** Extract strings, including obfuscated ones; Decode, decrypt or transform encoded data **Version:** numbers-to-string.py 0.0.11 **Captured:** cyberlab-aio via --help on 2026-08-02 **raw Docs:** <https://blog.didierstevens.com/2020/12/12/update-numbers-to-string-py-version-0-0-11/>

[← Capability index](#) · [Kit tool list](#)

Purpose

Translate number sequences into ASCII characters.

Synopsis

```
numbers-to-string.py [options] [expression [[@]file ...]]
```

Options

All 31 options parsed from the captured help text. The final column is filled in by review.

Flag	Argument	What it does	When you would use it
--version	—	show program's version number and exit	
-h	—	show this help message and exit	
--help	—	show this help message and exit	
-m	—	Print manual	
--man	—	Print manual	
-o	OUTPUT	Output to file	
--output	OUTPUT	Output to file	
-e	—	Generate error when error occurs in Python expression	
--error	—	Generate error when error occurs in Python expression	
-i	—	Ignore numbers greater than 255	
--ignore	—	Ignore numbers greater than 255	
-n	NUMBER	Minimum number of numbers (3 by default)	
--number	NUMBER	Minimum number of numbers (3 by default)	
-v	—	Verbose	
--verbose	—	Verbose	
-j	—	Join output	
--join	—	Join output	
-S	—	Generate statistics	
--statistics	—	Generate statistics	
-t	TABLE	Translation table	
--table	TABLE	Translation table	
-T	BEGINTABLE	Begin translation table	
--begintable	BEGINTABLE	Begin translation table	
-b	—	Produce binary output	
--binary	—	Produce binary output	
-l	LINE	Select line with given number	
--line	LINE	Select line with given number	
--grep	GREP	Grep expression	
--grepoptions	GREPOPTIONS	Grep options	
--begin	BEGIN	Begin substring	
--end	END	End substring	

See also

[floss](#), [strings](#), [base64dump.py](#), [cyberchef](#), [rax2](#), [xxd](#), [openssl](#)

objdump

Kit: REMnux **Capability:** Inspect PE / ELF structure; Disassemble and explore a binary **Version:** GNU objdump (GNU Binutils for Debian) 2.40 **Captured:** cyberlab-aio
via `--help` on 2026-08-02 **raw Docs:** <https://en.wikipedia.org/wiki/Objdump>

[← Capability index](#) · [Kit tool list](#)

Purpose

Disassemble and dump the contents of an object file or executable — sections, symbols, relocations and code.

Synopsis

```
objdump <option(s)> <file(s)>
```

Options

All 78 options parsed from the captured help text; 19 reviewed with usage guidance.

Flag	Argument	What it does	When you would use it
<code>-a</code>	—	Display archive header information	
<code>--archive-headers</code>	—	Display archive header information	
<code>-f</code>	—	Display the contents of the overall file header	The file header alone: format, architecture, entry point.
<code>--file-headers</code>	—	Display the contents of the overall file header	The file header alone: format, architecture, entry point.
<code>-p</code>	—	Display object format specific file header contents	Format-specific private headers. On a PE this is where the import table and data directories live.
<code>--private-headers</code>	—	Display object format specific file header contents	Format-specific private headers. On a PE this is where the import table and data directories live.
<code>-x</code>	—	Display the contents of all headers	All headers at once — the fastest orientation on an unknown object.
<code>--all-headers</code>	—	Display the contents of all headers	All headers at once — the fastest orientation on an unknown object.
<code>-d</code>	—	Display assembler contents of executable sections	Disassemble the executable sections. The common case.
<code>--disassemble</code>	—	Display assembler contents of executable sections	Disassemble the executable sections. The common case.
<code>-D</code>	—	Display assembler contents of all sections	Disassemble everything, including data. Use it when code has been hidden in a section not marked executable.
<code>--disassemble-all</code>	—	Display assembler contents of all sections	Disassemble everything, including data. Use it when code has been hidden in a section not marked executable.
<code>-S</code>	—	Intermix source code with disassembly --source-comment[=] Prefix lines of source code with	Interleave source with disassembly. Only useful when debug info survived, which for malware it has not.
<code>--source</code>	—	Intermix source code with disassembly --source-comment[=] Prefix lines of source code with	Interleave source with disassembly. Only useful when debug info survived, which for malware it has not.
<code>-s</code>	—	Display the full contents of all sections requested	Full contents of each section as a hex dump.
<code>--full-contents</code>	—	Display the full contents of all sections requested	Full contents of each section as a hex dump.
<code>-g</code>	—	Display debug information in object file	Debug information, if present.
<code>--debugging</code>	—	Display debug information in object file	Debug information, if present.
<code>-e</code>	—	Display debug information using ctags style	
<code>--debugging-tags</code>	—	Display debug information using ctags style	
<code>-G</code>	—	Display (in raw form) any STABS info in the file	
<code>--stabs</code>	—	Display (in raw form) any STABS info in the file	
<code>-L</code>	—	Display the contents of non-debug sections in separate debuginfo files. (Implies -WK)	
<code>--process-links</code>	—	Display the contents of non-debug sections in separate debuginfo files. (Implies -WK)	
<code>--ctf</code>	SECTION (optional)	Display CTF info from SECTION, (default '.ctf')	
<code>--sframe</code>	SECTION (optional)	Display SFrame info from SECTION, (default '.sframe')	
<code>-t</code>	—	Display the contents of the symbol table(s)	The symbol table, when the binary is not stripped.

Flag	Argument	What it does	When you would use it
<code>--syms</code>	—	Display the contents of the symbol table(s)	The symbol table, when the binary is not stripped.
<code>-T</code>	—	Display the contents of the dynamic symbol table	Dynamic symbols — what a shared object exports and imports.
<code>--dynamic-syms</code>	—	Display the contents of the dynamic symbol table	Dynamic symbols — what a shared object exports and imports.
<code>-r</code>	—	Display the relocation entries in the file	Relocations.
<code>--reloc</code>	—	Display the relocation entries in the file	Relocations.
<code>-R</code>	—	Display the dynamic relocation entries in the file	Dynamic relocations, which reveal the PLT/GOT layout.
<code>--dynamic-reloc</code>	—	Display the dynamic relocation entries in the file	Dynamic relocations, which reveal the PLT/GOT layout.
<code>-v</code>	—	Display this program's version number	
<code>--version</code>	—	Display this program's version number	
<code>-i</code>	—	List object formats and architectures supported	
<code>--info</code>	—	List object formats and architectures supported	
<code>-H</code>	—	Display this information	
<code>--help</code>	—	Display this information	
<code>-b</code>	BFDNAME	Specify the target object format as BFDNAME	Force the file format when detection fails.
<code>--target</code>	BFDNAME	Specify the target object format as BFDNAME	Force the file format when detection fails.
<code>-m</code>	MACHINE	Specify the target architecture as MACHINE	Force the architecture, needed for raw shellcode with no container to describe it.
<code>--architecture</code>	MACHINE	Specify the target architecture as MACHINE	Force the architecture, needed for raw shellcode with no container to describe it.
<code>-j</code>	NAME	Only display information for section NAME	Restrict the operation to one section, so a <code>-s</code> or <code>-d</code> on a large binary stays readable.
<code>--section</code>	NAME	Only display information for section NAME	Restrict the operation to one section, so a <code>-s</code> or <code>-d</code> on a large binary stays readable.
<code>--file-start-context</code>	—	Include context from start of file (with -S)	
<code>-I</code>	DIR	Add DIR to search list for source files	
<code>--include</code>	DIR	Add DIR to search list for source files	
<code>-l</code>	—	Include line numbers and filenames in output	Annotate with file and line numbers from debug info.
<code>--line-numbers</code>	—	Include line numbers and filenames in output	Annotate with file and line numbers from debug info.
<code>-F</code>	—	Include file offsets when displaying information	
<code>--file-offsets</code>	—	Include file offsets when displaying information	
<code>-C</code>	STYLE (optional)	Decode mangled/processed symbol names STYLE can be "none", "auto", "gnu-v3", "java", "gnat", "dlang", "rust"	Demangle C++ symbols into something readable.
<code>--demangle</code>	STYLE (optional)	Decode mangled/processed symbol names STYLE can be "none", "auto", "gnu-v3", "java", "gnat", "dlang", "rust"	Demangle C++ symbols into something readable.
<code>--recurse-limit</code>	—	Enable a limit on recursion whilst demangling (default)	
<code>--no-recurse-limit</code>	—	Disable a limit on recursion whilst demangling	
<code>-w</code>	—	Format output for more than 80 columns	Wide output that does not truncate long symbol names.
<code>--wide</code>	—	Format output for more than 80 columns	Wide output that does not truncate long symbol names.
<code>-z</code>	—	Do not skip blocks of zeroes when disassembling	Do not collapse blocks of zero bytes, when the padding itself matters.
<code>--disassemble-zeroes</code>	—	Do not skip blocks of zeroes when disassembling	Do not collapse blocks of zero bytes, when the padding itself matters.
<code>--start-address</code>	ADDR	Only process data whose address is >= ADDR	
<code>--stop-address</code>	ADDR	Only process data whose address is < ADDR	
<code>--no-addresses</code>	—	Do not print address alongside disassembly	
<code>--prefix-addresses</code>	—	Print complete address alongside disassembly	
<code>--insn-width</code>	WIDTH	Display WIDTH bytes on a single line for -d	
<code>--adjust-vma</code>	OFFSET	Add OFFSET to all displayed section addresses	
<code>--show-all-symbols</code>	—	When disassembling, display all symbols at a given address	
<code>--special-syms</code>	—	Include special symbols in symbol dumps	
<code>--inlines</code>	—	Print all inlines for source line (with -l)	
<code>--prefix</code>	PREFIX	Add PREFIX to absolute paths for -S	

Flag	Argument	What it does	When you would use it
<code>--prefix-strip</code>	LEVEL	Strip initial directory names for -S	
<code>--dwarf-depth</code>	N	Do not display DIEs at depth N or greater	
<code>--dwarf-start</code>	N	Display DIEs starting at offset N	
<code>--dwarf-check</code>	—	Make additional dwarf consistency checks.	
<code>--ctf-parent</code>	NAME	Use CTF archive member NAME as the CTF parent	
<code>--visualize-jumps</code>	—	Visualize jumps by drawing ASCII art lines	
<code>--disassembler-color</code>	off	Disable disassembler color output. (default)	

Gotchas

- Disassembly starts where the section says code starts. Packed and self-modifying binaries decrypt themselves at runtime, so a static pass shows the unpacking stub and nothing about the payload.
- Default output is AT&T syntax. Most Windows malware documentation is Intel, so `-M intel` avoids constant mental translation.
- `-d` skips sections not marked executable, which is precisely where code gets hidden. `-D` is slower and sees them.

See also

`rabin2`, `readelf`, `readelf.py`, `r2`, `rasm2`, `vivbin`, `vdbbin`

PE-Detective (GUI)

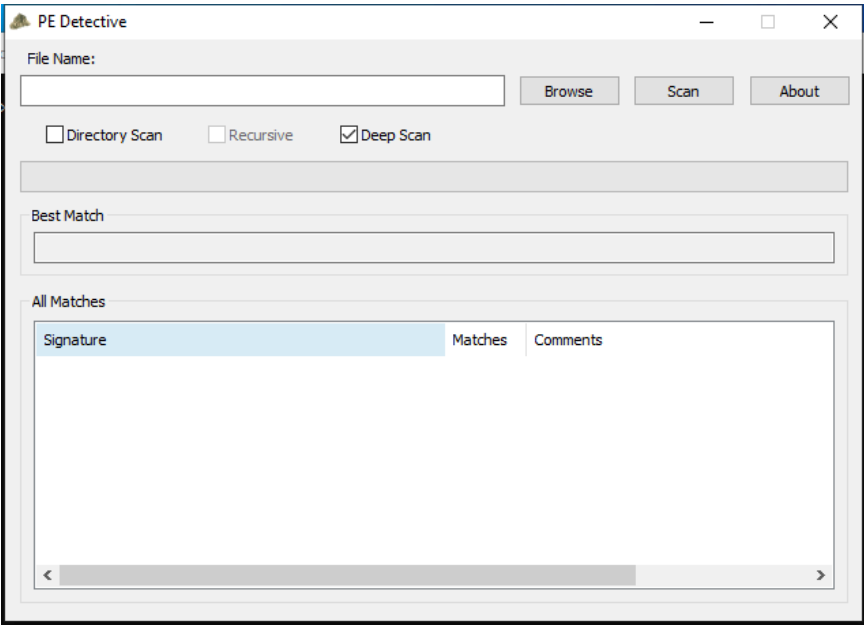
Capability: malware triage static **Window:** #32770 **Version:** — **Captured:** C:\Tools\Explorer Suite\PE Detective.exe on 2026-08-02 — control tree in [capture/gui/PE-Detective/PE-Detective.tree.txt](#)

[← Capability index](#) · [Kit tool list](#)

Purpose

PE packer and compiler signature scanner.

Window



Controls

All 15 nodes come from the capture; the 9 interactive controls are listed here.

Control	Type	AutomationId	What it does
File Name:	Pane	—	
Browse	Pane	1001	
Scan	Pane	1006	
About	Pane	1009	
Directory Scan	Pane	1012	
Recursive	Pane	1013	
Deep Scan	Pane	1002	
Best Match	Pane	1015	
All Matches	Pane	1014	

Using it

1. Press **Browse** and choose the binary.
2. Tick **Deep Scan** — the default pass only checks the entry point, and packers that relocate it are missed without this.
3. Choose **All Matches** rather than **Best Match** when triaging; a single best match hides the disagreement that tells you a signature is shaky.
4. Press **Scan**.
5. For a corpus, tick **Directory Scan** and **Recursive** instead of selecting one file.

Gotchas

- It reports signature matches, not facts. A packer name is a hypothesis; a custom or modified packer matches nothing at all, so silence never means clean.
- **Best Match** is the default and is the wrong setting for triage. Two signatures disagreeing is information.
- It shares a signature format with Signature Explorer, so an unrecognised sample can be turned into a new signature there.

rabin2

Kit: REMnux · Kali Linux · SIFT Workstation **Capability:** Inspect PE / ELF structure; Disassemble and explore a binary **Version:** rabin2 6.1.9 **Captured:** cyberlab-aio via `-help` on 2026-08-02 **raw Docs:** <https://www.radare.org/n/radare2.html>

[← Capability index](#) · [Kit tool list](#)

Purpose

Report the structure of a binary — headers, sections, imports, exports, strings, symbols — for any format radare2 understands.

Synopsis

```
rabin2 [-AcdeEghHiIjJlLMqrRsSuvVxzZ] [-@ at] [-a arch] [-b bits] [-B addr]
[-C F:C:D] [-f str] [-m addr] [-n str] [-N m:M] [-P[-P] pdb]
[-o str] [-O help] [-k query] [-D lang mangledsymbol] file
```

Common invocations

```
# from cyberlab 35-radare2-intro
rabin2 (shipped with radare2) lists strings from data sections; r2 disassembles main
# from cyberlab 35-radare2-intro
rabin2 -z exercise/hello
# from cyberlab 35-radare2-intro
rabin2 -z exercise/hello | grep r2rules
```

Options

All 47 options parsed from the captured help text; 20 reviewed with usage guidance.

Flag	Argument	What it does	When you would use it
<code>-@</code>	addr	show section, symbol or import at addr	
<code>-A</code>	—	list sub-binaries and their arch-bits pairs	List all sub-binaries in a fat/universal file.
<code>-a</code>	arch	set arch (x86, arm, .. or _)	Force the architecture when detection is wrong.
<code>-b</code>	bits	set bits (32, 64 ...)	Force the bit width, 32 or 64.
<code>-B</code>	addr	override base address (pie bins)	
<code>-c</code>	—	list classes	Class information, for Objective-C, Java and .NET binaries.
<code>-C</code>	fmt:C:D	create [elf,mach0,pe] with Code and Data hexpairs (see -a)	
<code>-d</code>	—	show debug/dwarf information	
<code>-e</code>	—	program entrypoint	Entrypoint address.
<code>-E</code>	—	globally exportable symbols	Exports — the entry points a DLL offers.
<code>-f</code>	str	select sub-bin named str	
<code>-F</code>	binfmt	force to use that bin plugin (ignore header check)	
<code>-g</code>	—	same as -SMZIHVResizcld -SS -SSS -ee (show all info)	
<code>-G</code>	addr	load address . address to header	

Flag	Argument	What it does	When you would use it
-h	—	this help message	
-H	—	header fields	Headers in full detail.
-i	—	imports (symbols imported from libraries)	Imports. What a binary asks the OS for is the fastest read on what it can do.
-I	—	binary info	The header summary: format, architecture, bits, endianness, stripped, and whether NX, PIE and canaries are on. The usual first command on an unknown binary.
-j	—	output in json	JSON output, for a pipeline.
-k	sdb-query	run sdb query. for example: '*'	Query a specific field rather than printing everything.
-K	algo	calculate checksums (md5, sha1, ..)	
-l	—	linked libraries	Linked libraries.
-L	plugin	list supported bin plugins or plugin details	
-m	addr	show source line at addr	
-M	—	main (show address of main symbol)	
-n	str	show section, symbol or import named str	
-N	min:max	force min:max number of chars per string (see -z and -zz)	
-o	str	output file or extraction directory for write operations	
-O	str	write/extract operations (-O help)	Patch the binary. Read-only work never needs this, and using it on evidence modifies it.
-p	—	show always physical addresses	
-P	—	show debug/pdb information	
-q	—	be quiet, just show fewer data	Quiet, minimal formatting — easier to parse in a shell.
-Q	—	show load address used by dlopen (non-aslr libs)	
-r	—	radare output	
-R	—	relocations	Relocations.
-s	—	symbols	Symbols, when the binary is not stripped.
-S	—	sections	Sections with their sizes and permissions. A section that is both writable and executable is worth a second look.
-t	—	display file hashes	
-T	—	display file signature	
-u	—	resources	
-v	—	display version and quit	Version.
-V	—	show binary version information	
-w	—	display try/catch blocks	
-x	—	extract sub-binaries (-xS sections, -xSS segments, -xu resources)	Extract the sub-binaries of a fat file.

Flag	Argument	What it does	When you would use it
<code>-y</code>	—	show types (structs, enums, function signatures)	
<code>-z</code>	—	strings (from data section)	Strings from the data sections.
<code>-Z</code>	—	guess size of binary program	

Gotchas

- This is the static, scriptable half of radare2. Nothing here executes the binary, which makes it safe to run over a corpus of samples — unlike opening them in a debugger.
- The security flags in `-I` describe the *binary*, not the running process. NX and ASLR also depend on the loader and the OS.
- `-z` misses strings outside the data sections, which is where packed and obfuscated samples hide theirs. Try `-zz` before concluding a sample has none.

See also

`readelf`, `readelf.py`, `objdump`, `r2`, `rasm2`, `vivbin`, `vdbbin`

radiff2

Kit: REMnux · Kali Linux · SIFT Workstation **Capability:** Compare or cluster samples; Diff two binaries **Version:** radiff2 6.1.9 **Captured:** cyberlab-aio via -h on 2026-08-02 **raw Docs:** <https://www.radare.org/n/radare2.html>

← [Capability index](#) · [Kit tool list](#)

Purpose

Examine binary files, including disassembling and debugging. Includes r2ai and decai plugins for LLM-powered analysis (API key or local Ollama required), plus the r2ghidra plugin for Ghidra decompilation via the pdg command.

Synopsis

```
radiff2 [-options] [-A[A]] [-B #] [-g sym] [-m graph_mode][-t %] [file] [file]
```

Options

All 29 options parsed from the captured help text. The final column is filled in by review.

Flag	Argument	What it does	When you would use it
-a	arch	specify architecture plugin to use (x86, arm, ..)	
-A	-A	run aaa or aaaa after loading each binary (see -C)	
-b	bits	specify register size for arch (16 (thumb), 32, 64, ..)	
-c	cmd	run given command on every RCore instance	
-C	—	graphdiff code (columns: off-A, match-ratio, off-B) (see -A)	
-d	—	use delta diffing	
-D	—	show disasm instead of hexpairs	
-e	k=v	set eval config var value for all RCore instances	
-f	help	select output format (see '-f help' for details)	
-g	arg	graph diff of [sym] or functions in [off1,off2]	
-i	help	compare bin information (symbols, strings, classes, ..)	
-j	—	output in json format (see -f json)	
-l	addr	specify final offset (length) for diffing, same format as -o	
-m	mode	choose the graph output mode (aditsjl)	
-n	—	count of changes	
-o	addr	specify initial offset for diffing, can use A:B format	
-O	—	code diffing with opcode bytes only	
-p	—	use physical addressing (io.va=false) (only for radiff2 -AC)	
-q	—	quiet mode (disable colors, reduce output)	
-r	—	output in radare commands	
-s	—	compute edit distance (no substitution, Eugene W. Myers O(ND) diff algorithm)	
-S	name	sort code diff (name, namelen, addr, size, type, dist) (only for -C or -g)	
-T	—	analyze files in threads (EXPERIMENTAL, 30% faster and crashy)	
-x	—	show two column hexdump diffing	
-X	—	use xpatch format for the diffing output	
-u	—	unified output (---+++)	
-U	—	unified output using system 'diff'	
-v	—	show version information	
-V	—	be verbose (current only for -s)	

See also

[ssdeep](#)

readelf

Kit: Base OS — present on every Linux image **Capability:** Inspect PE / ELF structure **Version:** GNU readelf (GNU Binutils for Debian) 2.40 **Captured:** cyberlab-aio via `-help` on 2026-08-02 [raw](#)

[← Capability index](#) · [Kit tool list](#)

Purpose

Display the structure of an ELF file — headers, sections, segments, symbols, notes and dynamic linkage.

Synopsis

```
readelf <option(s)> elf-file(s)
```

Options

All 68 options parsed from the captured help text; 17 reviewed with usage guidance.

Flag	Argument	What it does	When you would use it
<code>-a</code>	—	Equivalent to: <code>-h -l -S -s -r -d -V -A -l</code>	Everything. Verbose, and the right first move when you do not yet know what you are looking for.
<code>--all</code>	—	Equivalent to: <code>-h -l -S -s -r -d -V -A -l</code>	Everything. Verbose, and the right first move when you do not yet know what you are looking for.
<code>-h</code>	—	Display the ELF file header	The ELF header: type, architecture, entry point. First thing to run on an unknown Linux binary.
<code>--file-header</code>	—	Display the ELF file header	The ELF header: type, architecture, entry point. First thing to run on an unknown Linux binary.
<code>-l</code>	—	Display the program headers	Program headers, which describe what the loader maps. A section table can be stripped or faked; the program headers must be right or the binary will not run.
<code>--program-headers</code>	—	Display the program headers	Program headers, which describe what the loader maps. A section table can be stripped or faked; the program headers must be right or the binary will not run.
<code>--segments</code>	—	An alias for <code>--program-headers</code>	
<code>-S</code>	—	Display the sections' header	Section headers.
<code>--section-headers</code>	—	Display the sections' header	Section headers.
<code>--sections</code>	—	An alias for <code>--section-headers</code>	
<code>-g</code>	—	Display the section groups	
<code>--section-groups</code>	—	Display the section groups	
<code>-t</code>	—	Display the section details	Section details in full.
<code>--section-details</code>	—	Display the section details	Section details in full.
<code>-e</code>	—	Equivalent to: <code>-h -l -S</code>	Header summary: file, section and program headers together.
<code>--headers</code>	—	Equivalent to: <code>-h -l -S</code>	Header summary: file, section and program headers together.
<code>-s</code>	—	Display the symbol table	The symbol table.
<code>--syms</code>	—	Display the symbol table	The symbol table.
<code>--symbols</code>	—	An alias for <code>--syms</code>	
<code>--dyn-syms</code>	—	Display the dynamic symbol table	Dynamic symbols only — what the binary imports and exports at runtime.
<code>--lto-syms</code>	—	Display LTO symbol tables	
<code>--sym-base</code>	0 8 10 16	Force base for symbol sizes. The options are mixed (the default), octal, decimal, hexadecimal.	
<code>-C</code>	—	Decode mangled/processed symbol names STYLE can be "none", "auto", "gnu-v3", "java", "gnat", "dlang", "rust"	Demangle C++ symbols.
<code>--demangle</code>	STYLE (optional)	Decode mangled/processed symbol names STYLE can be "none", "auto", "gnu-v3", "java", "gnat", "dlang", "rust"	Demangle C++ symbols.
<code>--no-demangle</code>	—	Do not demangle low-level symbol names. (default)	

Flag	Argument	What it does	When you would use it
<code>--recurse-limit</code>	—	Enable a demangling recursion limit. (default)	
<code>--no-recurse-limit</code>	—	Disable a demangling recursion limit	
<code>-n</code>	—	Display the core notes (if present)	Notes, including the build ID that ties a stripped binary to its debug symbols.
<code>--notes</code>	—	Display the core notes (if present)	Notes, including the build ID that ties a stripped binary to its debug symbols.
<code>-r</code>	—	Display the relocations (if present)	Relocations.
<code>--relocs</code>	—	Display the relocations (if present)	Relocations.
<code>-u</code>	—	Display the unwind info (if present)	Unwind information.
<code>--unwind</code>	—	Display the unwind info (if present)	Unwind information.
<code>-d</code>	—	Display the dynamic section (if present)	The dynamic section — needed libraries, RPATH, and the init and fini arrays that run before and after <code>main</code> .
<code>--dynamic</code>	—	Display the dynamic section (if present)	The dynamic section — needed libraries, RPATH, and the init and fini arrays that run before and after <code>main</code> .
<code>-V</code>	—	Display the version sections (if present)	
<code>--version-info</code>	—	Display the version sections (if present)	
<code>-A</code>	—	Display architecture specific information (if any)	
<code>--arch-specific</code>	—	Display architecture specific information (if any)	
<code>-c</code>	—	Display the symbol/file index in an archive	
<code>--archive-index</code>	—	Display the symbol/file index in an archive	
<code>-D</code>	—	Use the dynamic section info when displaying symbols	Use the dynamic symbol table rather than the static one.
<code>--use-dynamic</code>	—	Use the dynamic section info when displaying symbols	Use the dynamic symbol table rather than the static one.
<code>-x</code>	—	Dump the contents of section as bytes	Hex dump of a named section.
<code>--hex-dump</code>	number name	Dump the contents of section as bytes	Hex dump of a named section.
<code>-p</code>	—	Dump the contents of section as strings	String dump of a named section — the targeted alternative to running <code>strings</code> over the whole file.
<code>--string-dump</code>	number name	Dump the contents of section as strings	String dump of a named section — the targeted alternative to running <code>strings</code> over the whole file.
<code>-R</code>	—	Dump the relocated contents of section	
<code>--relocated-dump</code>	number name	Dump the relocated contents of section	
<code>-z</code>	—	Decompress section before dumping it	
<code>--decompress</code>	—	Decompress section before dumping it	
<code>-P</code>	—	Display the contents of non-debug sections in separate debuginfo files. (Implies -wK)	
<code>--process-links</code>	—	Display the contents of non-debug sections in separate debuginfo files. (Implies -wK)	
<code>--dwarf-depth</code>	N	Do not display DIEs at depth N or greater	
<code>--dwarf-start</code>	N	Display DIEs starting at offset N	
<code>--ctf</code>	number name	Display CTF info from section	
<code>--ctf-parent</code>	name	Use CTF archive member as the CTF parent	
<code>--ctf-symbols</code>	number name	Use section as the CTF external symtab	
<code>--ctf-strings</code>	number name	Use section as the CTF external strtab	
<code>--sframe</code>	NAME (optional)	Display SFrame info from section NAME, (default '.sframe')	
<code>-I</code>	—	Display histogram of bucket list lengths	
<code>--histogram</code>	—	Display histogram of bucket list lengths	
<code>-W</code>	—	Allow output width to exceed 80 characters	Wide output that does not truncate.
<code>--wide</code>	—	Allow output width to exceed 80 characters	Wide output that does not truncate.
<code>-H</code>	—	Display this information	
<code>--help</code>	—	Display this information	
<code>-v</code>	—	Display the version number of readelf	

Flag	Argument	What it does	When you would use it
<code>--version</code>	—	Display the version number of readelf	

Gotchas

- The section header table is optional at runtime. Malware strips or corrupts it to break tools, and the binary still runs — when sections look wrong, read `-l` instead, because the loader must be able to.
- `DT_INIT` and `DT_INIT_ARRAY` in `-d` run before `main`. Code placed there executes even if `main` looks harmless.
- An `RPATH` or `RUNPATH` pointing somewhere writable is a hijack waiting to happen, and it shows up here.

See also

[rabin2](#), [readelf.py](#), [objdump](#)

readelf.py

Kit: REMnux **Capability:** Inspect PE / ELF structure **Version:** based on pyelftools 0.33 **Captured:** cyberlab-aio via `--help` on 2026-08-02 [raw Docs: https://github.com/eliben/pyelftools](https://github.com/eliben/pyelftools)

[← Capability index](#) · [Kit tool list](#)

Purpose

Python library for parsing and analyzing ELF files and DWARF debugging information.

Synopsis

```
usage: readelf.py [options] <elf-file>
```

Options

All 33 options parsed from the captured help text. The final column is filled in by review.

Flag	Argument	What it does	When you would use it
<code>-v</code>	—	show program's version number and exit	
<code>--version</code>	—	show program's version number and exit	
<code>-d</code>	—	Display the dynamic section	
<code>--dynamic</code>	—	Display the dynamic section	
<code>-H</code>	—	Display this information	
<code>--help</code>	—	Display this information	
<code>-h</code>	—	Display the ELF file header	
<code>--file-header</code>	—	Display the ELF file header	
<code>-l</code>	—	Display the program headers	
<code>--program-headers</code>	—	Display the program headers	
<code>--segments</code>	—	Display the program headers	
<code>-S</code>	—	Display the sections' headers	
<code>--section-headers</code>	—	Display the sections' headers	
<code>--sections</code>	—	Display the sections' headers	
<code>-e</code>	—	Equivalent to: <code>-h -l -S</code>	
<code>--headers</code>	—	Equivalent to: <code>-h -l -S</code>	
<code>-s</code>	—	Display the symbol table	
<code>--symbols</code>	—	Display the symbol table	
<code>--syms</code>	—	Display the symbol table	
<code>-n</code>	—	Display the core notes (if present)	
<code>--notes</code>	—	Display the core notes (if present)	
<code>-r</code>	—	Display the relocations (if present)	
<code>--relocs</code>	—	Display the relocations (if present)	
<code>-x</code>	number name	Dump the contents of section as bytes	
<code>--hex-dump</code>	number name	Dump the contents of section as bytes	
<code>-p</code>	number name	Dump the contents of section as strings	
<code>--string-dump</code>	number name	Dump the contents of section as strings	
<code>-V</code>	—	Display the version sections (if present)	
<code>--version-info</code>	—	Display the version sections (if present)	
<code>-A</code>	—	Display the architecture-specific information (if present)	
<code>--arch-specific</code>	—	Display the architecture-specific information (if present)	

Flag	Argument	What it does	When you would use it
<code>--debug-dump</code>	what	Display the contents of DWARF debug sections. can one of {info,decodedline,frames,frames-interp,aranges,pubtypes,pubnames,loc,Ranges}	
<code>--traceback</code>	—	Dump the Python traceback on ELFError exceptions from elftools	

See also

`rabin2`, `readelf`, `objdump`

Signature-Explorer (GUI)

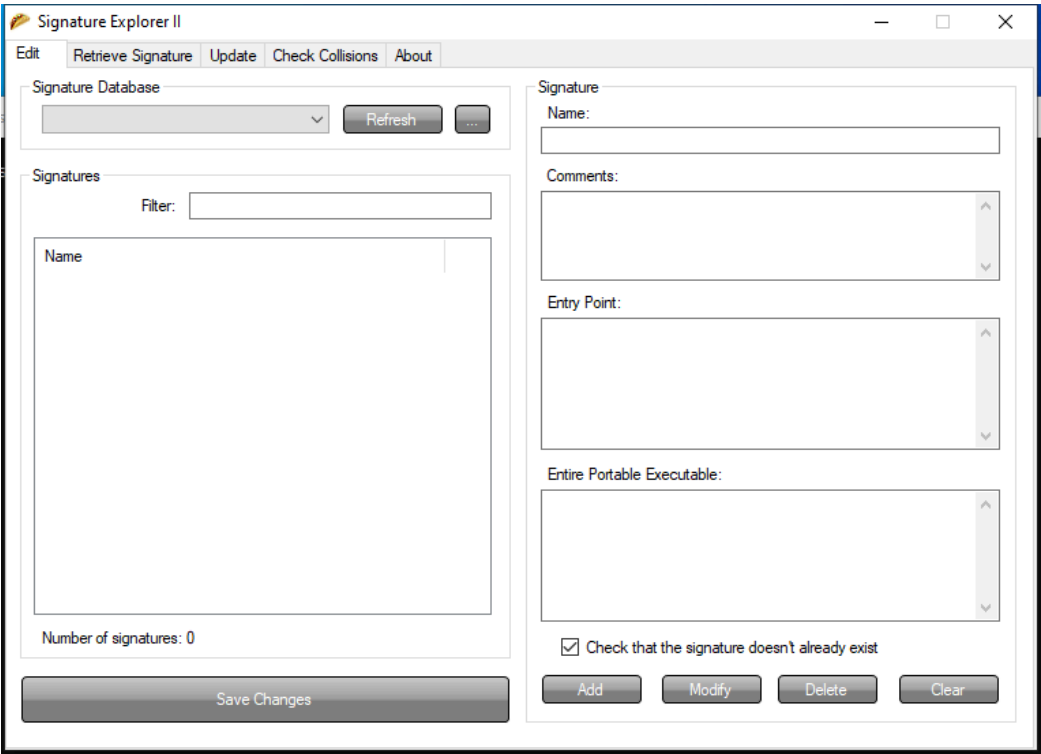
Capability: malware triage static **Window:** WindowsForms10.Window.8.app.0.378734a **Version:** — **Captured:** C:\Tools\Explorer Suite\Signature Explorer.exe on 2026-08-02 — control tree in [capture/gui/Signature-Explorer/Signature-Explorer.tree.txt](#)

[← Capability index](#) · [Kit tool list](#)

Purpose

Browse and edit packer signature databases.

Window



Controls

All 27 nodes come from the capture; the 18 interactive controls are listed here.

Control	Type	AutomationId	What it does
Edit	Pane	328216	
Save Changes	Pane	393754	
Signature	Pane	328224	
Check that the signature doesn't already exist	Pane	393636	
Clear	Pane	524690	
Delete	Pane	328078	
Modify	Pane	262508	
Add	Pane	327936	
Entire Portable Executable:	Pane	655436	
Entry Point:	Pane	393282	
Comments:	Pane	131374	
Name:	Pane	131370	
Signatures	Pane	589926	
Number of signatures: 0	Pane	458834	

Control	Type	AutomationId	What it does
Filter:	Pane	131476	
Signature Database	Pane	131454	
...	Pane	197130	
Refresh	Pane	328214	

Using it

1. Open the signature database from the **Signatures** pane.
2. Use **Check that the signature doesn't already exist** before adding anything — duplicates are the usual way a database rots.
3. Fill **Name:**, **Entry Point:** and, where the pattern is not at the entry point, **Entire Portable Executable:**.
4. Record why the signature exists in **Comments:** — a pattern with no rationale cannot be reviewed later.
5. Press **Add**, then **Save Changes**.

Gotchas

- This edits the database that PE Detective and CFF Explorer read. A bad signature here produces confident wrong answers in both.
- Entry-point signatures miss anything that relocates the entry point, which is exactly what many packers do. Prefer a whole-file pattern when the sample allows it.
- **Save Changes** is a separate action. Adding a signature and closing the window loses it.

unzip

Kit: REMnux **Capability:** Detect and reverse packing; Analyse other container formats **Version:** UnZip 6.00 **Captured:** cyberlab-aio via --help on 2026-08-02 [raw](#)
Docs: <http://infozip.sourceforge.net>

[← Capability index](#) · [Kit tool list](#)

Purpose

Compress and decompress files using the zip algorithm.

Synopsis

```
unzip [-Z] [-opts[modifiers]] file[.zip] [list] [-x xlist] [-d exdir]
Default action is to extract files in list, except those in xlist, to exdir;
file[.zip] may be a wildcard. -Z => ZipInfo mode ("unzip -Z" for usage).
```

Options

All 12 options parsed from the captured help text. The final column is filled in by review.

Flag	Argument	What it does	When you would use it
-p	—	extract files to pipe, no messages -l list files (short format)	
-f	—	freshen existing files, create none -t test compressed archive data	
-u	—	update files, create if necessary -z display archive comment only	
-v	—	list verbosely/show version info -T timestamp archive to latest	
-x	—	exclude files that follow (in xlist) -d extract files into exdir	
-n	—	never overwrite existing files -q quiet mode (-qq => quieter)	
-o	—	overwrite files WITHOUT prompting -a auto-convert any text files	
-j	—	junk paths (do not make directories) -aa treat ALL files as text	
-U	—	use escapes for all non-ASCII Unicode -UU ignore any Unicode fields	
-C	—	match filenames case-insensitively -L make (some) names lowercase	
-X	—	restore UID/GID info -V retain VMS version numbers	
-K	—	keep setuid/setgid/tacky permissions -M pipe through "more" pager	

See also

[upx](#) , [die](#) , [diec](#) , [binwalk](#) , [7za](#) , [msoffcrypto-tool](#)

upx

Kit: REMnux · Kali Linux · FLARE-VM · SIFT Workstation **Capability:** Detect and reverse packing **Version:** upx 4.2.4 **Captured:** cyberlab-aio via --help on 2026-08-02 [raw Docs: https://upx.github.io](https://upx.github.io)

← [Capability index](#) · [Kit tool list](#)

Purpose

Pack and unpack executables with the UPX compressor.

Synopsis

```
upx [-123456789dlthVL] [-qvfk] [-o file] file..
```

Common invocations

```
# from cyberlab 28-x64dbg-workflow
upx --best .\exercise\packed_hello.exe
# from cyberlab 28-x64dbg-workflow
upx -d .\exercise\packed_hello.exe -o .\exercise\unpacked_ref.exe
# from cyberlab 43-pe-bear-deep
upx --best -o C:\work\pe-lab\exercise\packed_hello.exe C:\work\pe-lab\exercise\hello.exe
# from cyberlab 52-unpacking-case
upx --best .\exercise\sample_packed.exe
```

Options

All 37 options parsed from the captured help text; 5 reviewed with usage guidance.

Flag	Argument	What it does	When you would use it
-1	—	compress faster -9 compress better	
-d	—	decompress -l list compressed file	Decompress — the only flag you normally want in analysis.
-t	—	test compressed file -V display version number	Test the file's integrity without writing anything.
-h	—	give this help -L display software license	
-q	—	be quiet -v be verbose	Quiet output.
-f	—	force compression of suspicious files	Force the operation on a file UPX considers questionable.
--no-color	—	change look	
--mono	—	change look	
--color	—	change look	
--no-progress	—	change look	
--lzma	—	try LZMA [slower but tighter than NRV]	
--brute	—	try all available compression methods & filters [slow]	
--ultra-brute	—	try even more compression variants [very slow]	
-k	—	keep backup files	Keep a backup of the original file.
--backup	—	keep backup files	Keep a backup of the original file.
--no-backup	—	no backup files [default]	
--overlay	copy	copy any extra data attached to the file [default]	
--force-overwrite	—	force overwrite of output files	
--link	—	preserve hard links (Unix only) [USE WITH CARE]	
--no-link	—	do not preserve hard links but rename files [default]	
--no-mode	—	do not preserve file mode (aka permissions)	
--no-owner	—	do not preserve file ownership	
--no-time	—	do not preserve file timestamp	
--coff	—	produce COFF output [default: EXE]	

Flag	Argument	What it does	When you would use it
<code>--8086</code>	—	make compressed com work on any 8086	
<code>--no-reloc</code>	—	put no relocations in to the exe header	
<code>--8-bit</code>	—	uses 8 bit size compression [default: 32 bit]	
<code>--8mib-ram</code>	—	8 megabyte memory limit [default: 2 MiB]	
<code>--boot-only</code>	—	disables client/host transfer compatibility	
<code>--no-align</code>	—	don't align to 2048 bytes [enables: <code>--console-run</code>]	
<code>--le</code>	—	produce LE output [default: EXE]	
<code>--compress-exports</code>	0	do not compress the export section	
<code>--compress-icons</code>	0	do not compress any icons	
<code>--compress-resources</code>	0	do not compress any resources at all	
<code>--keep-resource</code>	list	do not compress resources specified by list	
<code>--strip-relocs</code>	0	do not strip relocations	
<code>--preserve-build-id</code>	—	copy .gnu.note.build-id to compressed output	

Gotchas

- **Work on a copy.** `upx -d` rewrites the file in place, mutating your sample and invalidating its hash.
- Malware routinely uses a modified UPX header so stock `upx -d` refuses. Failure to unpack is itself an indicator, not a dead end.

See also

`die`, `diec`, `binwalk`, `7za`, `unzip`

yara

Kit: REMnux · Kali Linux · FLARE-VM **Capability:** Scan with signatures for known-bad **Version:** 4.2.3 **Captured:** cyberlab-aio via --help on 2026-08-02 **raw Docs:** <https://virustotal.github.io/yara/>

← [Capability index](#) · [Kit tool list](#)

Purpose

Scan files, directories or live processes against YARA rules.

Synopsis

```
yara [OPTION]... [NAMESPACE:]RULES_FILE... FILE | DIR | PID
```

Common invocations

```
# from cyberlab 08-malware-static-triage
yara --version
# from cyberlab 08-malware-static-triage
yara -r exercise/triage.yar exercise/sample.bin
# from cyberlab 21-yara-authoring
yara --help | head -n 20
# from cyberlab 21-yara-authoring
yara -w /tmp/hunt_eicar.yar /tmp/hunt_eicar.yar
# from cyberlab 21-yara-authoring
yara -r -s /tmp/hunt_eicar.yar /tmp
# from cyberlab 21-yara-authoring
yara -r -s /tmp/hunt_eicar.yar exercise/
# from cyberlab 21-yara-authoring
yara your_rule.yar exercise/eicar_sample.txt
# from cyberlab 34-clamav-scanning
yara -s exercise/lab34.yar exercise/sample.txt
```

Options

All 55 options parsed from the captured help text; 12 reviewed with usage guidance.

Flag	Argument	What it does	When you would use it
--atom-quality-table	FILE	path to a file with the atom quality table	
-C	—	load compiled rules	Load a pre-compiled ruleset (from yarac) instead of source.
--compiled-rules	—	load compiled rules	Load a pre-compiled ruleset (from yarac) instead of source.
-c	—	print only number of matches	Print only the count of matches per file, for a quick sweep.
--count	—	print only number of matches	Print only the count of matches per file, for a quick sweep.
-d	VAR=VALUE	define external variable	Define an external variable a rule expects, e.g. -d filename=x .
--define	VAR=VALUE	define external variable	Define an external variable a rule expects, e.g. -d filename=x .
--fail-on-warnings	—	fail on warnings	
-f	—	fast matching mode	Fast matching mode; faster but can miss some overlapping matches.
--fast-scan	—	fast matching mode	Fast matching mode; faster but can miss some overlapping matches.
-h	—	show this help and exit	
--help	—	show this help and exit	
-i	IDENTIFIER	print only rules named IDENTIFIER	Scan only the rule with the given identifier.
--identifier	IDENTIFIER	print only rules named IDENTIFIER	Scan only the rule with the given identifier.
--max-process-memory-chunk	NUMBER	set maximum chunk size while reading process memory (default=1073741824)	
-l	NUMBER	abort scanning after matching a NUMBER of rules	
--max-rules	NUMBER	abort scanning after matching a NUMBER of rules	

Flag	Argument	What it does	When you would use it
<code>--max-strings-per-rule</code>	NUMBER	set maximum number of strings per rule (default=10000)	
<code>-x</code>	MODULE=FILE	pass FILE's content as extra data to MODULE	
<code>--module-data</code>	MODULE=FILE	pass FILE's content as extra data to MODULE	
<code>-n</code>	—	print only not satisfied rules (negate)	Print files that do NOT match — useful for finding the odd one out.
<code>--negate</code>	—	print only not satisfied rules (negate)	Print files that do NOT match — useful for finding the odd one out.
<code>-N</code>	—	do not follow symlinks when scanning	
<code>--no-follow-symlinks</code>	—	do not follow symlinks when scanning	
<code>-w</code>	—	disable warnings	Suppress warnings, which are noisy on large third-party rulesets.
<code>--no-warnings</code>	—	disable warnings	Suppress warnings, which are noisy on large third-party rulesets.
<code>-m</code>	—	print metadata	Print rule metadata, which usually carries author and reference.
<code>--print-meta</code>	—	print metadata	Print rule metadata, which usually carries author and reference.
<code>-D</code>	—	print module data	
<code>--print-module-data</code>	—	print module data	
<code>-e</code>	—	print rules' namespace	
<code>--print-namespace</code>	—	print rules' namespace	
<code>-S</code>	—	print rules' statistics	
<code>--print-stats</code>	—	print rules' statistics	
<code>-s</code>	—	print matching strings	Print the matching strings and their offsets. Essential for verifying a hit is real rather than trusting the rule name.
<code>--print-strings</code>	—	print matching strings	Print the matching strings and their offsets. Essential for verifying a hit is real rather than trusting the rule name.
<code>-L</code>	—	print length of matched strings	
<code>--print-string-length</code>	—	print length of matched strings	
<code>-g</code>	—	print tags	
<code>--print-tags</code>	—	print tags	
<code>-r</code>	—	recursively search directories	Recurse into directories — the usual mode when triaging a tree.
<code>--recursive</code>	—	recursively search directories	Recurse into directories — the usual mode when triaging a tree.
<code>--scan-list</code>	—	scan files listed in FILE, one per line	
<code>-z</code>	NUMBER	skip files larger than the given size when scanning a directory	
<code>--skip-larger</code>	NUMBER	skip files larger than the given size when scanning a directory	
<code>-k</code>	SLOTS	set maximum stack size (default=16384)	
<code>--stack-size</code>	SLOTS	set maximum stack size (default=16384)	
<code>-t</code>	TAG	print only rules tagged as TAG	Scan only rules carrying a given tag.
<code>--tag</code>	TAG	print only rules tagged as TAG	Scan only rules carrying a given tag.
<code>-p</code>	NUMBER	use the specified NUMBER of threads to scan a directory	Use N threads to speed up a large scan.
<code>--threads</code>	NUMBER	use the specified NUMBER of threads to scan a directory	Use N threads to speed up a large scan.
<code>-a</code>	SECONDS	abort scanning after the given number of SECONDS	
<code>--timeout</code>	SECONDS	abort scanning after the given number of SECONDS	
<code>-v</code>	—	show version information	
<code>--version</code>	—	show version information	

Gotchas

- A rule that references an external variable fails to compile unless you supply it with `-d`. This is the most common 'the rule is broken' false alarm.
- Scanning a live process needs a PID rather than a path, and root.
- Compile large rulesets once with `yarac` and scan with `-C`; recompiling thousands of rules per scan dominates runtime.

See also

`yarac`, `clamscan`, `freshclam`, `sigtool`

yarac

Kit: REMnux · Kali Linux · FLARE-VM **Capability:** Scan with signatures for known-bad **Version:** 4.2.3 **Captured:** cyberlab-ai0 via --help on 2026-08-02 [raw](#) **Docs:** <https://virustotal.github.io/yara/>

[← Capability index](#) · [Kit tool list](#)

Purpose

Identify and classify malware samples using Yara rules.

Synopsis

```
yarac [OPTION]... [NAMESPACE:]SOURCE_FILE... OUTPUT_FILE
```

Options

All 11 options parsed from the captured help text. The final column is filled in by review.

Flag	Argument	What it does	When you would use it
--atom-quality-table	FILE	path to a file with the atom quality table	
-d	VAR=VALUE	define external variable	
--define	VAR=VALUE	define external variable	
--fail-on-warnings	—	fail on warnings	
-h	—	show this help and exit	
--help	—	show this help and exit	
--max-strings-per-rule	NUMBER	set maximum number of strings per rule (default=10000)	
-w	—	disable warnings	
--no-warnings	—	disable warnings	
-v	—	show version information	
--version	—	show version information	

See also

[yara](#), [clamscan](#), [freshclam](#), [sigtool](#)

aeskeyfind

Kit: REMnux · Kali Linux · SIFT Workstation **Capability:** Recover encryption keys from memory **Captured:** cyberlab-aio via --help on 2026-08-02 [raw](#) **Docs:** <https://citp.princeton.edu/our-work/memory/>

[← Capability index](#) · [Kit tool list](#)

Purpose

Find 128-bit and 256-bit AES keys in a memory image.

Synopsis

```
aeskeyfind [OPTION]... MEMORY-IMAGE
```

Common invocations

```
# from cyberlab 02-memory-forensics
aeskeyfind 2>&1 | head -n 1
# from cyberlab 02-memory-forensics
aeskeyfind sample.mem
```

Options

All 3 options parsed from the captured help text. The final column is filled in by review.

Flag	Argument	What it does	When you would use it
-v	—	verbose output -- prints the extended keys and the constraints on the rows of the key schedule	
-q	—	don't display a progress bar	
-h	—	displays this help message	

See also

[rsakeyfind](#), [bulk_extractor](#)

rsakeyfind

Kit: REMnux · Kali Linux · SIFT Workstation **Capability:** Recover encryption keys from memory **Captured:** `cyberlab-ai0` via `(no args)` on 2026-08-02 [raw](#) **Docs:** <https://citp.princeton.edu/our-work/memory/>

[← Capability index](#) · [Kit tool list](#)

Purpose

Find BER-encoded RSA private keys in a memory image.

Synopsis

```
rsakeyfind MEMORY-IMAGE [MODULUS-FILE]
```

Common invocations

```
# from cyberlab 02-memory-forensics
rsakeyfind 2>&1 | head -n 1
# from cyberlab 02-memory-forensics
rsakeyfind sample.mem
```

Options

No option definitions could be parsed from this tool's help output. It may be subcommand-driven or have no flags; needs manual review.

See also

`aeskeyfind`, `bulk_extractor`

vol

Kit: SIFT Workstation (Volatility 3) **Capability:** Analyse a memory image **Captured:** cyberlab-aio via --help on 2026-08-02 raw

[← Capability index](#) · [Kit tool list](#)

Purpose

Extract and analyse artifacts from a memory image using Volatility 3 plugins.

Synopsis

```
vol [-h] [-c CONFIG] [--parallelism [{processes,threads,off}]]
[-e EXTEND] [-p PLUGIN_DIRS] [-s SYMBOL_DIRS] [-v] [-l LOG]
[-o OUTPUT_DIR] [-q] [-f FILE] [--write-config]
[--save-config SAVE_CONFIG] [--clear-cache]
[--cache-path CACHE_PATH] [--offline | -u URL] [--filters FILTERS]
[--hide-columns [HIDE_COLUMNS ...]] [-r RENDERER]
```

Common invocations

```
# from cyberlab 02-memory-forensics
vol --help | head -n 3
# from cyberlab 02-memory-forensics
vol -h | grep -i -E "pslist|netscan|windows.info" | head -n 10
# from cyberlab 02-memory-forensics
vol -f $IMAGE windows.pslist | head -n 20
# from cyberlab 20-volatility-deep
vol --version
# from cyberlab 20-volatility-deep
vol -f exercise/memdump.raw windows.info
# from cyberlab 20-volatility-deep
vol -f exercise/memdump.raw windows.pslist
# from cyberlab 20-volatility-deep
vol -f exercise/memdump.raw windows.pstree
# from cyberlab 20-volatility-deep
vol -f exercise/memdump.raw windows.psscan
```

Options

All 35 options parsed from the captured help text; 9 reviewed with usage guidance.

Flag	Argument	What it does	When you would use it
-h	—	Show this help message and exit, for specific plugin options use 'vol --help'	
--help	—	Show this help message and exit, for specific plugin options use 'vol --help'	
-c	CONFIG	Load the configuration from a json file	Load saved configuration from a JSON file.
--config	CONFIG	Load the configuration from a json file	Load saved configuration from a JSON file.
--parallelism	{processes,threads,off}	Enables parallelism (defaults to off if no argument given)	
-e	EXTEND	Extend the configuration with a new (or changed) setting	
--extend	EXTEND	Extend the configuration with a new (or changed) setting	
-p	PLUGIN_DIRS	Semi-colon separated list of paths to find plugins	Add a directory of custom plugins.
--plugin-dirs	PLUGIN_DIRS	Semi-colon separated list of paths to find plugins	Add a directory of custom plugins.
-s	SYMBOL_DIRS	Semi-colon separated list of paths to find symbols	Point at a local symbol-table directory — the fix for an air-gapped host that cannot download symbols.
--symbol-dirs	SYMBOL_DIRS	Semi-colon separated list of paths to find symbols	Point at a local symbol-table directory — the fix for an air-gapped host that cannot download symbols.
-v	—	Increase output verbosity	Increase verbosity while diagnosing a symbol-table failure.
--verbosity	—	Increase output verbosity	Increase verbosity while diagnosing a symbol-table failure.

Flag	Argument	What it does	When you would use it
<code>-l</code>	LOG	Log output to a file as well as the console	
<code>--log</code>	LOG	Log output to a file as well as the console	
<code>-o</code>	OUTPUT_DIR	Directory in which to output any generated files	Directory for files the plugin dumps (processes, DLLs, files).
<code>--output-dir</code>	OUTPUT_DIR	Directory in which to output any generated files	Directory for files the plugin dumps (processes, DLLs, files).
<code>-q</code>	—	Remove progress feedback	Quiet — suppress the progress and INFO banner when scripting.
<code>--quiet</code>	—	Remove progress feedback	Quiet — suppress the progress and INFO banner when scripting.
<code>-f</code>	FILE	Shorthand for <code>--single-location=file://</code> if single- location is not defined	The memory image to analyse — required for almost every plugin.
<code>--file</code>	FILE	Shorthand for <code>--single-location=file://</code> if single- location is not defined	The memory image to analyse — required for almost every plugin.
<code>--write-config</code>	—	Write configuration JSON file out to config.json	
<code>--save-config</code>	SAVE_CONFIG	Save configuration JSON file to a file	
<code>--clear-cache</code>	—	Clears out all short-term cached items	
<code>--cache-path</code>	CACHE_PATH	Change the default path (<code>/root/.cache/volatility3</code>) used to store the cache	
<code>--offline</code>	—	Do not search online for additional JSON files	Never attempt a network fetch for symbols. Use this on evidence networks so a scan cannot stall on a download.
<code>-u</code>	URL	Search online for ISF json files	
<code>--remote-isf-url</code>	URL	Search online for ISF json files	
<code>--filters</code>	FILTERS	List of filters to apply to the output (in the form of <code>[+ -]columnname,pattern[!]</code>)	
<code>--hide-columns</code>	HIDE_COLUMNS ...	Case-insensitive space separated list of prefixes to determine which columns to hide in the output if provided	
<code>-r</code>	RENDERER	Determines how to render the output (quick, none, csv, pretty, json, jsonl, arrow, parquet)	Choose the renderer: <code>csv</code> / <code>json</code> when feeding another tool.
<code>--renderer</code>	RENDERER	Determines how to render the output (quick, none, csv, pretty, json, jsonl, arrow, parquet)	Choose the renderer: <code>csv</code> / <code>json</code> when feeding another tool.
<code>--single-location</code>	SINGLE_LOCATION	Specifies a base location on which to stack	
<code>--stackers</code>	STACKERS ...	List of stackers	
<code>--single-swap-locations</code>	SINGLE_SWAP_LOCATIONS ...	Specifies a list of swap layer URIs for use with single-location	

Gotchas

- **Timeout-guard `vol` in any harness.** On a synthetic or truncated memory image it has pinned a core at 99% indefinitely on this project. Never run it unbounded in an automated pass.
- Volatility 3 takes plugin names, not v2 syntax: `windows.pslist`, not `--profile=... pslist`. Most 'plugin not found' errors are pasted v2 commands.
- The first run against an unfamiliar image downloads symbol tables. On an isolated network that hangs — pre-stage symbols and use `--offline`.

See also

`volatility3`, `volshell`

volatility3

Kit: SIFT Workstation (Volatility 3) **Capability:** Analyse a memory image **Captured:** cyberlab-aio via --help on 2026-08-02 raw

[← Capability index](#) · [Kit tool list](#)

Purpose

Analyse a memory image. These are the framework-wide options; each plugin adds its own, shown by `volatility3 <plugin> --help`.

Synopsis

```
volatility3 [-h] [-c CONFIG] [--parallelism [{processes,threads,off}]]
[-e EXTEND] [-p PLUGIN_DIRS] [-s SYMBOL_DIRS] [-v] [-l LOG]
[-o OUTPUT_DIR] [-q] [-f FILE] [--write-config]
[--save-config SAVE_CONFIG] [--clear-cache]
[--cache-path CACHE_PATH] [--offline | -u URL]
[--filters FILTERS] [--hide-columns [HIDE_COLUMNS ...]]
```

Options

All 35 options parsed from the captured help text; 33 reviewed with usage guidance.

Flag	Argument	What it does	When you would use it
-h	—	Show this help message and exit, for specific plugin options use 'volatility3 --help'	
--help	—	Show this help message and exit, for specific plugin options use 'volatility3 --help'	
-c	CONFIG	Load the configuration from a json file	Load options from a JSON config.
--config	CONFIG	Load the configuration from a json file	Load options from a JSON config.
--parallelism	{processes,threads,off}	Enables parallelism (defaults to off if no argument given)	Enable process or thread parallelism. Off by default, and worth turning on for a long scan.
-e	EXTEND	Extend the configuration with a new (or changed) setting	Override a single configuration setting.
--extend	EXTEND	Extend the configuration with a new (or changed) setting	Override a single configuration setting.
-p	PLUGIN_DIRS	Semi-colon separated list of paths to find plugins	Additional plugin directories, for plugins outside the tree.
--plugin-dirs	PLUGIN_DIRS	Semi-colon separated list of paths to find plugins	Additional plugin directories.
-s	SYMBOL_DIRS	Semi-colon separated list of paths to find symbols	Where to find symbol tables. The usual fix when a Linux or macOS image will not resolve: the ISF for that exact kernel has to be reachable.
--symbol-dirs	SYMBOL_DIRS	Semi-colon separated list of paths to find symbols	Where to find symbol tables — the usual fix when a Linux or macOS image will not resolve.
-v	—	Increase output verbosity	More verbose output; repeat for more.
--verbosity	—	Increase output verbosity	More verbose output; repeat for more.
-l	LOG	Log output to a file as well as the console	Also write output to a log file.
--log	LOG	Log output to a file as well as the console	Also write output to a log file.
-o	OUTPUT_DIR	Directory in which to output any generated files	Directory for files the plugin writes — dumped processes, extracted DLLs. Required before any --dump plugin option does anything useful.
--output-dir	OUTPUT_DIR	Directory in which to output any generated files	Directory for files the plugin writes. Required before any --dump plugin option is useful.
-q	—	Remove progress feedback	Suppress progress feedback, for scripted runs.
--quiet	—	Remove progress feedback	Suppress progress feedback, for scripted runs.
-f	FILE	Shorthand for --single-location=file:// if single- location is not defined	The memory image. Almost every invocation starts here.
--file	FILE	Shorthand for --single-location=file:// if single- location is not defined	The memory image. Almost every invocation starts here.
--write-config	—	Write configuration JSON file out to config.json	Write the resolved configuration to config.json — useful for making a complex run reproducible.
--save-config	SAVE_CONFIG	Save configuration JSON file to a file	Write the resolved configuration to a named file.
--clear-cache	—	Clears out all short-term cached items	Drop cached items. First thing to try when results look stale or an image was replaced in place.

Flag	Argument	What it does	When you would use it
<code>--cache-path</code>	CACHE_PATH	Change the default path (/root/.cache/volatility3) used to store the cache	Move the cache somewhere with room; it grows.
<code>--offline</code>	—	Do not search online for additional JSON files	Never fetch symbols online. Use it on an analysis host that must not touch the network, and expect failures unless the ISFs are already local.
<code>-u</code>	URL	Search online for ISF json files	Point at an alternative ISF repository.
<code>--remote-isf-url</code>	URL	Search online for ISF json files	Point at an alternative ISF repository.
<code>--filters</code>	FILTERS	List of filters to apply to the output (in the form of [+ -]columnname,pattern[!])	Filter rows as <code>[+ -]column,pattern</code> , so a plugin that returns thousands of rows can be narrowed without a second pass.
<code>--hide-columns</code>	HIDE_COLUMNS ...	Case-insensitive space separated list of prefixes to determine which columns to hide in the output if provided	Drop columns from the output to keep a wide table readable.
<code>-r</code>	RENDERER	Determines how to render the output (quick, none, csv, pretty, json, jsonl, arrow, parquet)	Output renderer. <code>csv</code> or <code>jsonl</code> when the result feeds another tool; <code>pretty</code> is for reading, not for parsing.
<code>--renderer</code>	RENDERER	Determines how to render the output (quick, none, csv, pretty, json, jsonl, arrow, parquet)	Output renderer. <code>csv</code> or <code>jsonl</code> when the result feeds another tool; <code>pretty</code> is for reading.
<code>--single-location</code>	SINGLE_LOCATION	Specifies a base location on which to stack	The image URI, when it is not a plain local file (<code>-f</code> is shorthand for this).
<code>--stackers</code>	STACKERS ...	List of stackers	Control the layer stackers used to interpret the image.
<code>--single-swap-locations</code>	SINGLE_SWAP_LOCATIONS ...	Specifies a list of swap layer URIs for use with single-location	Supply swap files alongside the image, so paged-out memory can be resolved.

Gotchas

- Symbols are the usual failure, not the image. Windows profiles are generated automatically, but Linux and macOS need an ISF matching the exact kernel build — same version is not enough.
- Volatility 3 dropped the Volatility 2 profile system entirely, so `--profile` from older notes and blog posts does not exist here.
- It has pinned CPU indefinitely on a malformed image before. Run long analyses under a timeout rather than assuming progress.

See also

`vol`, `volshell`

volshell

Kit: SIFT Workstation (Volatility 3) **Capability:** Analyse a memory image **Captured:** cyberlab-aio via --help on 2026-08-02 [raw](#)

[← Capability index](#) · [Kit tool list](#)

Purpose

An interactive Python shell over a memory image, with Volatility's object model loaded — for questions no plugin answers.

Synopsis

```
volshell [-h] [-c CONFIG] [-e EXTEND] [-p PLUGIN_DIRS] [-s SYMBOL_DIRS]
[-v] [-o OUTPUT_DIR] [-q] [--log LOG] [-f FILE]
[--write-config] [--save-config SAVE_CONFIG] [--clear-cache]
[--cache-path CACHE_PATH] [--offline | -u URL] [-w | -l | -m]
[--single-location SINGLE_LOCATION]
[--stackers [STACKERS ...]]
```

Options

All 38 options parsed from the captured help text; 21 reviewed with usage guidance.

Flag	Argument	What it does	When you would use it
-h	—	show this help message and exit	
--help	—	show this help message and exit	
-c	CONFIG	Load the configuration from a json file	
--config	CONFIG	Load the configuration from a json file	
-e	EXTEND	Extend the configuration with a new (or changed) setting	
--extend	EXTEND	Extend the configuration with a new (or changed) setting	
-p	PLUGIN_DIRS	Semi-colon separated list of paths to find plugins	
--plugin-dirs	PLUGIN_DIRS	Semi-colon separated list of paths to find plugins	
-s	SYMBOL_DIRS	Semi-colon separated list of paths to find symbols	Where to find symbol tables — the usual fix when a Linux or macOS image will not resolve.
--symbol-dirs	SYMBOL_DIRS	Semi-colon separated list of paths to find symbols	Where to find symbol tables.
-v	—	Increase output verbosity	More verbose output.
--verbosity	—	Increase output verbosity	More verbose output.
-o	OUTPUT_DIR	Directory in which to output any generated files	Directory for files written out of the shell.
--output-dir	OUTPUT_DIR	Directory in which to output any generated files	Directory for files written out of the shell.
-q	—	Remove progress feedback	Quiet.
--quiet	—	Remove progress feedback	Quiet.
--log	LOG	Log output to a file as well as the console	
-f	FILE	Shorthand for --single-location=file:// if single-location is not defined	The memory image.
--file	FILE	Shorthand for --single-location=file:// if single-location is not defined	The memory image.
--write-config	—	Write configuration JSON file out to config.json	
--save-config	SAVE_CONFIG	Save configuration JSON file to a file	
--clear-cache	—	Clears out all short-term cached items	Drop cached items when results look stale.
--cache-path	CACHE_PATH	Change the default path (/root/.cache/volatility3) used to store the cache	
--offline	—	Do not search online for additional JSON files	Never fetch symbols online.
-u	URL	Search online for ISF json files	
--remote-isf-url	URL	Search online for ISF json files	
-w	—	Run a Windows volshell	Treat the image as Windows.

Flag	Argument	What it does	When you would use it
<code>--windows</code>	—	Run a Windows volshell	Treat the image as Windows.
<code>-l</code>	—	Run a Linux volshell	Treat the image as Linux.
<code>--linux</code>	—	Run a Linux volshell	Treat the image as Linux.
<code>-m</code>	—	Run a Mac volshell	Treat the image as macOS.
<code>--mac</code>	—	Run a Mac volshell	Treat the image as macOS.
<code>--single-location</code>	SINGLE_LOCATION	Specifies a base location on which to stack	
<code>--stackers</code>	STACKERS ...	List of stackers	
<code>--single-swap-locations</code>	SINGLE_SWAP_LOCATIONS ...	Specifies a list of swap layer URIs for use with single-location	
<code>--script</code>	SCRIPT	File to load and execute at start	Run a Python script against the image and drop into the shell afterwards — the repeatable form of an interactive session.
<code>--script-only</code>	—	Exit volshell after the script specified in <code>--script</code> completes	Run the script and exit. This is how an exploratory session becomes an automated one.
<code>--pid</code>	PID	Process ID	Enter with a process context already selected, so <code>cc()</code> is not the first thing you type.

Gotchas

- Reach for this when a plugin nearly answers the question but not quite. If a plugin exists, use the plugin — it is tested and this is not.
- Findings from an interactive session are not reproducible by default. Move anything that matters into `--script` so the result can be regenerated and reviewed.
- The same symbol requirement as [volatility3](#) applies: without an ISF matching the exact kernel build, a Linux or macOS image will not resolve at all.

See also

[vol](#), [volatility3](#)

arp-scan

Kit: SIFT Workstation **Capability:** Probe or scan hosts and services **Version:** arp-scan 1.10.0 **Captured:** `cyberlab-aio` via `--help` on 2026-08-02 [raw](#)

[← Capability index](#) · [Kit tool list](#)

Purpose

Target hosts must be specified on the command line unless the `--file` or

Synopsis

```
arp-scan [options] [hosts...]
```

Options

No option definitions could be parsed from this tool's help output. It may be subcommand-driven or have no flags; needs manual review.

See also

`nmap`, `nping`

capinfos

Kit: REMnux · Kali Linux · FLARE-VM · SIFT Workstation **Capability:** Read and filter packet captures **Version:** Git v4.0.17 **Captured:** cyberlab-aio via --help on 2026-08-02 **raw Docs:** <https://www.wireshark.org>

[← Capability index](#) · [Kit tool list](#)

Purpose

Capture and analyze network traffic with this sniffer.

Synopsis

```
capinfos [options] <infile> ...
```

Common invocations

```
# from cyberlab 24-wireshark-deep
capinfos exercise/sample.pcap
```

Options

All 4 options parsed from the captured help text. The final column is filled in by review.

Flag	Argument	What it does	When you would use it
-h	—	display this help and exit	
--help	—	display this help and exit	
-v	—	display version info and exit	
--version	—	display version info and exit	

See also

[tshark](#), [ngrep](#), [tcpflow](#)

editcap

Kit: REMnux · Kali Linux · FLARE-VM · SIFT Workstation **Capability:** Split, merge or repair capture files **Version:** Git v4.0.17 **Captured:** cyberlab-aio via `--help` on 2026-08-02 **raw Docs:** <https://www.wireshark.org>

← [Capability index](#) · [Kit tool list](#)

Purpose

Cut, split, deduplicate and convert capture files — the tool that makes an unmanageable pcap workable before analysis starts.

Synopsis

```
editcap [options] ... <infile> <outfile> [ <packet#>[<-packet#>] ... ]
```

Options

All 25 options parsed from the captured help text; 21 reviewed with usage guidance.

Flag	Argument	What it does	When you would use it
<code>-r</code>	—	keep the selected packets; default is to delete them.	Invert the selection: keep the specified packets instead of deleting them. Easy to forget, and it reverses the meaning of the whole command.
<code>-A</code>	start time	only read packets whose timestamp is after (or equal to) the given time.	Keep only packets at or after this timestamp.
<code>-B</code>	stop time	only read packets whose timestamp is before the given time. Time format for <code>-A/-B</code> options is YYYY-MM-DDThh:mm:ss[.nnnnnnnnn][Z + -hh:mm] Unix epoch timestamps are also supported.	Keep only packets before this timestamp. With <code>-A</code> , this is how a multi-gigabyte capture becomes the incident window.
<code>--novlan</code>	—	remove vlan info from packets before checking for duplicates.	Ignore VLAN tags when comparing for duplicates, so the same frame seen on two VLANs collapses to one.
<code>-d</code>	—	remove packet if duplicate (window == 5).	Drop duplicate packets using the default 5-packet window. Captures taken from a SPAN port routinely see each packet twice, which distorts every count downstream.
<code>-D</code>	dup window	remove packet if duplicate; configurable . Valid values are 0 to 1000000. NOTE: A of 0 with <code>-V</code> (verbose option) is useful to print MD5 hashes.	Drop duplicates with an explicit window, when <code>-d</code> 's default is too narrow.
<code>-w</code>	dup time window	remove packet if duplicate packet is found EQUAL TO OR LESS THAN prior to current packet. A is specified in relative seconds (e.g. 0.000001). NOTE: The use of the '	Drop duplicates within a time window rather than a packet count.
<code>-s</code>	snaplen	truncate each packet to max. bytes of data.	Truncate each packet to N bytes. Strips payload while keeping headers — the usual way to share a capture without its contents.
<code>-L</code>	—	adjust the frame (i.e. reported) length when chopping and/or snapping.	Adjust the recorded frame length to match after truncating, so the file is not self-inconsistent.
<code>-t</code>	time adjustment	adjust the timestamp of each packet. is in relative seconds (e.g. -0.5).	Shift every timestamp by a relative amount. This is how a capture from a host with a skewed clock is aligned to the rest of the timeline.
<code>-o</code>	change offset	When used in conjunction with <code>-E</code> , skip some bytes from the beginning of the packet. This allows one to preserve some bytes, in order to have some headers untouched.	With <code>-E</code> , skip bytes before introducing errors.
<code>--seed</code>	seed	When used in conjunction with <code>-E</code> , set the seed to use for the pseudo-random number generator. This allows one to repeat a particular sequence of errors.	With <code>-E</code> , fix the random seed so a corrupted-capture test is reproducible.
<code>-I</code>	bytes to ignore	ignore the specified number of bytes at the beginning of the frame during MD5 hash calculation, unless the frame is too short, then the full frame is used. Useful to remove duplicated packets taken on	Ignore N leading bytes when comparing for duplicates.
<code>-c</code>	packets per file	split the packet output to different files based on uniform packet counts with a maximum of each.	Split into files of N packets each. The standard fix for a capture too large for Wireshark to open.
<code>-i</code>	seconds per file	split the packet output to different files based on uniform time intervals with a maximum of each.	Split into files covering N seconds each — the same fix, when time is the natural unit.
<code>-F</code>	capture type	set the output file type; default is pcapng. An empty <code>"-F"</code> option will list the file types.	Output file format; pcapng by default. An empty <code>-F</code> lists the choices.
<code>-T</code>	encap type	set the output file encapsulation type; default is the same as the input file. An empty <code>"-T"</code> option will list the encapsulation types.	Output encapsulation type, when the link type must change.
<code>--discard-all-secrets</code>	—	Discard all decryption secrets from the input file when writing the output file. Does not discard secrets added by <code>"--inject-secrets"</code> in the same command line.	Strip embedded decryption secrets before handing the file to someone else.
<code>--capture-comment</code>	comment	Add a capture file comment, if supported.	Attach a comment to the file — a place to record provenance that travels with the capture.

Flag	Argument	What it does	When you would use it
<code>--discard-capture-comment</code>	—	Discard capture file comments from the input file when writing the output file. Does not discard comments added by " <code>--capture-comment</code> " in the same command line.	Remove existing comments on output.
<code>-h</code>	—	display this help and exit.	
<code>--help</code>	—	display this help and exit.	
<code>-V</code>	—	verbose output. If <code>-V</code> is used with any of the 'Duplicate Packet Removal' options (<code>-d</code> , <code>-D</code> or <code>-w</code>) then Packet lengths and MD5 hashes are printed to standard-error.	Verbose; with the duplicate options it reports what was removed rather than silently dropping packets.
<code>-v</code>	—	print version information and exit.	
<code>--version</code>	—	print version information and exit.	

Gotchas

- `-r` inverts the selection. Without it the named packets are **deleted**, which is the opposite of what most people intend the first time.
- Deduplication is a heuristic over a window, not a proof. A genuine retransmission looks like a duplicate, and dropping it destroys the evidence that a retransmission occurred.
- Splitting rennumbers packets per output file. Frame numbers cited from a split file do not refer to the original capture.

See also

`mergecap`, `reordercap`

inetsim

Kit: REMnux · Kali Linux **Capability:** Simulate network services for detonation **Version:** INetSim 1.3.2 **Captured:** cyberlab-aio via --help on 2026-08-02 [raw](#) **Docs:** <https://www.inetsim.org/>

← [Capability index](#) · [Kit tool list](#)

Purpose

Emulate common network services and interact with malware.

Synopsis

```
/usr/bin/inetsim [options]
```

Common invocations

```
# from cyberlab 38-network-emulation
inetsim --version 2>&1 | head -n 3
# from cyberlab 38-network-emulation
sudo inetsim --config ~/lab38/inetsim.conf --data-dir /var/lib/inetsim --log-dir ~/lab38/log &
```

Options

All 14 options parsed from the captured help text. The final column is filled in by review.

Flag	Argument	What it does	When you would use it
--help	—	Print this help message.	
--version	—	Show version information.	
--config	filename	Configuration file to use.	
--log-dir	directory	Directory logfiles are written to.	
--data-dir	directory	Directory containing service data.	
--report-dir	directory	Directory reports are written to.	
--bind-address	IP address	Default IP address to bind services to. Overrides configuration option 'default_bind_address'.	
--max-childs	num	Default maximum number of child processes per service. Overrides configuration option 'default_max_childs'.	
--user	username	Default user to run services. Overrides configuration option 'default_run_as_user'.	
--faketime-init-delta	secs	Initial faketime delta (seconds). Overrides configuration option 'faketime_init_delta'.	
--faketime-auto-delay	secs	Delay for auto incrementing faketime (seconds). Overrides configuration option 'faketime_auto_delay'.	
--faketime-auto-incr	secs	Delta for auto incrementing faketime (seconds). Overrides configuration option 'faketime_auto_increment'.	
--session	id	Session id to use. Defaults to main process id.	
--pidfile	filename	Pid file to use. Defaults to '/var/run/inetsim.pid'.	

mergecap

Kit: REMnux · Kali Linux · FLARE-VM · SIFT Workstation **Capability:** Split, merge or repair capture files **Version:** Git v4.0.17 **Captured:** cyberlab-aio via --help on 2026-08-02 **raw Docs:** <https://www.wireshark.org>

← [Capability index](#) · [Kit tool list](#)

Purpose

Capture and analyze network traffic with this sniffer.

Synopsis

```
mergecap [options] -w <outfile>|- <infile> [<infile> ...]
```

Options

All 7 options parsed from the captured help text. The final column is filled in by review.

Flag	Argument	What it does	When you would use it
-a	—	concatenate rather than merge files. default is to merge based on frame timestamps.	
-s	snaplen	truncate packets to bytes of data.	
-h	—	display this help and exit.	
--help	—	display this help and exit.	
-V	—	verbose output.	
-v	—	print version information and exit.	
--version	—	print version information and exit.	

See also

[editcap](#) , [reordercap](#)

ngrep

Kit: REMnux · SIFT Workstation **Capability:** Read and filter packet captures **Version:** V1.47.1 **Captured:** cyberlab-aio via --help on 2026-08-02 [raw Docs:](#) <https://github.com/jpr5/ngrep/>

[← Capability index](#) · [Kit tool list](#)

Purpose

Grep packet payloads, live or from a capture, with BPF filtering — pattern matching on the wire.

Synopsis

```
ngrep <--hNXViwqpevxlDtTRM> <--IO pcap_dump> <--n num> <--d dev> <--A num>
<--s snaplen> <--S limitlen> <--W normal|byline|single|none> <--c cols>
<--P char> <--F file> <--K count>
<match expression> <bpf filter>
```

Common invocations

```
# from cyberlab 07-network-pcap
ngrep -V 2>&1 | head -n 1
# from cyberlab 07-network-pcap
ngrep -I exercise/sample.pcap -q -W byline 'User-Agent'
# from cyberlab 07-network-pcap
ngrep -I exercise/sample.pcap -q -W byline 'User-Agent' | grep -i 'User-Agent'
# from cyberlab 24-wireshark-deep
ngrep -I exercise/sample.pcap -q -W byline "User-Agent"
# from cyberlab 24-wireshark-deep
ngrep -I exercise/sample.pcap -q -W byline "User-Agent" | grep -i "User-Agent"
```

Options

All 29 options parsed from the captured help text; 27 reviewed with usage guidance.

Flag	Argument	What it does	When you would use it
-h	—	is help/usage	
-V	—	is version information	
-q	—	is be quiet (don't print packet reception hash marks)	Suppress the reception hash marks.
-e	—	is show empty packets	Show empty packets, which are otherwise hidden.
-i	—	is ignore case	Case-insensitive match — usually what you want for protocol keywords and hostnames.
-v	—	is invert match	Invert the match, to see everything that is <i>not</i> the known traffic.
-R	—	is don't do privilege revocation logic	Skip privilege revocation.
-x	—	is print in alternate hexdump format	Print payloads as a hexdump — necessary when the protocol is not text.
-X	—	is interpret match expression as hexadecimal	Treat the pattern as hexadecimal, for matching binary signatures rather than text.
-w	—	is word-regex (expression must match as a word)	Match the pattern as a whole word, to stop a short string matching inside longer ones.
-p	—	is don't go into promiscuous mode	Do not enter promiscuous mode — capture only traffic addressed to this host.
-l	—	is make stdout line buffered	Line-buffer stdout, so output appears when piped.

Flag	Argument	What it does	When you would use it
-D	—	is replay pcap_dumps with their recorded time intervals	Replay a pcap at its recorded timing rather than as fast as possible.
-t	—	is print timestamp every time a packet is matched	Print a timestamp on every match.
-T	—	is print delta timestamp every time a packet is matched specify twice for delta from first match	Print the delta since the previous match; twice for delta from the first. Useful for spotting beaconing intervals.
-M	—	is don't do multi-line match (do single-line match instead)	Single-line matching instead of multi-line.
-I	—	is read packet stream from pcap format file pcap_dump	Read from a pcap file instead of an interface. The safe mode: no capture privileges and no risk of touching live traffic.
-O	—	is dump matched packets in pcap format to pcap_dump	Write matched packets to a pcap, turning a search into a smaller capture that Wireshark can open.
-n	—	is look at only num packets	Stop after N packets.
-A	—	is dump num packets after a match	Also print N packets following each match, for the response to the request that matched.
-s	—	is set the bpf caplen	Set the BPF capture length.
-S	—	is set the limitlen on matched packets	Limit how much of a matched packet is shown.
-W	—	is set the dump format (normal, byline, single, none)	Output format: <code>byline</code> is far more readable for text protocols than the default.
-c	—	is force the column width to the specified size	Force the column width.
-P	—	is set the non-printable display char to what is specified	Set the character shown for non-printable bytes.
-F	—	is read the bpf filter from the specified file	Read the BPF filter from a file, when it is too long to be comfortable on a command line.
-N	—	is show sub protocol number	Show sub-protocol numbers.
-d	—	is use specified device instead of the pcap default	Choose the capture interface rather than the pcap default.
-K	—	is send N packets to kill observed connections	Send packets to kill matched connections. This writes to the network — it is not an analysis option, and it does not belong anywhere near evidence handling.

Gotchas

- It matches within individual packets. A string split across TCP segments will not be found — reassembly is `tshark`'s job, not this one's.
- `-K` is the one flag here that changes the world instead of observing it. Everything else reads; that one transmits.
- Matching payload on a live interface needs capture privileges, and on a busy link `ngrep` drops packets silently. Capture first, search the file afterwards, when the answer has to be complete.

See also

`tshark`, `capinfos`, `tcpflow`

nmap

Kit: Kali Linux · FLARE-VM **Capability:** Probe or scan hosts and services **Version:** Nmap version 7.93 **Captured:** cyberlab-aio via --help on 2026-08-02 [raw](#)

[← Capability index](#) · [Kit tool list](#)

Purpose

Discover hosts, ports and services, and fingerprint what is listening.

Synopsis

```
nmap [Scan Type(s)] [Options] {target specification}
```

Common invocations

```
# from cyberlab 11-offensive-kali
nmap --version
# from cyberlab 11-offensive-kali
nmap -sT -Pn -p 1-1024 127.0.0.1
# from cyberlab 26-metasploit-workflow
nmap -sV -Pn -p 1-1000 -oN scan.txt "$TARGET"
# from cyberlab 26-metasploit-workflow
nmap -sV -Pn -oX scan.xml "$TARGET"
# from cyberlab 41-web-app-testing
nmap -sV -p 80,443,8080 "$TARGET"
# from cyberlab 41-web-app-testing
nmap -p 80,443 --script http-title,http-headers,http-server-header "$TARGET"
# from cyberlab 41-web-app-testing
nmap -p 8000 --script http-title 127.0.0.1
```

Options

All 66 options parsed from the captured help text; 24 reviewed with usage guidance.

Flag	Argument	What it does	When you would use it
--exclude	host1[,host2] [,host3],...	Exclude hosts/networks	Skip these hosts. The safety flag: use it for anything fragile before starting a range scan.
--excludefile	exclude_file	Exclude list from file	Skip the hosts listed in a file.
--dns-servers	serv1[,serv2],...	Specify custom DNS servers	
--system-dns	—	Use OS's DNS resolver	
--traceroute	—	Trace hop path to each host	Trace the path to each host.
--scanflags	flags	Customize TCP scan flags	
-b	FTP relay host	FTP bounce scan	
-p	port ranges	Only scan specified ports Ex: -p22; -p1-65535; -p U:53,111,137,T:21-25,80,139,8080,S:9	Which ports. -p- is all 65535 and takes far longer than people expect.
--exclude-ports	port ranges	Exclude the specified ports from scanning	
-F	—	Fast mode - Scan fewer ports than the default scan	Fast scan of the top 100 ports.
-r	—	Scan ports sequentially - don't randomize	
--top-ports	number	Scan most common ports	Scan the N most common ports — the usual time/coverage compromise.
--port-ratio	ratio	Scan ports more common than	
--version-intensity	level	Set from 0 (light) to 9 (try all probes)	How hard -sV tries, 0 to 9.
--version-light	—	Limit to most likely probes (intensity 2)	Intensity 2 — much faster, misses more.
--version-all	—	Try every single probe (intensity 9)	Intensity 9.
--version-trace	—	Show detailed version scan activity (for debugging)	
--script	Lua scripts	is a comma separated list of directories, script-files or script-categories	Run NSE scripts. The category matters — vuln and exploit scripts actively test, and exploit can change the target.

Flag	Argument	What it does	When you would use it
<code>--script-args</code>	<code>n1=v1,[n2=v2,...]</code>	provide arguments to scripts	Arguments for those scripts.
<code>--script-args-file</code>	filename	provide NSE script args in a file	
<code>--script-trace</code>	—	Show all data sent and received	
<code>--script-updatedb</code>	—	Update the script database.	
<code>--script-help</code>	Lua scripts	Show help about scripts. is a comma-separated list of script-files or script-categories.	Explain what a script does before running it, which is worth doing for anything outside <code>safe</code> .
<code>-O</code>	—	Enable OS detection	OS fingerprint from the TCP/IP stack. A guess with a confidence, not a fact.
<code>--osscan-limit</code>	—	Limit OS detection to promising targets	
<code>--osscan-guess</code>	—	Guess OS more aggressively	Report near matches rather than staying silent.
<code>-T</code>	0-5	Set timing template (higher is faster)	Timing template 0-5. <code>-T4</code> is the usual choice on a LAN; <code>-T0</code> and <code>-T1</code> exist for evading rate-based detection and take hours.
<code>--min-hostgroup</code>	—	Parallel host scan group sizes	
<code>--min-parallelism</code>	—	Probe parallelization	
<code>--min-rtt-timeout</code>	—	Specifies probe round trip time.	
<code>--max-retries</code>	tries	Caps number of port scan probe retransmissions.	Cap retransmissions on a lossy link.
<code>--host-timeout</code>	time	Give up on target after this long	Give up on a host after this long, so one dead host cannot stall a range.
<code>--min-rate</code>	number	Send packets no slower than per second	
<code>--max-rate</code>	number	Send packets no faster than per second	
<code>-f</code>	—	fragment packets (optionally w/given MTU)	
<code>--mtu</code>	val	fragment packets (optionally w/given MTU)	
<code>-D</code>	decoy1,decoy2[,ME],...	Cloak a scan with decoys	Decoy scan.
<code>-S</code>	IP_Address	Spoof source address	Spoof the source address.
<code>-e</code>	iface	Use specified interface	Choose the interface to scan from.
<code>--proxies</code>	url1,[url2],...	Relay connections through HTTP/SOCKS4 proxies	
<code>--data</code>	hex string	Append a custom payload to sent packets	
<code>--data-string</code>	string	Append a custom ASCII string to sent packets	
<code>--data-length</code>	num	Append random data to sent packets	
<code>--ip-options</code>	options	Send packets with specified ip options	
<code>--ttl</code>	val	Set IP time-to-live field	
<code>--spoof-mac</code>	mac address	Spoof your MAC address	
<code>--badsum</code>	—	Send packets with a bogus TCP/UDP/SCTP checksum	
<code>-v</code>	—	Increase verbosity level (use -vv or more for greater effect)	Verbose; repeat for more.
<code>-d</code>	—	Increase debugging level (use -dd or more for greater effect)	
<code>--reason</code>	—	Display the reason a port is in a particular state	
<code>--open</code>	—	Only show open (or possibly open) ports	Show only open ports, cutting the closed-port noise.
<code>--packet-trace</code>	—	Show all packets sent and received	
<code>--iflist</code>	—	Print host interfaces and routes (for debugging)	
<code>--append-output</code>	—	Append to rather than clobber specified output files	
<code>--resume</code>	filename	Resume an aborted scan	
<code>--noninteractive</code>	—	Disable runtime interactions via keyboard	
<code>--stylesheet</code>	path	XSL stylesheet to transform XML output to HTML	
<code>--webxml</code>	—	Reference stylesheet from Nmap.Org for more portable XML	
<code>--no-stylesheet</code>	—	Prevent associating of XSL stylesheet w/XML output	
<code>-6</code>	—	Enable IPv6 scanning	Scan IPv6. Hosts frequently expose more on v6 than v4 because the firewall rules were never mirrored.
<code>-A</code>	—	Enable OS detection, version detection, script scanning, and traceroute	Aggressive: version, OS, scripts and traceroute together. Convenient and unmistakably noisy.
<code>--datadir</code>	dirname	Specify custom Nmap data file location	

Flag	Argument	What it does	When you would use it
<code>--privileged</code>	—	Assume that the user is fully privileged	
<code>--unprivileged</code>	—	Assume the user lacks raw socket privileges	
<code>-V</code>	—	Print version number	
<code>-h</code>	—	Print this help summary page.	

Gotchas

- Scanning is not passive. `-sV` and NSE talk to services properly, `--script exploit` may change the target, and everything here is recorded by anything watching. Have authorisation before running it, and use `--exclude` for hosts that must not be touched.
- `-p` on a /24 is a very different job from the default scan. Scope the ports before scoping the hosts.
- A closed port and a filtered port are different findings. 'Filtered' means something dropped the probe, which is information about the network rather than about the host.

See also

`nping`, `arp-scan`

nping

Kit: Kali Linux · FLARE-VM **Capability:** Probe or scan hosts and services **Captured:** cyberlab-aio via --help on 2026-08-02 [raw](#)

[← Capability index](#) · [Kit tool list](#)

Purpose

Nping 0.7.93 (<https://nmap.org/nping>)

Synopsis

```
nping [Probe mode] [Options] {target specification}
```

Options

All 85 options parsed from the captured help text. The final column is filled in by review.

Flag	Argument	What it does	When you would use it
--tcp-connect	—	Unprivileged TCP connect probe mode.	
--tcp	—	TCP probe mode.	
--udp	—	UDP probe mode.	
--icmp	—	ICMP probe mode.	
--arp	—	ARP/RARP probe mode.	
--tr	—	Traceroute mode (can only be used with TCP/UDP/ICMP modes).	
--traceroute	—	Traceroute mode (can only be used with TCP/UDP/ICMP modes).	
-p	port spec	Set destination port(s).	
--dest-port	port spec	Set destination port(s).	
-g	portnumber	Try to use a custom source port.	
--source-port	portnumber	Try to use a custom source port.	
--seq	seqnumber	Set sequence number.	
--flags	flag list	Set TCP flags (ACK,PSH,RST,SYN,FIN...)	
--ack	acknumber	Set ACK number.	
--win	size	Set window size.	
--badsum	—	Use a random invalid checksum.	
--icmp-type	type	ICMP type.	
--icmp-code	code	ICMP code.	
--icmp-id	id	Set identifier.	
--icmp-seq	n	Set sequence number.	
--icmp-redirect-addr	addr	Set redirect address.	
--icmp-param-pointer	pnt	Set parameter problem pointer.	
--icmp-advert-lifetime	time	Set router advertisement lifetime.	
--icmp-advert-entry	IP,pref	Add router advertisement entry.	
--icmp-orig-time	—	: Set originate timestamp.	
--icmp-recv-time	—	: Set receive timestamp.	
--icmp-trans-time	timestamp	Set transmit timestamp.	
--arp-type	type	Type: ARP, ARP-reply, RARP, RARP-reply.	
--arp-sender-mac	mac	Set sender MAC address.	
--arp-sender-ip	—	: Set sender IP address.	
--arp-target-mac	mac	Set target MAC address.	
--arp-target-ip	—	: Set target IP address.	
-S	—	Set source IP address.	
--source-ip	—	Set source IP address.	
--dest-ip	addr	Set destination IP address (used as an alternative to {target specification}).	

Flag	Argument	What it does	When you would use it
<code>--tos</code>	tos	Set type of service field (8bits).	
<code>--id</code>	—	: Set identification field (16 bits).	
<code>--df</code>	—	Set Don't Fragment flag.	
<code>--mf</code>	—	Set More Fragments flag.	
<code>--evil</code>	—	Set Reserved / Evil flag.	
<code>--ttl</code>	hops	Set time to live [0-255].	
<code>--badsum-ip</code>	—	Use a random invalid checksum.	
<code>--ip-options</code>	S[R [route]]L [route]]T]U ...	Set IP options	
<code>--mtu</code>	size	Set MTU. Packets get fragmented if MTU is small enough.	
<code>-6</code>	—	Use IP version 6.	
<code>--IPv6</code>	—	Use IP version 6.	
<code>--hop-limit</code>	—	Set hop limit (same as IPv4 TTL).	
<code>--traffic-class</code>	class	: Set traffic class.	
<code>--flow</code>	label	Set flow label.	
<code>--dest-mac</code>	mac	Set destination mac address. (Disables ARP resolution)	
<code>--source-mac</code>	mac	Set source MAC address.	
<code>--ether-type</code>	type	Set EtherType value.	
<code>--data</code>	hex string	Include a custom payload.	
<code>--data-string</code>	text	Include a custom ASCII text.	
<code>--data-length</code>	len	Include len random bytes as payload.	
<code>--echo-client</code>	passphrase	Run Nping in client mode.	
<code>--echo-server</code>	passphrase	Run Nping in server mode.	
<code>--echo-port</code>	port	Use custom to listen or connect.	
<code>--no-crypto</code>	—	Disable encryption and authentication.	
<code>--once</code>	—	Stop the server after one connection.	
<code>--safe-payloads</code>	—	Erase application data in echoed packets.	
<code>--delay</code>	time	Adjust delay between probes.	
<code>--rate</code>	—	: Send num packets per second.	
<code>-h</code>	—	Display help information.	
<code>--help</code>	—	Display help information.	
<code>-V</code>	—	Display current version number.	
<code>--version</code>	—	Display current version number.	
<code>-c</code>	n	Stop after rounds.	
<code>--count</code>	n	Stop after rounds.	
<code>-e</code>	name	Use supplied network interface.	
<code>--interface</code>	name	Use supplied network interface.	
<code>-H</code>	—	Do not display sent packets.	
<code>--hide-sent</code>	—	Do not display sent packets.	
<code>-N</code>	—	Do not try to capture replies.	
<code>--no-capture</code>	—	Do not try to capture replies.	
<code>--privileged</code>	—	Assume user is fully privileged.	
<code>--unprivileged</code>	—	Assume user lacks raw socket privileges.	
<code>--send-eth</code>	—	Send packets at the raw Ethernet layer.	
<code>--send-ip</code>	—	Send packets using raw IP sockets.	
<code>--bpf-filter</code>	filter spec	Specify custom BPF filter.	
<code>-v</code>	—	Increment verbosity level by one.	
<code>-d</code>	—	Increment debugging level by one.	
<code>-q</code>	—	Decrease verbosity level by one.	
<code>--quiet</code>	—	Set verbosity and debug level to minimum.	
<code>--debug</code>	—	Set verbosity and debug to the max level.	

See also

`nmap` , `arp-scan`

reordercap

Kit: REMnux · Kali Linux · FLARE-VM · SIFT Workstation **Capability:** Split, merge or repair capture files **Version:** Git v4.0.17 **Captured:** cyberlab-aio via --help on 2026-08-02 **raw Docs:** <https://www.wireshark.org>

[← Capability index](#) · [Kit tool list](#)

Purpose

Capture and analyze network traffic with this sniffer.

Synopsis

```
reordercap [options] <infile> <outfile>
```

Options

All 3 options parsed from the captured help text. The final column is filled in by review.

Flag	Argument	What it does	When you would use it
-n	—	don't write to output file if the input file is ordered.	
-h	—	display this help and exit.	
-v	—	print version information and exit.	

See also

[editcap](#), [mergecap](#)

tcpflow

Kit: REMnux · Kali Linux · SIFT Workstation **Capability:** Read and filter packet captures; Extract files and payloads from traffic **Version:** TCPFLOW 1.6.1 **Captured:** cyberlab-aio via --help on 2026-08-02 **raw Docs:** <https://downloads.digitalcorpora.org/downloads/tcpflow/>

← [Capability index](#) · [Kit tool list](#)

Purpose

Analyze the flow of network traffic.

Synopsis

```
tcpflow [-aBcDhIpsvVZ] [-b max_bytes] [-d debug_level]
[-[eE] scanner] [-f max_fds] [-F[ctTXMkmg]] [-h|--help] [-i iface]
[-l files...] [-L semlock] [-m min_bytes] [-o outdir] [-r file] [-R file]
[-S name=value] [-T template] [-U|--relinquish-privileges user] [-v|--verbose]
[-w file] [-x scanner] [-X xmlfile] [-z|--chroot dir] [expression]
```

Common invocations

```
# from cyberlab 07-network-pcap
tcpflow --version | head -n 1
```

Options

All 33 options parsed from the captured help text. The final column is filled in by review.

Flag	Argument	What it does	When you would use it
-a	—	do ALL post-processing.	
-b	max_bytes	max number of bytes per flow to save	
-d	debug_level	debug level; default is 1	
-f	—	maximum number of file descriptors to use	
-h	—	print this help message (-hh for more help)	
-H	—	print detailed information about each scanner	
-i	—	network interface on which to listen	
-I	—	write for each flow another file *.findx to provide byte-indexed timestamps	
-g	—	output each flow in alternating colors (note change!)	
-l	—	treat non-flag arguments as input files rather than a pcap expression	
-L	—	semlock - specifies that writes are locked using a named semaphore	
-p	—	don't use promiscuous mode	
-q	—	quiet mode - do not print warnings	
-r	file	read packets from tcpdump pcap file (may be repeated)	
-R	file	read packets from tcpdump pcap file TO FINISH CONNECTIONS	
-v	—	verbose operation equivalent to -d 10	
-V	—	print version number and exit	
-w	—	file : write packets not processed to file	
-o	—	outdir : specify output directory (default '.')	
-X	—	filename : DFXML output to filename	
-m	—	bytes : specifies skip that starts a new stream (default 16777216).	
-Z	—	do not decompress gzip-compressed HTTP transactions	
-K	—	output keep pcap flow structure.	
-U	user	relinquish privileges and become user (if running as root)	
-z	dir	chroot to dir (requires that -U be used).	
-E	scanner	- turn off all scanners except scanner	

Flag	Argument	What it does	When you would use it
<code>-B</code>	—	binary output, even with <code>-c</code> or <code>-C</code> (normally <code>-c</code> or <code>-C</code> turn it off)	
<code>-c</code>	—	console print only (don't create files)	
<code>-C</code>	—	console print only, but without the display of source/dest header	
<code>-0</code>	—	don't print newlines after packets when printing to console	
<code>-s</code>	—	strip non-printable characters (change to '.')	
<code>-J</code>	—	output json format.	
<code>-D</code>	—	output in hex (useful to combine with <code>-c</code> or <code>-C</code>)	

See also

[tshark](#), [capinfos](#), [ngrep](#), [tcpxtract](#), [foremost](#)

ezhexviewer

Kit: REMnux **Capability:** Inspect files by hand **Captured:** cyberlab-aio via --help on 2026-08-02 raw **Docs:** <https://www.decalage.info/python/oletools>

[← Capability index](#) · [Kit tool list](#)

Purpose

Microsoft Office OLE2 compound documents.

Options

No option definitions could be parsed from this tool's help output. It may be subcommand-driven or have no flags; needs manual review.

See also

`xxd`, `less`

dnSpy (GUI)

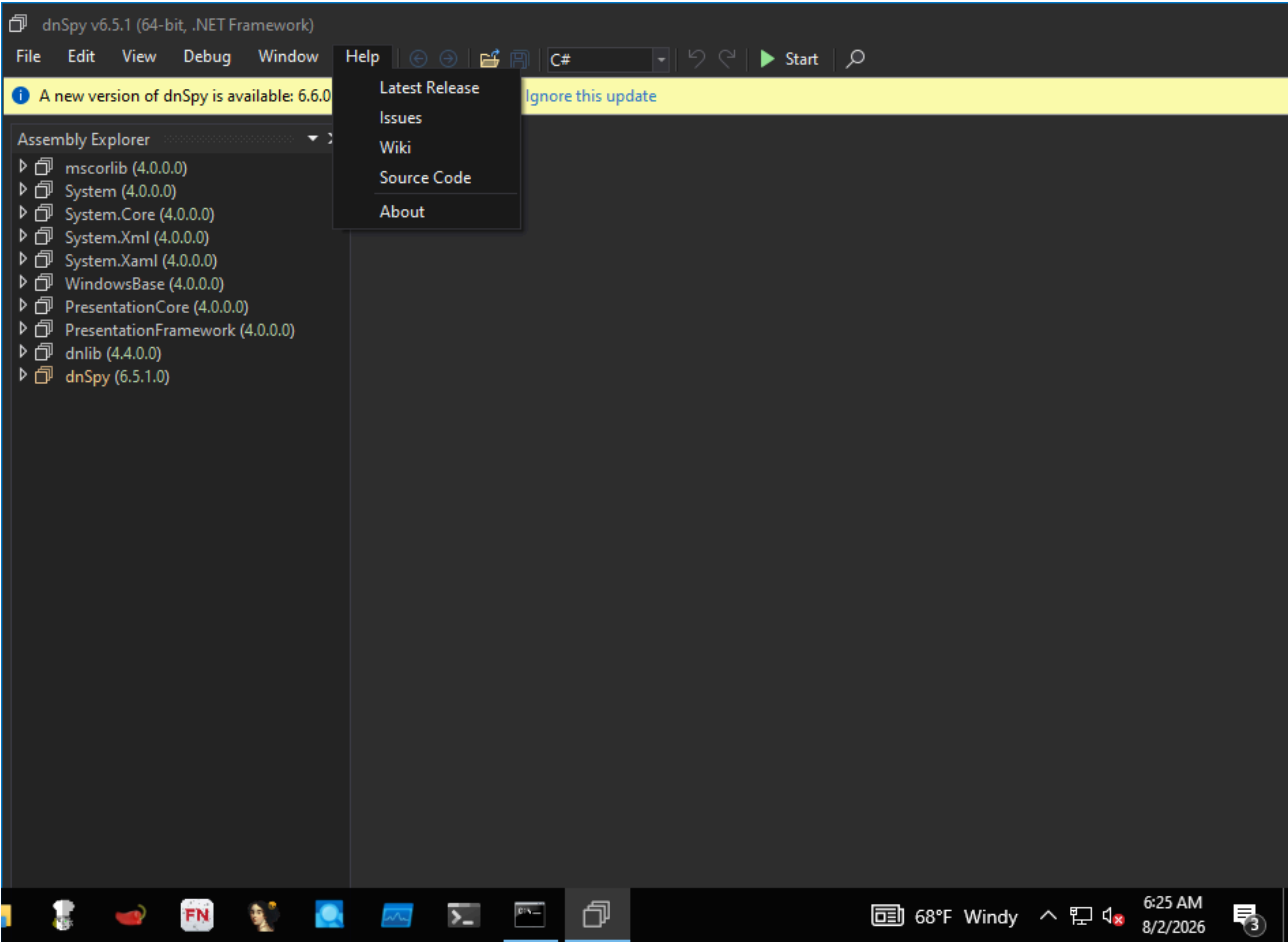
Capability: reverse engineering **Window:** Window **Version:** v6.5.1 **Captured:** C:\Tools\dnSpy\dnSpy.exe on 2026-08-02 — control tree in [capture/gui/dnSpy/dnSpy.tree.txt](#)

← [Capability index](#) · [Kit tool list](#)

Purpose

.NET assembly browser and debugger.

Window



Controls

All 295 nodes come from the capture; the 183 interactive controls are listed here.

Control	Type	AutomationId	What it does
File	MenuItem	-	
Export to Project...	MenuItem	-	
Export to Pro_ject...	Text	-	
Save...	MenuItem	-	
_Save...	Text	-	
Save Module...	MenuItem	-	
Save _Module...	Text	-	

Control	Type	AutomationId	What it does
Save All...	MenuItem	-	
Save A ll...	Text	-	
Open...	MenuItem	-	
_Open...	Text	-	
Open from GAC...	MenuItem	-	
Open from _GAC...	Text	-	
Open List...	MenuItem	-	
Open L_ist...	Text	-	
Recent Files	MenuItem	-	
Recent _Files	Text	-	
Reload All Assemblies	MenuItem	-	
_Reload All Assemblies	Text	-	
Close All	MenuItem	-	
Close All	Text	-	
Close Old In-Memory Modules	MenuItem	-	
Close Old In-Memory Modules	Text	-	
Close All Framework Assemblies	MenuItem	-	
Close All Framework Assemblies	Text	-	
Close All Missing Files	MenuItem	-	
Close All Missing Files	Text	-	
Sort Assemblies	MenuItem	-	
Sor_t Assemblies	Text	-	
Restart as Administrator	MenuItem	-	
Restart as Administrator	Text	-	
Exit	MenuItem	-	
E_xit	Text	-	
Edit	MenuItem	-	
Undo	MenuItem	-	
Undo	Text	-	
Redo	MenuItem	-	
Redo	Text	-	
Find	MenuItem	-	
_Find	Text	-	
Search Assemblies	MenuItem	-	
_Search Assemblies	Text	-	
Create Assembly...	MenuItem	-	
Create Assembly...	Text	-	
Create NetModule...	MenuItem	-	
Create NetModule...	Text	-	
View	MenuItem	-	
Word Wrap	MenuItem	-	
_Word Wrap	Text	-	
Highlight Current Line	MenuItem	-	
_Highlight Current Line	Text	-	
Full Screen	MenuItem	-	
F_ull Screen	Text	-	
Collapse TreeView Nodes	MenuItem	-	
Collapse Tree_View Nodes	Text	-	
Theme	MenuItem	-	
Blue	MenuItem	-	
_Blue	Text	-	

Control	Type	AutomationId	What it does
Dark	MenuItem	-	
_Dark	Text	-	
Light	MenuItem	-	
_Light	Text	-	
Language	MenuItem	-	
System	MenuItem	-	
System	Text	-	
čeština	MenuItem	-	
čeština	Text	-	
Deutsch	MenuItem	-	
Deutsch	Text	-	
English	MenuItem	-	
English	Text	-	
español (España, alfabetización internacional)	MenuItem	-	
español (España, alfabetización internacional)	Text	-	
français	MenuItem	-	
français	Text	-	
italiano	MenuItem	-	
italiano	Text	-	
magyar	MenuItem	-	
magyar	Text	-	
português (Brasil)	MenuItem	-	
português (Brasil)	Text	-	
português (Portugal)	MenuItem	-	
português (Portugal)	Text	-	
Tiếng Việt	MenuItem	-	
Tiếng Việt	Text	-	
Türkçe	MenuItem	-	
Türkçe	Text	-	
русский	MenuItem	-	
русский	Text	-	
українська	MenuItem	-	
українська	Text	-	
فارسی	MenuItem	-	
فارسی	Text	-	
中文(中国)	MenuItem	-	
中文(中国)	Text	-	
Code	MenuItem	-	
_Code	Text	-	
Assembly Explorer	MenuItem	-	
Assembly Ex_plorer	Text	-	
Output	MenuItem	-	
_Output	Text	-	
C# Interactive	MenuItem	-	
C# Interactive	Text	-	
Bookmarks	MenuItem	-	
Bookmark Window	MenuItem	-	
_Bookmark Window	Text	-	
Toggle Bookmark	MenuItem	-	
_Toggle Bookmark	Text	-	
Enable All Bookmarks	MenuItem	-	

Control	Type	AutomationId	What it does
Enable _All Bookmarks	Text	-	
Enable Bookmark	MenuItem	-	
Enable Bookmark	Text	-	
Previous Bookmark	MenuItem	-	
_Previous Bookmark	Text	-	
Next Bookmark	MenuItem	-	
Next _Bookmark	Text	-	
Clear Bookmarks	MenuItem	-	
_Clear Bookmarks	Text	-	
Previous Bookmark With Same Label	MenuItem	-	
Previous Bookmark With Same Label	Text	-	
Next Bookmark With Same Label	MenuItem	-	
Next Bookmark With Same Label	Text	-	
Previous Bookmark In Document	MenuItem	-	
P_revious Bookmark In Document	Text	-	
Next Bookmark In Document	MenuItem	-	
N_ext Bookmark In Document	Text	-	
Clear All Bookmarks In Document	MenuItem	-	
Clear All Bookmarks In Document	Text	-	
Options...	MenuItem	-	
_Options...	Text	-	
Debug	MenuItem	-	
Windows	MenuItem	-	
Breakpoints	MenuItem	-	
_Breakpoints	Text	-	
Module Breakpoints	MenuItem	-	
Mod_ule Breakpoints	Text	-	
Exception Settings	MenuItem	-	
E_xception Settings	Text	-	
Start Debugging...	MenuItem	-	
_Start Debugging...	Text	-	
Start Without Debugging	MenuItem	-	
Start Wit_hout Debugging	Text	-	
Attach to Process...	MenuItem	-	
Attach to _Process...	Text	-	
Attach to Process (Unity)...	MenuItem	-	
Attach to Process (_Unity)...	Text	-	
Toggle Breakpoint	MenuItem	-	
To_ggle Breakpoint	Text	-	
Window	MenuItem	-	
New Window	MenuItem	-	
_New Window	Text	-	
Close All Tabs	MenuItem	-	
C_lose All Tabs	Text	-	
Windows...	MenuItem	-	
_Windows...	Text	-	
Help	MenuItem	-	
Latest Release	MenuItem	-	
_Latest Release	Text	-	
Issues	MenuItem	-	
_Issues	Text	-	

Control	Type	AutomationId	What it does
Wiki	MenuItem	–	
_Wiki	Text	–	
Source Code	MenuItem	–	
_Source Code	Text	–	
About	MenuItem	–	
_About	Text	–	
Back	Button	BrowseBack	
Forward	Button	BrowseForward	
Open	Button	Open	
SaveAll	Button	SaveAll	
Undo	Button	Undo	
Redo	Button	Redo	
Start	Text	–	
SearchRoutedCommand	Button	SearchRoutedCommand	
closeBtn	Button	closeBtn	
A new version of dnSpy is available: 6.6.0.0	Text	–	
Open download page	Text	text	
Ignore this update	Text	text	
Assembly Explorer	Text	–	

Using it

1. **Open** the .NET assembly.
2. Read the decompiled C# rather than the IL first — .NET decompiles cleanly enough that the source is usually the fastest route.
3. Use **Search Assemblies** for the strings and API names that matter, instead of browsing the namespace tree.
4. Set a breakpoint and start debugging when static reading stalls — this is a debugger as well as a decompiler, which is the reason to choose it.
5. **Export to Project...** when the assembly is worth reading as a whole in an editor.

Gotchas

- Obfuscated assemblies decompile into something that looks like code and is not. Names are meaningless after obfuscation; the control flow may be too. De-obfuscate first with `de4dot`.
- Debugging runs the sample. Do it only on the isolated VM, with no network path out.
- It edits and recompiles assemblies. That is useful and it is also how evidence gets modified by accident — work on a copy.

frida

Kit: REMnux **Capability:** Emulate or instrument execution **Version:** 17.16.3 **Captured:** cyberlab-aio via --help on 2026-08-02 **raw Docs:** <https://frida.re>

[← Capability index](#) · [Kit tool list](#)

Purpose

Trace the execution of a process to analyze its behavior.

Synopsis

```
frida [options] target
```

Common invocations

```
# from cyberlab 39-frida-tracing
frida --version
# from cyberlab 39-frida-tracing
frida -f ./target -l /dev/stdin --no-pause <<'EOF'
```

Options

All 66 options parsed from the captured help text. The final column is filled in by review.

Flag	Argument	What it does	When you would use it
-h	—	show this help message and exit	
--help	—	show this help message and exit	
-D	ID	connect to device with the given ID	
--device	ID	connect to device with the given ID	
-U	—	connect to USB device	
--usb	—	connect to USB device	
-R	—	connect to remote frida-server	
--remote	—	connect to remote frida-server	
-H	HOST	connect to remote frida-server on HOST	
--host	HOST	connect to remote frida-server on HOST	
--certificate	CERTIFICATE	speak TLS with HOST, expecting CERTIFICATE	
--origin	ORIGIN	connect to remote server with "Origin" header set to ORIGIN	
--token	TOKEN	authenticate with HOST using TOKEN	
--keepalive-interval	INTERVAL	set keepalive interval in seconds, or 0 to disable (defaults to -1 to auto-select based on transport)	
--device-option	option	override a backend-specific option, such as "control- endpoint=(string)localabstract/my-frida-server" (supported types are: string, bool, int)	
--p2p	—	establish a peer-to-peer connection with target	
--stun-server	ADDRESS	set STUN server ADDRESS to use with --p2p	
-f	TARGET	spawn FILE	
--file	TARGET	spawn FILE	
-F	—	attach to frontmost application	
--attach-frontmost	—	attach to frontmost application	
-n	NAME	attach to NAME	
--attach-name	NAME	attach to NAME	
-N	IDENTIFIER	attach to IDENTIFIER	
--attach-identifier	IDENTIFIER	attach to IDENTIFIER	
-p	PID	attach to PID	
--attach-pid	PID	attach to PID	

Flag	Argument	What it does	When you would use it
<code>-W</code>	PATTERN	await spawn matching PATTERN	
<code>--await</code>	PATTERN	await spawn matching PATTERN	
<code>--stdio</code>	inherit,pipe	stdio behavior when spawning (defaults to "inherit")	
<code>--aux</code>	option	set aux option when spawning, such as "uid=(int)42" (supported types are: string, bool, int)	
<code>--realm</code>	native,emulated	realm to attach in	
<code>--exceptor</code>	full,handler-only,off	configure the exception handling mode	
<code>--disable-unwind-broker</code>	—	disable the unwind broker	
<code>--disable-exit-monitor</code>	—	disable the exit monitor	
<code>--disable-thread-suspend-monitor</code>	—	disable the thread suspend monitor	
<code>--linker-notifier-offset</code>	OFFSET	add a linker notifier OFFSET (may be specified multiple times)	
<code>--runtime</code>	qjs,v8	script runtime to use	
<code>--debug</code>	—	enable the Node.js compatible script debugger	
<code>--squelch-crash</code>	—	if enabled, will not dump crash report to console	
<code>-O</code>	FILE	text file containing additional command line options	
<code>--options-file</code>	FILE	text file containing additional command line options	
<code>--version</code>	—	show program's version number and exit	
<code>-l</code>	SCRIPT	load SCRIPT	
<code>--load</code>	SCRIPT	load SCRIPT	
<code>-P</code>	PARAMETERS_JSON	parameters as JSON, same as Gadget	
<code>--parameters</code>	PARAMETERS_JSON	parameters as JSON, same as Gadget	
<code>-C</code>	USER_CMODULE	load CMODULE	
<code>--cmodule</code>	USER_CMODULE	load CMODULE	
<code>--toolchain</code>	any,internal,external	CModule toolchain to use when compiling from source code	
<code>-c</code>	CODESHARE_URI	load CODESHARE_URI	
<code>--codeshare</code>	CODESHARE_URI	load CODESHARE_URI	
<code>-e</code>	CODE	evaluate CODE	
<code>--eval</code>	CODE	evaluate CODE	
<code>-q</code>	—	quiet mode (no prompt) and quit after -l and -e	
<code>-t</code>	TIMEOUT	seconds to wait before terminating in quiet mode (or 'inf' to run forever)	
<code>--timeout</code>	TIMEOUT	seconds to wait before terminating in quiet mode (or 'inf' to run forever)	
<code>--pause</code>	—	leave main thread paused after spawning program	
<code>-o</code>	LOGFILE	output to log file	
<code>--output</code>	LOGFILE	output to log file	
<code>--eternalize</code>	—	eternalize the script before exit	
<code>--exit-on-error</code>	—	exit with code 1 after encountering any exception in the SCRIPT	
<code>--kill-on-exit</code>	—	kill the spawned program when Frida exits	
<code>--auto-perform</code>	—	wrap entered code with Java.perform	
<code>--auto-reload</code>	—	Enable auto reload of provided scripts and c module (on by default, will be required in the future)	
<code>--no-auto-reload</code>	—	Disable auto reload of provided scripts and c module	

See also

[frida-trace](#), [frida-ps](#), [frida-discover](#), [frida-kill](#), [frida-ls-devices](#)

frida-discover

Kit: REMnux **Capability:** Emulate or instrument execution **Version:** 17.16.3 **Captured:** cyberlab-aio via --help on 2026-08-02 **raw Docs:** <https://frida.re>

[← Capability index](#) · [Kit tool list](#)

Purpose

Trace the execution of a process to analyze its behavior.

Synopsis

```
frida-discover [options] target
```

Options

All 43 options parsed from the captured help text. The final column is filled in by review.

Flag	Argument	What it does	When you would use it
-h	—	show this help message and exit	
--help	—	show this help message and exit	
-D	ID	connect to device with the given ID	
--device	ID	connect to device with the given ID	
-U	—	connect to USB device	
--usb	—	connect to USB device	
-R	—	connect to remote frida-server	
--remote	—	connect to remote frida-server	
-H	HOST	connect to remote frida-server on HOST	
--host	HOST	connect to remote frida-server on HOST	
--certificate	CERTIFICATE	speak TLS with HOST, expecting CERTIFICATE	
--origin	ORIGIN	connect to remote server with "Origin" header set to ORIGIN	
--token	TOKEN	authenticate with HOST using TOKEN	
--keepalive-interval	INTERVAL	set keepalive interval in seconds, or 0 to disable (defaults to -1 to auto-select based on transport)	
--device-option	option	override a backend-specific option, such as "control- endpoint=(string)localabstract:/my-frida-server" (supported types are: string, bool, int)	
--p2p	—	establish a peer-to-peer connection with target	
--stun-server	ADDRESS	set STUN server ADDRESS to use with --p2p	
-f	TARGET	spawn FILE	
--file	TARGET	spawn FILE	
-F	—	attach to frontmost application	
--attach-frontmost	—	attach to frontmost application	
-n	NAME	attach to NAME	
--attach-name	NAME	attach to NAME	
-N	IDENTIFIER	attach to IDENTIFIER	
--attach-identifier	IDENTIFIER	attach to IDENTIFIER	
-p	PID	attach to PID	
--attach-pid	PID	attach to PID	
-W	PATTERN	await spawn matching PATTERN	
--await	PATTERN	await spawn matching PATTERN	
--stdio	inherit,pipe	stdio behavior when spawning (defaults to "inherit")	
--aux	option	set aux option when spawning, such as "uid=(int)42" (supported types are: string, bool, int)	
--realm	native,emulated	realm to attach in	
--exceptor	full,handler-only,off	configure the exception handling mode	

Flag	Argument	What it does	When you would use it
<code>--disable-unwind-broker</code>	—	disable the unwind broker	
<code>--disable-exit-monitor</code>	—	disable the exit monitor	
<code>--disable-thread-suspend-monitor</code>	—	disable the thread suspend monitor	
<code>--linker-notifier-offset</code>	OFFSET	add a linker notifier OFFSET (may be specified multiple times)	
<code>--runtime</code>	qjs,v8	script runtime to use	
<code>--debug</code>	—	enable the Node.js compatible script debugger	
<code>--squelch-crash</code>	—	if enabled, will not dump crash report to console	
<code>-O</code>	FILE	text file containing additional command line options	
<code>--options-file</code>	FILE	text file containing additional command line options	
<code>--version</code>	—	show program's version number and exit	

See also

`frida`, `frida-trace`, `frida-ps`, `frida-kill`, `frida-ls-devices`

frida-kill

Kit: REMnux **Capability:** Emulate or instrument execution **Version:** 17.16.3 **Captured:** cyberlab-aio via --help on 2026-08-02 **raw Docs:** <https://frida.re>

[← Capability index](#) · [Kit tool list](#)

Purpose

Trace the execution of a process to analyze its behavior.

Synopsis

```
frida-kill [options] process
```

Options

All 20 options parsed from the captured help text. The final column is filled in by review.

Flag	Argument	What it does	When you would use it
-h	—	show this help message and exit	
--help	—	show this help message and exit	
-D	ID	connect to device with the given ID	
--device	ID	connect to device with the given ID	
-U	—	connect to USB device	
--usb	—	connect to USB device	
-R	—	connect to remote frida-server	
--remote	—	connect to remote frida-server	
-H	HOST	connect to remote frida-server on HOST	
--host	HOST	connect to remote frida-server on HOST	
--certificate	CERTIFICATE	speak TLS with HOST, expecting CERTIFICATE	
--origin	ORIGIN	connect to remote server with "Origin" header set to ORIGIN	
--token	TOKEN	authenticate with HOST using TOKEN	
--keepalive-interval	INTERVAL	set keepalive interval in seconds, or 0 to disable (defaults to -1 to auto-select based on transport)	
--device-option	option	override a backend-specific option, such as "control- endpoint=(string)localabstract/my-frida-server" (supported types are: string, bool, int)	
--p2p	—	establish a peer-to-peer connection with target	
--stun-server	ADDRESS	set STUN server ADDRESS to use with --p2p	
-O	FILE	text file containing additional command line options	
--options-file	FILE	text file containing additional command line options	
--version	—	show program's version number and exit	

See also

[frida](#), [frida-trace](#), [frida-ps](#), [frida-discover](#), [frida-ls-devices](#)

frida-ls-devices

Kit: REMnux **Capability:** Emulate or instrument execution **Version:** 17.16.3 **Captured:** cyberlab-aio via --help on 2026-08-02 **raw Docs:** <https://frida.re>

[← Capability index](#) · [Kit tool list](#)

Purpose

Trace the execution of a process to analyze its behavior.

Synopsis

```
frida-ls-devices [options]
```

Options

All 5 options parsed from the captured help text. The final column is filled in by review.

Flag	Argument	What it does	When you would use it
-h	—	show this help message and exit	
--help	—	show this help message and exit	
-O	FILE	text file containing additional command line options	
--options-file	FILE	text file containing additional command line options	
--version	—	show program's version number and exit	

See also

[frida](#), [frida-trace](#), [frida-ps](#), [frida-discover](#), [frida-kill](#)

frida-ps

Kit: REMnux **Capability:** Emulate or instrument execution **Version:** 17.16.3 **Captured:** cyberlab-aio via --help on 2026-08-02 **raw Docs:** <https://frida.re>

[← Capability index](#) · [Kit tool list](#)

Purpose

Trace the execution of a process to analyze its behavior.

Synopsis

```
frida-ps [options]
```

Options

All 28 options parsed from the captured help text. The final column is filled in by review.

Flag	Argument	What it does	When you would use it
-h	—	show this help message and exit	
--help	—	show this help message and exit	
-D	ID	connect to device with the given ID	
--device	ID	connect to device with the given ID	
-U	—	connect to USB device	
--usb	—	connect to USB device	
-R	—	connect to remote frida-server	
--remote	—	connect to remote frida-server	
-H	HOST	connect to remote frida-server on HOST	
--host	HOST	connect to remote frida-server on HOST	
--certificate	CERTIFICATE	speak TLS with HOST, expecting CERTIFICATE	
--origin	ORIGIN	connect to remote server with "Origin" header set to ORIGIN	
--token	TOKEN	authenticate with HOST using TOKEN	
--keepalive-interval	INTERVAL	set keepalive interval in seconds, or 0 to disable (defaults to -1 to auto-select based on transport)	
--device-option	option	override a backend-specific option, such as "control- endpoint=(string)localabstract/my-frida-server" (supported types are: string, bool, int)	
--p2p	—	establish a peer-to-peer connection with target	
--stun-server	ADDRESS	set STUN server ADDRESS to use with --p2p	
-O	FILE	text file containing additional command line options	
--options-file	FILE	text file containing additional command line options	
--version	—	show program's version number and exit	
-a	—	list only applications	
--applications	—	list only applications	
-i	—	include all installed applications	
--installed	—	include all installed applications	
-j	—	output results as JSON	
--json	—	output results as JSON	
-e	—	exclude icons in output	
--exclude-icons	—	exclude icons in output	

See also

[frida](#), [frida-trace](#), [frida-discover](#), [frida-kill](#), [frida-ls-devices](#)

frida-trace

Kit: REMnux **Capability:** Emulate or instrument execution **Version:** 17.16.3 **Captured:** cyberlab-aio via --help on 2026-08-02 **raw Docs:** <https://frida.re>

[← Capability index](#) · [Kit tool list](#)

Purpose

Trace the execution of a process to analyze its behavior.

Synopsis

```
frida-trace [options] target
```

Common invocations

```
# from cyberlab 39-frida-tracing
frida-trace --version
# from cyberlab 39-frida-tracing
timeout 8 frida-trace -f ./target -i getenv
```

Options

All 82 options parsed from the captured help text. The final column is filled in by review.

Flag	Argument	What it does	When you would use it
-h	—	show this help message and exit	
--help	—	show this help message and exit	
-D	ID	connect to device with the given ID	
--device	ID	connect to device with the given ID	
-U	—	connect to USB device	
--usb	—	connect to USB device	
-R	—	connect to remote frida-server	
--remote	—	connect to remote frida-server	
-H	HOST	connect to remote frida-server on HOST	
--host	HOST	connect to remote frida-server on HOST	
--certificate	CERTIFICATE	speak TLS with HOST, expecting CERTIFICATE	
--origin	ORIGIN	connect to remote server with "Origin" header set to ORIGIN	
--token	TOKEN	authenticate with HOST using TOKEN	
--keepalive-interval	INTERVAL	set keepalive interval in seconds, or 0 to disable (defaults to -1 to auto-select based on transport)	
--device-option	option	override a backend-specific option, such as "control- endpoint=(string)localabstract/my-frida-server" (supported types are: string, bool, int)	
--p2p	—	establish a peer-to-peer connection with target	
--stun-server	ADDRESS	set STUN server ADDRESS to use with --p2p	
-f	TARGET	spawn FILE	
--file	TARGET	spawn FILE	
-F	—	attach to frontmost application	
--attach-frontmost	—	attach to frontmost application	
-n	NAME	attach to NAME	
--attach-name	NAME	attach to NAME	
-N	IDENTIFIER	attach to IDENTIFIER	
--attach-identifier	IDENTIFIER	attach to IDENTIFIER	
-p	PID	attach to PID	
--attach-pid	PID	attach to PID	

Flag	Argument	What it does	When you would use it
<code>-W</code>	PATTERN	await spawn matching PATTERN	
<code>--await</code>	PATTERN	await spawn matching PATTERN	
<code>--stdio</code>	inherit,pipe	stdio behavior when spawning (defaults to "inherit")	
<code>--aux</code>	option	set aux option when spawning, such as "uid=(int)42" (supported types are: string, bool, int)	
<code>--realm</code>	native,emulated	realm to attach in	
<code>--exceptor</code>	full,handler-only,off	configure the exception handling mode	
<code>--disable-unwind-broker</code>	—	disable the unwind broker	
<code>--disable-exit-monitor</code>	—	disable the exit monitor	
<code>--disable-thread-suspend-monitor</code>	—	disable the thread suspend monitor	
<code>--linker-notifier-offset</code>	OFFSET	add a linker notifier OFFSET (may be specified multiple times)	
<code>--runtime</code>	qjs,v8	script runtime to use	
<code>--debug</code>	—	enable the Node.js compatible script debugger	
<code>--squelch-crash</code>	—	if enabled, will not dump crash report to console	
<code>-O</code>	FILE	text file containing additional command line options	
<code>--options-file</code>	FILE	text file containing additional command line options	
<code>--version</code>	—	show program's version number and exit	
<code>-I</code>	MODULE	include MODULE	
<code>--include-module</code>	MODULE	include MODULE	
<code>-X</code>	MODULE	exclude MODULE	
<code>--exclude-module</code>	MODULE	exclude MODULE	
<code>-i</code>	FUNCTION	include [MODULE!]FUNCTION	
<code>--include</code>	FUNCTION	include [MODULE!]FUNCTION	
<code>-x</code>	FUNCTION	exclude [MODULE!]FUNCTION	
<code>--exclude</code>	FUNCTION	exclude [MODULE!]FUNCTION	
<code>-T</code>	INCLUDE_IMPORTS	include program's imports	
<code>--include-imports</code>	INCLUDE_IMPORTS	include program's imports	
<code>-t</code>	MODULE	include MODULE imports	
<code>--include-module-imports</code>	MODULE	include MODULE imports	
<code>-m</code>	OBJC_METHOD	include OBJC_METHOD	
<code>--include-objc-method</code>	OBJC_METHOD	include OBJC_METHOD	
<code>-M</code>	OBJC_METHOD	exclude OBJC_METHOD	
<code>--exclude-objc-method</code>	OBJC_METHOD	exclude OBJC_METHOD	
<code>-y</code>	SWIFT_FUNC	include SWIFT_FUNC	
<code>--include-swift-func</code>	SWIFT_FUNC	include SWIFT_FUNC	
<code>-Y</code>	SWIFT_FUNC	exclude SWIFT_FUNC	
<code>--exclude-swift-func</code>	SWIFT_FUNC	exclude SWIFT_FUNC	
<code>-j</code>	JAVA_METHOD	include JAVA_METHOD	
<code>--include-java-method</code>	JAVA_METHOD	include JAVA_METHOD	
<code>-J</code>	JAVA_METHOD	exclude JAVA_METHOD	
<code>--exclude-java-method</code>	JAVA_METHOD	exclude JAVA_METHOD	
<code>-s</code>	DEBUG_SYMBOL	include DEBUG_SYMBOL	
<code>--include-debug-symbol</code>	DEBUG_SYMBOL	include DEBUG_SYMBOL	
<code>-q</code>	—	do not format output messages	
<code>--quiet</code>	—	do not format output messages	
<code>-d</code>	—	add module name to generated onEnter log statement	
<code>--decorate</code>	—	add module name to generated onEnter log statement	
<code>-S</code>	PATH	path to JavaScript file used to initialize the session	
<code>--init-session</code>	PATH	path to JavaScript file used to initialize the session	
<code>-P</code>	PARAMETERS_JSON	parameters as JSON, exposed as a global named 'parameters'	
<code>--parameters</code>	PARAMETERS_JSON	parameters as JSON, exposed as a global named 'parameters'	

Flag	Argument	What it does	When you would use it
<code>-o</code>	OUTPUT	dump messages to file	
<code>--output</code>	OUTPUT	dump messages to file	
<code>--ui-host</code>	UI_HOST	the host to serve the UI on (default localhost)	
<code>--ui-port</code>	UI_PORT	the TCP port to serve the UI on	
<code>--ui-allow-origin</code>	ORIGIN	allow browser requests from ORIGIN; may be specified multiple times	

See also

`frida`, `frida-ps`, `frida-discover`, `frida-kill`, `frida-ls-devices`

idr (GUI)

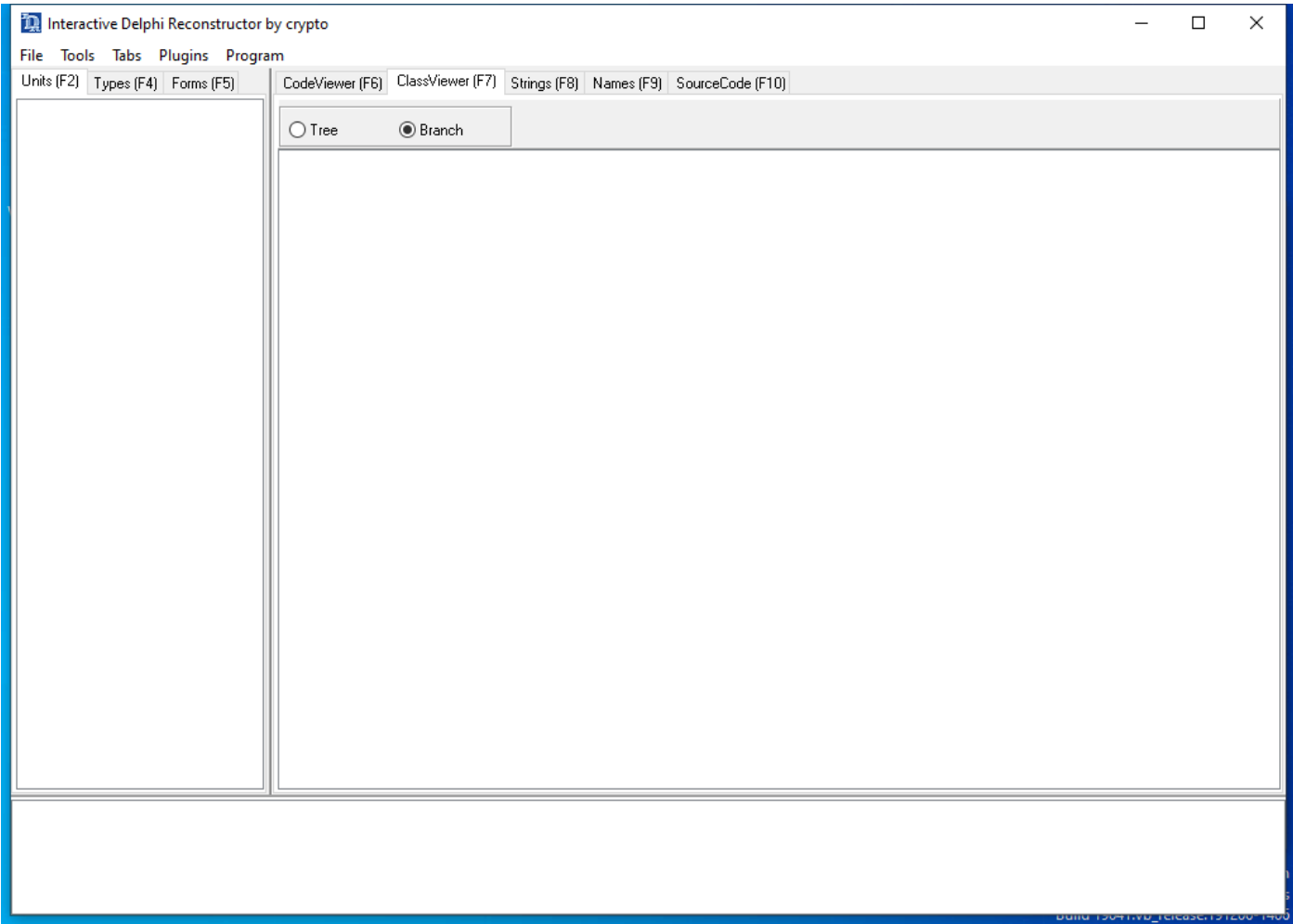
Capability: reverse engineering **Window:** TFMMain_11011981 **Version:** — **Captured:** C:\Tools\idr\idr.exe on 2026-08-02 — control tree in [capture/gui/idr/idr.tree.txt](#)

[← Capability index](#) · [Kit tool list](#)

Purpose

Delphi decompiler and form reconstructor.

Window



Controls

All 12 nodes come from the capture; the 4 interactive controls are listed here.

Control	Type	AutomationId	What it does
Units (F2)	Pane	590436	
ClassViewer (F7)	Pane	328448	
Branch	Pane	328414	
Tree	Pane	328428	

Using it

1. Open the Delphi executable.

2. Let it match the runtime library first; without that, most of the disassembly is Delphi's own code rather than the author's.
3. Read the reconstructed forms and event handlers, which is where Delphi programs keep their logic.

Gotchas

- Delphi binaries are mostly runtime. Library signature matching is what separates the few hundred lines that matter from the tens of thousands that do not.
- The knowledge base is version-specific. A mismatch against the Delphi version used to build the sample leaves the runtime unidentified and the output far less useful.

Kit: REMnux · Kali Linux · SIFT Workstation **Capability:** Disassemble and explore a binary **Version:** 6.1.9-124 r2 **Captured:** cyberlab-ai0 via -h on 2026-08-02 [raw Docs: https://www.radare.org/n/radare2.html](https://www.radare.org/n/radare2.html)

[← Capability index](#) · [Kit tool list](#)

Purpose

Examine binary files, including disassembling and debugging. Includes r2ai and decal plugins for LLM-powered analysis (API key or local Ollama required), plus the r2ghidra plugin for Ghidra decompilation via the pdg command.

Synopsis

```
r2 [-ACdfjLMnQsTuvwzX] [-P patch] [-p prj] [-a arch] [-b bits] [-c cmd]
[-s addr] [-B baddr] [-m maddr] [-i script] [-e k=v] file|pid|--|=
```

Options

All 31 options parsed from the captured help text. The final column is filled in by review.

Flag	Argument	What it does	When you would use it
-0	—	print \x00 after init and every command	
-1	—	redirect stderr to stdout	
-2	—	close stderr file descriptor (silent warning messages)	
-a	arch	set asm.arch	
-A	—	run 'aaa' command to analyze all referenced code	
-b	bits	set asm.bits	
-B	baddr	set base address for PIE binaries	
-C	—	file is host:port (alias for -c+=http://%/s/cmd/)	
-d	—	debug the executable 'file' or running process 'pid'	
-E	—	open cfg.editor with the user rc file	
-f	—	block size = file size	
-i	file	run rlang program, r2script file or load plugin	
-I	file	run script file before the file is opened	
-j	—	use json for -v, -L and maybe others	
-m	addr	map file at given address (loadaddr)	
-M	—	do not demangle symbol names	
-N	—	do not load user settings and scripts	
-q	—	quiet mode (no prompt) and quit after -i	
-Q	—	quiet mode (no prompt) and quit faster (quickLeak=true)	
-p	prj	use project, list if no arg, load if no file	
-P	file	apply rapatch file and quit	
-r	rarun2	specify rarun2 profile to load (same as -e dbg.profile=X)	
-s	addr	initial seek	
-S	—	start r2 in sandbox mode	
-t	—	load rabin2 info in thread	
-u	—	set bin.filter=false to get raw sym/sec/cls names	
-v	—	show radare2 version (-V show lib versions)	
-V	—	show radare2 version (-V show lib versions)	
-w	—	open file in write mode	
-x	—	open without exec-flag (asm.emu will not work), See io.exec	
-X	—	same as -e bin.usextr=false (useful for dyldcache)	

See also

[rabin2](#), [rasm2](#), [objdump](#), [vivbin](#), [vdbbin](#)

rasm2

Kit: REMnux **Capability:** Disassemble and explore a binary; Analyse shellcode **Version:** rasm2 6.1.9 **Captured:** cyberlab-ai0 via --help on 2026-08-02 [raw Docs:](#) <https://www.radare.org/n/radare2.html>

[← Capability index](#) · [Kit tool list](#)

Purpose

Examine binary files, including disassembling and debugging. Includes r2ai and decal plugins for LLM-powered analysis (API key or local Ollama required), plus the r2ghidra plugin for Ghidra decompilation via the pdg command.

Synopsis

```
rasm2 [-ACdDehHLBvw] [-a arch] [-b bits] [-s addr] [-S syntax]
[-f file] [-o file] [-F fil:ter] [-i skip] [-l len] 'code'|hex|0101b|-
```

Options

All 25 options parsed from the captured help text. The final column is filled in by review.

Flag	Argument	What it does	When you would use it
-a	arch	set architecture to assemble/disassemble (see -L)	
-A	—	show Analysis information from given hexpairs	
-b	bits	set cpu register size (8, 16, 32, 64) (RASM2_BITS)	
-B	—	binary input/output (-l is mandatory for binary input)	
-c	cpu	select specific CPU (depends on arch)	
-C	—	output in C format	
-d	—	disassemble from hexpair bytes (-D show hexpairs)	
-D	—	disassemble from hexpair bytes (-D show hexpairs)	
-e	—	use big endian instead of little endian	
-E	—	display ESIL expression (same input as in -d)	
-f	file	read data from file	
-F	parser	specify which parse filter use (see -LL)	
-i	len	ignore/skip N bytes of the input buffer	
-j	—	output in json format	
-k	kernel	select operating system (linux, windows, darwin, android, ios, ..)	
-l	len	input/Output length	
-N	—	same as r2 -N (or R2_NOPLUGINS) (not load any plugin)	
-o	file	output file name (rasm2 -Bf a.asm -o a)	
-p	—	run SPP over input for assembly	
-q	—	quiet mode	
-r	—	output in radare commands	
-S	syntax	select syntax (intel, att)	
-v	—	show version information	
-x	—	use hex dwords instead of hex pairs when assembling.	
-w	—	what's this instruction for? describe opcode	

See also

[r2](#), [rabin2](#), [objdump](#), [vivbin](#), [vdbbin](#), [xortool](#)

VB-Decompiler (GUI)

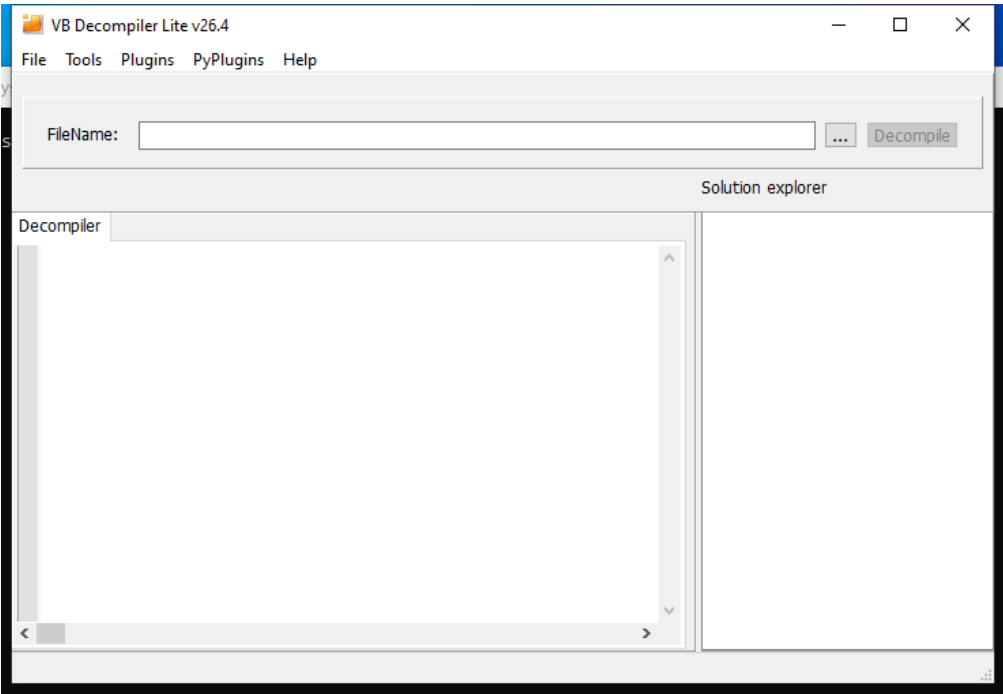
Capability: reverse engineering **Window:** t0000000 **Version:** v26.4 **Captured:** C:\Tools\VB Decompiler\VB Decompiler.exe on 2026-08-02 — control tree in [capture/gui/VB-Decompiler/VB-Decompiler.tree.txt](#)

[← Capability index](#) · [Kit tool list](#)

Purpose

Visual Basic decompiler.

Window



Controls

All 12 nodes come from the capture; the 5 interactive controls are listed here.

Control	Type	AutomationId	What it does
Decompile	Pane	131836	
...	Pane	655904	
Panel3	Pane	459348	
Decompiler	Pane	786932	
Panel2	Pane	393782	

Using it

1. Open the Visual Basic executable.
2. Check whether it is p-code or native — the tool tells you, and it decides how much you get back.
3. Read the form definitions: VB malware often carries its logic in event handlers rather than in a main routine.

Gotchas

- P-code decompiles close to source. Native-compiled VB gives you disassembly with VB runtime calls, which is a different and much slower job.
- The Lite edition omits much of the analysis. Check which build you are on before concluding a feature is missing.

vbdec (GUI)

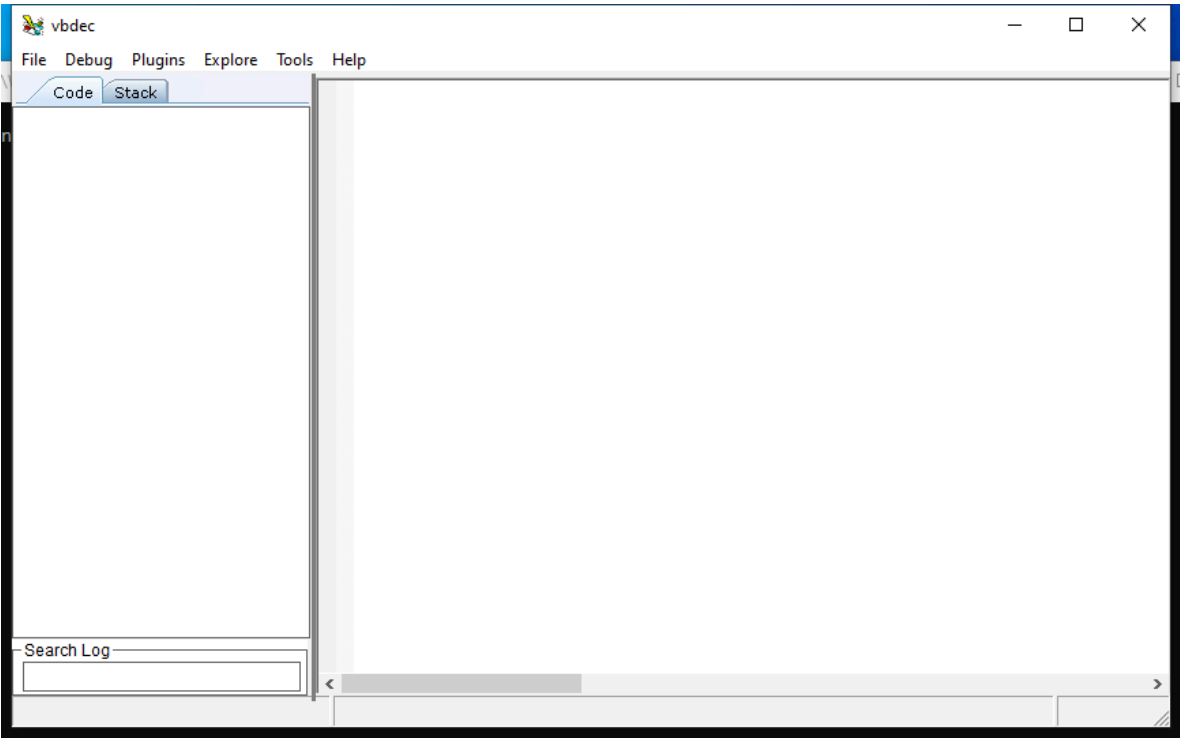
Capability: reverse engineering **Window:** ThunderRT6FormDC **Version:** — **Captured:** C:\Tools\vbdec\vbdec.exe on 2026-08-02 — control tree in [capture/gui/vbdec/vbdec.tree.txt](#)

[← Capability index](#) · [Kit tool list](#)

Purpose

Visual Basic 5/6 decompiler.

Window



Controls

All 8 nodes come from the capture; the 1 interactive controls are listed here.

Control	Type	AutomationId	What it does
Search Log	Pane	10	

Using it

1. Open the VB5/VB6 binary.
2. Work from the form and control names, which usually survive and describe the program's intent.

Gotchas

- Scope is VB5 and VB6 only. VB.NET is a .NET assembly — use [dnSpy](#) for those.

vdbbin

Kit: REMnux **Capability:** Disassemble and explore a binary **Captured:** cyberlab-aio via --help on 2026-08-02 **raw Docs:** <https://github.com/vivisect/vivisect>

[← Capability index](#) · [Kit tool list](#)

Purpose

Statically examine and emulate binary files.

Synopsis

```
vdbbin [options] [platformopt=foo, ...]
```

Options

All 27 options parsed from the captured help text. The final column is filled in by review.

Flag	Argument	What it does	When you would use it
-h	—	show this help message and exit	
--help	—	show this help message and exit	
-c	COMMAND	Debug a fired command	
--cmd	COMMAND	Debug a fired command	
-p	PROCESS	Attach to process by name or pid	
--process	PROCESS	Attach to process by name or pid	
-Q	—	Run the QT gui	
--qt	—	Run the QT gui	
-R	REMOTEHOST	Attach to remote VDB server	
--remote	REMOTEHOST	Attach to remote VDB server	
-r	—	Do not stop on attach	
--run	—	Do not stop on attach	
-s	SNAPSHOT	Load a vtrace snapshot file	
--snapshot	SNAPSHOT	Load a vtrace snapshot file	
-S	—	—	
--server	—	—	
-v	—	—	
--verbose	—	—	
-t	TARGET	Activate special vdb target (-t ? for list)	
--target	TARGET	Activate special vdb target (-t ? for list)	
--android	—	Debug Android with ADB!	
-e	EVENTID	Used for Windows JIT	
--eventid	EVENTID	Used for Windows JIT	
-w	WAITFOR	Wait for process with name	
--waitfor	WAITFOR	Wait for process with name	
--LI	—	Breakpoint on Library initialization	
--LL	—	Breakpoint on Library load time	

See also

[r2](#), [rabin2](#), [rasm2](#), [objdump](#), [vivbin](#)

vivbin

Kit: REMnux **Capability:** Disassemble and explore a binary **Captured:** cyberlab-aio via --help on 2026-08-02 **raw Docs:** <https://github.com/vivisect/vivisect>

[← Capability index](#) · [Kit tool list](#)

Purpose

Statically examine and emulate binary files.

Synopsis

```
vivbin [options] <workspace|binaries...>
```

Options

All 34 options parsed from the captured help text. The final column is filled in by review.

Flag	Argument	What it does	When you would use it
-h	—	show this help message and exit	
--help	—	show this help message and exit	
-M	MODNAME	run the file listed as an analysis module in non-gui mode and exit	
--module	MODNAME	run the file listed as an analysis module in non-gui mode and exit	
-A	—	Do <i>not</i> do an initial auto-analysis pass	
--skip-analysis	—	Do <i>not</i> do an initial auto-analysis pass	
-B	—	Do <i>not</i> start the gui, just load, analyze and save	
--bulk	—	Do <i>not</i> start the gui, just load, analyze and save	
-C	—	Output vivisect performace profiling (cProfile) info	
--cprofile	—	Output vivisect performace profiling (cProfile) info	
-E	ENTRYPOINTS	Add Entry Point for bulk analysis (can have multiple "-E " args	
--entrypoint	ENTRYPOINTS	Add Entry Point for bulk analysis (can have multiple "-E " args	
-O	OPTION	.= (optval must be json syntax)	
--option	OPTION	.= (optval must be json syntax)	
-p	PARSEMOD	Manually specify the parser module (pe/elf/blob/...)	
--parser	PARSEMOD	Manually specify the parser module (pe/elf/blob/...)	
-s	STORAGE_NAME	Specify a storage module by name	
--storage	STORAGE_NAME	Specify a storage module by name	
-S	—	Run Vivisect Server. Last argument should be the VivWorkspaces directory.	
--server	—	Run Vivisect Server. Last argument should be the VivWorkspaces directory.	
-P	SERVER_PORT	Port to run Vivisect Server on (only meaningful with --server)	
--port	SERVER_PORT	Port to run Vivisect Server on (only meaningful with --server)	
-v	—	Enable verbose mode (multiples matter: -vvv)	
--verbose	—	Enable verbose mode (multiples matter: -vvv)	
-V	VERSION	Add file version (if available) to save file name	
--version	VERSION	Add file version (if available) to save file name	
-c	CONFIG	Path to a directory to use for config data	
--config	CONFIG	Path to a directory to use for config data	
-a	—	Autosave configuration data	
--autosave	—	Autosave configuration data	
-o	OUTFILE	Name of VivWorkspace file to create (useful for loading multiple binaries into one workspace)	
--outfile	OUTFILE	Name of VivWorkspace file to create (useful for loading multiple binaries into one workspace)	
-m	—	List Architectures and their version/maturity	
--archmaturity	—	List Architectures and their version/maturity	

See also

[r2](#), [rabin2](#), [rasm2](#), [objdump](#), [vdbbin](#)

xortool

Kit: REMnux **Capability:** Analyse shellcode; Break simple obfuscation **Version:** 1.1.0 **Captured:** cyberlab-aio via --help on 2026-08-02 [raw Docs:](#) <https://github.com/hellman/xortool>

[← Capability index](#) · [Kit tool list](#)

Purpose

Recover the key length and key of an XOR-encrypted file by frequency analysis.

Synopsis

```
xortool [-x] [-m MAX-LEN] [-f] [-t CHARSET] [FILE]
xortool [-x] [-l LEN] [-c CHAR | -b | -o] [-f] [-t CHARSET] [-p PLAIN] [-r PERCENT] [FILE]
xortool [-x] [-m MAX-LEN] -l LEN [-c CHAR | -b | -o] [-f] [-t CHARSET] [-p PLAIN] [-r PERCENT] [FILE]
xortool [-h | --help]
xortool --version
```

Common invocations

```
# from cyberlab 09-deobfuscation
xortool --version
# from cyberlab 09-deobfuscation
xortool -c 20 exercise/encoded_payload.bin
```

Options

All 20 options parsed from the captured help text; 4 reviewed with usage guidance.

Flag	Argument	What it does	When you would use it
-x	—	input is hex-encoded str	
--hex	—	input is hex-encoded str	
-l	LEN	length of the key	Fix the key length when you already know it.
--key-length	LEN	length of the key	Fix the key length when you already know it.
-m	MAX-LEN	maximum key length to probe [default: 65]	Maximum key length to consider.
--max-keylen	MAX-LEN	maximum key length to probe [default: 65]	Maximum key length to consider.
-c	CHAR	most frequent char (one char or hex code)	Give the most frequent character of the plaintext — usually 20 (space) for text, 00 for binaries. This is the flag that makes or breaks the attack.
--char	CHAR	most frequent char (one char or hex code)	Give the most frequent character of the plaintext — usually 20 (space) for text, 00 for binaries. This is the flag that makes or breaks the attack.
-b	—	brute force all possible most frequent chars	Brute-force the most frequent character rather than guessing.
--brute-chars	—	brute force all possible most frequent chars	Brute-force the most frequent character rather than guessing.
-o	—	same as -b but will only check printable chars	
--brute-printable	—	same as -b but will only check printable chars	
-f	—	filter outputs based on the charset	
--filter-output	—	filter outputs based on the charset	
-p	PLAIN	use known plaintext for decoding	
--known-plaintext	PLAIN	use known plaintext for decoding	
-r	PERCENT	threshold validity percentage [default: 95]	

Flag	Argument	What it does	When you would use it
<code>--threshold</code>	PERCENT	threshold validity percentage [default: 95]	
<code>-h</code>	—	show this help	
<code>--help</code>	—	show this help	

Gotchas

- Output lands in an `xortool_out/` directory, not stdout. People routinely think it did nothing.
- `-c 00` is right far more often than the default for packed binaries, because null padding dominates.

See also

`rasm2`, `floss`, `xlmedeobfuscator`

AccessEnum (GUI)

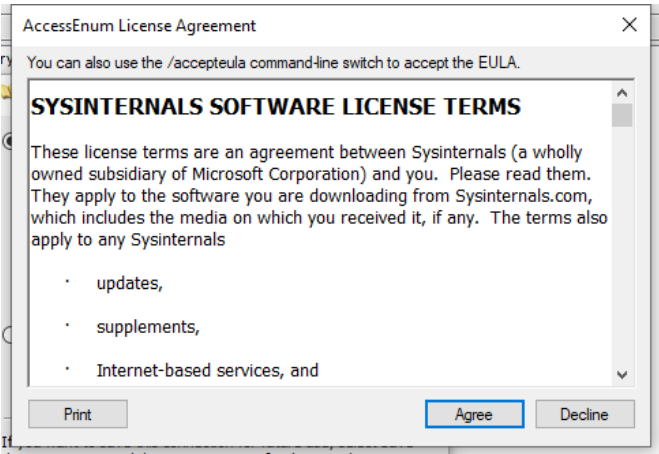
Capability: windows artifacts **Window:** #32770 **Version:** — **Captured:** C:\Tools\sysinternals\AccessEnum.exe on 2026-08-02 — control tree in `capture/gui/AccessEnum/AccessEnum.tree.txt`

[← Capability index](#) · [Kit tool list](#)

Purpose

Enumerate filesystem and registry permissions.

Window



Controls

All 30 nodes come from the capture; the 4 interactive controls are listed here.

Control	Type	AutomationId	What it does
You can also use the /accepteula command-line switch to accept the EULA.	Pane	502	
Agree	Pane	1	
Decline	Pane	2	
Print	Pane	501	

Using it

1. Accept the licence on first run; until that is done the tool never reaches its main window.
2. Choose the directory or registry key to enumerate.
3. Run the scan, then sort by the permissions column rather than by path — the outliers are the finding, and they do not cluster by location.
4. Save the results as the before-state when you are about to change anything.

Gotchas

- It reports what the ACLs say, not what is reachable. Group membership, inheritance and share permissions all sit on top of this.
- The capture behind this page is the licence dialog, not the tool: a first run shows the EULA and nothing else. Sysinternals binaries need `-accepteula` before they can be driven automatically, so the control table here documents that dialog rather than the main window.

AmcacheParser

Kit: FLARE-VM / SIFT (Eric Zimmerman tools) **Capability:** Parse ESE / SRUM / Amcache databases; Parse execution and persistence artifacts **Version:** 2026.5.0+76dc8354aa98ce1e1c6f942abcfb09f583f411dd **Captured:** cyberlab-aio via --help on 2026-08-02 [raw](#)

[← Capability index](#) · [Kit tool list](#)

Purpose

Parse Amcache.hve — the record of programs present on a host, with SHA-1 hashes, including binaries that have since been deleted.

Synopsis

```
AmcacheParser [options]
```

Options

All 15 options parsed from the captured help text; 9 reviewed with usage guidance.

Flag	Argument	What it does	When you would use it
-f	f	Amcache.hve file to parse	The Amcache.hve to parse.
-i	—	Include file entries for Programs entries	Include file entries associated with Programs entries. More complete, and noisier.
-w	w	Path to file containing SHA-1 hashes to <i>exclude</i> from the results. Blacklisting overrides whitelisting	Blacklist of SHA-1 hashes to exclude. Blacklisting overrides whitelisting, so a hash in both is dropped — the safe default when suppressing known-good noise.
-b	b	Path to file containing SHA-1 hashes to <i>include</i> from the results. Blacklisting overrides whitelisting	Whitelist of SHA-1 hashes to include.
--csv	csv	Directory to save CSV formatted results to. Be sure to include the full path in double quotes	Write CSV to a directory. The usual output.
--csvf	csvf	File name to save CSV formatted results to. When present, overrides default name	Override the generated CSV filename.
--dt	dt	The custom date/time format to use when displaying time stamps. See https://goo.gl/CNVq0k for options. Default is: yyyy-MM-dd HH:mm:ss [default: yyyy-MM-dd HH:mm:ss]	Custom timestamp format for the output.
--mp	—	When true, display higher precision for timestamps	Higher-precision timestamps.
--nl	—	When true, ignore transaction log files for dirty hives. Default is FALSE	Ignore transaction logs for a dirty hive. Leave this off unless you know why you want it: skipping the logs means parsing a hive that is missing its most recent changes.
--debug	—	Show debug information during processing	
--trace	—	Show trace information during processing	
-?	—	Show help and usage information	
-h	—	Show help and usage information	
--help	—	Show help and usage information	
--version	—	Show version information	

Gotchas

- Amcache records that a binary was **present**, not that it ran. It is evidence of existence; Prefetch and event logs are evidence of execution. Conflating the two is the standard error with this artifact.
- It carries SHA-1 for entries, which makes it the fastest way to tie a deleted binary to threat intelligence long after the file is gone.
- The hive is usually dirty when collected from a live host. Let the transaction logs replay — the entries only in the logs are the most recent, which is normally the part you care about.

See also

[esedbexport](#), [esedbinfo](#), [SrumsECmd](#), [PECmd](#), [AppCompatCacheParser](#), [MFTECmd](#)

AppCompatCacheParser

Kit: FLARE-VM / SIFT (Eric Zimmerman tools) **Capability:** Parse execution and persistence artifacts **Version:** 2026.5.0+0cf059f40c2f7b31acdccb142461945402217398
Captured: cyberlab-aio via --help on 2026-08-02 [raw](#)

[← Capability index](#) · [Kit tool list](#)

Purpose

Description:

Synopsis

```
AppCompatCacheParser [options]
```

Options

All 13 options parsed from the captured help text. The final column is filled in by review.

Flag	Argument	What it does	When you would use it
-f	f	Full path to SYSTEM hive to process. If this option is not specified, the live Registry will be used	
--csv	csv	Directory to save CSV formatted results to. Be sure to include the full path in double quotes	
--csvf	csvf	File name to save CSV formatted results to. When present, overrides default name	
-c	c	The ControlSet to parse. Default is to extract all control sets [default: -1]	
-t	—	Sorts last modified timestamps in descending order	
--dt	dt	The custom date/time format to use when displaying time stamps. See https://goo.gl/CNVq0k for options [default: yyyy-MM-dd HH:mm:ss]	
--nl	—	When true, ignore transaction log files for dirty hives	
--debug	—	Show debug information during processing	
--trace	—	Show trace information during processing	
-?	—	Show help and usage information	
-h	—	Show help and usage information	
--help	—	Show help and usage information	
--version	—	Show version information	

See also

[PECmd](#), [MFTECmd](#), [AmcacheParser](#)

chainsaw

Kit: SIFT / Security Onion (Sigma-based log hunting) **Capability:** Parse Windows event logs **Version:** chainsaw 2.16.0 **Captured:** cyberlab-aio via --help on 2026-08-02 [raw](#)

[← Capability index](#) · [Kit tool list](#)

Purpose

Hunt through Windows event logs with Sigma rules and built-in detection logic, at speed.

Synopsis

```
chainsaw [OPTIONS] <COMMAND>
```

Common invocations

```
# from cyberlab 58-eventlog-hunting
chainsaw hunt Security_sample.evtx -s sigma/ --mapping mapping.yml
```

Options

All 6 options parsed from the captured help text; 2 reviewed with usage guidance.

Flag	Argument	What it does	When you would use it
--no-banner	—	Hide Chainsaw's banner	Suppress the banner, for clean output in a report or a pipeline.
--num-threads	NUM_THREADS	Limit the thread number (default: num of CPUs)	Cap the thread count. Defaults to every core, which is usually right on a dedicated analysis box and rude on a shared one.
-h	—	Print help	
--help	—	Print help	
-V	—	Print version	
--version	—	Print version	

Gotchas

- The interesting options live on the subcommands — `hunt`, `search`, `dump` — not at the top level captured here. Run `chainsaw hunt --help` for the ones that matter.
- Rules are not bundled with the binary. Without a Sigma rule set and the mapping file, `hunt` runs and finds nothing, which looks identical to a clean host.

See also

[evtexport](#), [evtxinfo](#), [EvtxECmd](#), [hayabusa](#)

esedbexport

Kit: SIFT Workstation (libyal) **Capability:** Parse ESE / SRUM / Amcache databases **Captured:** cyberlab-aio via --help on 2026-08-02 [raw](#)

[← Capability index](#) · [Kit tool list](#)

Purpose

Invalid argument: esedbexport

Synopsis

```
esedbexport [ -c codepage ] [ -l logfile ] [ -m mode ] [ -t target ]
[ -T table_name ] [ -hvV ] source
```

Common invocations

```
# from cyberlab 06-windows-artifact-libs
esedbexport -V
# from cyberlab 06-windows-artifact-libs
esedbexport -t /tmp/edb_out exercise/Current.edb
```

Options

All 8 options parsed from the captured help text. The final column is filled in by review.

Flag	Argument	What it does	When you would use it
-c	—	codepage of ASCII strings, options: ascii, windows-874, windows-932, windows-936, windows-1250, windows-1251, windows-1252 (default), windows-1253, windows-1254 windows-1255, windows-1256, windows-125	
-h	—	shows this help	
-l	—	logs information about the exported items	
-m	—	export mode, option: all, tables (default) 'all' exports all the tables or a single specified table with indexes, 'tables' exports all the tables or a single specified table	
-t	—	specify the basename of the target directory to export to (default is the source filename) esedbexport will add the suffix .export to the basename	
-T	—	exports only a specific table	
-v	—	verbose output to stderr	
-V	—	print version	

See also

[esedbinfo](#), [SrumECmd](#), [AmcacheParser](#)

esedbinfo

Kit: SIFT Workstation (libyal) **Capability:** Parse ESE / SRUM / Amcache databases **Captured:** cyberlab-aio via --help on 2026-08-02 [raw](#)

[← Capability index](#) · [Kit tool list](#)

Purpose

Invalid argument: esedbinfo

Synopsis

```
esedbinfo [ -hvV ] source
```

Options

All 3 options parsed from the captured help text. The final column is filled in by review.

Flag	Argument	What it does	When you would use it
-h	—	shows this help	
-v	—	verbose output to stderr	
-V	—	print version	

See also

[esedbexport](#), [SrumsCmd](#), [AmcacheParser](#)

EvtxECmd

Kit: FLARE-VM / SIFT (Eric Zimmerman tools) **Capability:** Parse Windows event logs **Version:** 2026.5.0+bfc7f47ccbf65ffc9a3777cde5498db2fdd94664 **Captured:** cyberlab-aio via `--help` on 2026-08-02 [raw](#)

[← Capability index](#) · [Kit tool list](#)

Purpose

Parse Windows event logs into a normalised, filterable CSV, mapping the useful fields out of the XML payload.

Synopsis

```
EvtxECmd [options]
```

Options

All 26 options parsed from the captured help text; 20 reviewed with usage guidance.

Flag	Argument	What it does	When you would use it
<code>-f</code>	f	File to process. Either this or -d is required	A single log, when you already know which one matters.
<code>-d</code>	d	Directory to recursively process. Either this or -f is required	Recurse a directory of .evt files — the usual mode when working from a collected <code>winevt\Logs</code> .
<code>--csv</code>	csv	Directory to save CSV formatted results to. Be sure to include the full path in double quotes	Write CSV to a directory. The normal output, and the form the rest of a timeline workflow expects.
<code>--csvf</code>	csvf	File name to save CSV formatted results to. When present, overrides default name	Override the generated CSV filename.
<code>--json</code>	json	Directory to save JSON formatted results to. Be sure to include the full path in double quotes	JSON output, for feeding another tool.
<code>--jsonf</code>	jsonf	File name to save JSON formatted results to. When present, overrides default name	Override the generated JSON filename.
<code>--xml</code>	xml	Directory to save XML formatted results to	XML output.
<code>--xmlf</code>	xmlf	File name to save XML formatted results to. When present, overrides default name	Override the generated XML filename.
<code>--dt</code>	dt	The custom date/time format to use when displaying time stamps. See https://goo.gl/CNVq0k for options [default: yyyy-MM-dd HH:mm:ss.ffffff]	Custom timestamp format for the output.
<code>--inc</code>	inc	List of Event IDs to process. All others are ignored. Overrides --exc Format is 4624,4625,5410,5500-5600	Process only these Event IDs. The fastest way to cut a multi-gigabyte log down to the question being asked — ranges are allowed (<code>4624,4625,5410-5500</code>).
<code>--exc</code>	exc	List of Event IDs to IGNORE. All others are included. Format is 4624,4625,5410,5500-5600	Process everything except these Event IDs. <code>--inc</code> wins if both are given.
<code>--sd</code>	sd	Start date for including events (UTC). Anything OLDER than this is dropped. Format should match --dt	Drop events older than this date (UTC).
<code>--ed</code>	ed	End date for including events (UTC). Anything NEWER than this is dropped. Format should match --dt	Drop events newer than this date (UTC). With <code>--sd</code> , scopes the parse to the incident window instead of all history.
<code>--fj</code>	—	When true, export all available data when using --json	Export all available data in JSON rather than the mapped subset.
<code>--tdt</code>	tdt	The number of seconds to use for time discrepancy detection. Default is 1 [default: 1]	Seconds of tolerance for time-discrepancy detection — flags records whose timestamps disagree, a clock-tampering signal.
<code>--met</code>	—	When true, show metrics about processed event log. Default is true	Show per-log metrics about what was processed.
<code>--maps</code>	maps	The path where event maps are located. Defaults to 'Maps' folder where program was executed [default: /opt/eztools/EvtxECmd/Maps]	Where the event maps live. The maps are what turn raw XML into named columns; without the right ones, useful fields stay buried in the payload.
<code>--vss</code>	—	Process all Volume Shadow Copies that exist on drive specified by -f or -d	Also parse every Volume Shadow Copy on the drive, which is where cleared or rotated logs may survive.
<code>--dedupe</code>	—	Deduplicate -f or -d & VSCs based on SHA-1. First file found wins	Drop duplicates by SHA-1 across the source and shadow copies. Use it whenever <code>--vss</code> is on.
<code>--sync</code>	—	If true, the latest maps from https://github.com/EricZimmerman/evt/tree/master/evt/Maps are downloaded and local maps updated	Pull the latest maps from upstream. Worth doing before a big parse — map coverage improves continuously.
<code>--debug</code>	—	Show debug information during processing	
<code>--trace</code>	—	Show trace information during processing	
<code>-?</code>	—	Show help and usage information	
<code>-h</code>	—	Show help and usage information	
<code>--help</code>	—	Show help and usage information	

Flag	Argument	What it does	When you would use it
<code>--version</code>	—	Show version information	

Gotchas

- Without a map for an Event ID, the interesting values stay inside the XML payload rather than becoming columns. If an expected field is missing, the map is usually the reason, not the log.
- A cleared log is itself the finding: Security 1102 and System 104 record the clearing. Include them explicitly when hunting anti-forensics.
- Event log timestamps are recorded in UTC but `--sd / --ed` are only as good as your assumption about the host's clock. Corroborate before building a timeline on them.

See also

`evtxexport`, `evtxinfo`, `chainsaw`, `hayabusa`

evtxexport

Kit: SIFT Workstation (libyal) **Capability:** Parse Windows event logs **Captured:** cyberlab-aio via --help on 2026-08-02 [raw](#)

[← Capability index](#) · [Kit tool list](#)

Purpose

Invalid argument: evtxexport

Synopsis

```
evtxexport [ -c codepage ] [ -f format ] [ -l log_file ]
[ -m mode ] [ -p resource_files_path ]
[ -r registry_files_path ] [ -s system_file ]
[ -S software_file ] [ -t event_log_type ]
[ -hTvV ] source
```

Common invocations

```
# from cyberlab 06-windows-artifact-libs
evtxexport -V
# from cyberlab 06-windows-artifact-libs
evtxexport -h
# from cyberlab 06-windows-artifact-libs
evtxexport -f text exercise/Security.evtx > /tmp/security_events.txt
```

Options

All 13 options parsed from the captured help text. The final column is filled in by review.

Flag	Argument	What it does	When you would use it
-c	—	codepage of ASCII strings, options: ascii, windows-874, windows-932, windows-936, windows-949, windows-950, windows-1250, windows-1251, windows-1252 (default), windows-1253, windows-1254, windows-1255	
-f	—	output format, options: xml, text (default)	
-h	—	shows this help	
-l	—	logs information about the exported items	
-m	—	export mode, option: all, items (default), recovered 'all' exports the (allocated) items and recovered items, 'items' exports the (allocated) items and 'recovered' exports the recovered items	
-p	—	search PATH for the resource files	
-r	—	name of the directory containing the SOFTWARE and SYSTEM (Windows) Registry file	
-s	—	filename of the SYSTEM (Windows) Registry file. This option overrides the path provided by -r	
-S	—	filename of the SOFTWARE (Windows) Registry file. This option overrides the path provided by -r	
-t	—	event log type, options: application, security, system if not specified the event log type is determined based on the filename.	
-T	—	use event template definitions to parse the event record data	
-v	—	verbose output to stderr	
-V	—	print version	

See also

[evtxinfo](#), [EvtxECmd](#), [chainsaw](#), [hayabusa](#)

evtxinfo

Kit: SIFT Workstation (libyal) **Capability:** Parse Windows event logs **Captured:** cyberlab-aio via --help on 2026-08-02 [raw](#)

[← Capability index](#) · [Kit tool list](#)

Purpose

Invalid argument: evtxinfo

Synopsis

```
evtxinfo [ -c codepage ] [ -hvV ] source
```

Options

All 4 options parsed from the captured help text. The final column is filled in by review.

Flag	Argument	What it does	When you would use it
-c	—	codepage of ASCII strings, options: ascii, windows-874, windows-932, windows-936, windows-949, windows-950, windows-1250, windows-1251, windows-1252 (default), windows-1253, windows-1254, windows-1255	
-h	—	shows this help	
-v	—	verbose output to stderr	
-V	—	print version	

See also

[evtxexport](#) , [EvtxECmd](#) , [chainsaw](#) , [hayabusa](#)

hayabusa

Kit: SIFT / Security Onion (Sigma-based log hunting) **Capability:** Parse Windows event logs **Version:** error: unexpected argument '--version' found **Captured:** cyberlab-aio via help on 2026-08-02 [raw](#)

[← Capability index](#) · [Kit tool list](#)

Purpose

Hayabusa v3.9.0 - Showa Day Release

Synopsis

```
hayabusa.exe <COMMAND> [OPTIONS]
hayabusa.exe help <COMMAND> or hayabusa.exe <COMMAND> -h
```

Options

No option definitions could be parsed from this tool's help output. It may be subcommand-driven or have no flags; needs manual review.

See also

[evtexport](#), [evtxinfo](#), [EvtxECmd](#), [chainsaw](#)

hivexsh

Kit: Kali Linux **Capability:** Parse registry hives **Captured:** `cyberlab-aio` via `help` on 2026-08-02 [raw](#)

[← Capability index](#) · [Kit tool list](#)

Purpose

hivexsh: failed to open hive file: help: No such file or directory

Options

No option definitions could be parsed from this tool's help output. It may be subcommand-driven or have no flags; needs manual review.

See also

`rip.pl`, `regripper`, `regfexport`, `regfinfo`, `regfmount`, `regipy-dump`, `regipy-parse-header`, `regipy-plugins-run`

MFTECmd

Kit: FLARE-VM / SIFT (Eric Zimmerman tools) **Capability:** Parse execution and persistence artifacts **Version:** 2026.5.0+4fd94a6bd12237e8501baff5fc9e5b1b01c53862
Captured: cyberlab-ai0 via --help on 2026-08-02 [raw](#)

[← Capability index](#) · [Kit tool list](#)

Purpose

Parse NTFS metadata files — \$MFT, \$J, \$Boot, \$SDS, \$I30 — into CSV or bodyfile, including entries for deleted files.

Synopsis

MFTECmd [options]

Options

All 32 options parsed from the captured help text; 26 reviewed with usage guidance.

Flag	Argument	What it does	When you would use it
<code>-f</code>	f	File to process (\$MFT \$J \$Boot \$SDS \$I30). Required	The metadata file to parse. Required, and the file type is detected from its contents rather than its name.
<code>-m</code>	m	\$MFT file to use when -f points to a \$J file (Use this to resolve parent path in \$J CSV output)	Supply the \$MFT alongside a \$J. Without it the journal shows file names with no path, because the parent directory only exists in the \$MFT.
<code>--csv</code>	csv	Directory to save CSV formatted results to. Be sure to include the full path in double quotes. This or --json required unless --de or --body is specified	Write CSV to a directory. The normal output.
<code>--csvf</code>	csvf	File name to save CSV formatted results to. When present, overrides default name	Override the generated CSV filename.
<code>--json</code>	json	Directory to save JSON formatted results to. Be sure to include the full path in double quotes. This or --csv required unless --de or --body is specified	JSON output, for a pipeline.
<code>--jsonf</code>	jsonf	File name to save JSON formatted results to. When present, overrides default name	Override the generated JSON filename.
<code>--body</code>	body	Directory to save bodyfile formatted results to. --bdl is also required when using this option	Bodyfile output, which is what <code>mactime</code> consumes — the bridge from NTFS metadata into a classic timeline.
<code>--bodyf</code>	bodyf	File name to save body formatted results to. When present, overrides default name	Override the generated bodyfile name.
<code>--bdl</code>	bdl	Drive letter (C, D, etc.) to use with bodyfile. Only the drive letter itself should be provided	Drive letter to record in the bodyfile. Required with <code>--body</code> , because a bodyfile path is meaningless without the volume it came from.
<code>--blf</code>	—	When true, use LF vs CRLF for newlines	Use LF rather than CRLF, when the output is going to a Unix toolchain.
<code>--dd</code>	dd	Directory to save exported \$MFT FILE record. --do is also required when using this option	Directory to write an exported FILE record to.
<code>--do</code>	do	Offset of the \$MFT FILE record to dump as decimal or hex. Ex: 5120 or 0x1400 Use --de or --debug to see offsets	Offset of the FILE record to dump, decimal or hex.
<code>--de</code>	de	Dump full details for \$MFT entry/sequence #. Format is 'Entry' or 'Entry-Seq' as decimal or hex. Example: 5, 624-5 or 0x270-0x5.	Dump full detail for one entry, as <code>Entry</code> or <code>Entry-Seq</code> . The flag for interrogating a single suspicious file.
<code>--dr</code>	—	When true, dump \$MFT resident files to dir specified by --csv or --json, in 'Resident' subdirectory. Files will be named '--_bin'	Dump resident files out of the \$MFT. Small files live entirely inside their MFT record, so this recovers content with no data runs to follow.
<code>--fls</code>	—	When true, displays contents of directory from \$MFT specified by -de. Ignored when --de points to a file	List a directory's contents from the \$MFT, for the entry given by <code>--de</code> .
<code>--ds</code>	ds	Dump full details for Security Id from \$SDS as decimal or hex. Example: 624 or 0x270	Dump a security descriptor from \$SDS by Id — how you get from a file to the ACL that was on it.
<code>--dt</code>	dt	The custom date/time format to use when displaying time stamps. See https://goo.gl/CNVq0k for options [default: yyyy-MM-dd HH:mm:ss.ffffff]	Custom timestamp format for the output.
<code>--sn</code>	—	Include DOS file name types in \$MFT output	Include DOS 8.3 short names.
<code>--fl</code>	—	Generate condensed file listing of parsed \$MFT contents. Requires --csv	Condensed file listing instead of the full attribute dump.
<code>--at</code>	—	When true, include all timestamps from 0x30 attribute vs only when they differ from 0x10 in the \$MFT	Include all \$STANDARD_INFORMATION timestamps rather than only those that differ from \$FILE_NAME. Differences between the two attribute sets are the classic timestamping signal, so include them when that is the question.

Flag	Argument	What it does	When you would use it
<code>--rs</code>	—	When true, recover slack space from FILE records when processing \$MFT files. This option has no effect for \$I30 files	Recover slack space from FILE records. Old entries survive in the unused tail of a record, so this reaches deleted metadata that a straight parse skips.
<code>--vss</code>	—	Process all Volume Shadow Copies that exist on drive specified by -f	Also parse every Volume Shadow Copy, which is where an older \$MFT still holds entries the live one has reused.
<code>--dedupe</code>	—	Deduplicate -f & VSCs based on SHA-1. First file found wins	Drop duplicates by SHA-1 across the source and shadow copies. Use it whenever <code>--vss</code> is on.
<code>--ir</code>	—	Include resident data in JSON/CSV output	Include resident data inline in the output rather than as separate files.
<code>--re</code>	re	Comma-separated list of extensions to include for resident data (e.g., 'txt,ps1,bat'). If omitted, includes all	Restrict resident extraction to these extensions.
<code>--rm</code>	rm	Maximum size in bytes for resident data to include (max: 1024000) [default: 1024]	Cap the size of resident data included.
<code>--debug</code>	—	Show debug information during processing	
<code>--trace</code>	—	Show trace information during processing	
<code>-?</code>	—	Show help and usage information	
<code>-h</code>	—	Show help and usage information	
<code>--help</code>	—	Show help and usage information	
<code>--version</code>	—	Show version information	

Gotchas

- \$MFT entries are reused. A deleted file's record is overwritten by the next file that needs it, so an entry describing a deleted file may already belong to something else — check the sequence number before asserting the two are the same file.
- \$STANDARD_INFORMATION timestamps are trivially forged; \$FILE_NAME timestamps are not, because they update only through the kernel. `--at` is what lets you compare them, and a mismatch is the strongest cheap indicator of timestamping.
- Parsing a live volume's \$MFT copied with a normal file copy will fail — it is locked. Extract it with a forensic tool or from a shadow copy.

See also

[PECmd](#), [AppCompatCacheParser](#), [AmcacheParser](#)

PECmd

Kit: FLARE-VM / SIFT (Eric Zimmerman tools) **Capability:** Parse execution and persistence artifacts **Version:** 2026.5.0+bde430c69ba4d97fea8b71fdddb6df7849419c10
Captured: cyberlab-aio via --help on 2026-08-02 [raw](#)

[← Capability index](#) · [Kit tool list](#)

Purpose

Parse Windows Prefetch files into evidence of what executed, when, how often, and which files each run touched.

Synopsis

```
PECmd [options]
```

Options

All 20 options parsed from the captured help text; 14 reviewed with usage guidance.

Flag	Argument	What it does	When you would use it
<code>-f</code>	<code>f</code>	File to process. Either this or <code>-d</code> is required	A single .pf file, when chasing one binary.
<code>-d</code>	<code>d</code>	Directory to recursively process. Either this or <code>-f</code> is required	Recurse a directory — the normal mode, since Prefetch is only meaningful as a set. Point it at <code>C:\Windows\Prefetch</code> .
<code>-k</code>	<code>k</code>	Comma separated list of keywords to highlight in output. By default, 'temp' and 'tmp' are highlighted. Any additional keywords will be added to these	Highlight extra keywords in the output. <code>temp</code> and <code>tmp</code> are highlighted by default; add the names you are hunting.
<code>-o</code>	<code>o</code>	When specified, save prefetch file bytes to the given path. Useful to look at decompressed Win10 files	Save the decompressed Prefetch bytes. Win10+ Prefetch is MAM-compressed, so this is how you get something another tool or a hex editor can read.
<code>-q</code>	—	Do not dump full details about each file processed. Speeds up processing when using <code>--json</code> or <code>--csv</code>	Suppress the per-file detail. Worth it on a large directory when the CSV is the real output.
<code>--csv</code>	<code>csv</code>	Directory to save CSV formatted results to. Be sure to include the full path in double quotes	Write CSV to a directory. This is the output that matters: Prefetch is a timeline source, and Timeline Explorer or a spreadsheet is where the pattern shows up.
<code>--csvf</code>	<code>csvf</code>	File name to save CSV formatted results to. When present, overrides default name	Override the generated CSV filename.
<code>--json</code>	<code>json</code>	Directory to save JSON formatted results to. Be sure to include the full path in double quotes	JSON output, when the results feed another tool.
<code>--jsonf</code>	<code>jsonf</code>	File name to save JSON formatted results to. When present, overrides default name	Override the generated JSON filename.
<code>--html</code>	<code>html</code>	Directory to save xhtml formatted results to. Be sure to include the full path in double quotes	XHTML report, for handing to someone who will not open a CSV.
<code>--dt</code>	<code>dt</code>	The custom date/time format to use when displaying time stamps. See https://goo.gl/CNVq0k for options [default: yyyy-MM-dd HH:mm:ss]	Custom timestamp format for the output.
<code>--mp</code>	—	When true, display higher precision for timestamps	Higher-precision timestamps, when ordering events within the same second matters.
<code>--vss</code>	—	Process all Volume Shadow Copies that exist on drive specified by <code>-f</code> or <code>-d</code>	Also parse every Volume Shadow Copy on the drive. Prefetch rolls over at 1024 entries on Win10+, so shadow copies are often the only place an older execution still exists.
<code>--dedupe</code>	—	Deduplicate <code>-f</code> or <code>-d</code> & VSCs based on SHA-1. First file found wins	Drop duplicates by SHA-1 across the source and the shadow copies. Effectively mandatory with <code>--vss</code> , which otherwise returns the same file many times over.
<code>--debug</code>	—	Show debug information during processing	
<code>--trace</code>	—	Show trace information during processing	
<code>-?</code>	—	Show help and usage information	
<code>-h</code>	—	Show help and usage information	
<code>--help</code>	—	Show help and usage information	
<code>--version</code>	—	Show version information	

Gotchas

- Prefetch proves a program **ran**; it does not prove who ran it or what it did. Pair it with event logs or [AmcacheParser](#) before attributing anything.
- Absence is not evidence of absence. Prefetch can be disabled, is commonly off on SSD-era server builds, and rolls over — a missing entry means nothing on its own.
- The last-run timestamps are the eight most recent executions only. Older runs are gone from the file even though the run count keeps counting them.

See also

[AppCompatCacheParser](#), [MFTECmd](#), [AmcacheParser](#)

RECmd

Kit: FLARE-VM / SIFT (Eric Zimmerman tools) **Capability:** Parse registry hives **Version:** 2026.5.0+bcd0ac33ed98de61ea6de551eef96052bddbbd49 **Captured:** cyberlab-aio via `--help` on 2026-08-02 [raw](#)

[← Capability index](#) · [Kit tool list](#)

Purpose

Description:

Synopsis

```
RECmd [options]
```

Options

All 33 options parsed from the captured help text. The final column is filled in by review.

Flag	Argument	What it does	When you would use it
<code>-f</code>	f	Hive to search. -f or -d is required	
<code>-d</code>	d	Directory to look for hives (recursively). -f or -d is required	
<code>--kn</code>	kn	Display details for key name. Includes subkeys and values	
<code>--vn</code>	vn	Value name. Only this value will be dumped	
<code>--bn</code>	bn	Use settings from supplied file to find keys/values. See included sample file for examples	
<code>--csv</code>	csv	Directory to save CSV formatted results to. Be sure to include the full path in double quotes	
<code>--csvf</code>	csvf	File name to save CSV formatted results to. When present, overrides default name	
<code>--saveTo</code>	saveTo	Saves --vn value data in binary form to file. Expects path to a FILE	
<code>--json</code>	json	Directory to save JSON formatted results to. Be sure to include the full path in double quotes	
<code>--jsonf</code>	jsonf	File name to save JSON formatted results to. When present, overrides default name	
<code>--details</code>	—	Show more details when displaying results	
<code>--base64</code>	base64	Find Base64 encoded values with size >= Base64 (specified in bytes)	
<code>--minSize</code>	minSize	Find values with data size >= MinSize (specified in bytes)	
<code>--sa</code>	sa	Search for in keys, values, data, and slack	
<code>--sk</code>	sk	Search for in value record's key names	
<code>--sv</code>	sv	Search for in value record's value names	
<code>--sd</code>	sd	Search for in value record's value data	
<code>--ss</code>	ss	Search for in value record's value slack	
<code>--literal</code>	—	If true, --sd and --ss search value will not be interpreted as ASCII or Unicode byte strings	
<code>--nd</code>	—	If true, do not show data when using --sd or --ss	
<code>--regex</code>	—	If present, treat in --sk, --sv, --sd, and --ss as a regular expression	
<code>--dt</code>	dt	The custom date/time format to use when displaying time stamps. See https://goo.gl/CNVq0k for options [default: yyyy-MM-dd HH:mm:ss.ffffff]	
<code>--nl</code>	—	When true, allow transaction log files to not exist for dirty hives	
<code>--recover</code>	—	If true, recover deleted keys/values. Default is true	
<code>--vss</code>	—	Process all Volume Shadow Copies that exist on drive specified by -f or -d	
<code>--dedupe</code>	—	Deduplicate -f or -d & VSCs based on SHA-1. First file found wins	
<code>--sync</code>	—	If true, the latest batch files from https://github.com/EricZimmerman/RECmd/tree/master/BatchExamples are downloaded and local files updated	
<code>--debug</code>	—	Show debug information during processing	
<code>--trace</code>	—	Show trace information during processing	

Flag	Argument	What it does	When you would use it
<code>-?</code>	—	Show help and usage information	
<code>-h</code>	—	Show help and usage information	
<code>--help</code>	—	Show help and usage information	
<code>-- version</code>	—	Show version information	

See also

`rip.pl`, `regripper`, `hivexsh`, `regfexport`, `regfinfo`, `regfmount`, `regipy-dump`, `regipy-parse-header`

regfexport

Kit: SIFT Workstation (libyal) **Capability:** Parse registry hives **Captured:** cyberlab-aio via --help on 2026-08-02 [raw](#)

[← Capability index](#) · [Kit tool list](#)

Purpose

Invalid argument: regfexport

Synopsis

```
regfexport [ -c codepage ] [ -K key_path ] [ -l logfile ]
[ -hvV ] source
```

Common invocations

```
# from cyberlab 04-registry-analysis
regfexport -V
# from cyberlab 04-registry-analysis
regfexport exercise/SYSTEM_sample.hive > /tmp/system_dump.txt
# from cyberlab 04-registry-analysis
regfexport exercise/SYSTEM_sample.hive | grep -i "ComputerName"
```

Options

All 6 options parsed from the captured help text. The final column is filled in by review.

Flag	Argument	What it does	When you would use it
-c	—	codepage of ASCII strings, options: ascii, windows-874, windows-932, windows-936, windows-949, windows-950, windows-1250, windows-1251, windows-1252 (default), windows-1253, windows-1254, windows-1255	
-h	—	shows this help	
-K	—	show information about a specific key path.	
-l	—	logs information about the exported items	
-v	—	verbose output to stderr	
-V	—	print version	

See also

[rip.pl](#), [regripper](#), [hivexsh](#), [regfinfo](#), [regfmount](#), [regipy-dump](#), [regipy-parse-header](#), [regipy-plugins-run](#)

regfinfo

Kit: SIFT Workstation (libyal) **Capability:** Parse registry hives **Captured:** cyberlab-ai0 via --help on 2026-08-02 [raw](#)

[← Capability index](#) · [Kit tool list](#)

Purpose

Invalid argument: regfinfo

Synopsis

```
regfinfo [ -B bodyfile ] [ -c codepage ] [ -hHvV ] source
```

Common invocations

```
# from cyberlab 04-registry-analysis
regfinfo -V
# from cyberlab 04-registry-analysis
regfinfo exercise/SYSTEM_sample.hive
# from cyberlab 04-registry-analysis
regfinfo /tmp/SYSTEM_recovered.hive
```

Options

All 6 options parsed from the captured help text. The final column is filled in by review.

Flag	Argument	What it does	When you would use it
-B	—	output key and value hierarchy as a bodyfile	
-c	—	codepage of ASCII strings, options: ascii, windows-874, windows-932, windows-936, windows-949, windows-950, windows-1250, windows-1251, windows-1252 (default), windows-1253, windows-1254, windows-1255	
-h	—	shows this help	
-H	—	shows the key and value hierarchy	
-v	—	verbose output to stderr	
-V	—	print version	

See also

[rip.pl](#), [regripper](#), [hivexsh](#), [regfexport](#), [regfmount](#), [regipy-dump](#), [regipy-parse-header](#), [regipy-plugins-run](#)

regfmount

Kit: SIFT Workstation (libyal) **Capability:** Parse registry hives **Captured:** cyberlab-aio via --help on 2026-08-02 [raw](#)

[← Capability index](#) · [Kit tool list](#)

Purpose

Invalid argument: regfmount

Synopsis

```
regfmount [ -c codepage ] [ -X extended_options ] [ -hvV ] file
mount_point
```

Options

All 5 options parsed from the captured help text. The final column is filled in by review.

Flag	Argument	What it does	When you would use it
-c	—	codepage of ASCII strings, options: ascii, windows-874, windows-932, windows-936, windows-949, windows-950, windows-1250, windows-1251, windows-1252 (default), windows-1253, windows-1254, windows-1255	
-h	—	shows this help	
-v	—	verbose output to stderr, while regfmount will remain running in the foreground	
-V	—	print version	
-X	—	extended options to pass to sub system	

See also

[rip.pl](#), [regripper](#), [hivexsh](#), [regfexport](#), [regfinfo](#), [regipy-dump](#), [regipy-parse-header](#), [regipy-plugins-run](#)

regipy-diff

Kit: SIFT Workstation (regipy) **Capability:** Parse registry hives **Captured:** cyberlab-aio via --help on 2026-08-02 [raw](#)

[← Capability index](#) · [Kit tool list](#)

Synopsis

```
regipy-diff [OPTIONS] FIRST_HIVE_PATH SECOND_HIVE_PATH
```

Options

All 4 options parsed from the captured help text. The final column is filled in by review.

Flag	Argument	What it does	When you would use it
-o	FILE	—	
-v	—	Verbosity	
--verbose	—	Verbosity	
--help	—	Show this message and exit.	

See also

[rip.pl](#), [regripper](#), [hivexsh](#), [regfexport](#), [regfinfo](#), [regfmount](#), [regipy-dump](#), [regipy-parse-header](#)

regipy-dump

Kit: SIFT Workstation (regipy) **Capability:** Parse registry hives **Captured:** cyberlab-aio via --help on 2026-08-02 [raw](#)

[← Capability index](#) · [Kit tool list](#)

Purpose

identified for some reason

Synopsis

```
regipy-dump [OPTIONS] HIVE_PATH
```

Options

All 18 options parsed from the captured help text. The final column is filled in by review.

Flag	Argument	What it does	When you would use it
-o	FILE	—	
-p	TEXT	A registry path to start iterating from	
--registry-path	TEXT	A registry path to start iterating from	
-t	—	Create a CSV timeline instead	
--timeline	—	Create a CSV timeline instead	
-l	TEXT	Specify a hive type, if it could not be identified for some reason	
--hive-type	TEXT	Specify a hive type, if it could not be identified for some reason	
-r	TEXT	The path from which the partial hive actually starts, for example: -t ntuser -r "/Software" would mean this is actually a HKCU hive, starting from HKCU/Software	
--partial_hive_path	TEXT	The path from which the partial hive actually starts, for example: -t ntuser -r "/Software" would mean this is actually a HKCU hive, starting from HKCU/Software	
-v	—	Verbosity	
--verbose	—	Verbosity	
-d	—	Not fetching the values for each subkey makes the iteration way faster. Values count will still be returned	
--do-not-fetch-values	—	Not fetching the values for each subkey makes the iteration way faster. Values count will still be returned	
-s	TEXT	If "-s" was specified, fetch only values for subkeys starting this timestamp in isoformat	
--start-date	TEXT	If "-s" was specified, fetch only values for subkeys starting this timestamp in isoformat	
-e	TEXT	If "-e" was specified, fetch only values for subkeys until this timestamp in isoformat	
--end-date	TEXT	If "-e" was specified, fetch only values for subkeys until this timestamp in isoformat	
--help	—	Show this message and exit.	

See also

[rip.pl](#), [regripper](#), [hivexsh](#), [regfexport](#), [regfinfo](#), [regfmount](#), [regipy-parse-header](#), [regipy-plugins-run](#)

regipy-parse-header

Kit: SIFT Workstation (regipy) **Capability:** Parse registry hives **Captured:** cyberlab-aio via --help on 2026-08-02 [raw](#)

[← Capability index](#) · [Kit tool list](#)

Synopsis

```
regipy-parse-header [OPTIONS] HIVE_PATH
```

Options

All 3 options parsed from the captured help text. The final column is filled in by review.

Flag	Argument	What it does	When you would use it
-v	—	Verbosity	
--verbose	—	Verbosity	
--help	—	Show this message and exit.	

See also

[rip.pl](#), [regripper](#), [hivexsh](#), [regfexport](#), [regfinfo](#), [regfmount](#), [regipy-dump](#), [regipy-plugins-run](#)

regipy-plugins-run

Kit: SIFT Workstation (regipy) **Capability:** Parse registry hives **Captured:** cyberlab-aio via --help on 2026-08-02 [raw](#)

[← Capability index](#) · [Kit tool list](#)

Purpose

identified for some reason

Synopsis

```
regipy-plugins-run [OPTIONS] HIVE_PATH
```

Options

All 11 options parsed from the captured help text. The final column is filled in by review.

Flag	Argument	What it does	When you would use it
-o	FILE	Output path for plugins result [required]	
-p	TEXT	A plugin or list of plugins to execute command separated	
--plugins	TEXT	A plugin or list of plugins to execute command separated	
-t	TEXT	Specify a hive type, if it could not be identified for some reason	
--hive-type	TEXT	Specify a hive type, if it could not be identified for some reason	
-r	TEXT	The path from which the partial hive actually starts, for example: -t ntuser -r "/Software" would mean this is actually a HKCU hive, starting from HKCU/Software	
--partial_hive_path	TEXT	The path from which the partial hive actually starts, for example: -t ntuser -r "/Software" would mean this is actually a HKCU hive, starting from HKCU/Software	
-v	—	Verbosity	
--verbose	—	Verbosity	
--include-unvalidated	—	Include plugins that don't have validation test cases. These plugins may return incomplete or inaccurate data. Use at your own risk.	
--help	—	Show this message and exit.	

See also

[rip.pl](#), [regripper](#), [hivexsh](#), [regfexport](#), [regfinfo](#), [regfmount](#), [regipy-dump](#), [regipy-parse-header](#)

regripper

Kit: Kali Linux **Capability:** Parse registry hives **Captured:** `cyberlab-aio` via `--help` on 2026-08-02 [raw](#)

[← Capability index](#) · [Kit tool list](#)

Purpose

Rip v.3.0 - CLI RegRipper tool

Options

No option definitions could be parsed from this tool's help output. It may be subcommand-driven or have no flags; needs manual review.

See also

[rip.pl](#), [hivexsh](#), [regfexport](#), [regfinfo](#), [regfmount](#), [regipy-dump](#), [regipy-parse-header](#), [regipy-plugins-run](#)

rip.pl

Kit: Kali Linux **Capability:** Parse registry hives **Captured:** `cyberlab-aio` via `--help` on 2026-08-02 [raw](#)

[← Capability index](#) · [Kit tool list](#)

Purpose

Rip v.3.0 - CLI RegRipper tool

Common invocations

```
# from cyberlab 04-registry-analysis
rip.pl -h
# from cyberlab 04-registry-analysis
rip.pl -r exercise/SYSTEM_sample.hive -p compname
# from cyberlab 49-intrusion-timeline-case
rip.pl -h | head -n 3
# from cyberlab 49-intrusion-timeline-case
rip.pl -r NTUSER.DAT -p run
# from cyberlab 49-intrusion-timeline-case
rip.pl -r SYSTEM -p usbstor
```

Options

No option definitions could be parsed from this tool's help output. It may be subcommand-driven or have no flags; needs manual review.

See also

[regripper](#), [hivexsh](#), [regfexport](#), [regfinfo](#), [regfmount](#), [regipy-dump](#), [regipy-parse-header](#), [regipy-plugins-run](#)

SrumECmd

Kit: FLARE-VM / SIFT (Eric Zimmerman tools) **Capability:** Parse ESE / SRUM / Amcache databases **Version:** 2026.5.0+880ad26bcb011976a8fc521eea63fc5e6e65ba02
Captured: cyberlab-aio via --help on 2026-08-02 [raw](#)

[← Capability index](#) · [Kit tool list](#)

Purpose

Description:

Synopsis

```
SrumECmd [options]
```

Options

All 11 options parsed from the captured help text. The final column is filled in by review.

Flag	Argument	What it does	When you would use it
-f	f	SRUDB.dat file to parse	
-r	r	SOFTWARE hive to process. This is optional, but recommended	
-d	d	Directory to recursively process, looking for SRUDB.dat and SOFTWARE hive. This mode is primarily used with KAPE so both SRUDB.dat and SOFTWARE hive can be located	
--csv	csv	Directory to save CSV formatted results to. Be sure to include the full path in double quotes	
--dt	dt	The custom date/time format to use when displaying time stamps. See https://goo.gl/CNVq0k for options [default: yyyy-MM-dd HH:mm:ss]	
--debug	—	Show debug information during processing	
--trace	—	Show trace information during processing	
-?	—	Show help and usage information	
-h	—	Show help and usage information	
--help	—	Show help and usage information	
--version	—	Show version information	

See also

[esedbexport](#), [esedbinfo](#), [AmcacheParser](#)

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